eemath

E math

Lower Sec E Math Concepts

Prime Factorisation

bro

$$\begin{array}{r|l} 2 & 162 \\ \hline 3 & 81 \\ \hline 3 & 27 \\ \hline 3 & 9 \\ \hline 3 & 3 \\ \hline & 1 \end{array}$$

$$162 = 2 \times 3^4$$

To find min value of k where 162k is a square number,

$$k = 2$$

$$162k = 2 \times 3^4 \times 2$$

$$162k = 2^2 \times 3^4$$

(make all powers even)

HCF & LCM

bro

Find HCF and LCM of 2520 and 350

$$\begin{array}{r|l} 2 & 2520 \\ \hline 2 & 1260 \\ \hline 2 & 630 \\ \hline 3 & 315 \\ \hline 3 & 105 \\ \hline 5 & 35 \\ \hline 7 & 7 \\ \hline & 1 \end{array} \qquad \begin{array}{r|l} 2 & 350 \\ \hline 5 & 175 \\ \hline 5 & 35 \\ \hline 7 & 7 \\ \hline & 1 \end{array}$$

$$2520 = 2^3 \times 3^2 \times 5 \times 7$$
$$350 = 2 \times (3^0) \times 5^2 \times 7$$

Dont include the 3^0 factor, shown here only for clarity

take lowest power of each factor and multiply together

$$2520 = 2^3 \times 3^2 \times \boxed{5} \times 7$$
$$350 = \boxed{2} \times (3^0) \times 5^2 \times \boxed{7}$$

$$HCF = 2 \times 5 \times 7$$
$$HCF = 70$$

take highest power of each factor and multiply together

$$2520 = \boxed{2^3} \times \boxed{3^2} \times 5 \times \boxed{7}$$
$$350 = 2 \times (3^0) \times \boxed{5^2} \times 7$$

$$LCM = 2^3 \times 3^2 \times 5^2 \times 7$$
$$LCM = 12600$$

Congruency & Similarity

bro

Congruency Tests

SSS

SAS

SAA

ASA

RHS

$ABC \equiv XYZ$

Invalid Tests

~~AAA~~

~~SSA~~

(a type of SSA, where angle is 90°)

Similarity Tests

SSS

SAS

AA

for tests with sides, check that corresponding sides have the same ratio

A has to similar to X, i.e corresponding

triangle ABC is similar to triangle XYZ

Similarity Ratios

For two similar 3d objects with side length ratio $1 : 2$,

Ratio of areas is

$$1^2 : 2^2$$
$$1 : 4$$

Ratio of volumes is

$$1^3 : 2^3$$
$$1 : 8$$

For two triangles with an equal base,

$$\text{Ratio of triangle areas} = \frac{\cancel{0.5bh_1}}{\cancel{0.5bh_2}}$$

$$\text{Ratio of triangle areas} = \frac{h_1}{h_2}$$

cancel out common terms to find ratio of sides

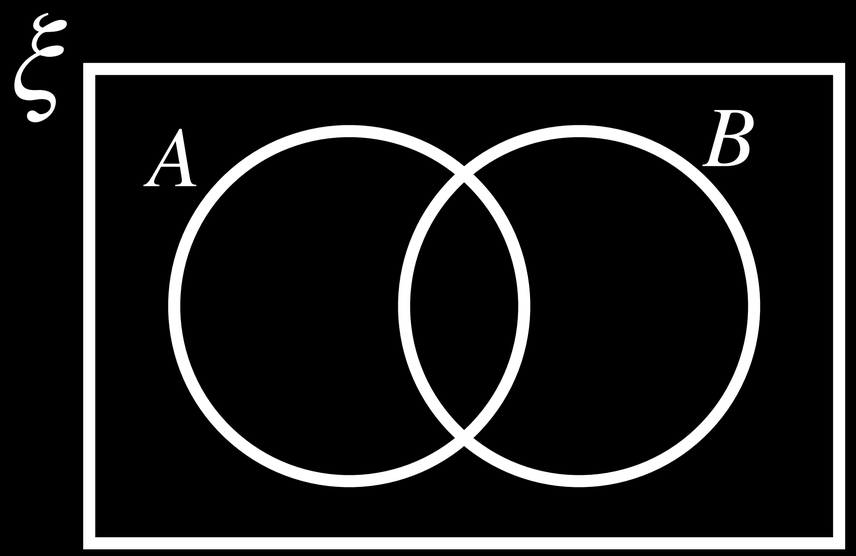
Set Notation

bro

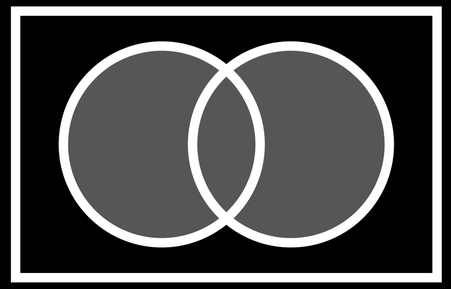
Symbol	Meaning	Usage
ξ or ε	The set that contains everything/universal set	For $\xi = \{2,3,5,7,4,1\}$, $\{2,4\} \subset \xi$
$\emptyset$	The set that contains nothing/null set/empty set	$\emptyset = \{\}$
$\in$	Is an element of	$3 \in \{3,6,1,2\}$
$\notin$	Is not an element of	$5 \notin \{3,6,1,2\}$
$\subset$	Is strictly a subset of	$\{3,2\} \subset \{3,6,1,2\}$
$\not\subset$	Is strictly not a subset of	$\{3,5,1\} \not\subset \{3,6,1,2\}$, $\{3,5,1\} \not\subset \{3,5,1\}$
$\subseteq$	Is a subset or equal to	$\{4,1,2\} \subseteq \{4,1,2\}$, $\{4,1\} \subseteq \{4,1,2\}$
$\not\subseteq$	Is not a subset or equal to	$\{4,7\} \not\subseteq \{4,1,2\}$
$\cup$	Union, aka joining two sets	$A \cup B$
$\cap$	Intersection, aka common terms in two sets	$A \cap B$
$\{x : C\}$	The set of all values of x that fit the condition C	$\{\text{integers } x : 3 \leq x \leq 7\} = \{3,4,5,6,7\}$
$n(S)$	Number of elements in the set S	$S = \{2,4,1,6\}, n(S) = 4$

Set Notation

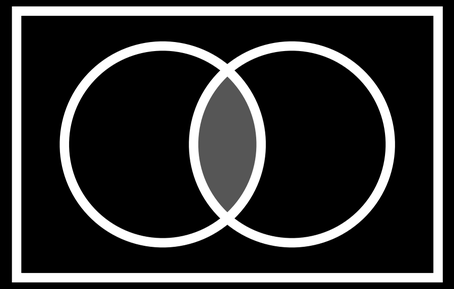
bro



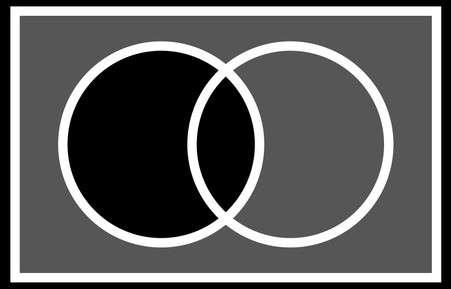
for two sets A and B,



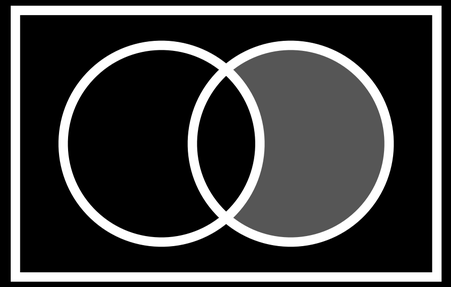
$A \cup B$



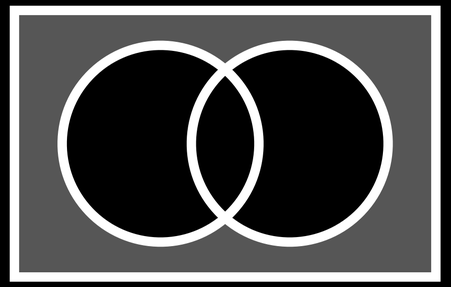
$A \cap B$



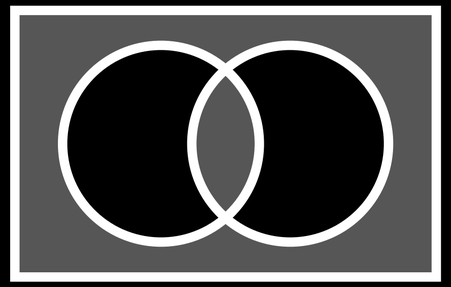
A'



$A' \cap B$



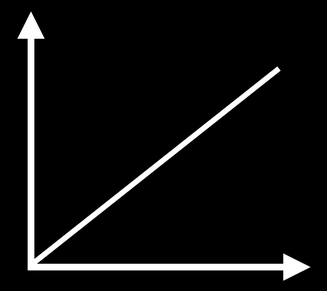
$A' \cap B'$



$(A' \cap B') \cup (A \cap B)$

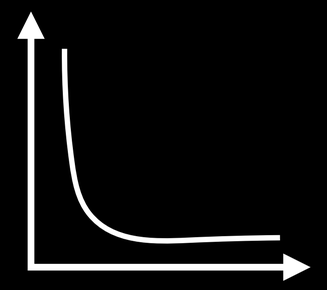
Direct/Inverse Proportion

bro



Direct Proportion

$y = kx,$ where k is a const



Inverse Proportion

$y = \frac{k}{x}$ where k is a const

y is directly proportional to x^2. When x=2, y=12. Thus, find x when y = 27.

$$y = kx^2 \qquad y = 3x^2$$
$$12 = k(2)^2 \qquad 27 = 3x^2$$
$$12 = 4k \qquad 9 = x^2$$
$$k = 3 \qquad x = 3$$

y is inversely proportional to x^2. For a value of x, y=0.5. When that value of x is halved, y=?

$0.5 = \frac{k}{X^2}$, where X represents specifc value of x

$$0.5X^2 = k$$
$$y = \frac{0.5X^2}{(\frac{X}{2})^2} \qquad y = \frac{0.5X^2}{1}\frac{4}{X^2}$$
$$y = \frac{0.5X^2}{\frac{X^2}{4}} \qquad y = 4(0.5)$$
$$y = 2$$

Number Sequences

bro

for a sequence 2,7,12,17,22, find the nth term

(second term – first term)n + first term

$(7-2)n+2$ **first term has n=0, not n=1

$5n+2$

only applies for linear sequences

other sequences may be non-linear, e,g, square numbers

Probability

the c

Basics of probability:

$$P(A) = \frac{\text{possible outcomes of } A}{\text{total outcomes}}$$

For two independent events A and B (i.e. outcome does not affect other event, like flipping two coins)

$P(A \text{ or } B) = P(A) + P(B)$

$P(A \text{ and } B) = P(A) \times P(B)$

Probability - Dependent Events

the c

For two dependent events A and B (i.e. outcome of one event affects outcome of the other, like picking out cards without replacement)

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$
$$P(A \text{ and } B) = P(A) \times P(B \text{ after A occured})$$

From a standard deck of 52 playing cards, 26 of them are red. What is the probability that 2 red cards are drawn without replacement?

$$P(\text{drawing 2 reds}) = \frac{26}{52} \times \frac{25}{51} = \frac{25}{102}$$

In one class, there are 15 boys and 11 girls, where 6 boys and 5 girls scored an A. What is the probability that a randomly picked student is either a boy or someone who scored an A?

$$P(\text{boy or A}) = \frac{15}{26} + \frac{11}{26} - \frac{6}{26} = \frac{10}{13}$$

subtract P(boy and A) as it does not fit the probability we want to find

**take note of order; lets say with a playing card deck and drawing 2 cards,
P(drawing one red and one black) = P(red first, black second) + P(black first, red second) as order of cards drawn does not matter

Probability

the c

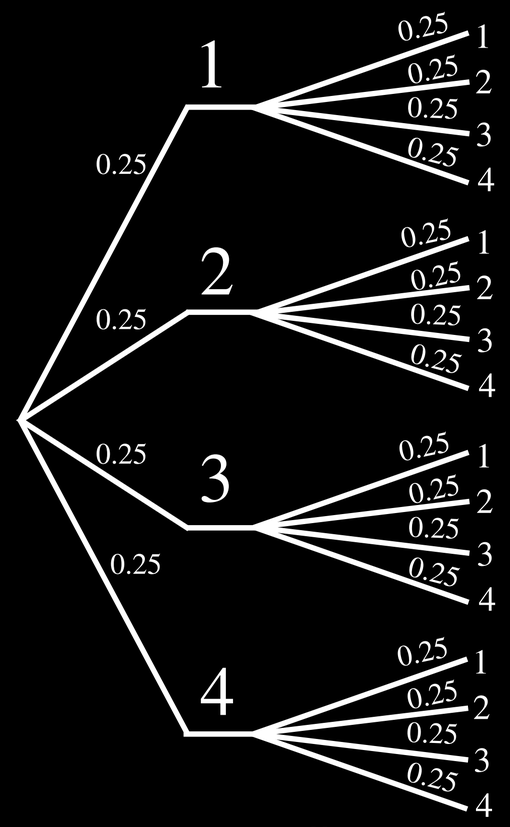
Probabilities can be represented with either a possibility diagram or a tree diagram
For example, we can represent the probabilities of rolling two four sided dice as so:

Dice B

Dice A

A,B	1	2	3	4
1	1,1	1,2	1,3	1,4
2	2,1	2,2	2,3	2,4
3	3,1	3,2	3,3	3,4
4	4,1	4,2	4,3	4,4

possibility diagram



tree diagram

Geometric Construction & Polygons

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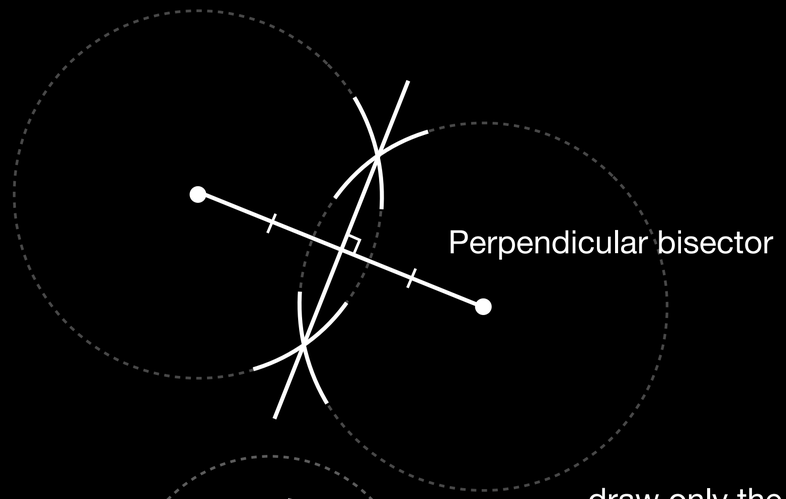
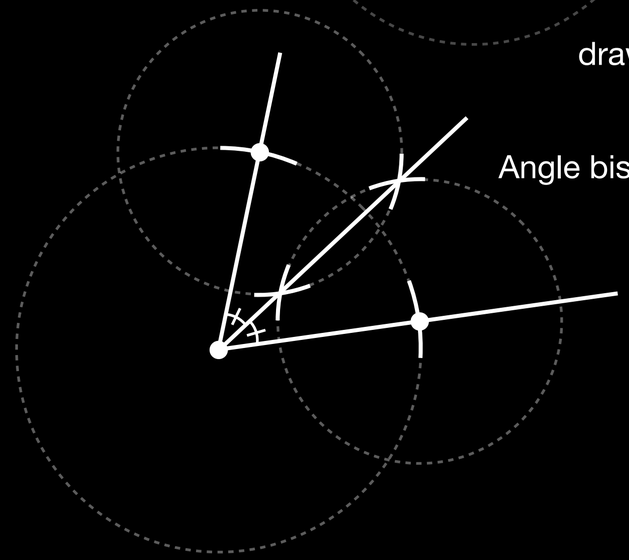
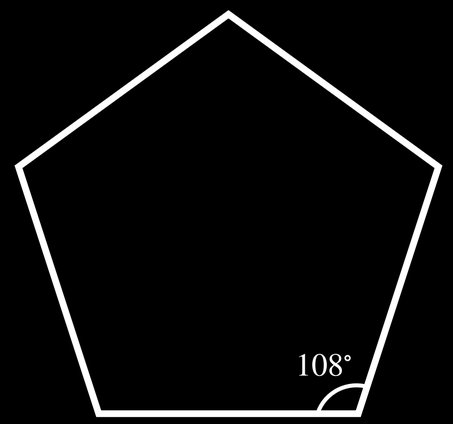


draw only the solid lines
include arcs!!!





For a polygon with n sides:

sum of interior angles $= (n-2)180^\circ$

interior angle (of regular polygon) $= \dfrac{(n-2)180^\circ}{n}$

Matrices

who

A matrix acts as a table with a number of rows and columns.
We describe a matrix by its order(rows x columns)
In this case, matrix A is a 2 x 3 matrix

$$A = \overbrace{\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}}^{\text{3 columns}} \Big\} \text{2 rows}$$

Matrices can only be added together if they have the same order

$$A = \begin{bmatrix} 6 & 3 \\ -2 & 1 \end{bmatrix} \qquad B = \begin{bmatrix} 2 & 4 \\ 5 & -3 \end{bmatrix}$$

$$A + B = \begin{bmatrix} 6+2 & 3+4 \\ -2+5 & 1-3 \end{bmatrix} = \begin{bmatrix} 8 & 7 \\ 3 & -2 \end{bmatrix}$$

A matrix can be multiplied to a scalar as so:

$$A = \begin{bmatrix} 6 & 3 \\ -2 & 1 \end{bmatrix}$$

$$3A = \begin{bmatrix} 18 & 9 \\ -6 & 3 \end{bmatrix} \quad kA = \begin{bmatrix} 6k & 3k \\ -2k & k \end{bmatrix}$$

Matrices

who

Two matrices can be multiplied together only if the number of columns in the first matrix is equal to the number of rows in the second matrix

$$A = \begin{bmatrix} 6 & 3 & 9 \\ 1 & 7 & 2 \end{bmatrix} \qquad B = \begin{bmatrix} 4 \\ 3 \\ 5 \end{bmatrix}$$

2×3 matrix $\qquad$ 3×1 matrix

has to be the same

resultant matrix's order (2×1)

The resultant matrix AB will have the same number of rows as the first matrix and the same number of columns as the second matrix

Multiply the values as so and add them together to find the resultant matrix:

multiply

multiply

multiply

			4
			3
			5
6	3	9	6(4)+3(3)+9(5) = 78
1	7	2	1(4)+7(3)+2(5)= 35

$$AB = \begin{bmatrix} 6 & 3 & 9 \\ 1 & 7 & 2 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \\ 5 \end{bmatrix} = \begin{bmatrix} 78 \\ 35 \end{bmatrix}$$

Matrices qns will ask you to form a matrix that represents a set of data (e.g. prices of each type of product)

Indices & Standard Form

Indices Laws

amn

$$a^m \times a^n = a^{(m+n)}$$

$$\frac{a^m}{a^n} = a^{(m-n)}$$

$$(a^m)^n = a^{mn}$$

Common base

$$a^m * b^m = (ab)^m$$

$$\frac{a^m}{b^m} = \left(\frac{a}{b}\right)^m$$

Common power

$$a^0 = 1$$

$$a^{-n} = \frac{1}{a^n}$$

$$a = a^1$$

Non positive power

$$a^{\frac{1}{n}} = \sqrt[n]{a}$$

$$a^{\frac{m}{n}} = \left(\sqrt[n]{a}\right)^m$$

$$= \sqrt[n]{a^m}$$

either works

Fractional powers

Questions may ask you to express in positive index, negative index, or surd form (roots)

Remember that equal signs are two way; given a^{m+n}, you can expand to $a^m \times a^n$

Indices Examples

Example

Simplify $\left(2^{n+1}+4^{n}\right)8^{-n}$ in postive index form

$$\begin{aligned}\left(2^{n+1}+4^{n}\right)8^{-n} &= \frac{2^{n+1}+4^{n}}{8^{n}}\\ &= \frac{2^{n}\times 2+2^{2n}}{2^{3n}}\\ &= \frac{2^{n}(2+2^{n})}{2^{3n}}\\ &= \frac{2+2^{n}}{4^{n}}\end{aligned}$$

***Almost every time when given an expression already containing $a^{m\pm n}$, you will have to expand it out to $a^{m}\times a^{n}$ or $a^{m}\div a^{n}$

Simplify $a^{\frac{1}{6}}\left(a^{2}b\right)^{\frac{2}{3}}$ in surd form

$$\begin{aligned}a^{\frac{1}{6}}\left(a^{2}b\right)^{\frac{2}{3}} &= a^{\frac{1}{6}}(a^{2})^{\frac{2}{3}}b^{\frac{2}{3}}\\ &= a^{\frac{1}{6}}a^{\frac{4}{3}}b^{\frac{2}{3}}\\ &= a^{\left(\frac{1}{6}+\frac{4}{3}\right)}b^{\frac{2}{3}}\\ &= a^{\frac{3}{2}}b^{\frac{2}{3}}\\ &= \sqrt{a^{3}}\sqrt[3]{b^{2}}\end{aligned}$$

Standard Form

Click to add

$$A \times 10^n$$

standard form notation

where $1 \leq A < 10$, and n is an integer

or $-1 \geq A > -10$

Standard Form Operations

Click to add

$$(A \times 10^n) \times (B \times 10^m) = AB \times 10^{m+n}$$

$$\frac{(A \times 10^n)}{(B \times 10^m)} = \frac{A}{B} \times 10^{n-m}$$

$$(A \times 10^n) \pm (B \times 10^m) = 10^n(A \pm \frac{B \times 10^m}{\cancel{10^n}})$$

This assumes that n < m

$$= 10^n(A \pm B \times 10^{m-n})$$

convert to standard form

$$(A \times 10^n)^m = (A)^m \times (10^n)^m$$
$$= A^m \times 10^{nm}$$

MAKE SURE FINAL ANSWER IS IN CORRECT FORMAT: VALUE MULTIPLIED BY POWER 10 HAS TO BE $1 \leq x < 10$ OR $-1 \geq A > -10$

Solving Indices Equations

Click to add

$$a^b = a^c$$
$$b = c$$

$$2^b = 32$$
$$2^b = 2^5$$
$$b = 5$$

Convert terms to same base to be able to compare indexes

base should be smallest possible number

you will not be needed to use logarithms !!!!

Solving Indices Equations

Click to add

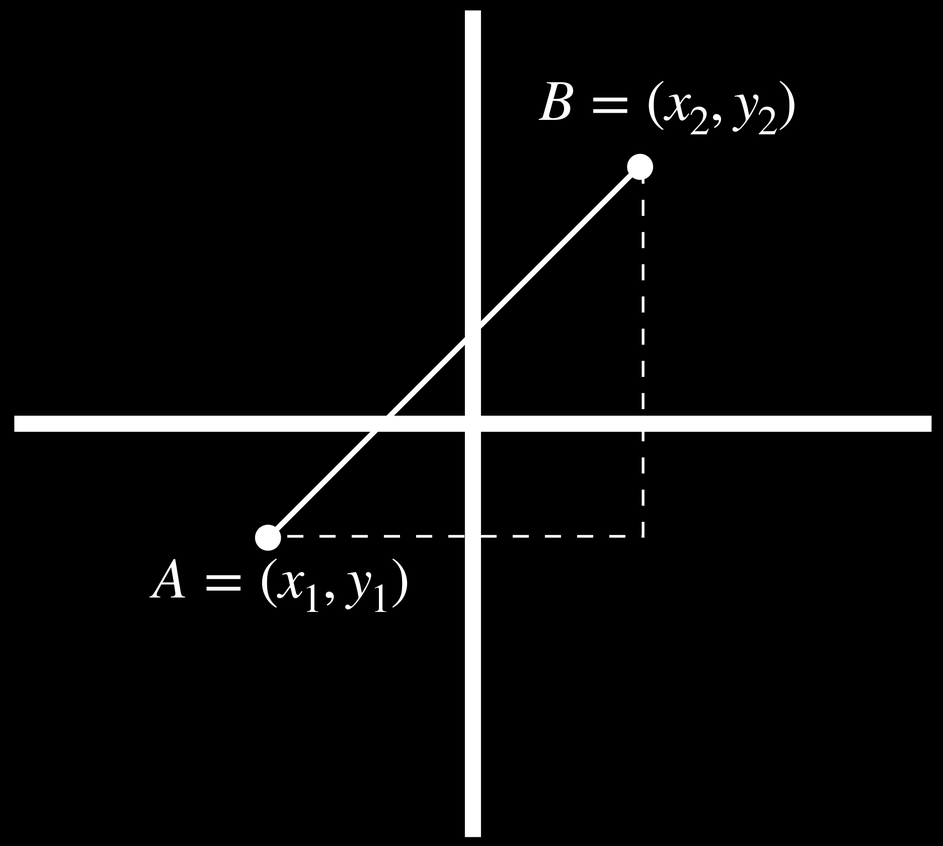
$$2^{x+1} = \frac{4^x}{8^x}$$
$$2^{x+1} = \left(\frac{4}{8}\right)^x$$
$$2^{x+1} = \left(\frac{1}{2}\right)^x$$
$$2^{x+1} = \frac{1}{2^x}$$
$$2^{x+1} = 2^{-x}$$
$$x + 1 = -x$$
$$1 = -2x$$
$$x = -0.5$$

$$\frac{(4^{2x})(10^x)}{5^x} = (8^{3x+2})(4^{2x-1})$$
$$\frac{(16^x)(10^x)}{5^x} = (2^{3(3x+2)})(2^{2(2x-1)})$$
$$\frac{160^x}{5^x} = (2^{3(3x+2)})(2^{2(2x-1)})$$
$$\left(\frac{160}{5}\right)^x = (2^{3(3x+2)})(2^{2(2x-1)})$$
$$32^x = (2^{3(3x+2)})(2^{2(2x-1)})$$
$$2^{5x} = 2^{3(3x+2)+2(2x-1)}$$
$$5x = 3(3x + 2) + 2(2x - 1)$$
$$5x = 9x + 6 + 4x - 2$$
$$-8x = 4$$
$$x = -0.5$$

Coordinate Geometry [E MATH]

Distance Formula

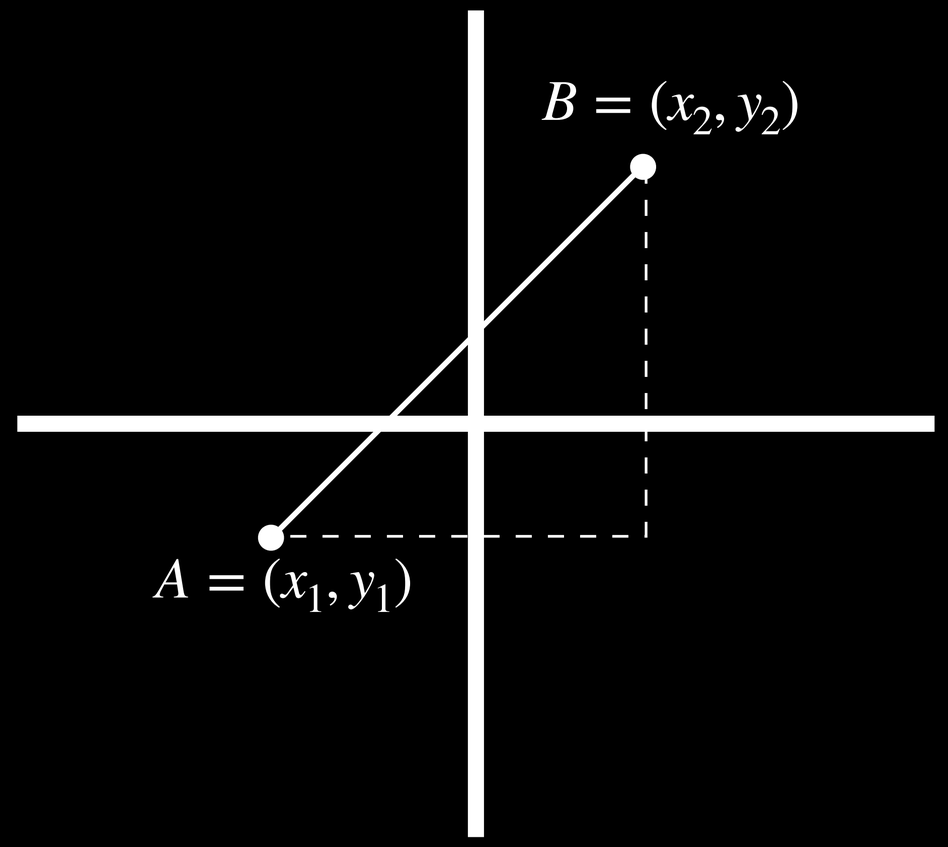
It only gets worse from here



Length of $AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Gradient Formula

It only gets worse from here

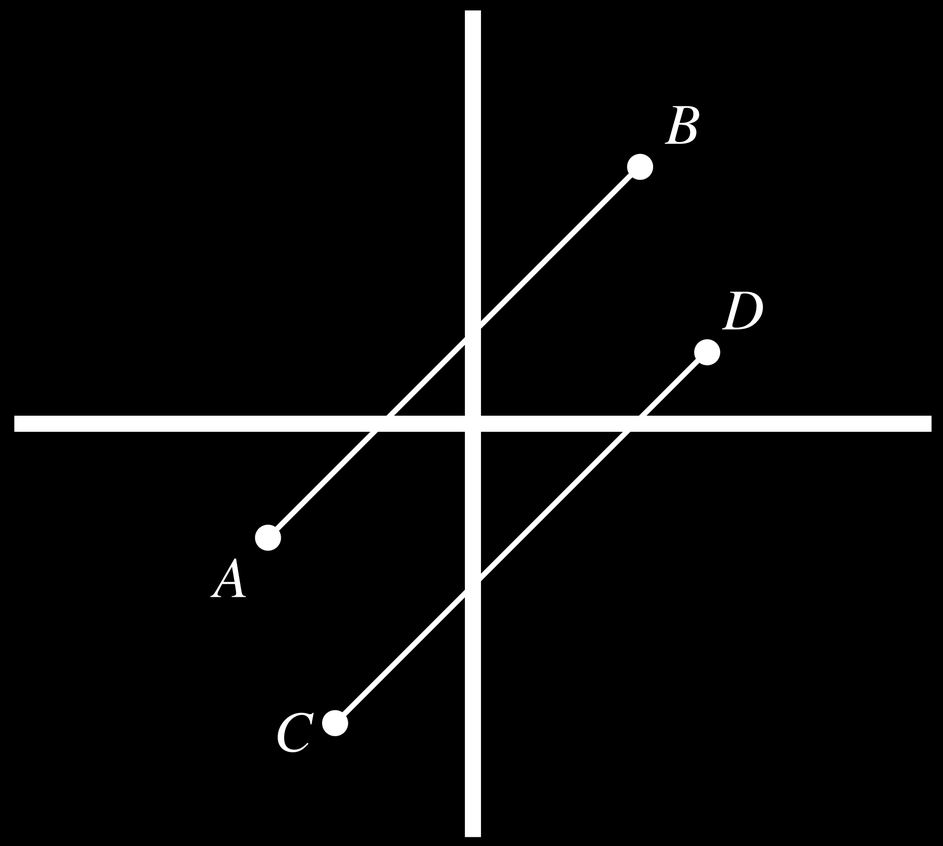


$$m_{AB} = \frac{\text{rise}}{\text{run}}$$

$$= \frac{y_2 - y_1}{x_2 - x_1}$$

Gradient Formula - Parallel Lines

It only gets worse from here

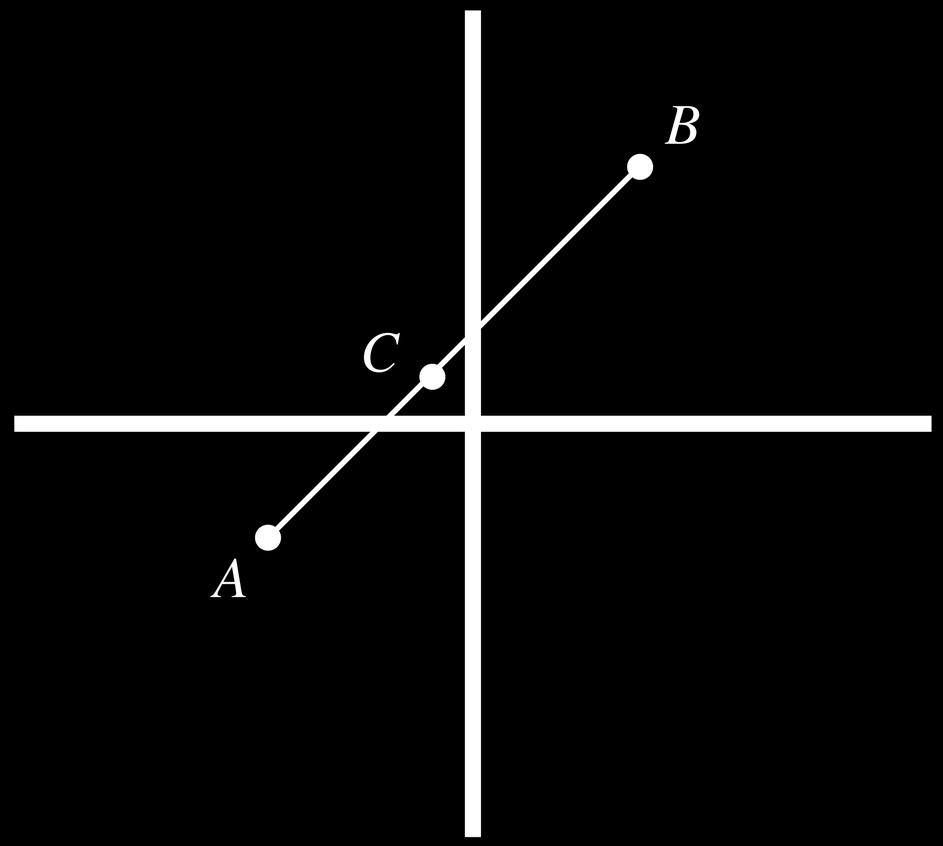


When AB is parallel to CD,

$$m_{AB} = m_{CD}$$

Gradient Formula - Parallel Lines

Go to slide -12.524

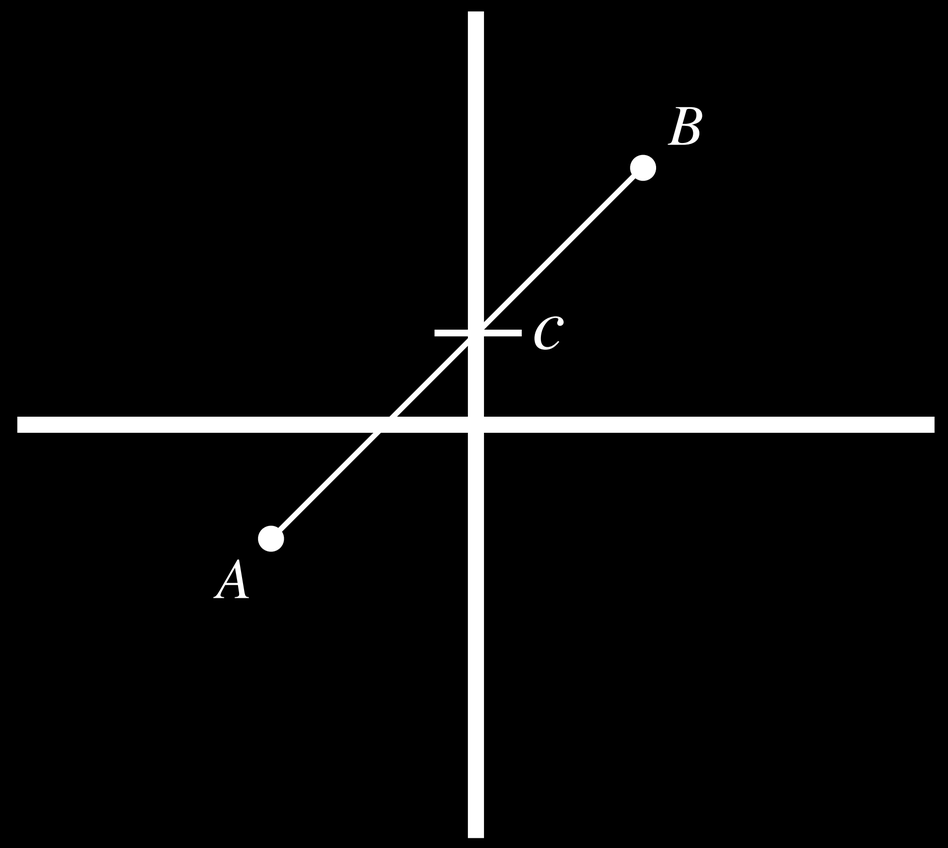


When the points A, B and C are collinear,

$$m_{AB} = m_{AC} = m_{BC}$$

Lines - Gradient Intercept Form

It only gets worse from here



y-intercept

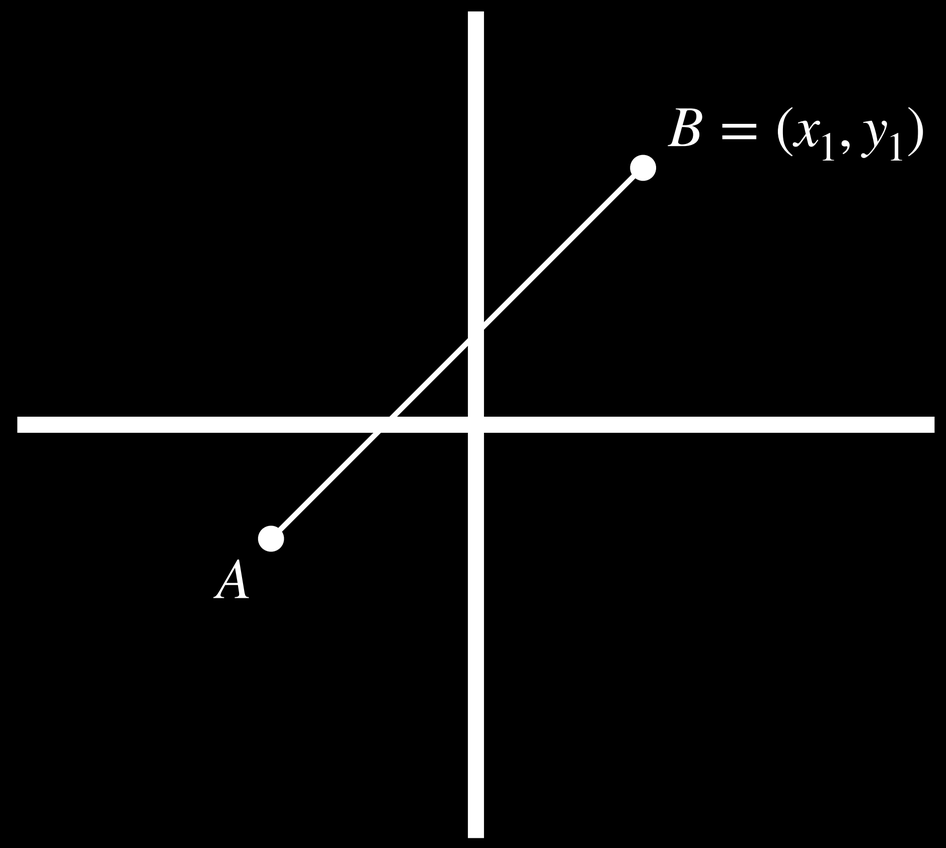
$$y = mx + c$$

gradient

To form gradient intercept form, make y the subject

Lines - Gradient Point Form

It only gets worse from here



$$y - y_1 = m(x - x_1)$$

Lines - Intersections

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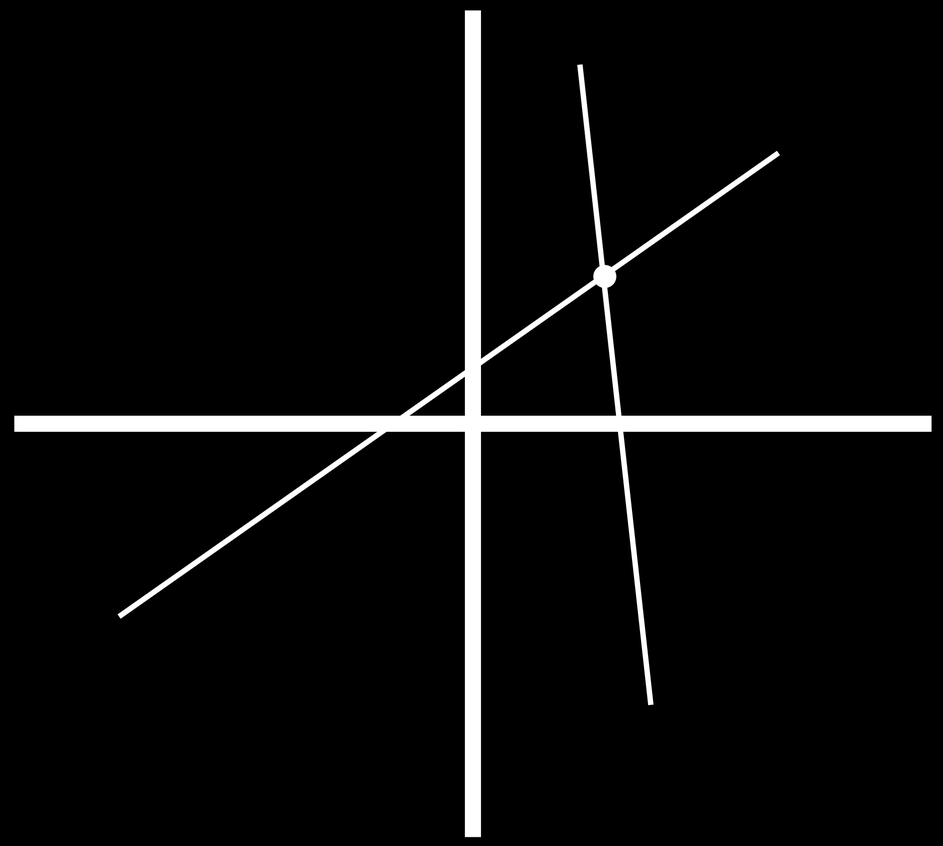

Solve line equations as simulateous equations

Lines - Areas of Regions

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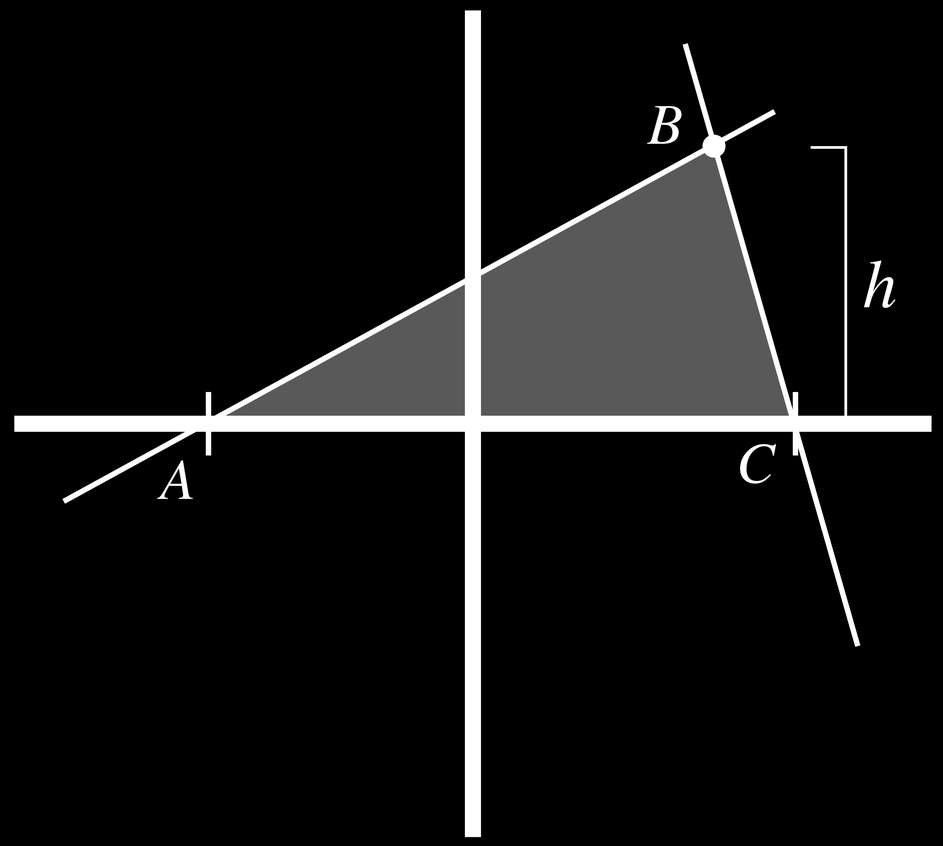


Find quantities that are needed to find area

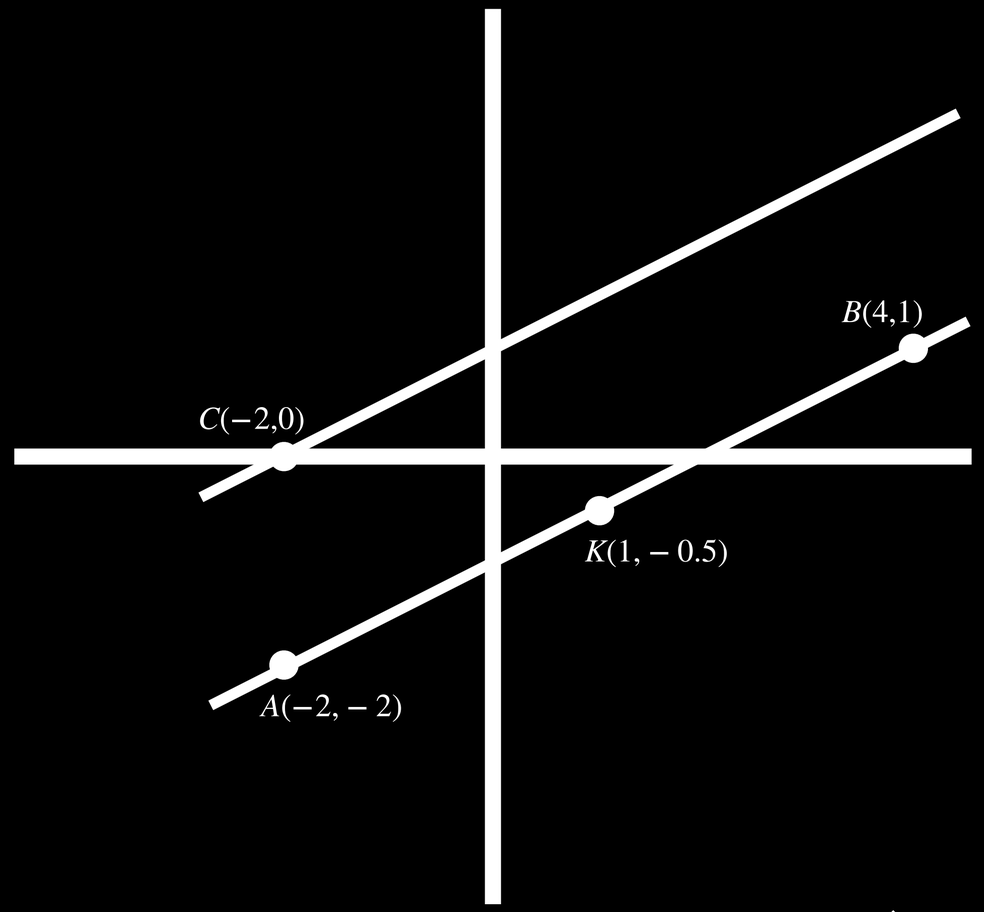
Example:

$$\frac{1}{2}b * h = \frac{1}{2}(AC) * (\text{y coordinate of } B)$$

Solving Questions

It only gets worse from here

Two points A and B have the coordinates (-2,-2) and (4,1) respectively.
a)find the equation of the line AB.
b) show that the point K(1,-0.5) lies on AB.
c) Another line passes through the point C(-2,0) that is parallel to AB. Find the equation of this line.
d) find the area of triangle ABC.



Solving Questions

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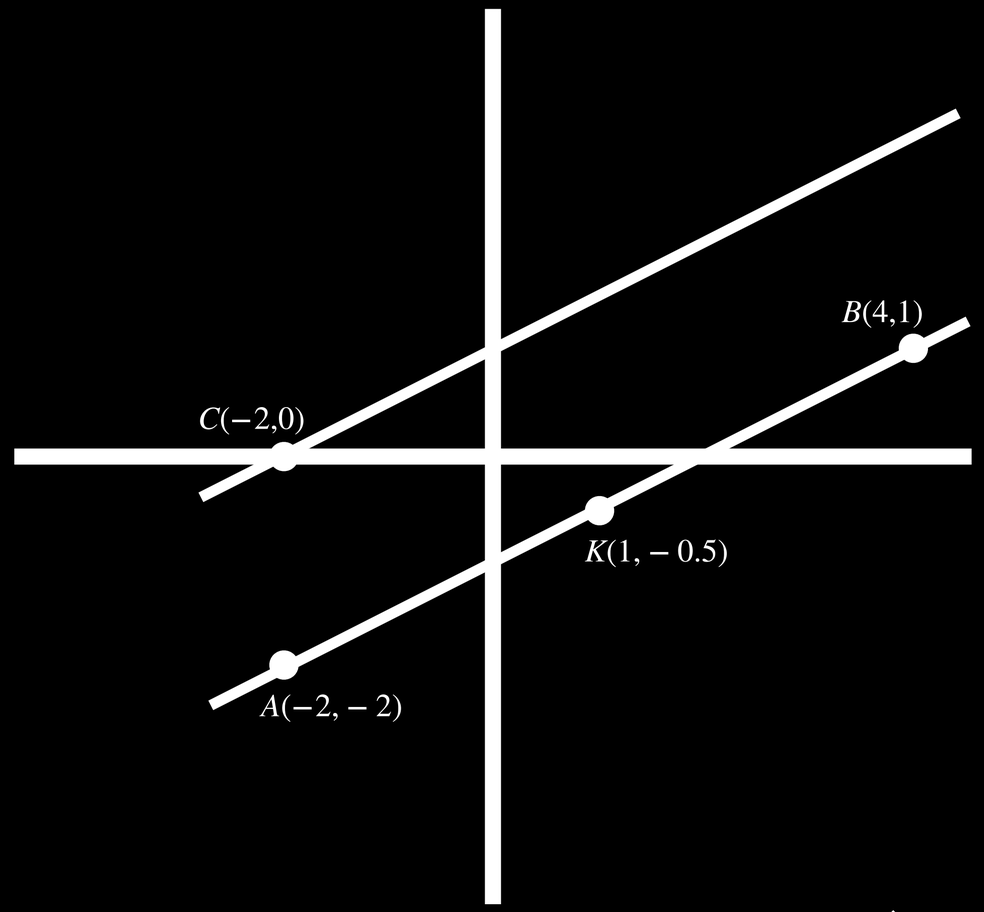
d) find the area of triangle ABC.

$$m_{AB} = \frac{1-(-2)}{4-(-2)}$$

$$m_{AB} = 0.5$$

$\therefore$ eqn of AB is

$$y - 1 = 0.5(x - 4)$$
$$y - 1 = 0.5x - 2$$
$$y = 0.5x - 1$$



Solving Questions

It only gets worse from here

Two points A and B have the coordinates (-2,-2) and (4,1) respectively.
a)find the equation of the line AB.
b) show that the point K(1,-0.5) lies on AB.
c) Another line passes through the point C(-2,0) that is parallel to AB. Find the equation of this line.
d) find the area of triangle ABC.

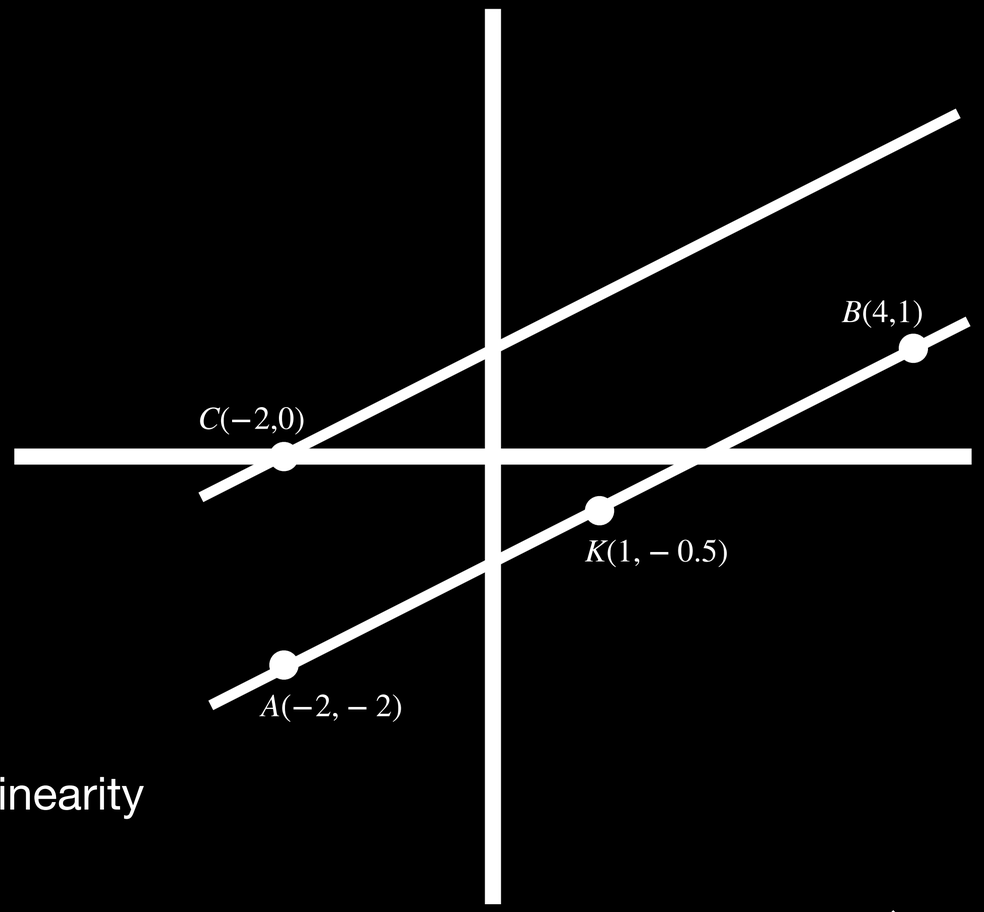
$$m_{AK} = \frac{-0.5-(-2)}{1-(-2)}$$
$$= 0.5$$

$m_{AB} = 0.5$(from (a))

As $m_{AB} = m_{BK}$ and A is a common point, points A, B and K are collinear

$\therefore$ K lies on AB

K need not be a common point to prove collinearity



Solving Questions

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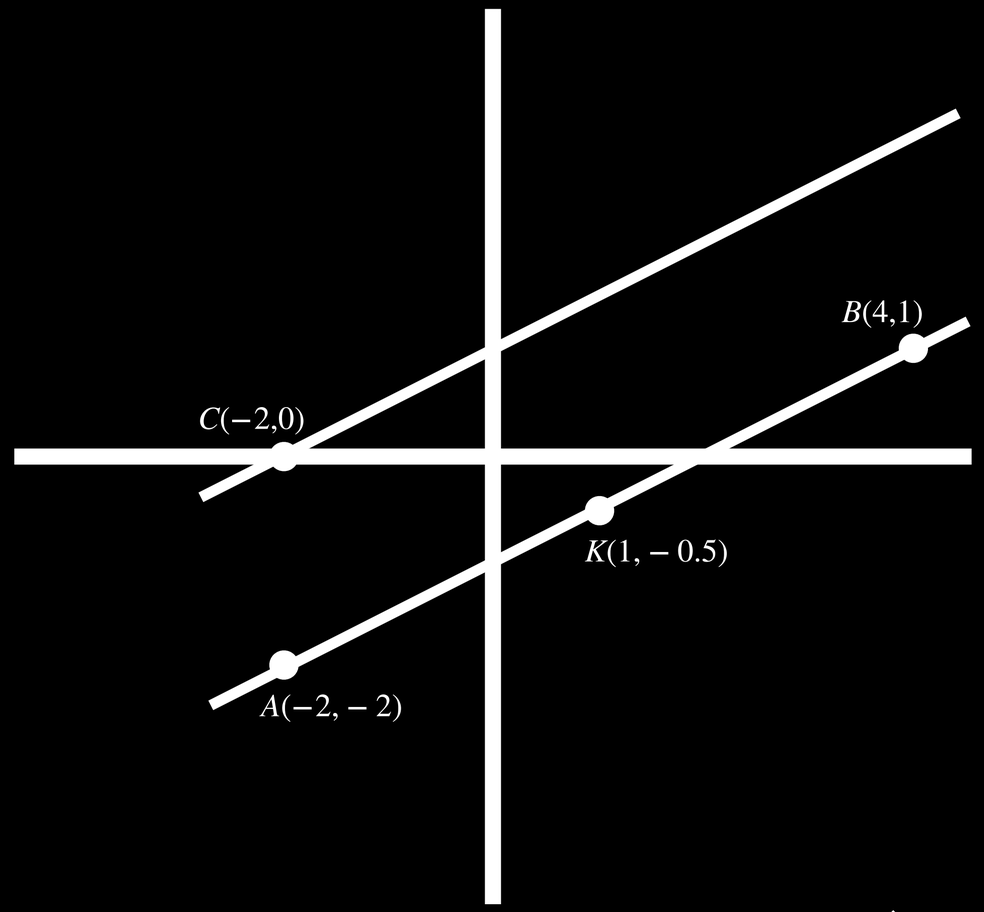
d) find the area of triangle ABC.

as line is parallel, gradient is 0.5

∴ eqn of line at C is

$$y - 0 = 0.5(x - (-2))$$
$$y = 0.5(x + 2)$$
$$y = 0.5x + 1$$



Solving Questions

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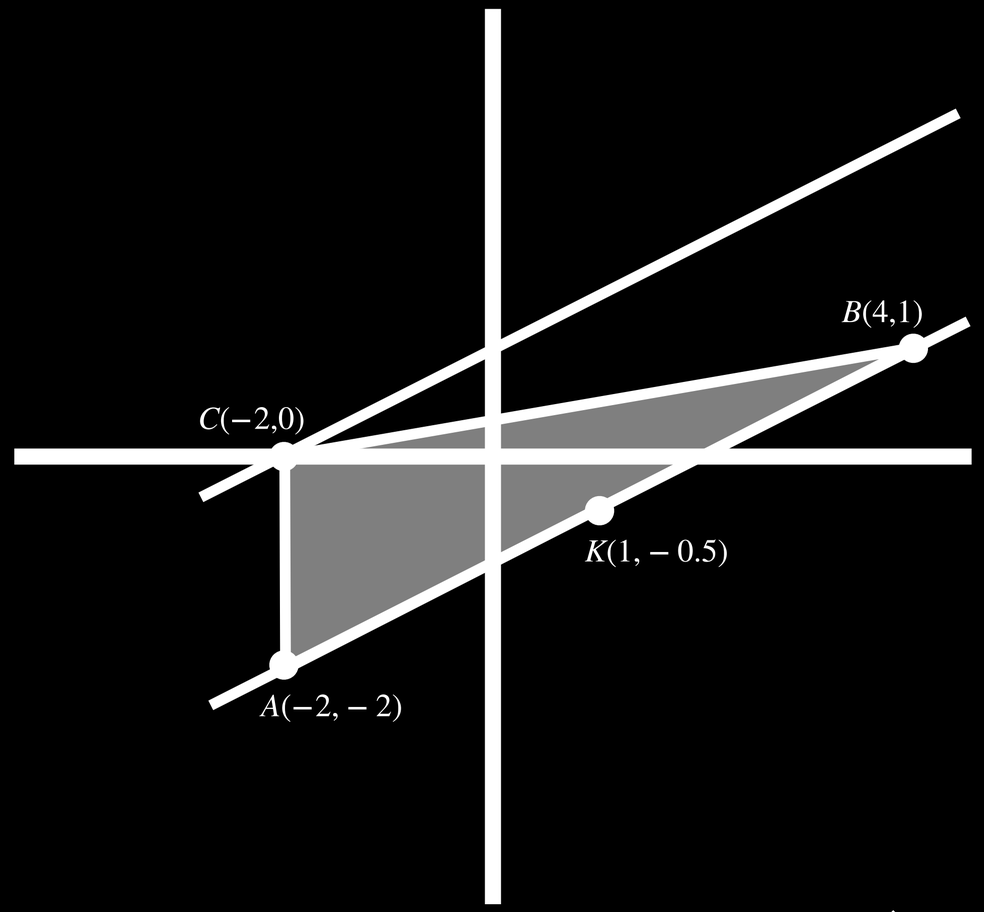
c) Another line passes through the point C(-2,0) that is parallel to AB.
Find the equation of this line.
d) find the area of triangle ABC.

$$\frac{1}{2}(0-(-2))(4-(-2))$$

$$=\frac{1}{2}(2)(6)$$

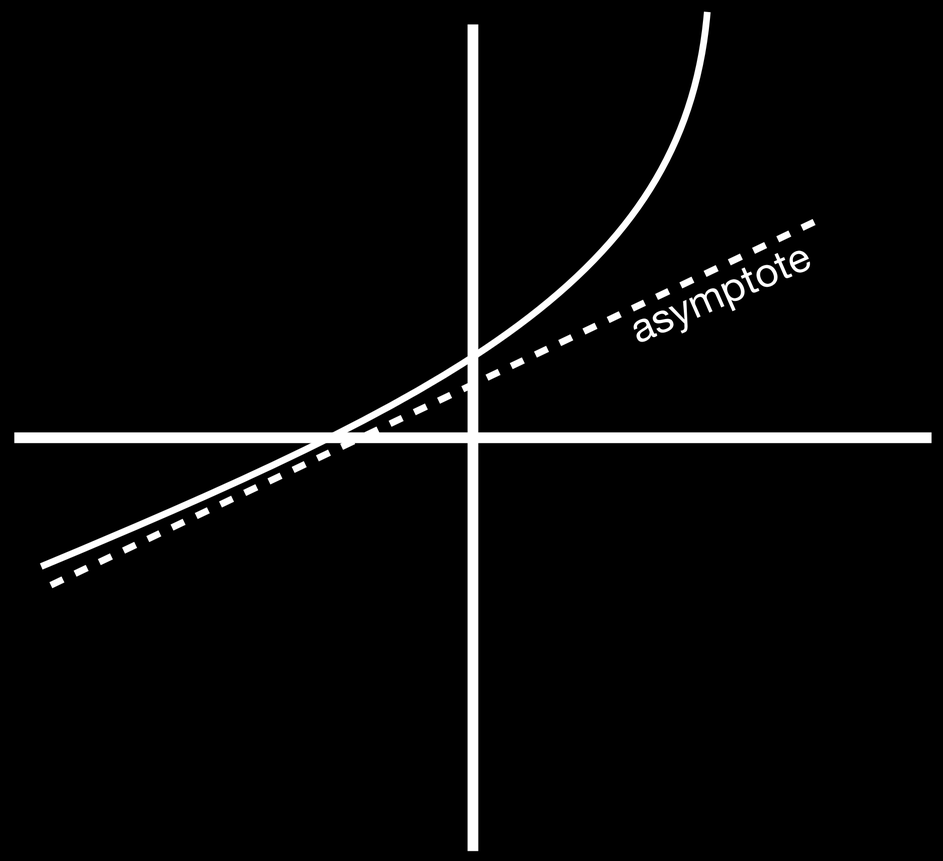
$$= 6 \text{ units}^2$$



Functions & Graphs

Asymptotes

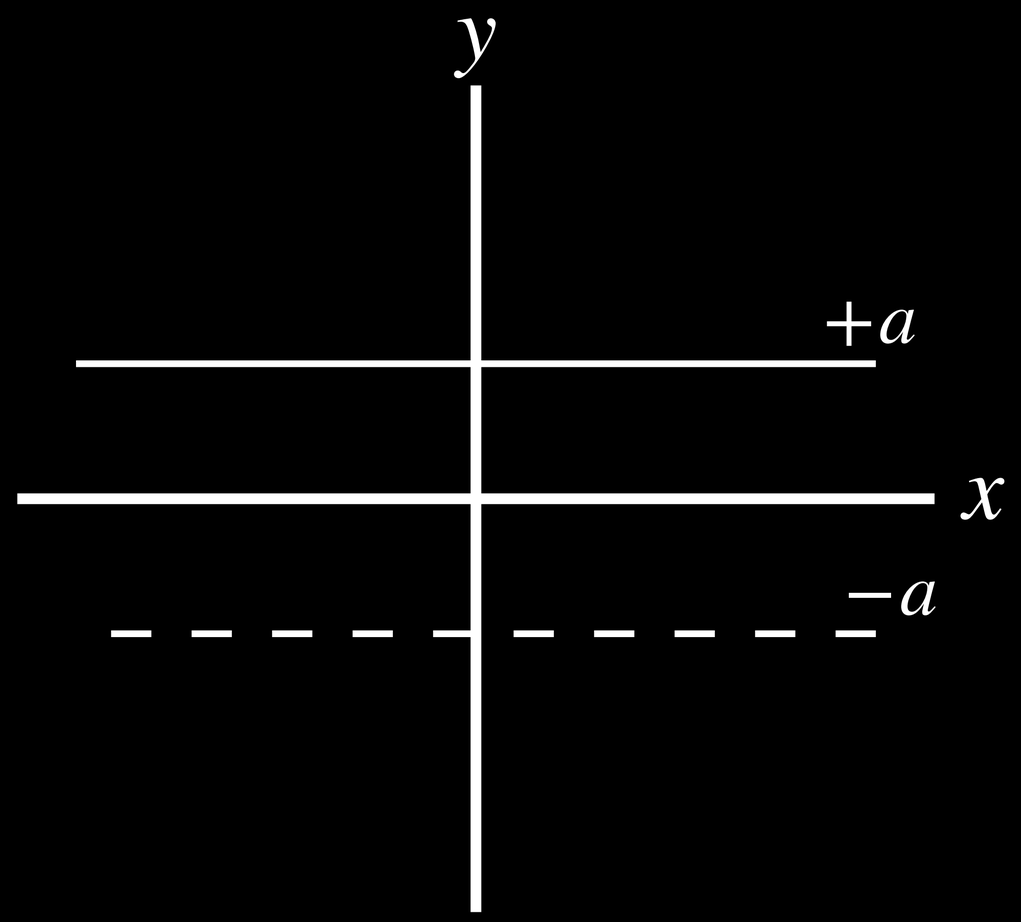
A sim toet



A line that approaches a curve arbitrarily closely

Graph Shapes

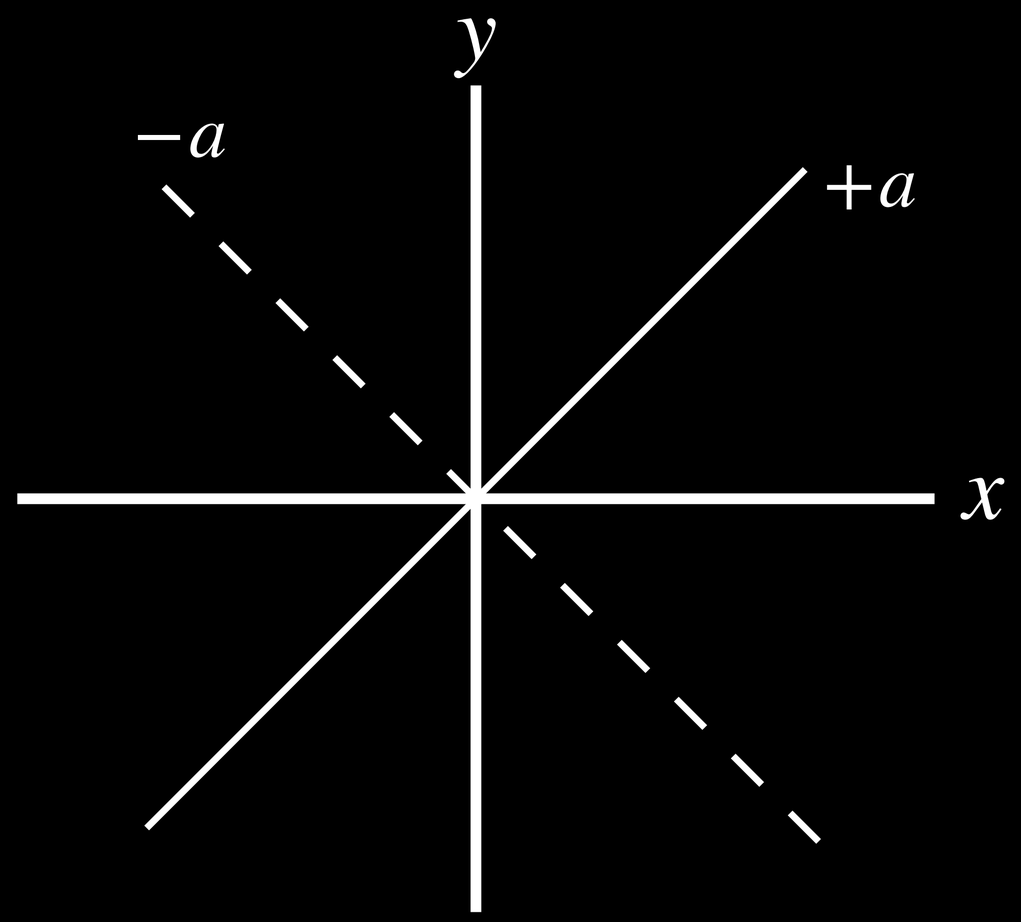
hr



$$y = ax^0 = a$$

Graph Shapes

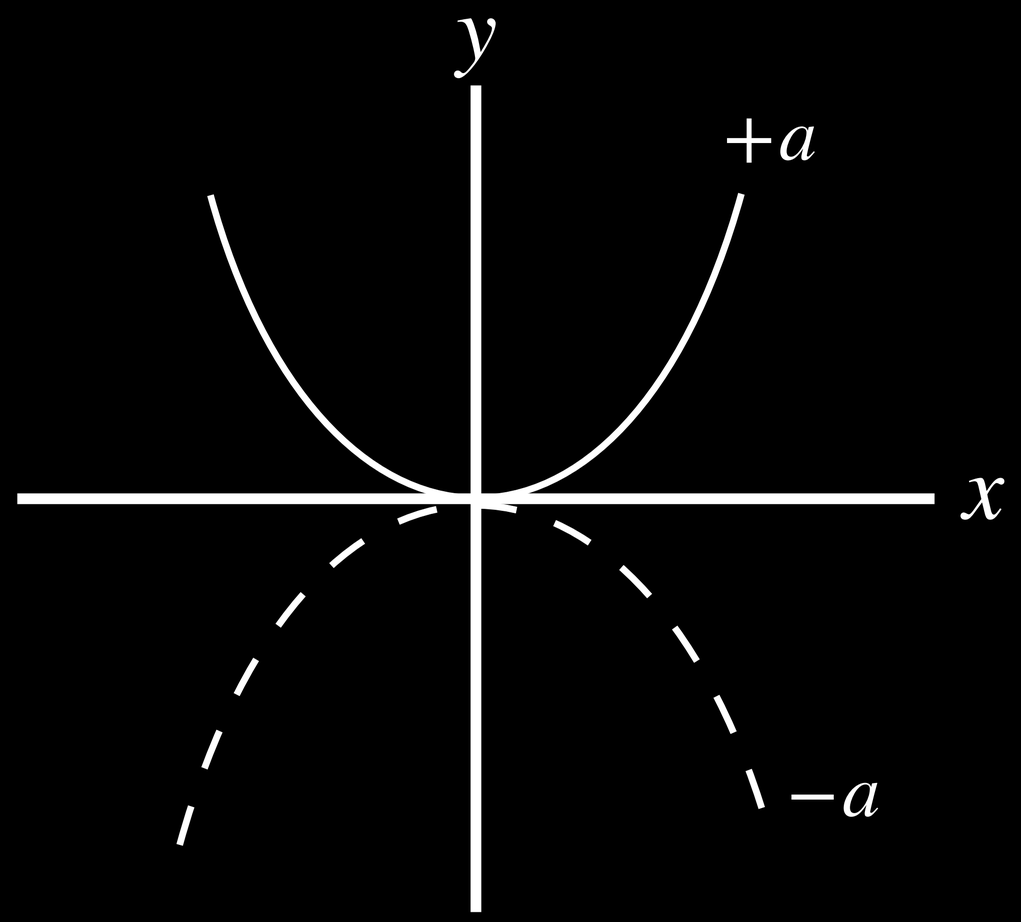
hr



$$y = ax$$

Graph Shapes

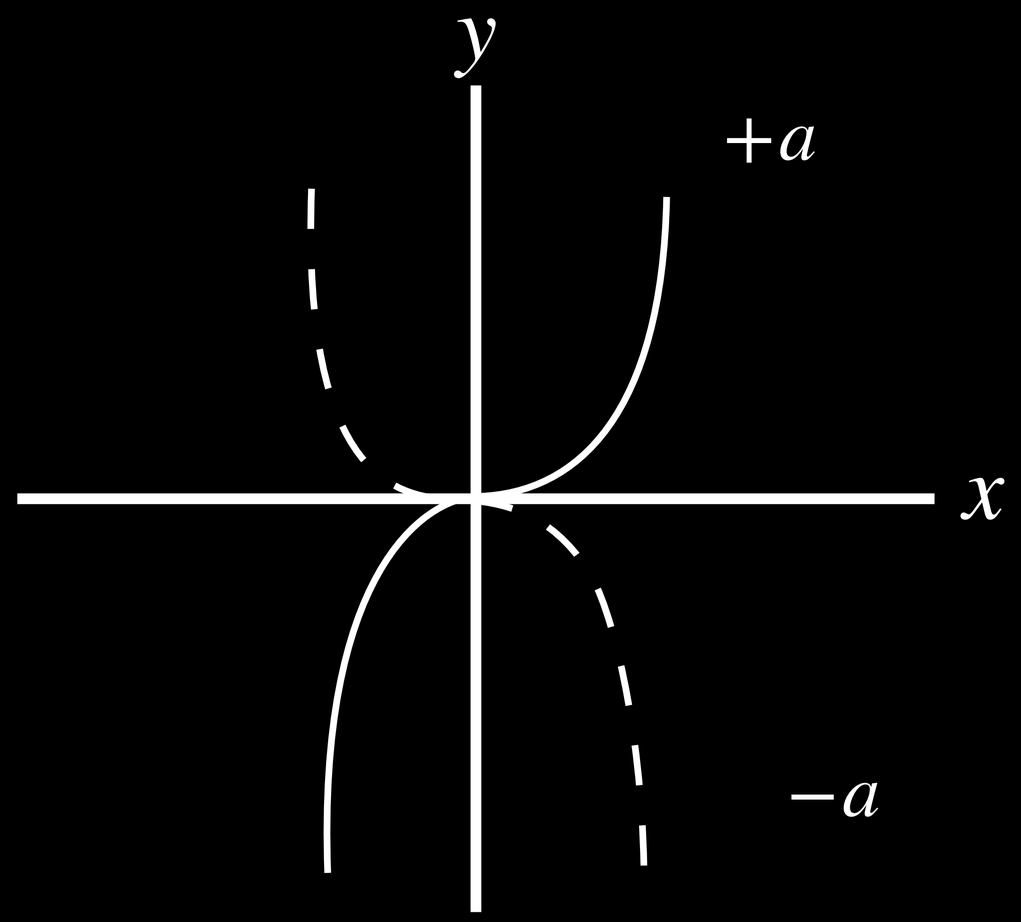
hr



$$y = ax^2$$

Graph Shapes

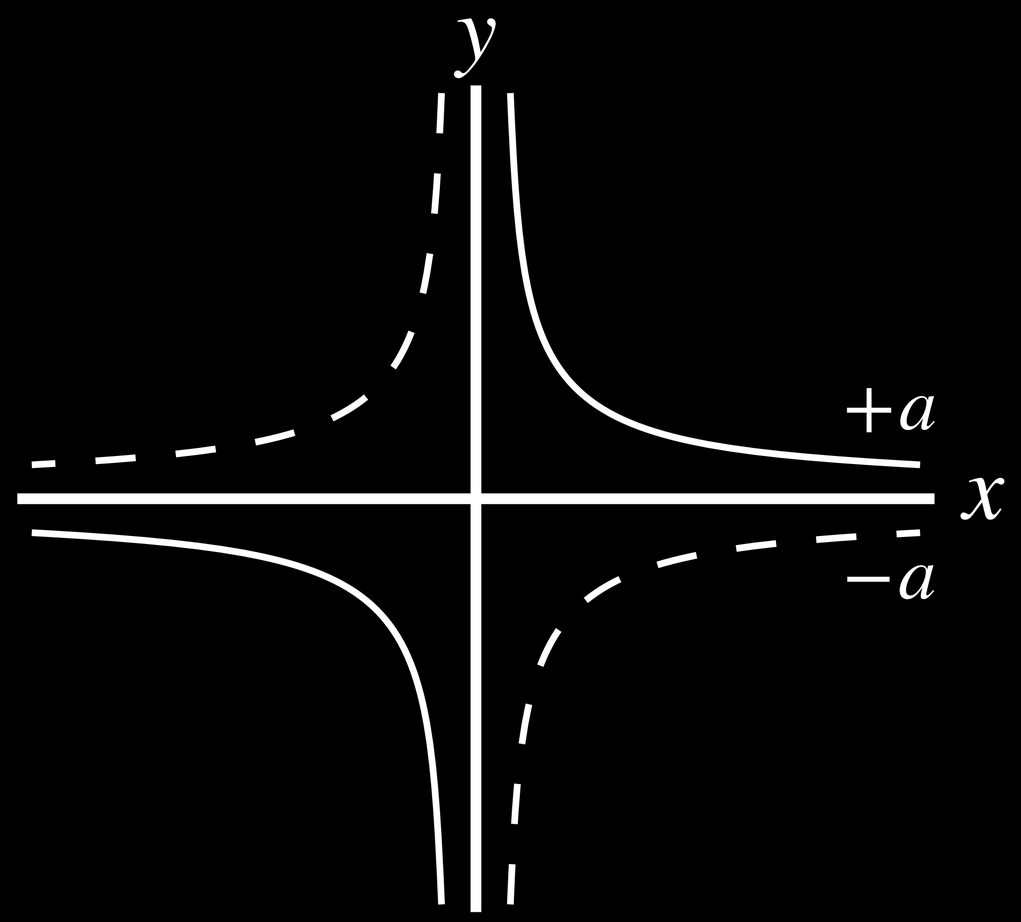
hr



$$y = ax^3$$

Graph Shapes

hr

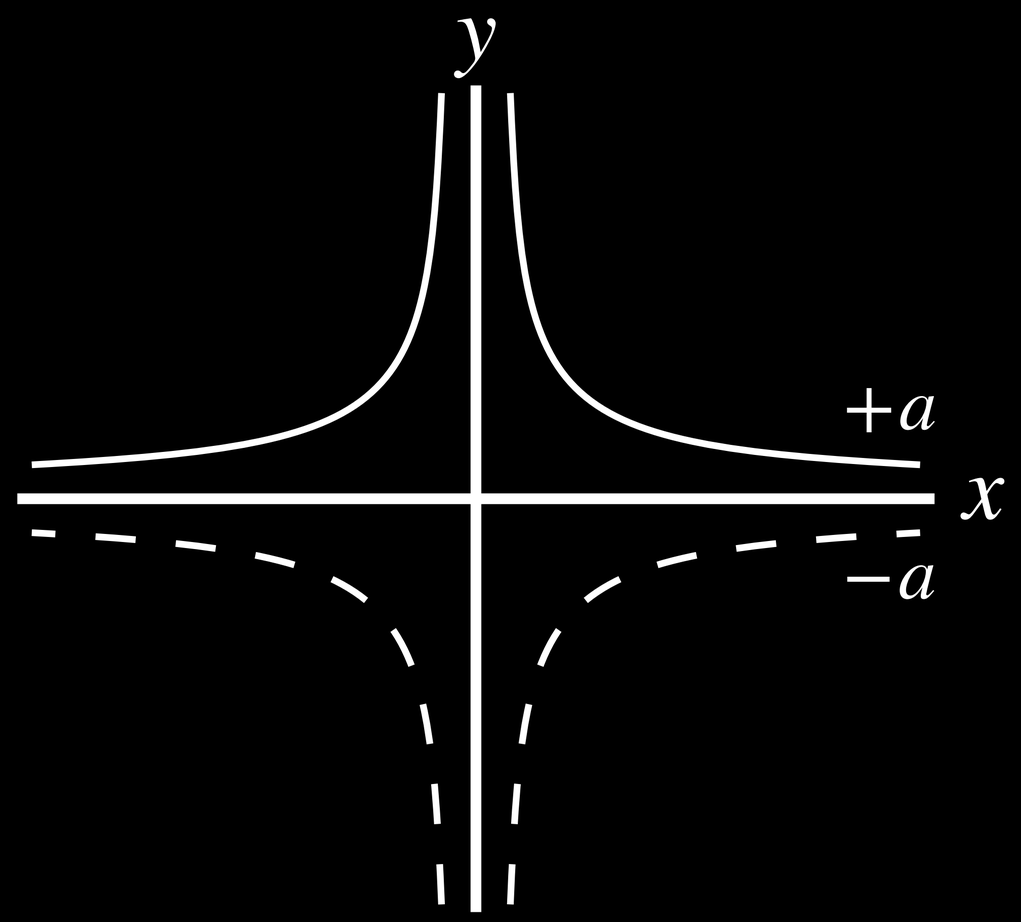


$$y = \frac{a}{x} = ax^{-1}$$

Has asymptote
$x = 0, y = 0$

Graph Shapes

hr

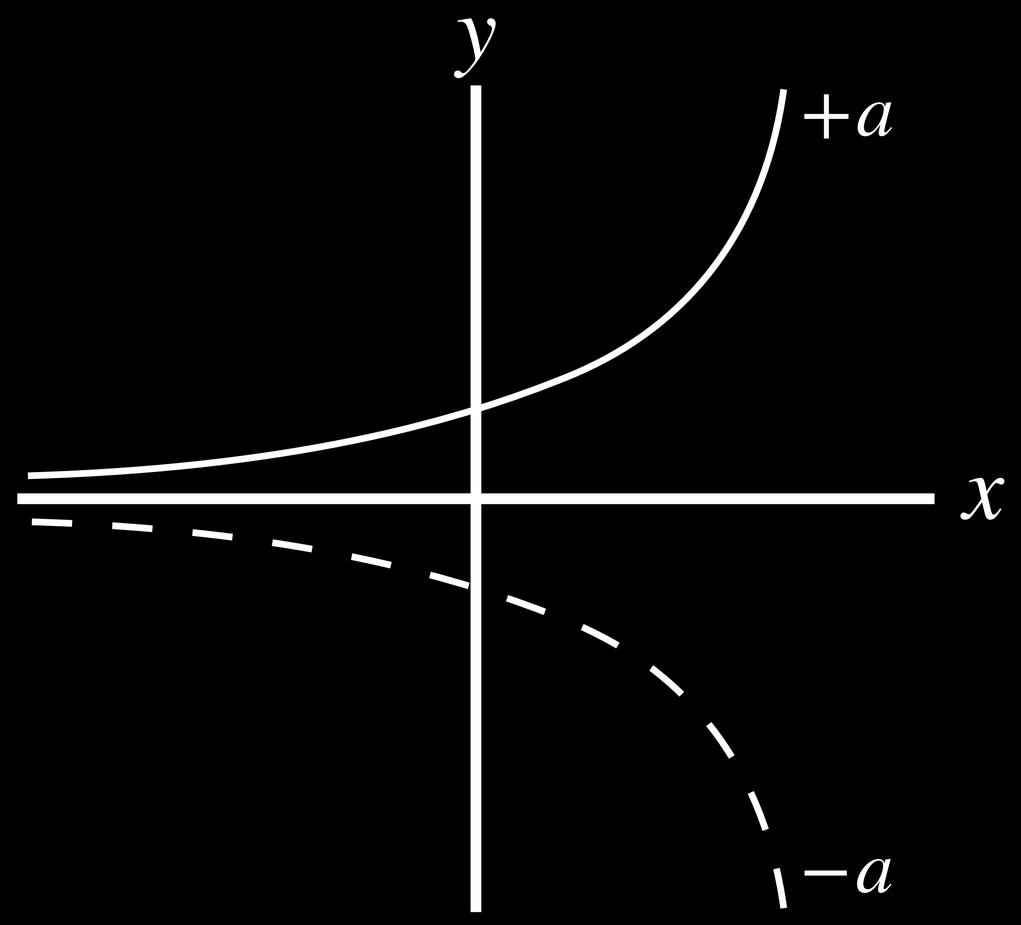


$$y = \frac{a}{x^2} = ax^{-2}$$

Has asymptote
$x = 0, y = 0$

Graph Shapes

hr

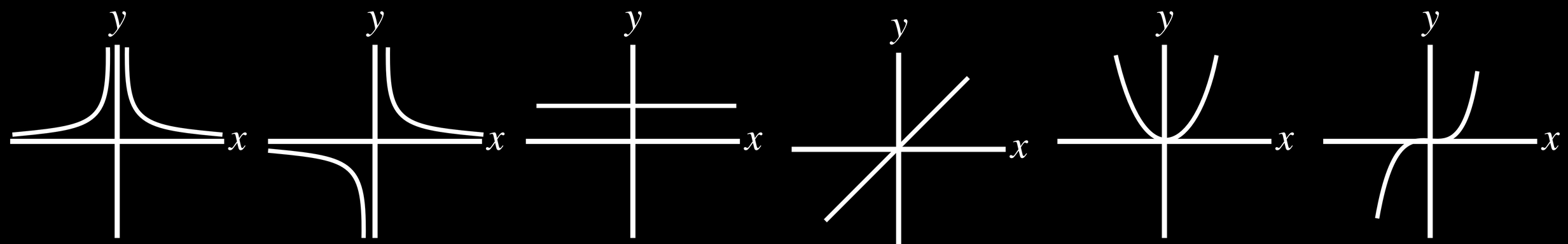


$$y = ab^x$$

Has asymptote
$y = 0$

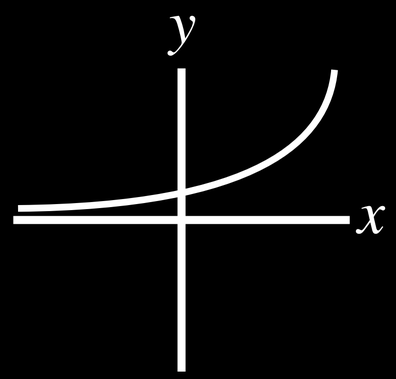
Graph Shapes

hr



$n = -2$ $n = -1$ $n = 0$ $n = 1$ $n = 2$ $n = 3$

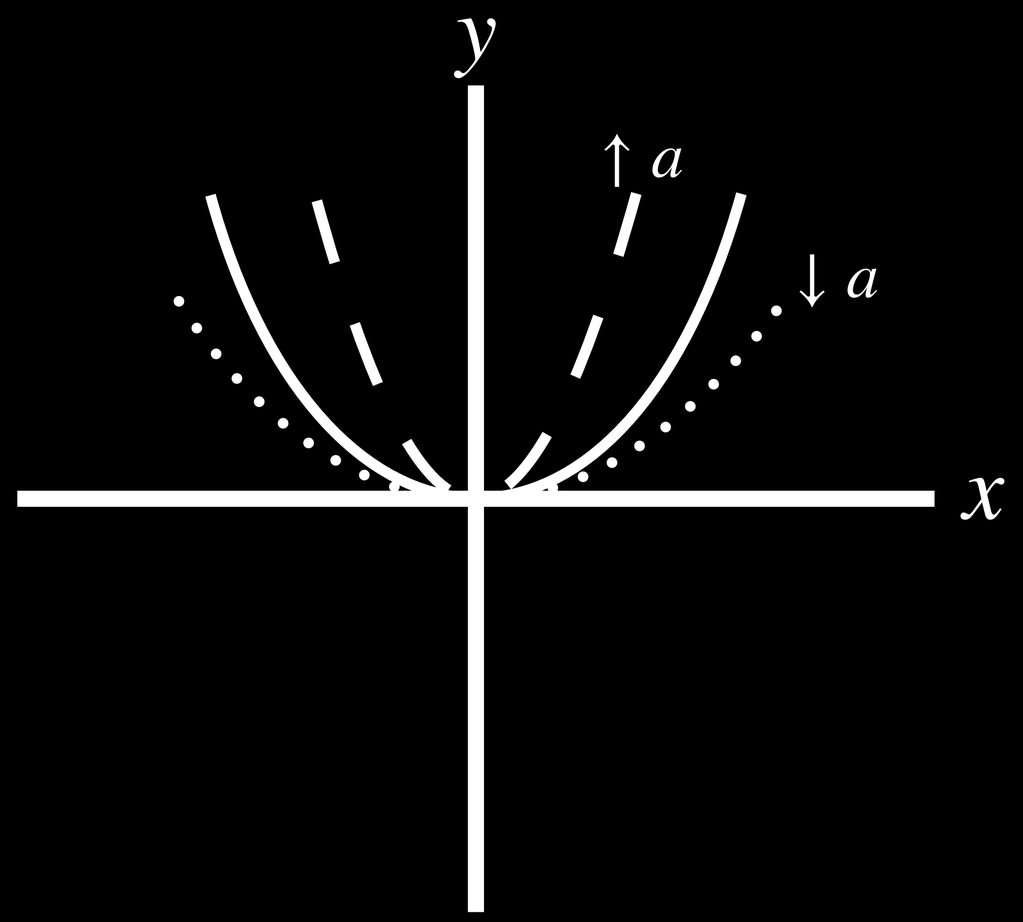
$$y = ax^n$$



$$y = ab^x$$

Graph Transformation

hr



For a function $f(x)$,

$$af(x)$$

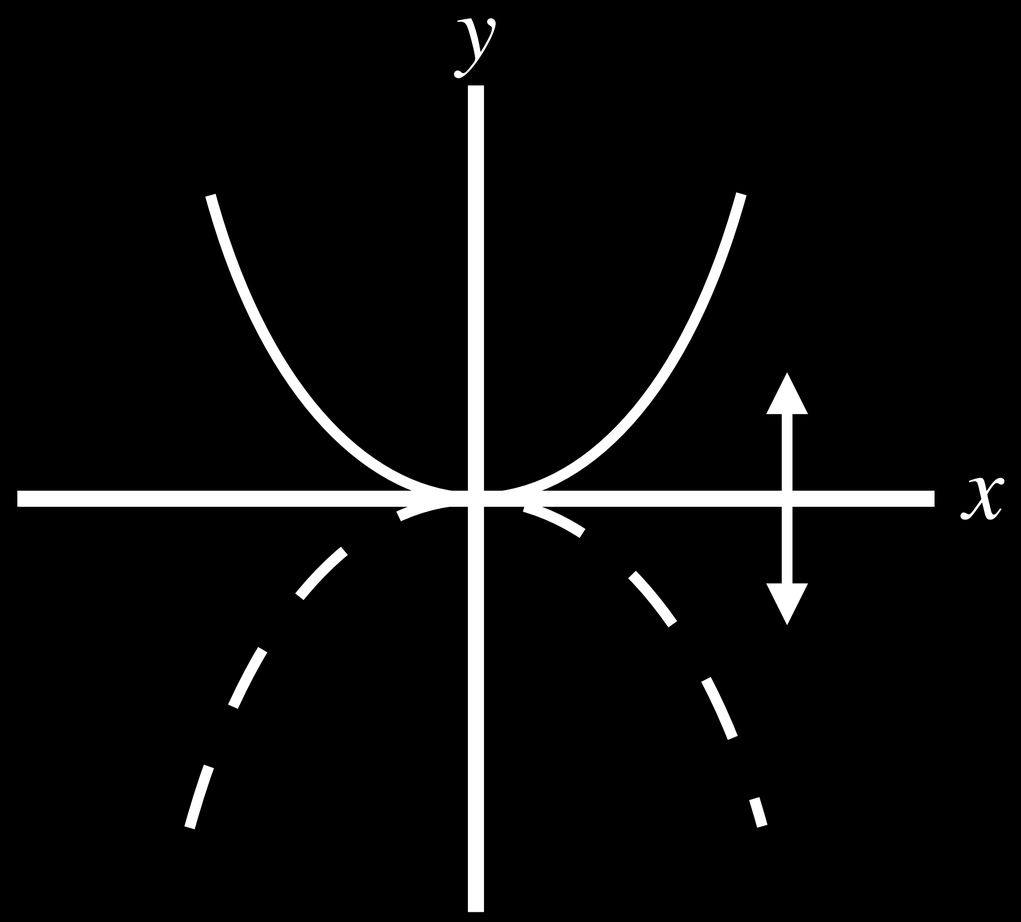
when a increases, graph gets steeper
when a decreases, graph gets gentler

(for the absolute value of a)

**if the graph has a non-0 y intercept of c,
$af(x)$ will have also have a y intercept of ac

Graph Transformation

hr



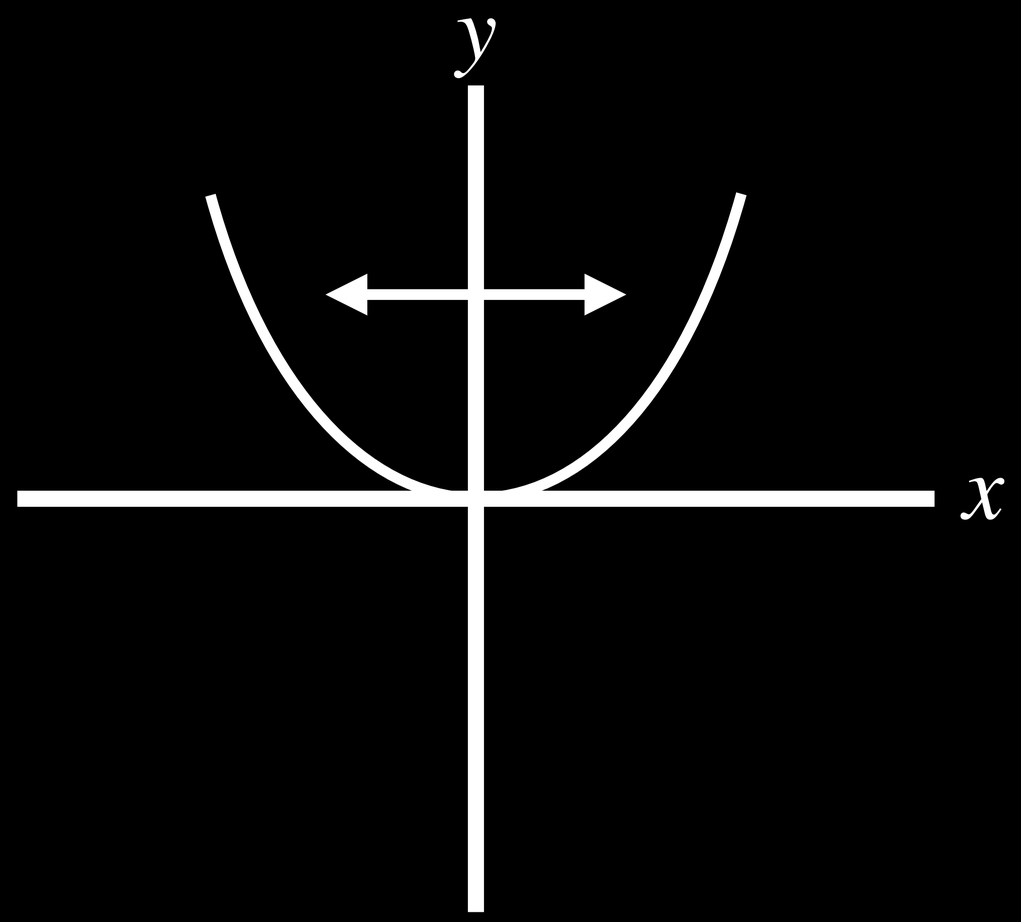
For a function $f(x)$,

$$\pm af(x)$$

graph is flipped about x axis

Graph Transformation

hr



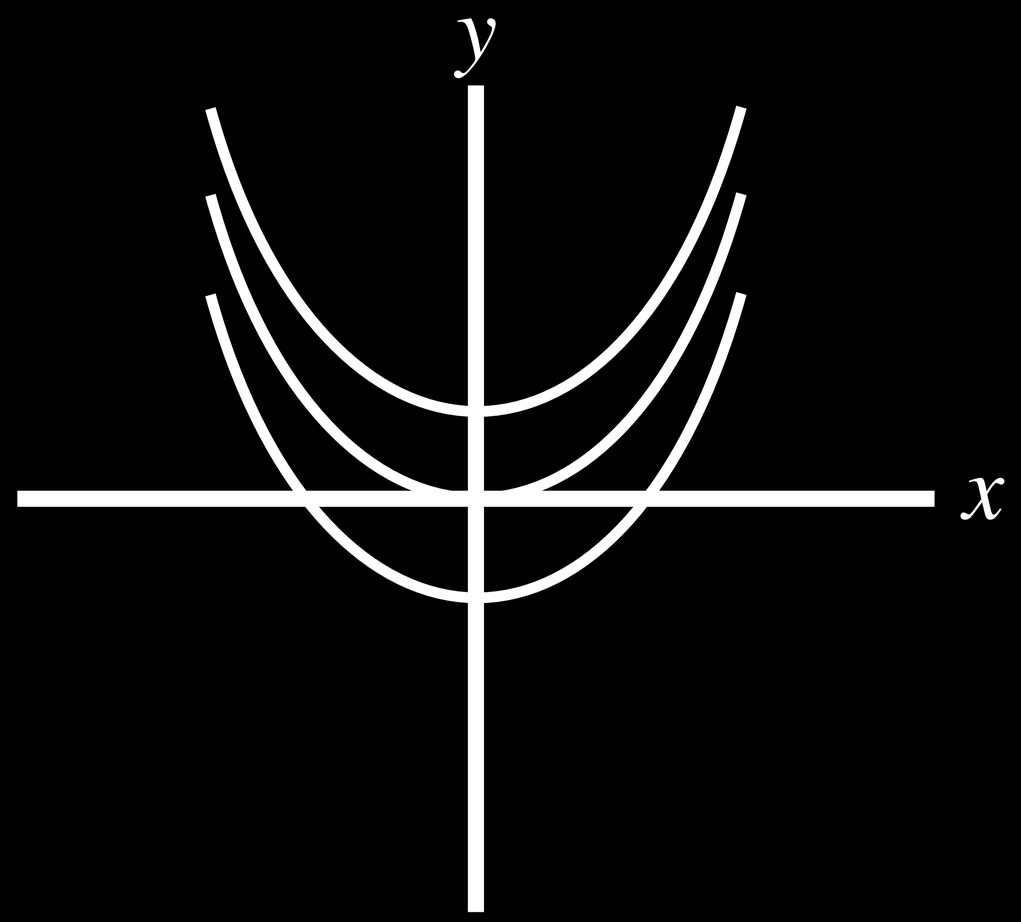
For a function $f(x)$,

$$f(\pm x)$$

graph is flipped about y axis

Graph Transformation

hr



For a function $f(x)$,

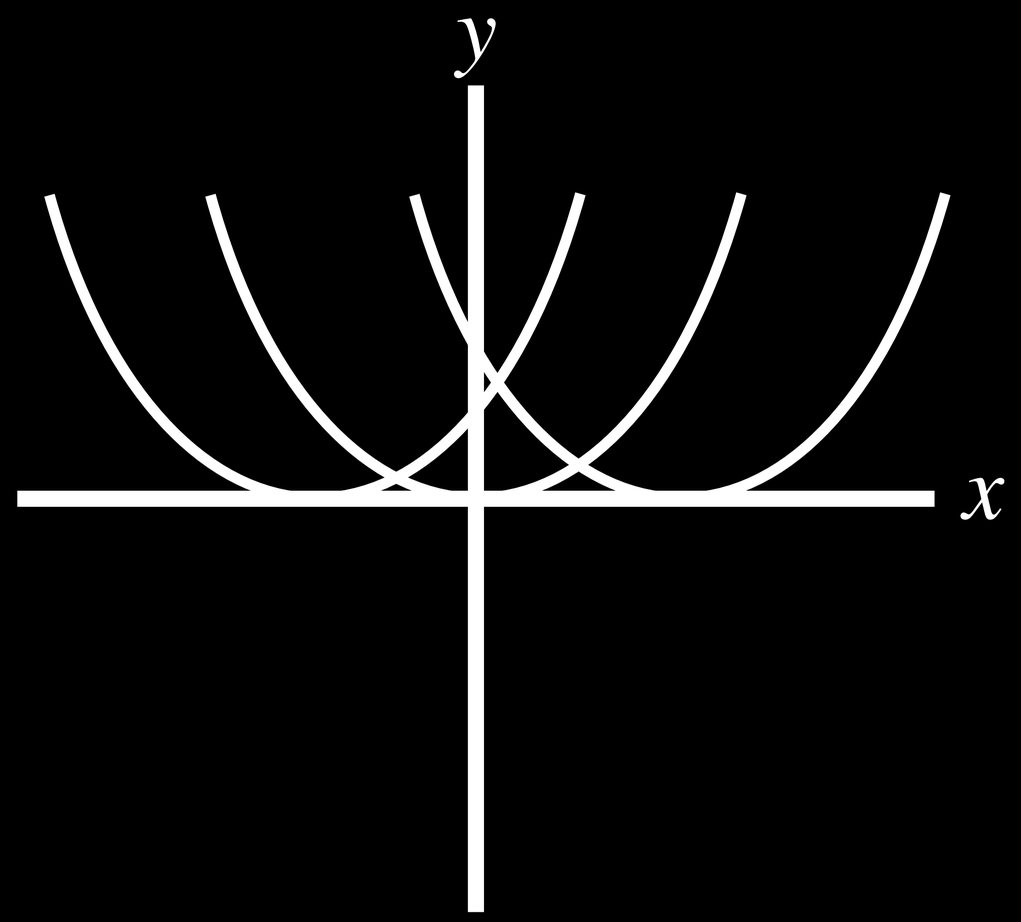
$$f(x) \pm c$$

moves graph up for $+c$

moves graph down for $-c$

Graph Transformation

hr



For a function $f(x)$,

$$f(x \pm h)$$

moves graph to the left for $+h$
moves graph to the right for $-h$

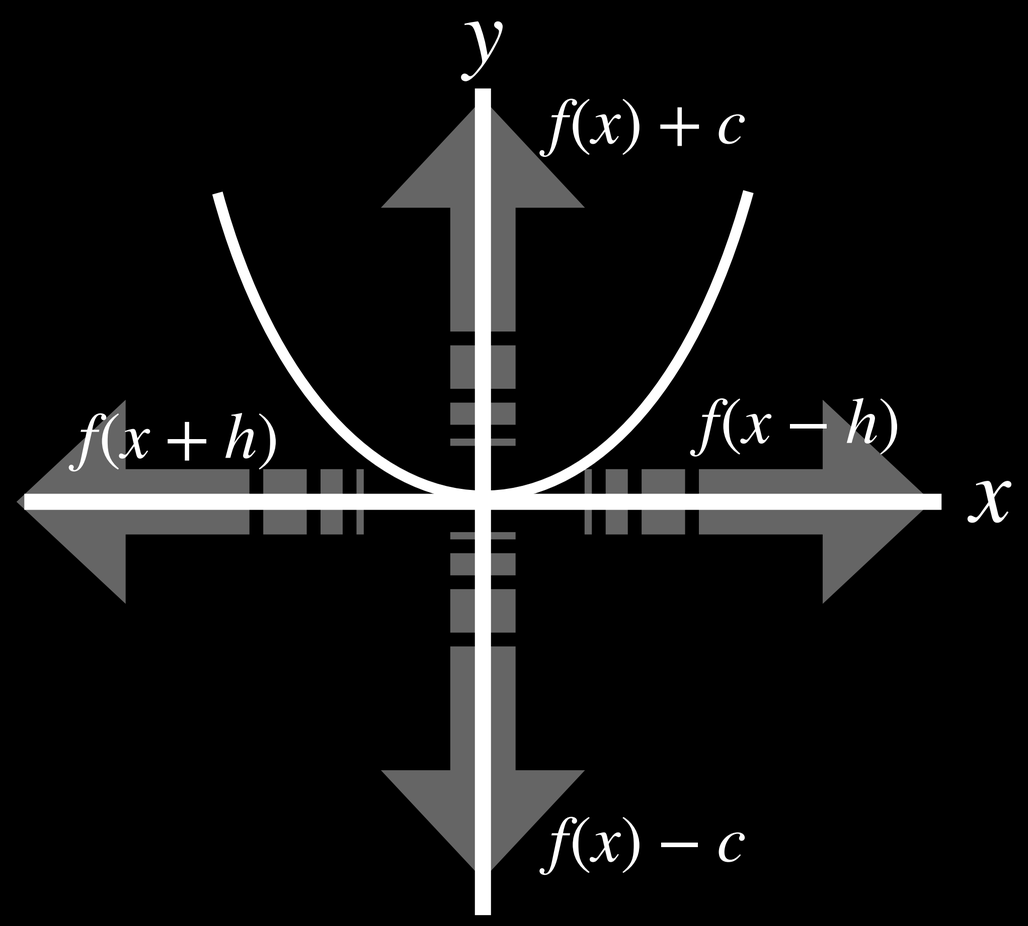
Graph Transformation

hr

In summary,

For a function $f(x)$:

to move $f(x)$,



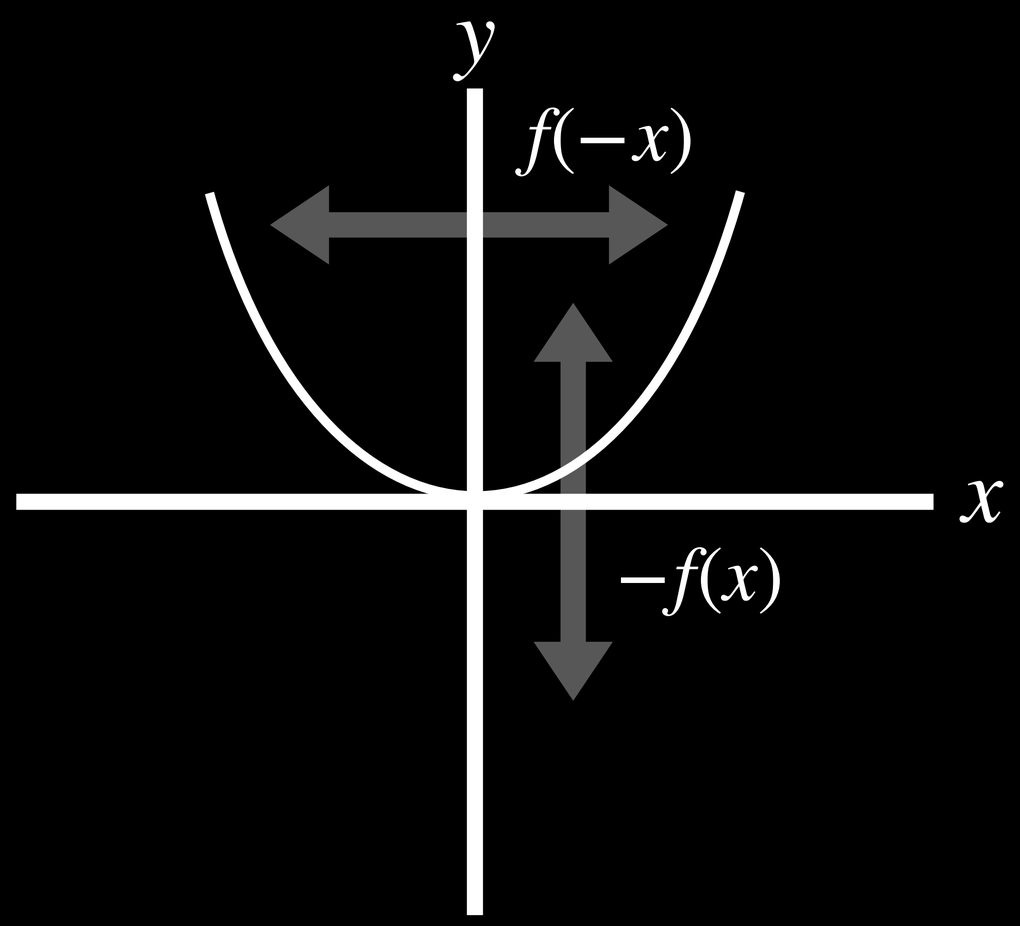
Graph Transformation

hr

In summary,

For a function $f(x)$:

to flip $f(x)$,



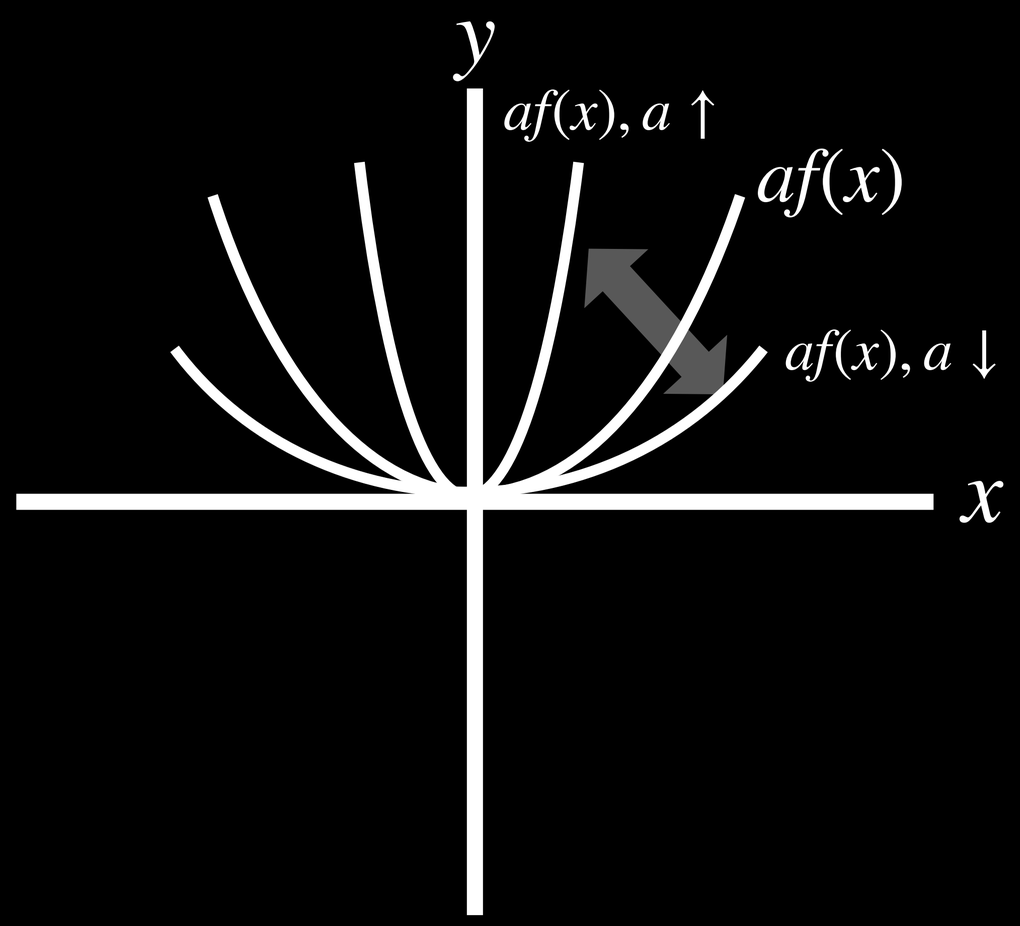
Graph Transformation

hr

In summary,

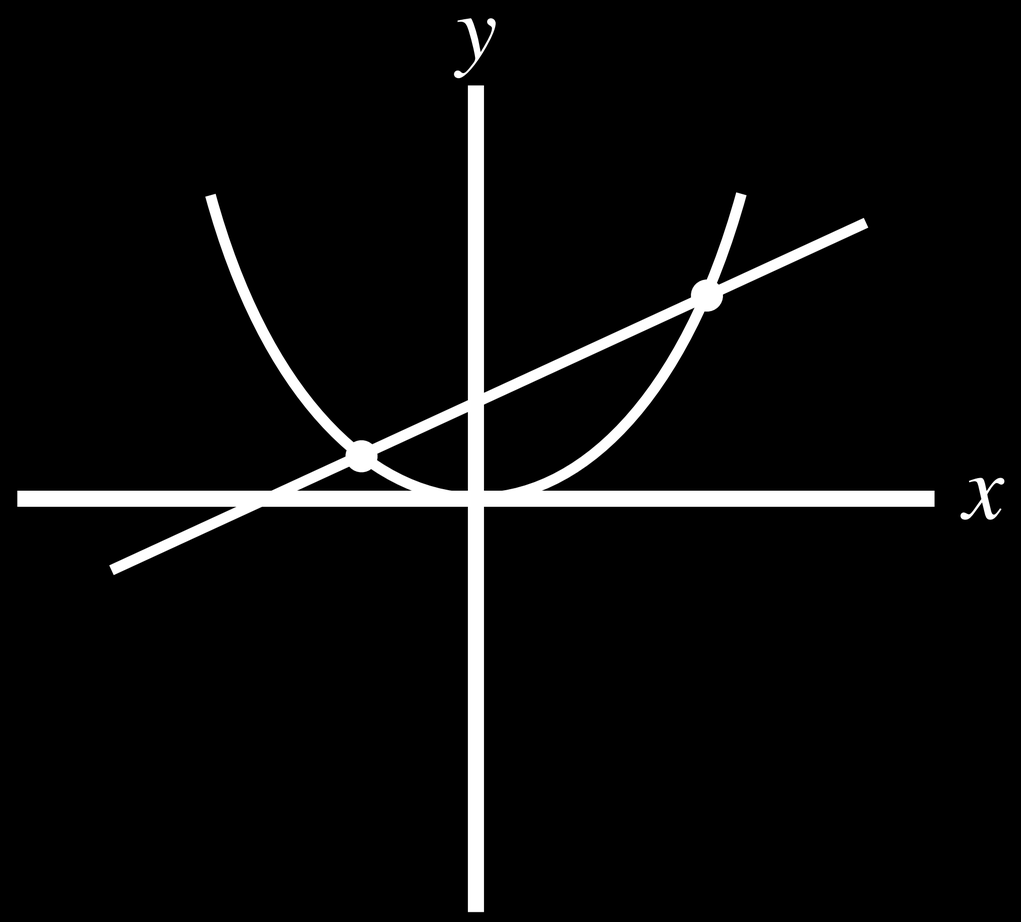
For a function $f(x)$:

to scale $f(x)$,



Graph Solutions

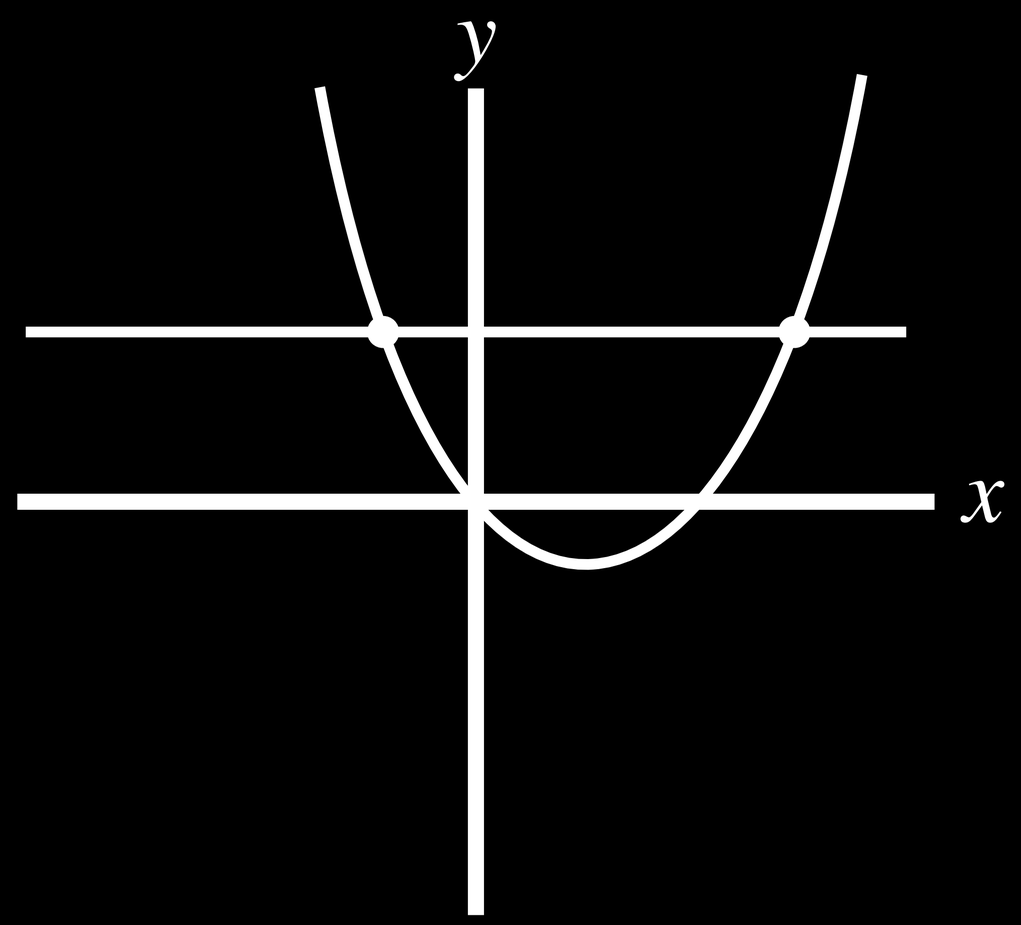
I like



Given two functions $f(x), g(x)$
intersections of graphs are solutions

Graph Solutions

I like

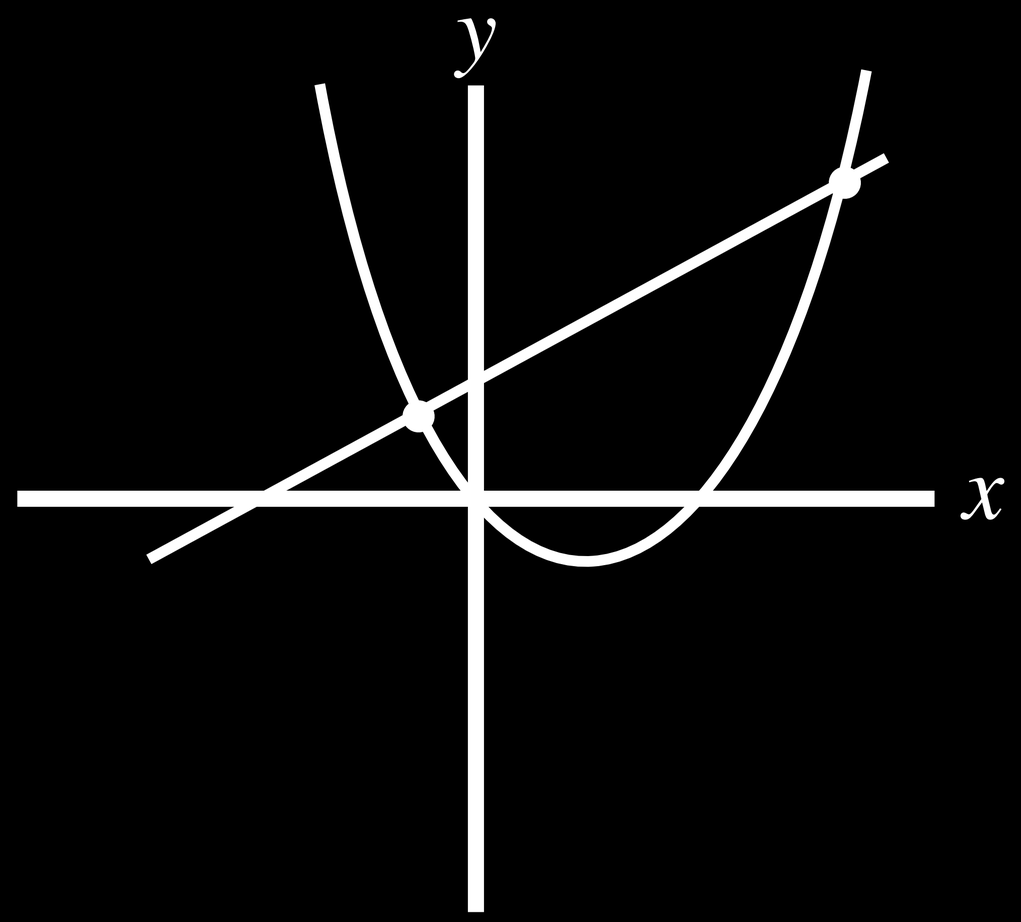


The graph $x^2 - 3x$ is plotted. Find $x^2 - 3x = 5$

plot out $y = 5$
find intersections

Graph Solutions

I like



The graph $x^2 - 3x$ is plotted. Find

$$-4 - \frac{5}{x} = -x\,.$$

$$-4 - \frac{5}{x} = -x$$

$$-4 - \frac{5}{x} + x = 0$$

$$x - 4 - \frac{5}{x} = 0$$

$\times x$ usually qns need you to multiply or divide x throughout

$$x^2 - 4x - 5 = 0$$

$+x$ $+5$ change to function given

$$x^2 - 3x + 0 = x + 5$$

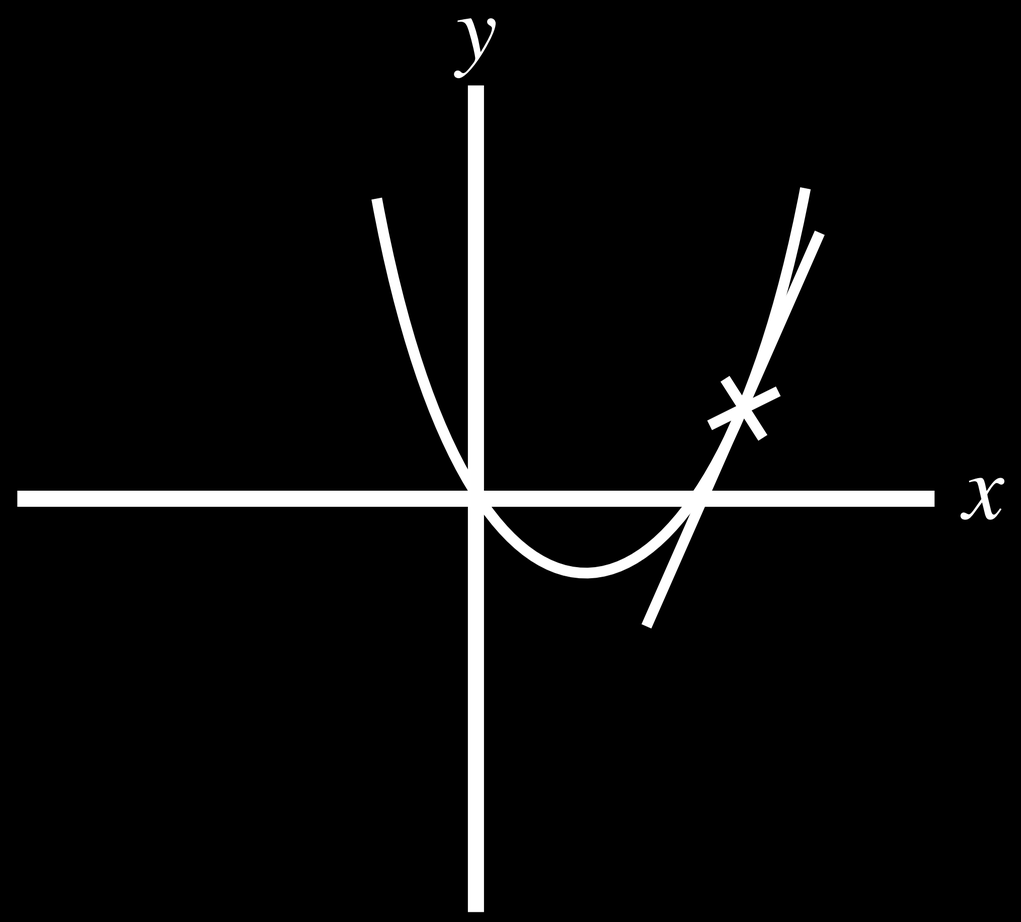
$$x^2 - 3x = x + 5$$

plot out $y = x + 5$

find intersections

Graph Tangents

differentiation but stupid

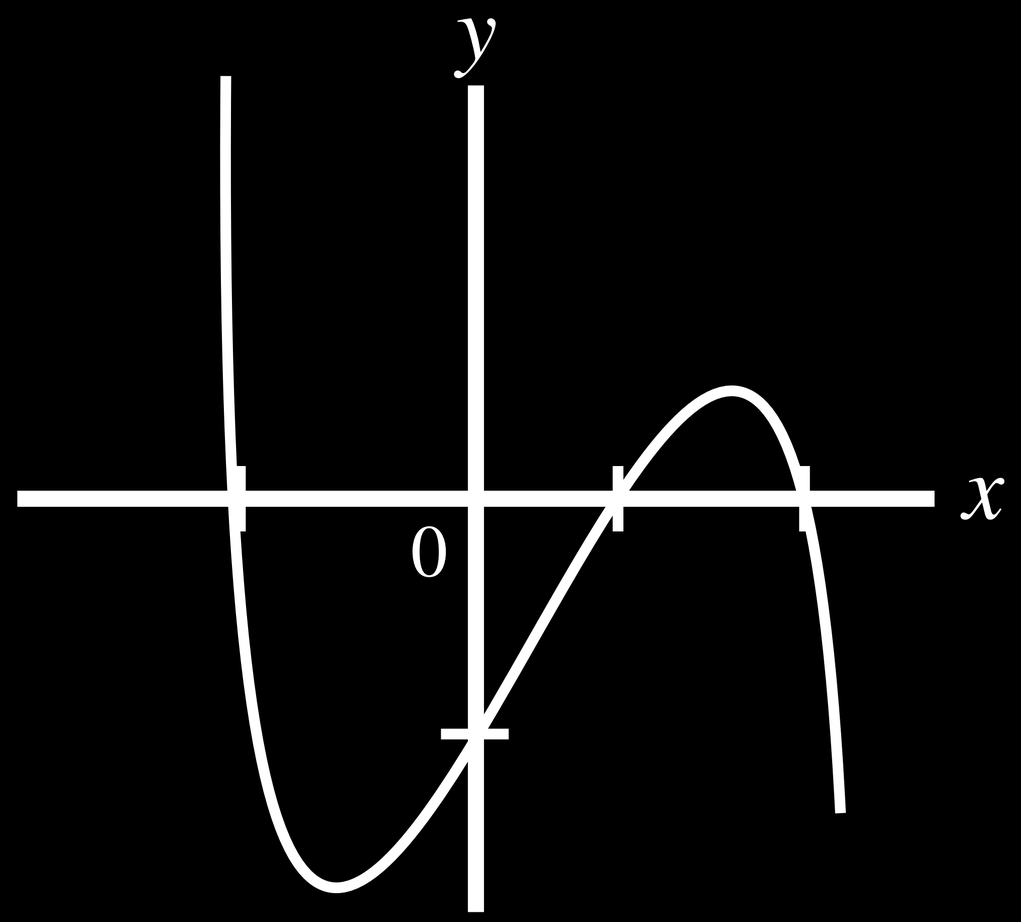


tangent = line that intersects the graph once

approximate tangent

Sketching & Plotting

I am having g fun

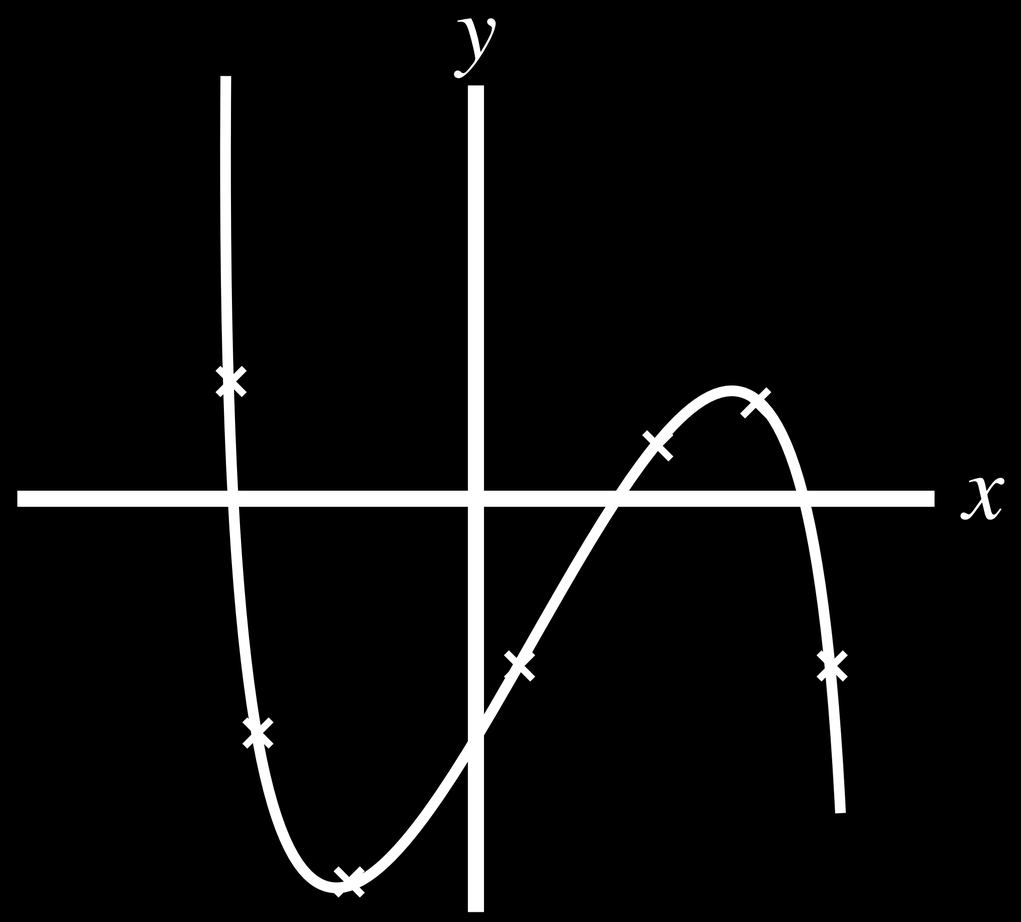


Sketching:

Label all x/y intercepts
axes
origin

Sketching & Plotting

I am having g fun



Plotting:

Scale is of correct size

usually stated in qn, if not, make appropriate scale

Line is smooth and continuous

no double line, no gaps, no sharp corners

Axes are labelled properly

x and y labelled, intervals are spaced evenly

Points are clearly plotted

Points are plotted with a clear "x", at the right coordinates

Trigonometry [E MATH]

Angle Presentation

cambrige

If the angle is in degrees, you are REQUIRED to put the degrees symbol
Else, it will be regarded as radians

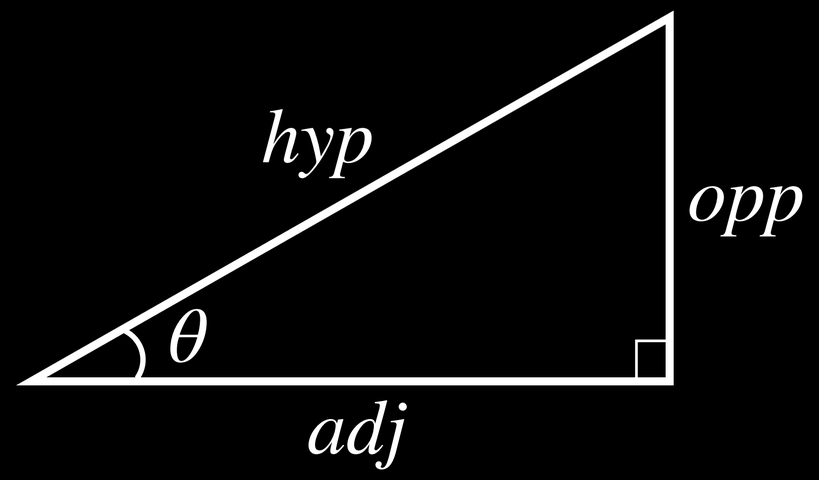
Trigonometric Ratios/SOH CAH TOA

Classic

$$\sin(\theta) = \frac{opp}{hyp}$$

$$\cos(\theta) = \frac{adj}{hyp}$$

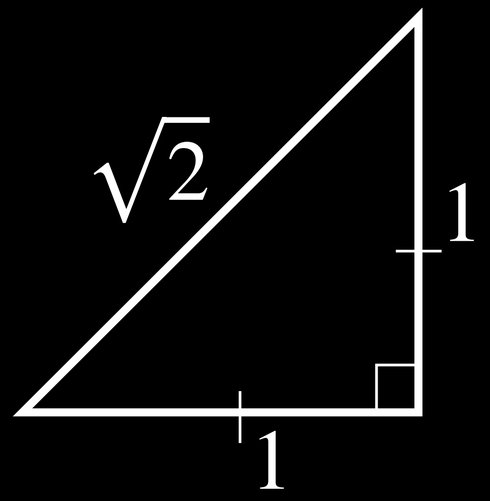
$$\tan(\theta) = \frac{opp}{adj} = \frac{\sin(\theta)}{\cos(\theta)}$$

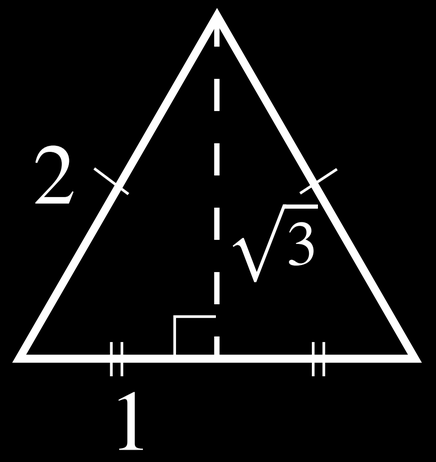


Special Angles

Classic

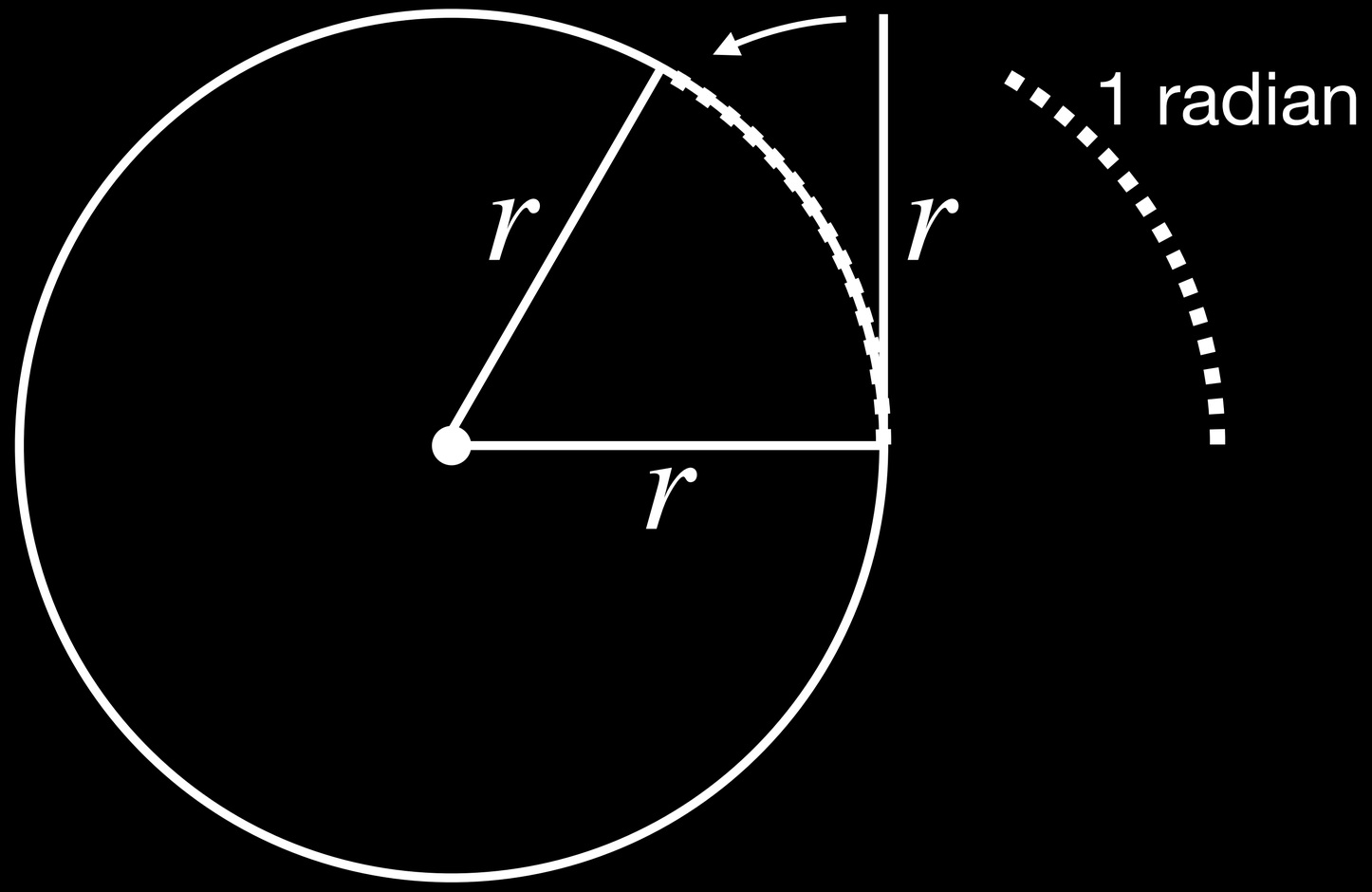
θ	0°	30°	45°	60°	90°
	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
$\sin(\theta)$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\cos(\theta)$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\tan(\theta)$	0	$\frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	undefined





Radians

(Hey guys radians are not really that cool)



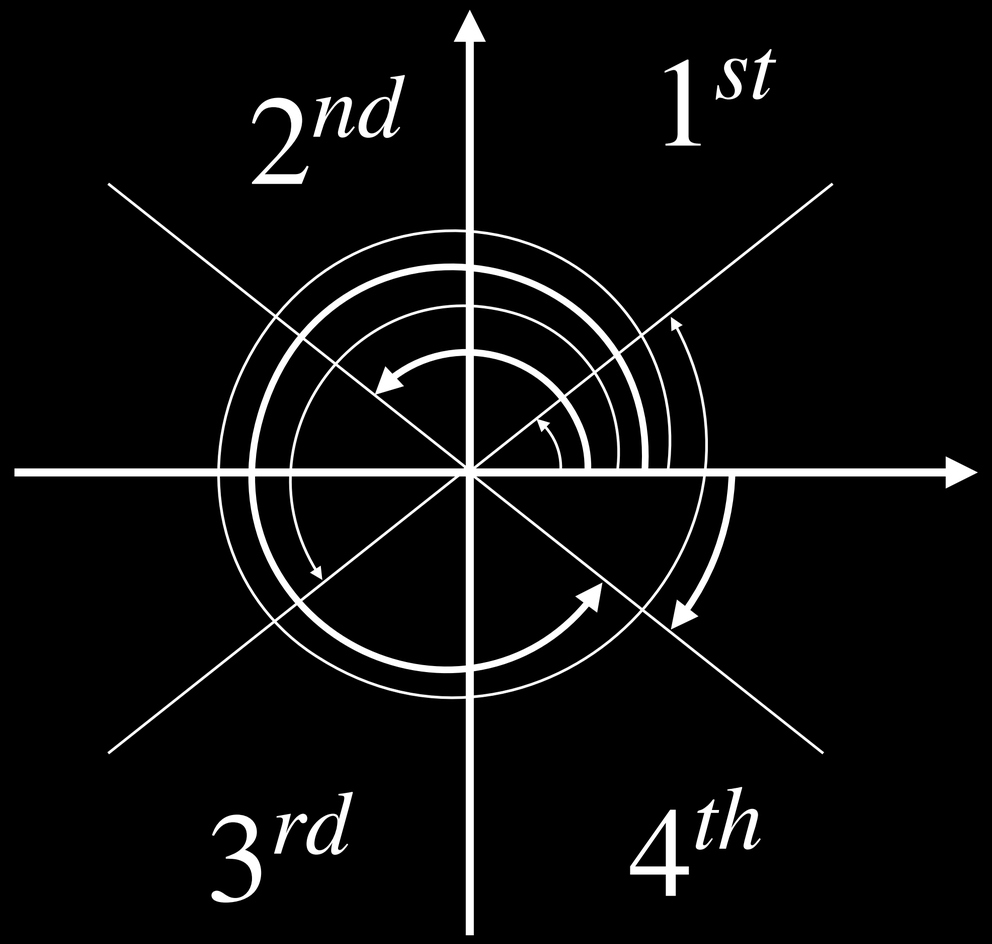
just another way to measure angles

$$2\pi\ rad = 360^\circ$$
$$\pi\ rad = 180^\circ$$
$$\therefore 1\ rad = \frac{180}{\pi}$$
$$1^\circ = \frac{\pi}{180}$$

you do not need to understand this visual

Quadrants

The beginning



From inside to outside:

$\theta = 30^\circ$, 1st quadrant

$\theta = 150^\circ$, 2nd quadrant

$\theta = 210^\circ$, 3rd quadrant

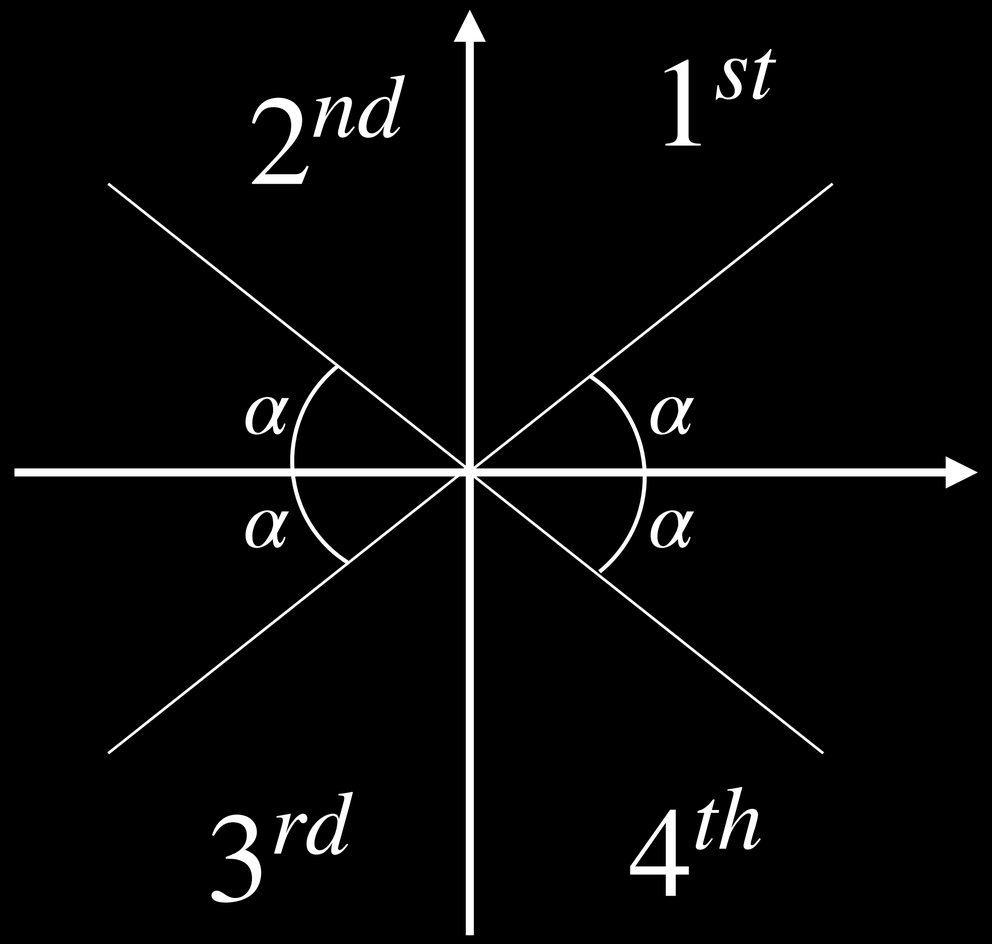
$\theta = 330^\circ$, 4th quadrant

$\theta = 390^\circ$, 1st quadrant

$\theta = -30^\circ$, 4th quadrant

Quadrants

The beginning

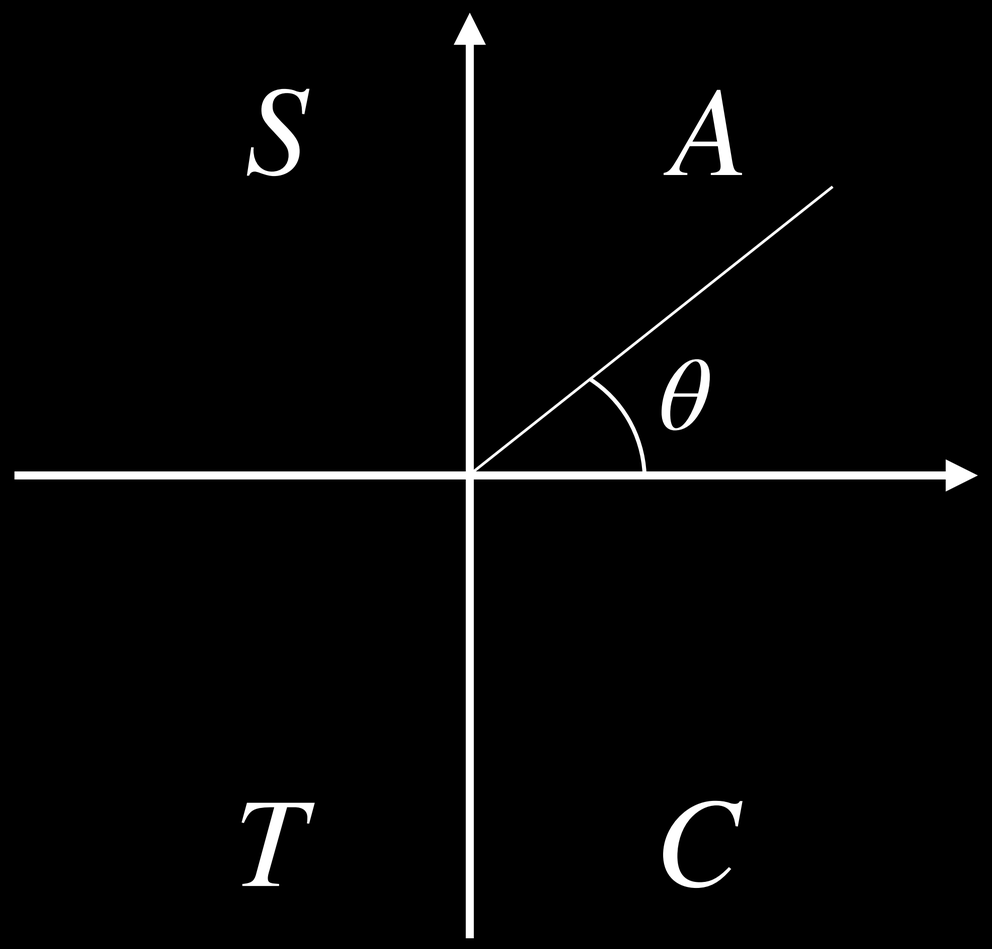


Basic angle: α
postive angle between x axis and angle

$$\theta = 30^\circ, \alpha = 30^\circ$$
$$\theta = 150^\circ, \alpha = 30^\circ(180^\circ - 150^\circ)$$
$$\theta = 210^\circ, \alpha = 30^\circ(210^\circ - 180^\circ)$$
$$\theta = 330^\circ, \alpha = 30^\circ(360^\circ - 330^\circ)$$
$$\theta = 390^\circ, \alpha = 30^\circ(390^\circ - 360^\circ)$$
$$\theta = -150^\circ, \alpha = 30^\circ(180^\circ - 150^\circ)$$

Quadrants

The beginning



For an angle θ,

If in 1st quadrant:

$$\sin(\theta) = \sin(\alpha)$$
$$\cos(\theta) = \cos(\alpha)$$
$$\tan(\theta) = \tan(\alpha)$$

all values +

If in 2nd quadrant:

$$\sin(\theta) = \sin(\alpha)$$
$$\cos(\theta) = -\cos(\alpha)$$
$$\tan(\theta) = -\tan(\alpha)$$

only sine +

If in 3rd quadrant:

$$\sin(\theta) = -\sin(\alpha)$$
$$\cos(\theta) = -\cos(\alpha)$$
$$\tan(\theta) = \tan(\alpha)$$

only tangent +

If in 4th quadrant:

$$\sin(\theta) = -\sin(\alpha)$$
$$\cos(\theta) = \cos(\alpha)$$
$$\tan(\theta) = -\tan(\alpha)$$

only cosine +

Trigonometry Unknowns

un knows

Given that $\tan(A) = \frac{3}{4}$, and that $180° \leq A \leq 360°$, find:

$\cos(A)$

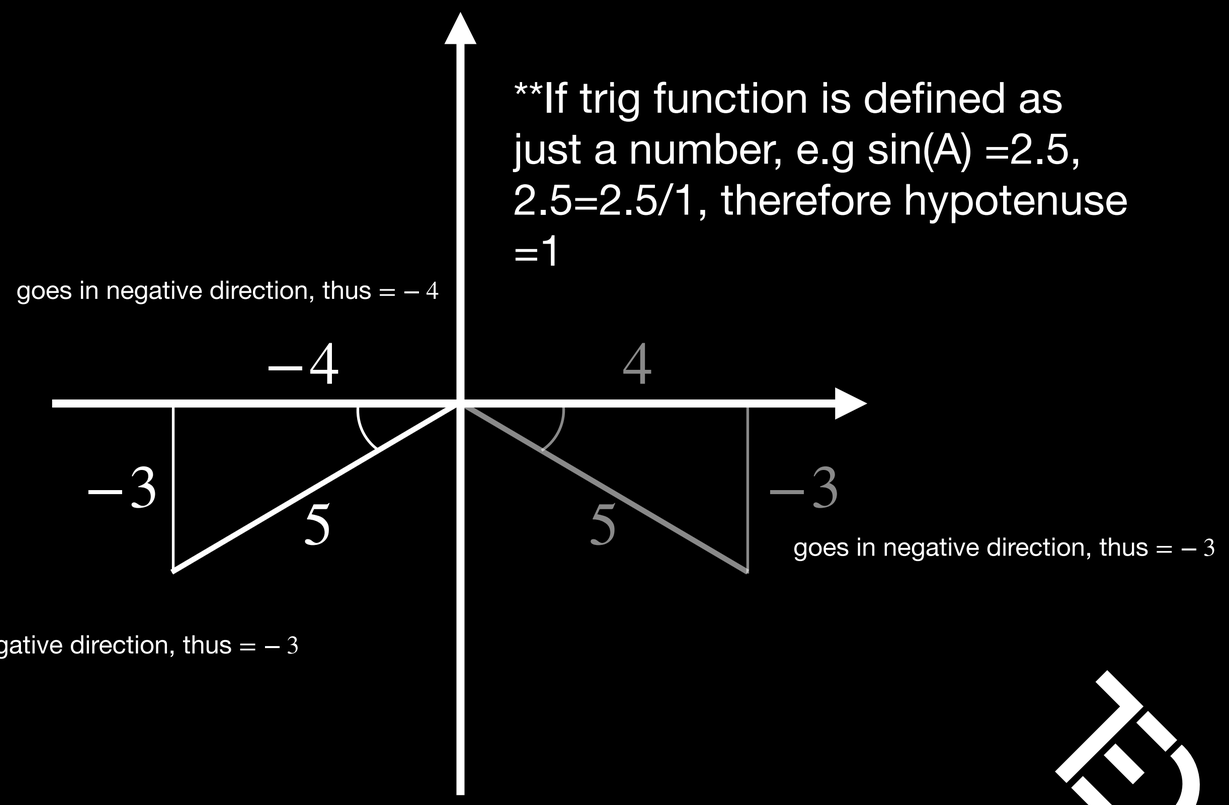
$$\tan(A) = \frac{-3}{-4} = \frac{opp}{adj}$$

$$\therefore hyp = \sqrt{(-3)^2 + (-4)^2}$$

$$hyp = 5$$

Since $\tan(A)$ is positive,
A lies in the third quadrant

**If trig function is defined as just a number, e.g sin(A) =2.5, 2.5=2.5/1, therefore hypotenuse =1



Trigonometry Unknowns

un knows

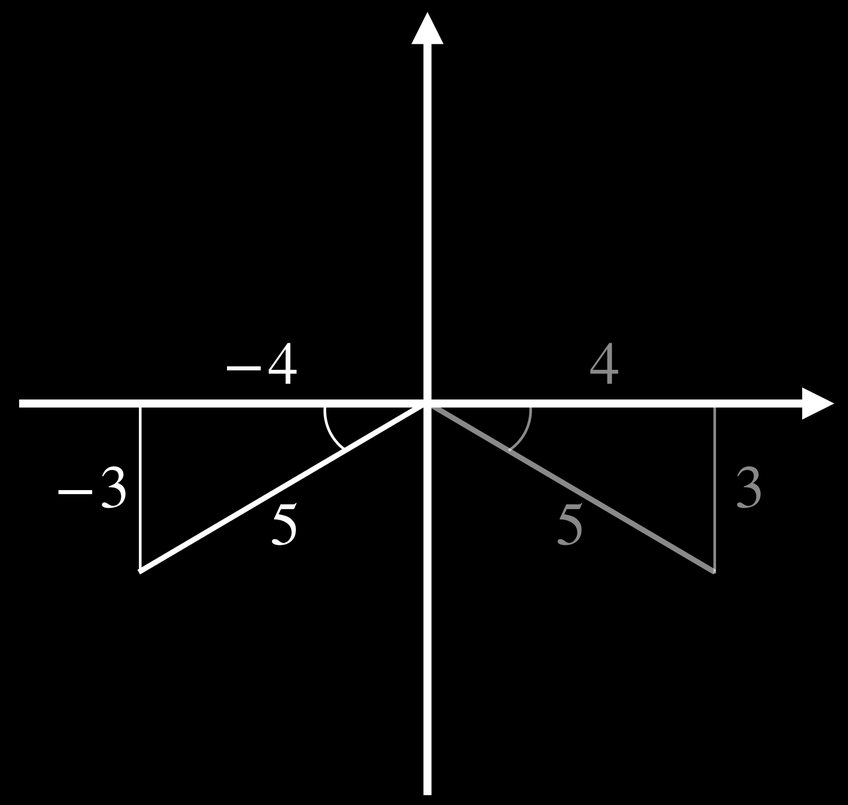
Given that $\tan(A) = \frac{3}{4}$, and that $180° \leq A \leq 360°$, find:

$\cos(A)$

$$\cos(A) = \frac{adj}{hyp}$$

$$\cos(A) = \frac{-4}{5}$$

$$\cos(A) = -0.8$$

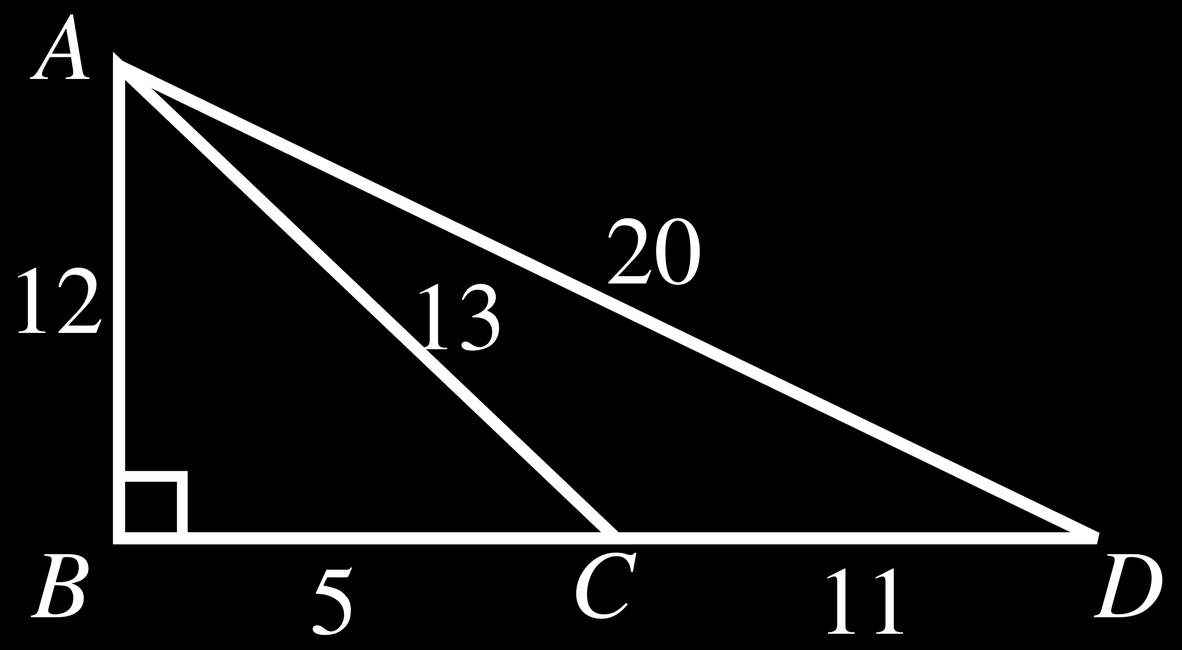


Geometric Diagrams & Trigonometry

trianlg

Express the following in exact form:

$\cos(BAC)$
$\tan(BDA)$
$\cos(ACD)$



Geometric Diagrams & Trigonometry

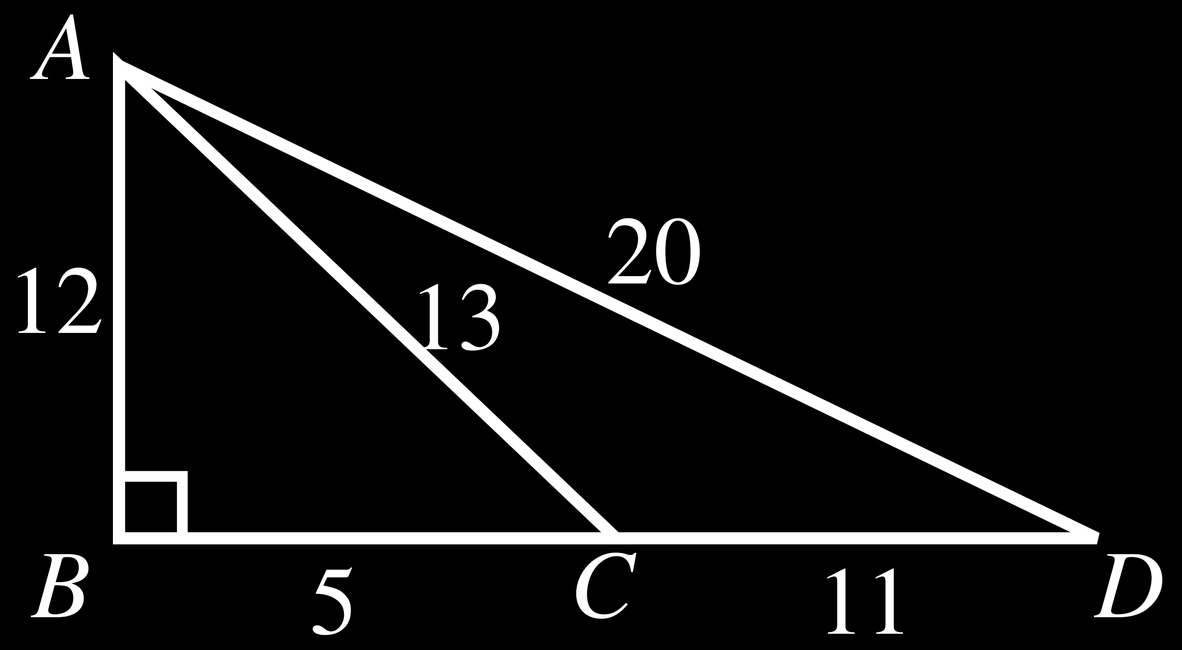
trianlg

Express the following in exact form:

$$\cos(BAC) = \frac{12}{13}$$

$$\tan(BDA) = \frac{12}{5+11}$$

$$= \frac{12}{16} = 0.75$$

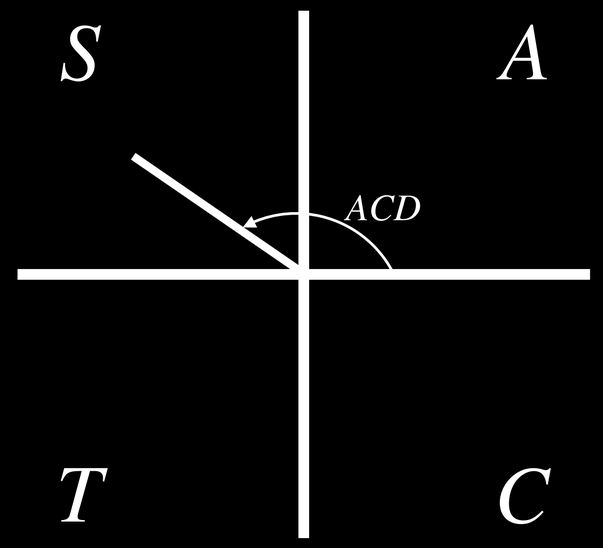


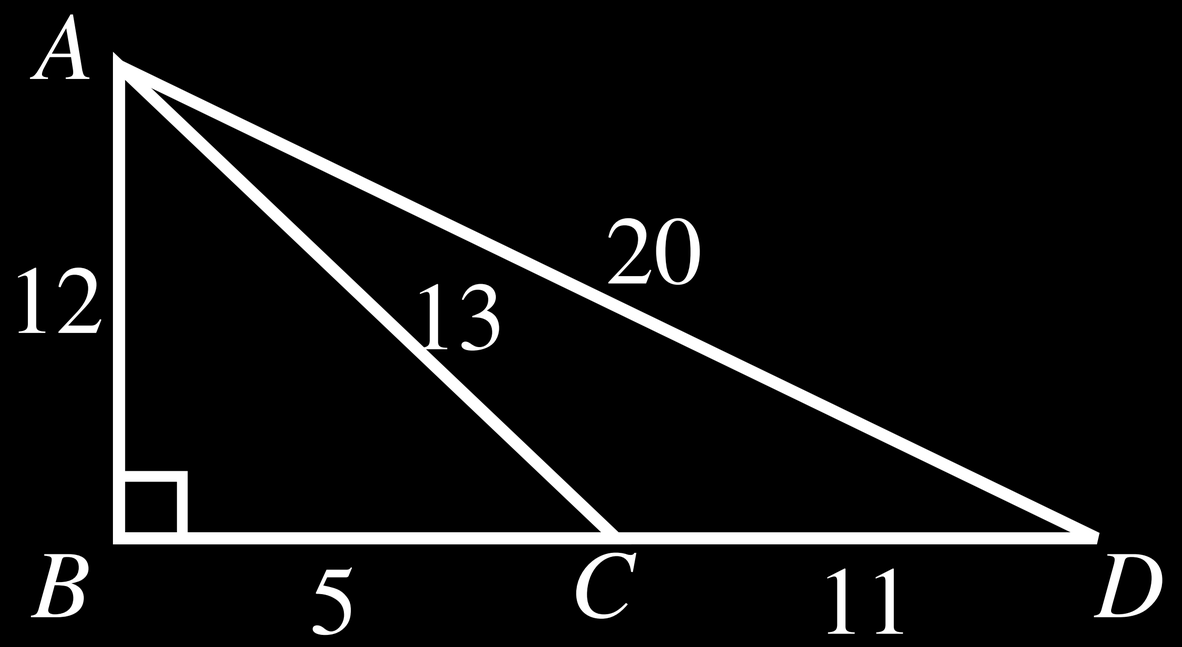
Geometric Diagrams & Trigonometry

trianlg

Express the following in exact form:

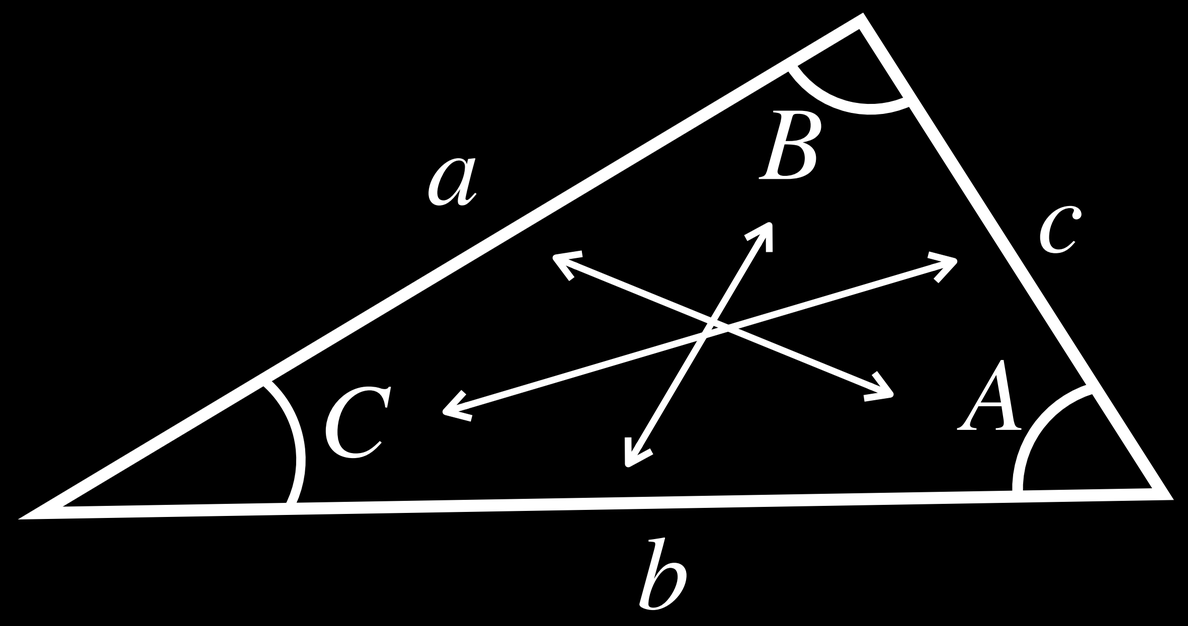
$$\cos(ACD) = \cos(180^\circ - ACB)$$
$$= -\cos(ACB)$$
$$= -\frac{5}{13}$$





Geometric Notation

trianlg



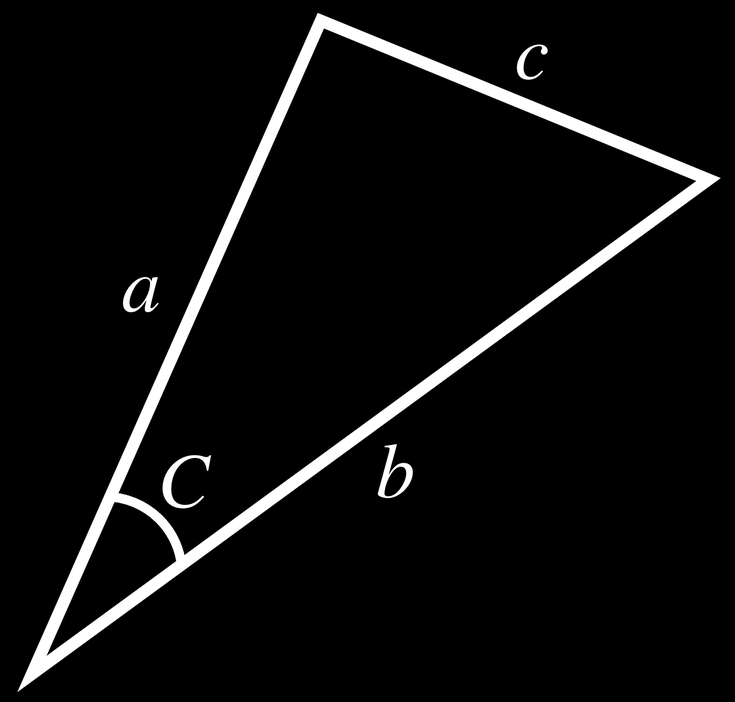
Capital Letters represent angles, small letters represent sides
For triangles, opposite lines/angles are labeled as the same letter

Area of Triangle Using Trigonometry

trianlg

For a triangle ABC,

$$A = \frac{1}{2}ab\sin(C)$$



Sine Rule

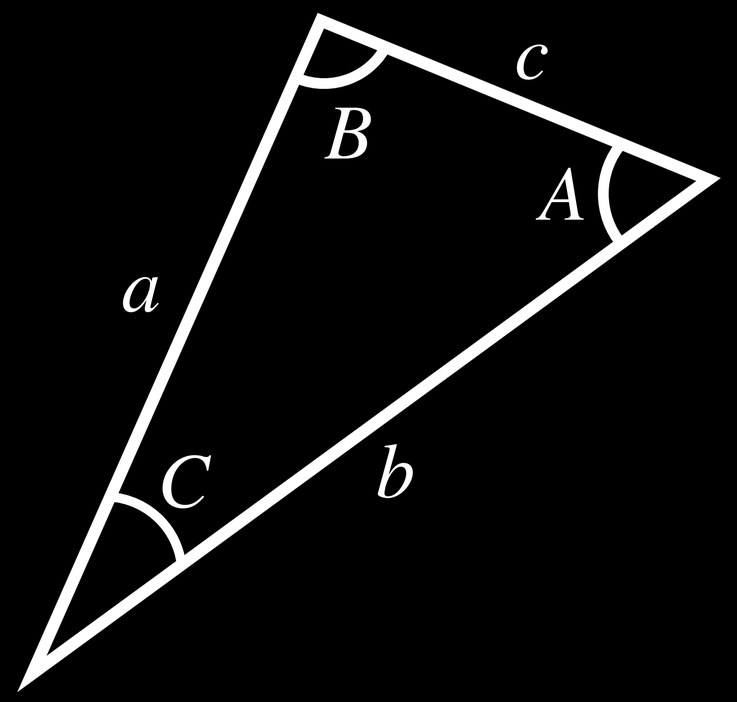
Sine rule

For a triangle ABC,

$$\frac{\sin(A)}{a} = \frac{\sin(B)}{b} = \frac{\sin(C)}{c}$$

or

$$\frac{a}{\sin(A)} = \frac{b}{\sin(B)} = \frac{c}{\sin(C)}$$



Sine Rule

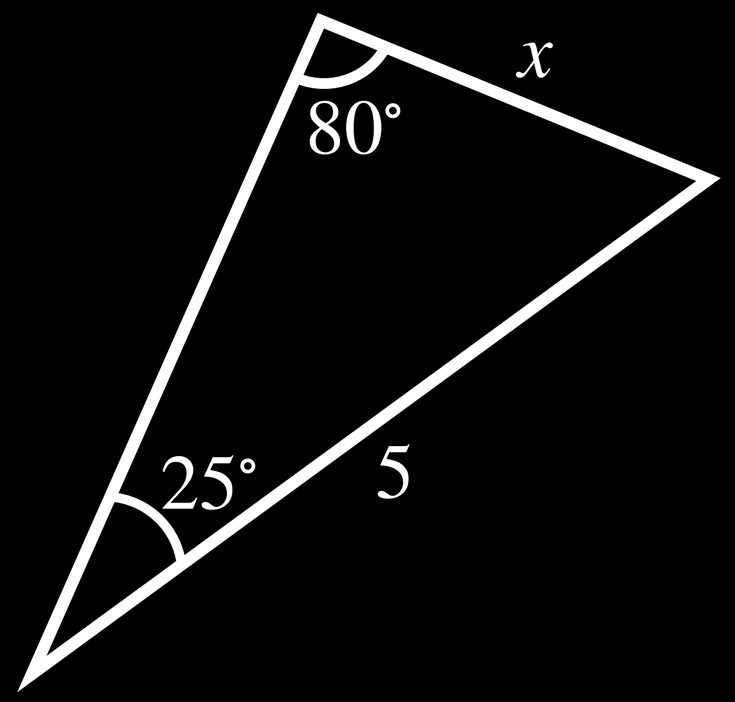
sien rule

Find side length x

$$\frac{x}{\sin(25^\circ)} = \frac{5}{\sin(80^\circ)}$$

$$x = \frac{5}{\sin(80^\circ)}(\sin(25^\circ))$$

$$x = 2.15(3.s.f)$$



When finding side with two pairs of opp angles and sides

Sine Rule

sien ruel

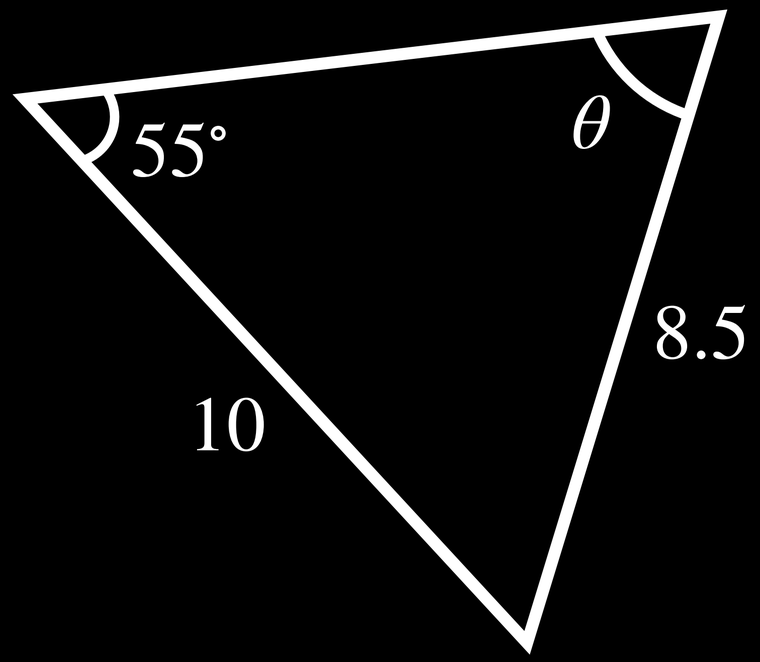
Find angle θ

$$\frac{\sin(\theta)}{10} = \frac{\sin(55^\circ)}{8.5}$$

$$\sin(\theta) = \frac{\sin(55^\circ)}{8.5} * 10$$

$$\theta = \sin^{-1}\left(\frac{\sin(55^\circ)}{8.5} * 10\right)$$

$$\theta = 74.5^\circ(1.d\,.\,p)$$

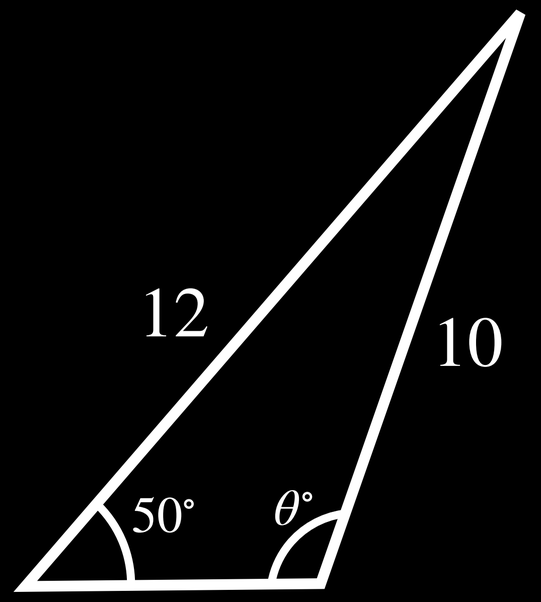


When finding angle with two pairs of opp angles and sides

Sine Rule - Ambiguity

SSA test

As sin(x) can have multiple positive values from 0 to 180 degrees, certain values found when using the sine rule may not be correct



Find angle θ

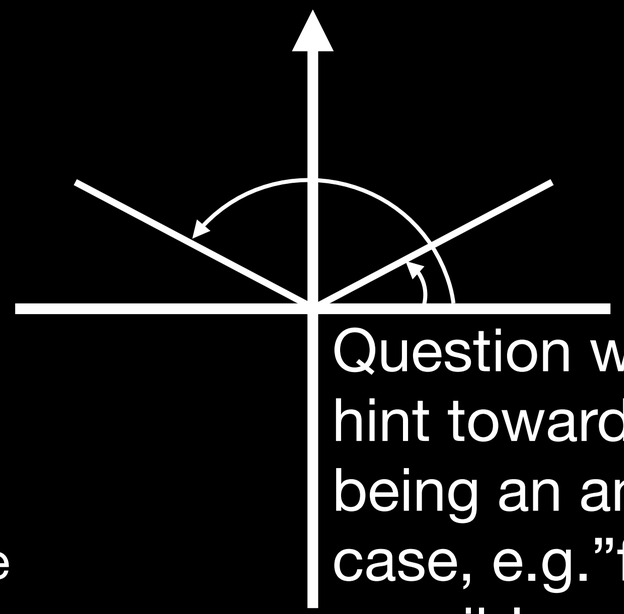
$$\frac{\sin(\theta)}{12} = \frac{\sin(50^\circ)}{10}$$

$$\sin(\theta) = \frac{12\sin(50^\circ)}{10}$$

$$\theta = \sin^{-1}\left(\frac{12\sin(50^\circ)}{10}\right) = 66.8^\circ$$

However, angle shown is obtuse

Thus, $\theta = 180 - \sin^{-1}\left(\frac{12\sin(50^\circ)}{10}\right)$
$= 113.2^\circ$



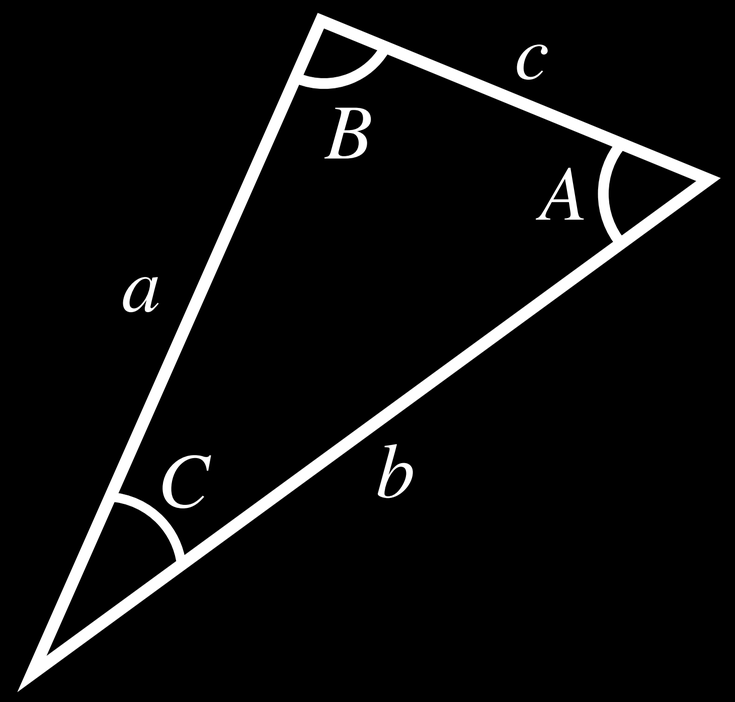
Question will usually hint towards there being an ambiguous case, e.g."find the possible values", "find two values", etc.

Cosine Rule

trianlg

For a triangle ABC,

$$a^2 = b^2 + c^2 - 2bc\cos(A)$$



Cosine Rule

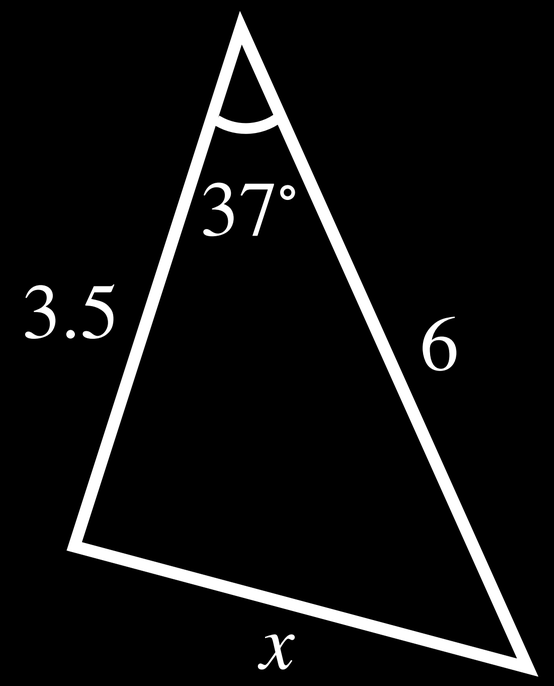
trianlg

Find side length x

$$x^2 = 3.5^2 + 6^2 - 2(3.5)(6)\cos(37^\circ)$$

$$x = \sqrt{3.5^2 + 6^2 - 2(3.5)(6)\cos(37^\circ)}$$

$$x = 3.84$$



find side length with opposite angle and adj. sides

Cosine Rule

trianlg

Find angle θ

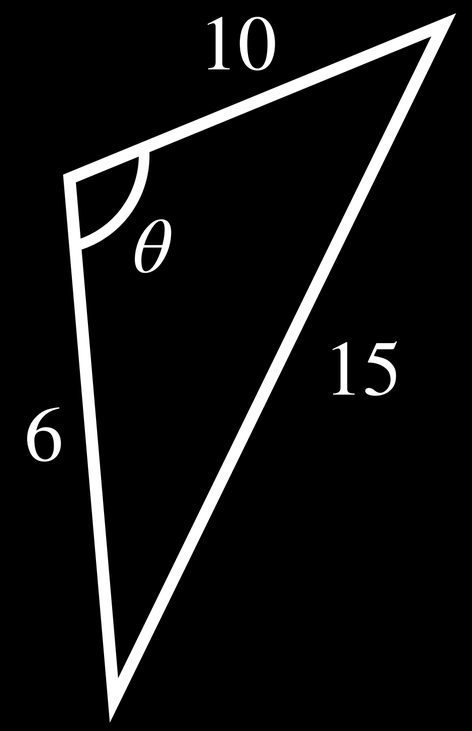
$$15^2 = 6^2 + 10^2 - 2(6)(10)\cos(\theta)$$

$$15^2 - 6^2 - 10^2 = -2(6)(10)\cos(\theta)$$

$$\frac{15^2 - 6^2 - 10^2}{-2(6)(10)} = \cos(\theta)$$

$$\theta = \cos^{-1}\left(\frac{15^2 - 6^2 - 10^2}{-2(6)(10)}\right)$$

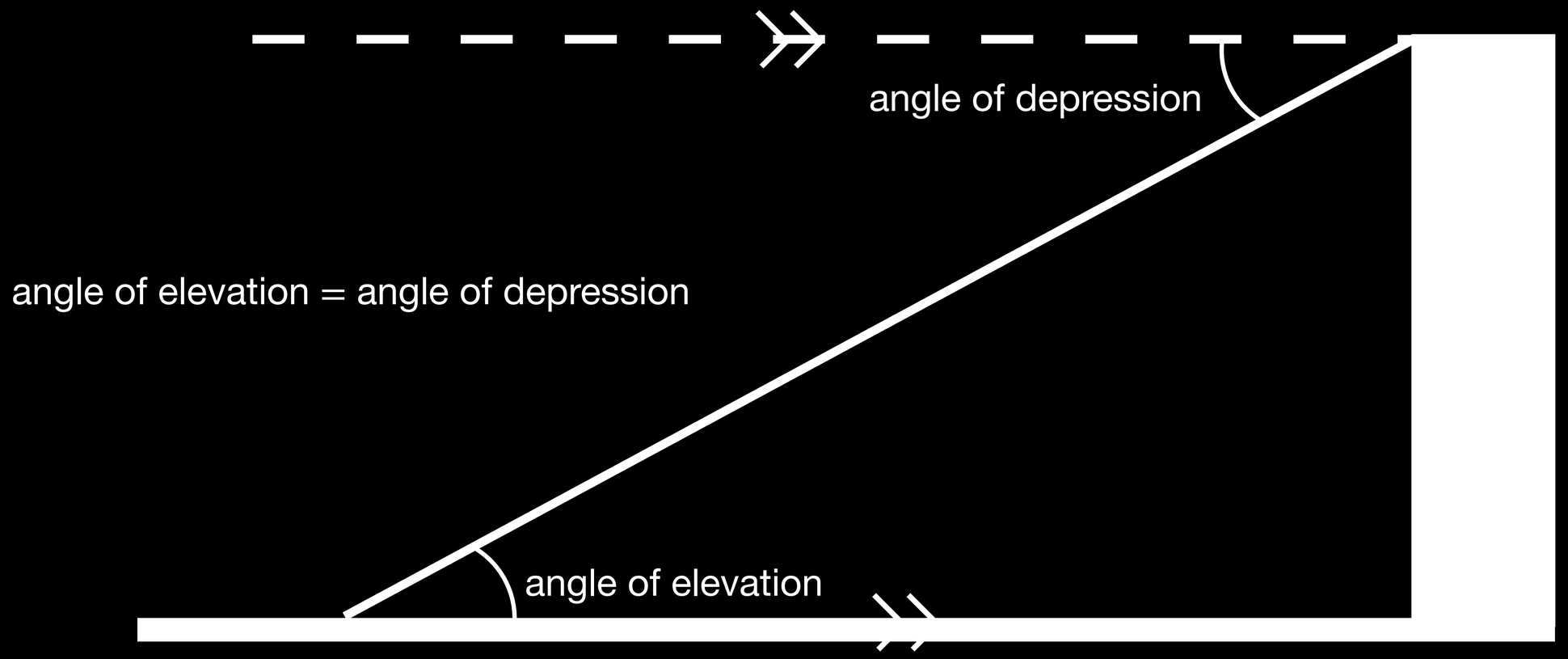
$$= 137.9^\circ$$



find angle with all 3 sides

Angle of Elevation/Depression

The big sad



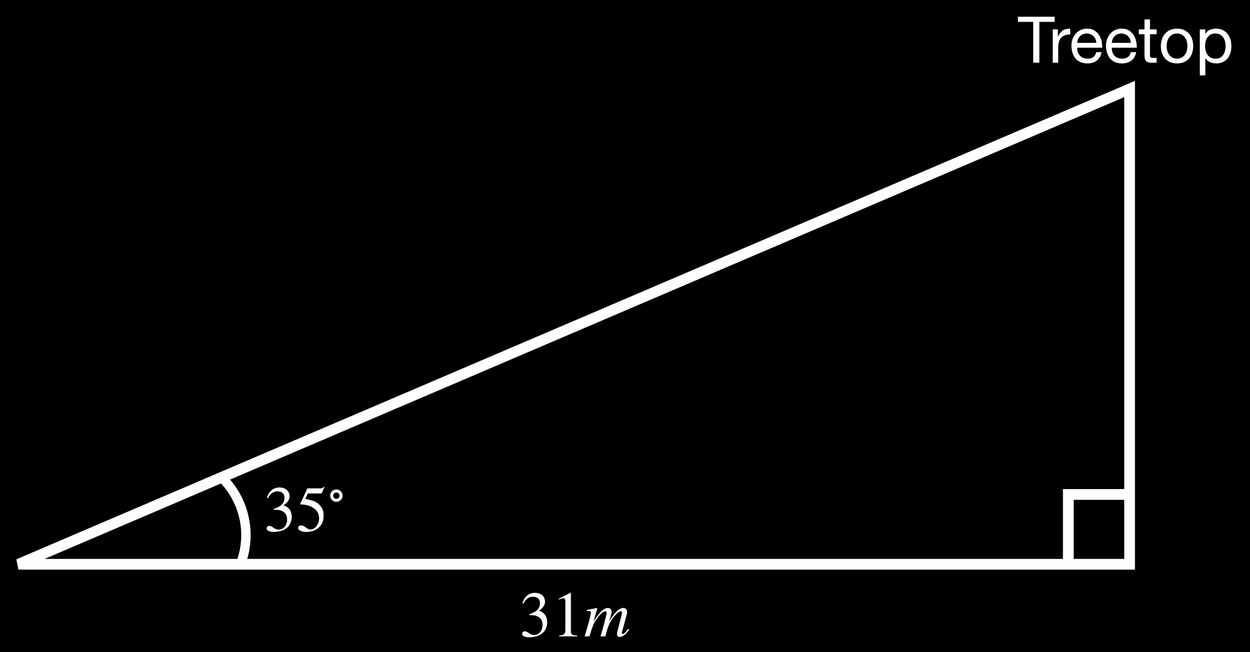
Angle of Elevation/Depression

The big sad

$\hat{\xi}$ looks up at a tree with an angle of elevation of 35° and is 31 meters away from the tree.

a) If $\hat{\xi}$ can see only the top of the tree, what is the height of said tree?
b) If $\hat{\xi}$ stands 42 meters away from the tree, what angle of elevation does $\hat{\xi}$ have to look at

**We always assume that height of $\hat{\xi}$ is 0, i.e $\hat{\xi}$'s eyes are on the ground



Angle of Elevation/Depression

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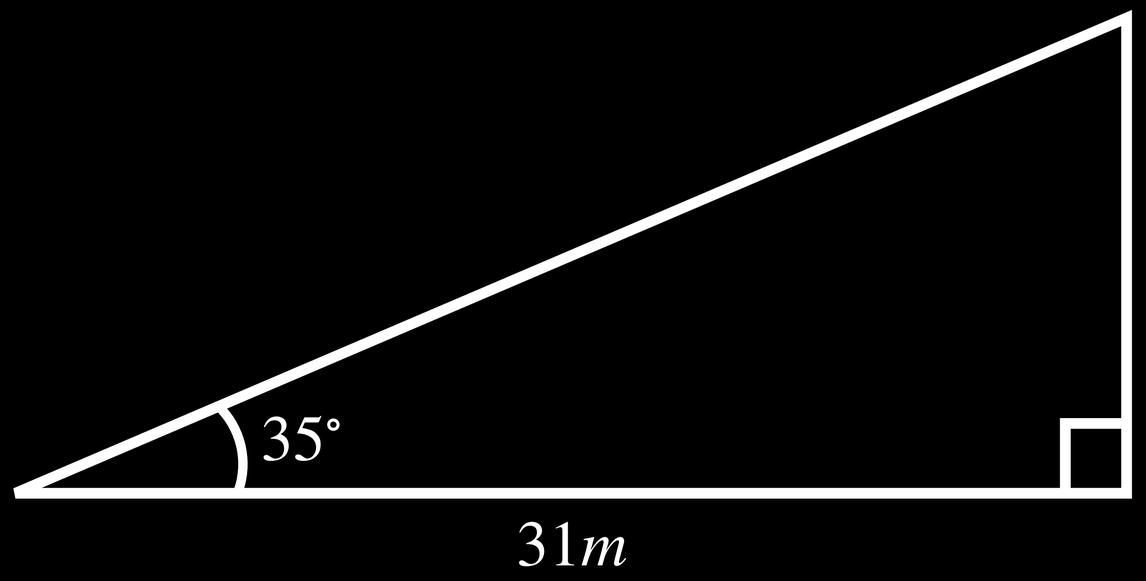
a) If $\hat{\xi}$ can see only the top of the tree, what is the height of said tree?

b) If $\hat{\xi}$ stands 42 meters away from the tree, what angle of elevation does $\hat{\xi}$ have to look at

$$\tan(35^\circ) = \frac{height}{31}$$

$$31\tan(35^\circ) = height$$

$$height = 21.7m(3.s.f)$$



Angle of Elevation/Depression

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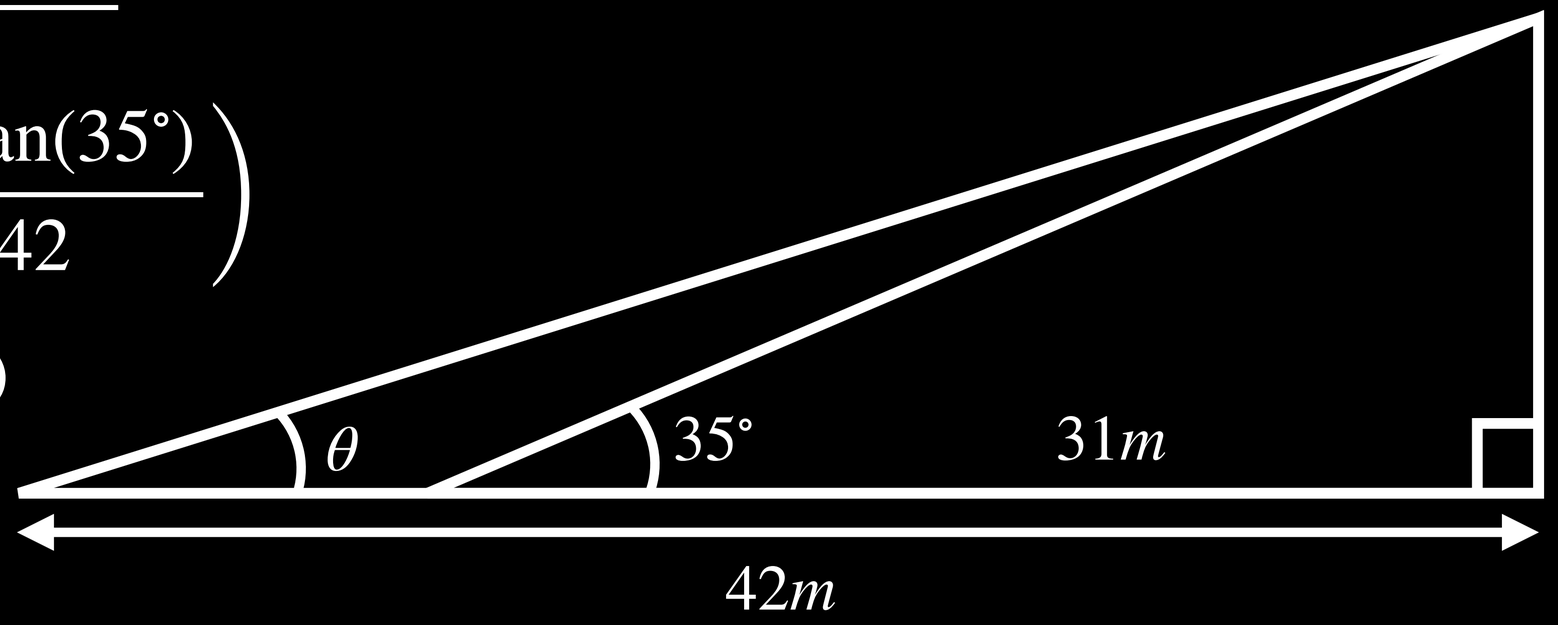
a) If $\hat{\xi}$ can see only the top of the tree, what is the height of said tree?

b) If $\hat{\xi}$ stands 42 meters away from the tree, what angle of elevation does $\hat{\xi}$ have to look at

$$\tan(\theta) = \frac{31\tan(35^\circ)}{42}$$

$$\theta = \tan^{-1}\left(\frac{31\tan(35^\circ)}{42}\right)$$

$$= 27.3^\circ(1.d.p)$$



Angle of Elevation/Depression

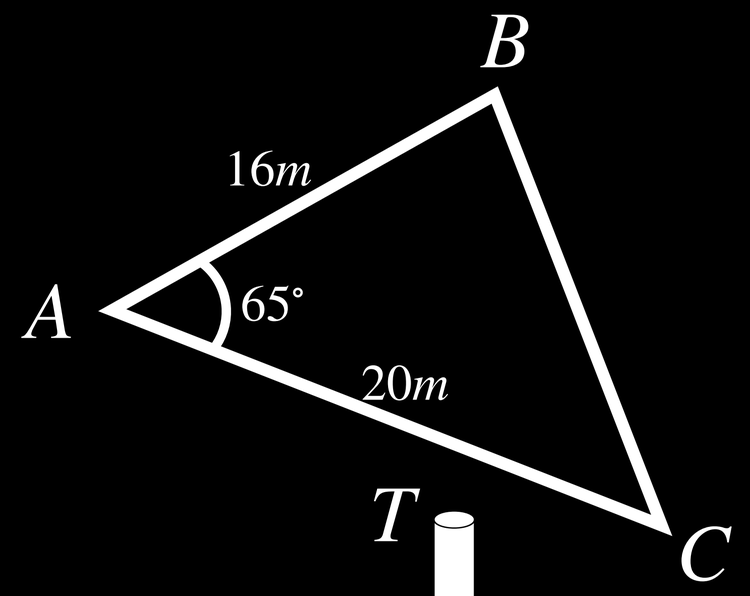
The big sad

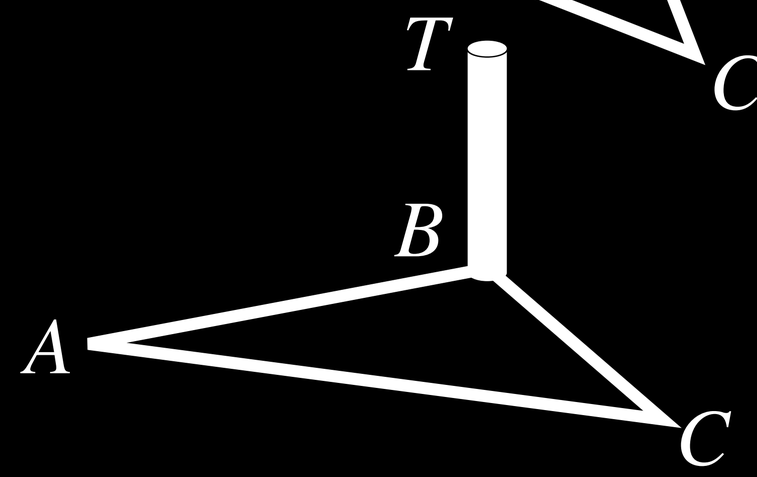
A triangular patch of land ABC has a tower BT at point B as shown.

a) Find the length of BC

b) Find the area of the patch of land

c) $\hat{\hbar}_{\zeta}$ walks along the line AC while looking at the top of the tower. If the maximum angle of elevation from $\hat{\hbar}_{\zeta}$ to T is 55°, what is the height of the tower?





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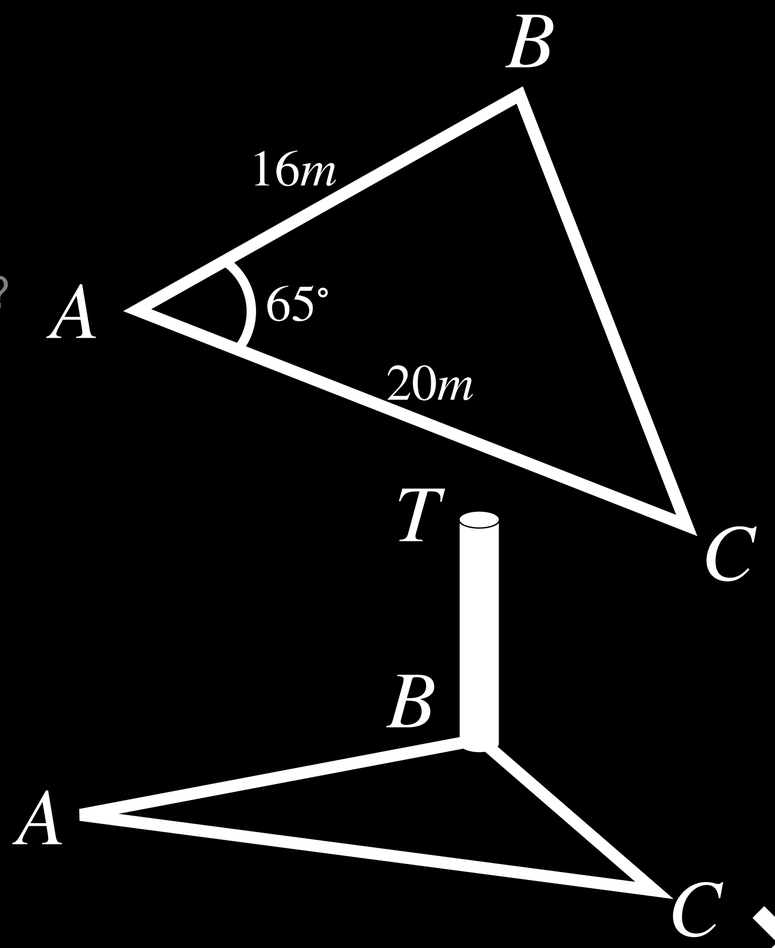

$$BC^2 = 16^2 + 20^2 - 2(16)(20)\cos(65^{\circ})$$

$$BC^2 = 656 - 640\cos(65^{\circ})$$

$$BC = \sqrt{656 - 640\cos(65^{\circ})}$$

$$BC = 19.6m(3.s.f)$$

Angle of Elevation/Depression

The big sad

A triangular patch of land ABC has a tower BT at point B as shown.

a) Find the length of BC

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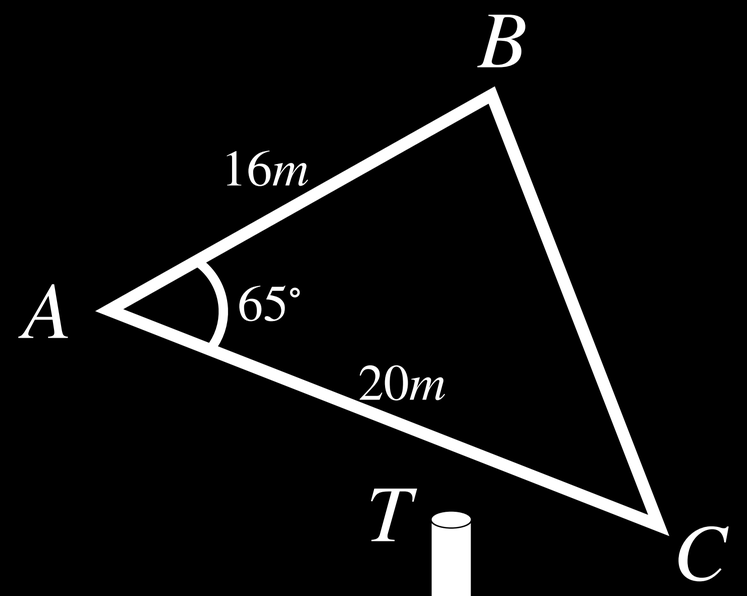
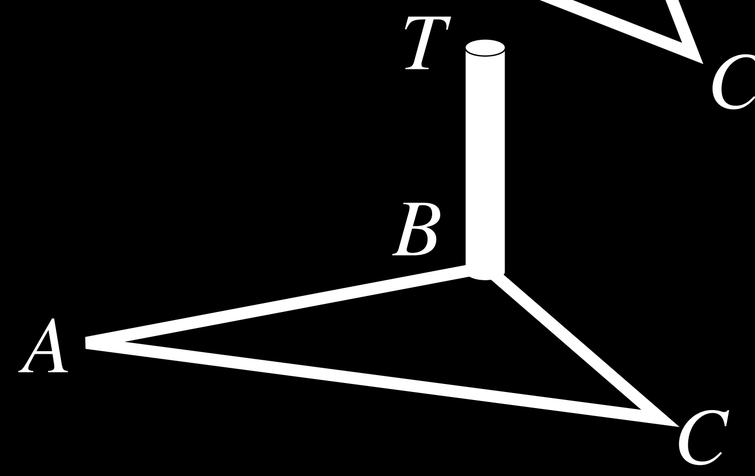


$$A = \frac{1}{2}(16)(20)\sin(65^{\circ})$$

$$A = 160\sin(65^{\circ})$$

$$A = 145m^2$$



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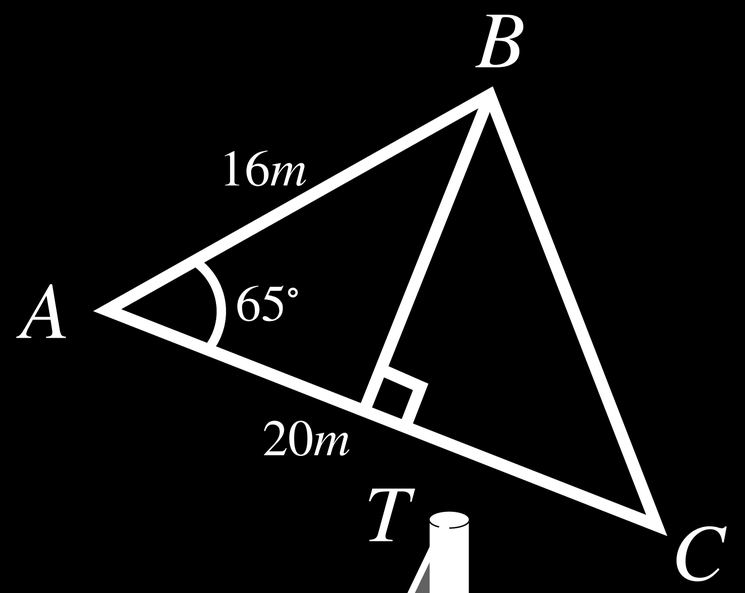


For the maximum angle of elevation, distance has to be shortest possible distance

Thus, distance between $\hat{\hbar}_{\zeta}$ and B at max angle is perpendicular distance between B and AC.

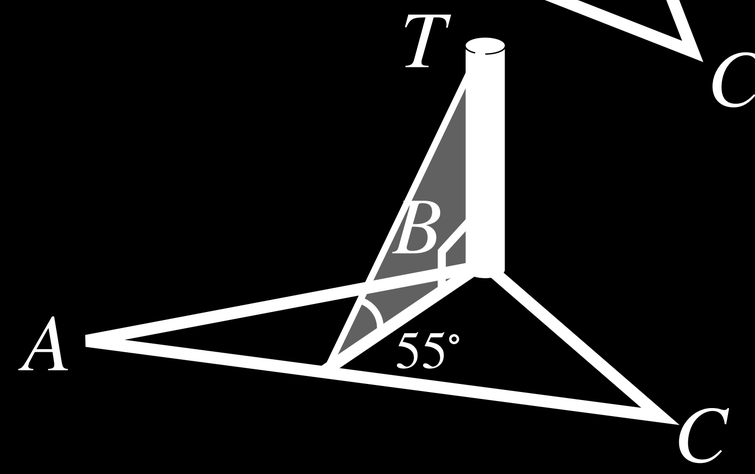
$$\sin(65^{\circ}) = \frac{\text{Perpedicular dist}}{16}$$

Perpendicular dist $= 16\sin(65^{\circ})$

$$\tan(55^{\circ}) = \frac{\text{tower height}}{16\sin(65^{\circ})}$$

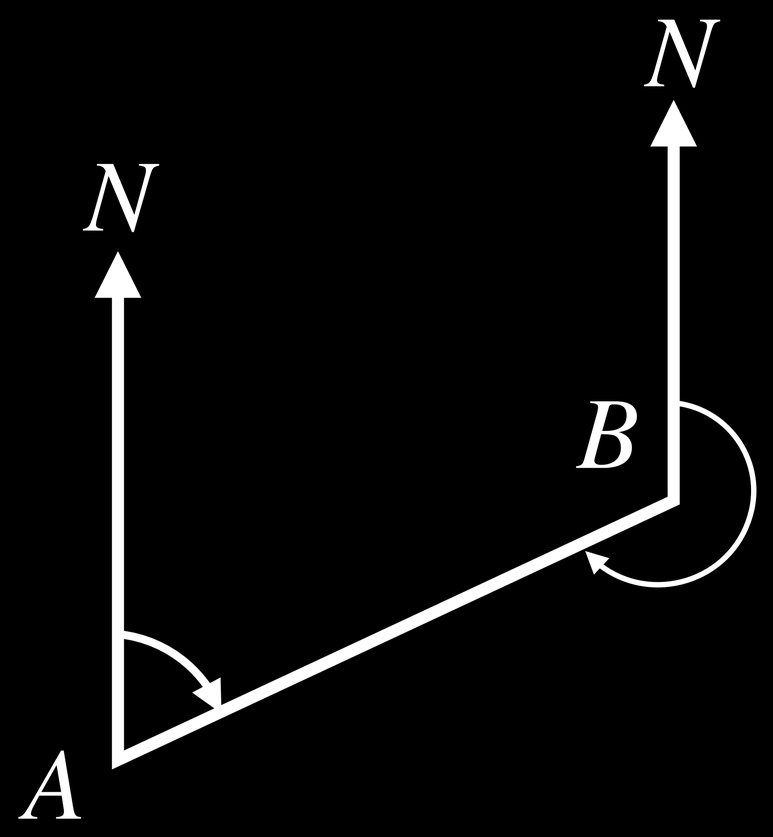
$16\tan(55^{\circ})\sin(65^{\circ}) =$ tower height

tower height $= 20.7m(3.s.f)$



Compass Bearings

goraphy



"Bearing of A from B":

Start from B
Rotate clockwise from north to A

"Bearing of B from A":

Start from A
Rotate clockwise from north to B

Bearing should have 3 digits before decimal point

$105^\circ = 105^\circ$
$35^\circ = 035^\circ$
$7^\circ = 007^\circ$
$242.4^\circ = 242.4^\circ$
$35.1^\circ = 035.1^\circ$
$3.4^\circ = 003.4^\circ$

3D Diagrams

2D+

A cube of side length 5cm is shown. Find the length of the diagonal corners shown.

diagonal is part of right triangle shaded

base is diagonal of one face, height is one side of cube

Face Diagonal $= \sqrt{2(5^2)}$

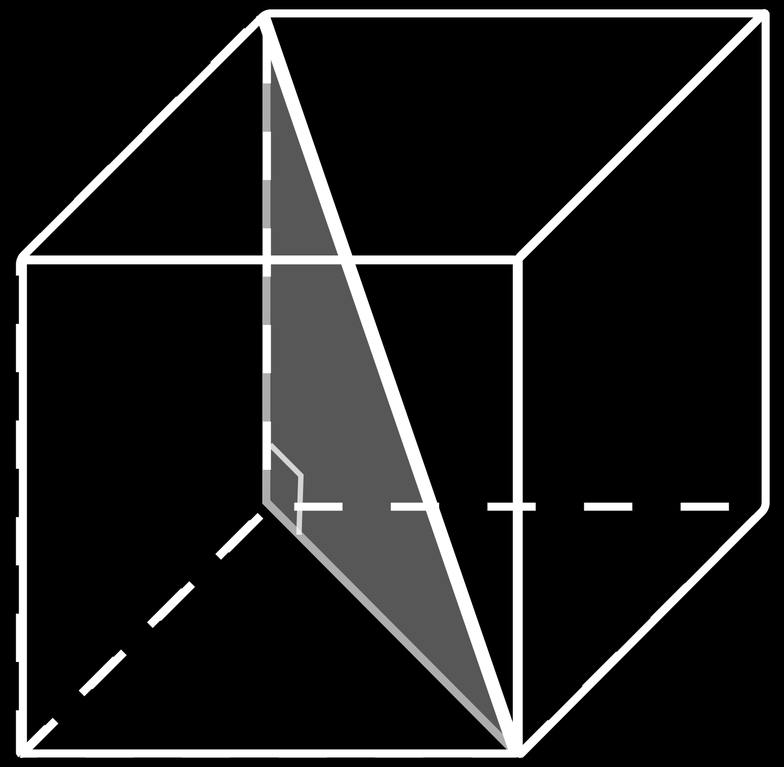
$= \sqrt{50}$

Diagonal $= \sqrt{5^2 + \sqrt{50}^2}$

$= \sqrt{75}$

$= 8.66(3.s.f)$

Main idea: form right angled triangles from sides/diagonals and find distances via pythagorean theorem/ SOHCAHTOA



Circle Properties

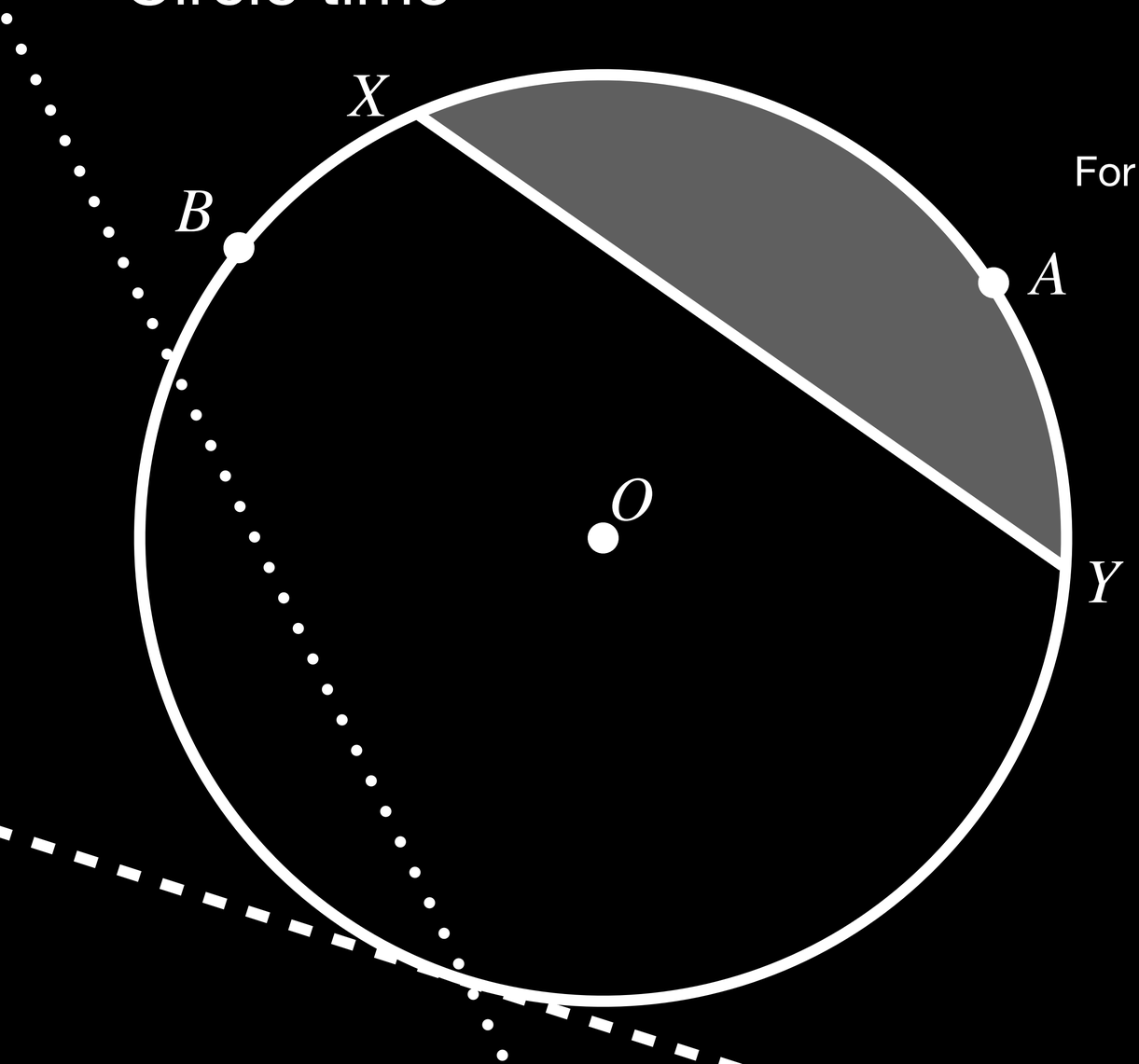
Angle Presentation

cambrige

If the angle is in degrees, you are REQUIRED to put the degrees symbol
Else, it will be regarded as radians

Parts of a Circle

Circle time



For a circle with center O, where A,B,X and Y lie on the circumference,

XY is a chord(line joining two points on the circumference)
XAY is a minor arc(smaller part of circumference)
XBY is a major arc(larger part of circumference)

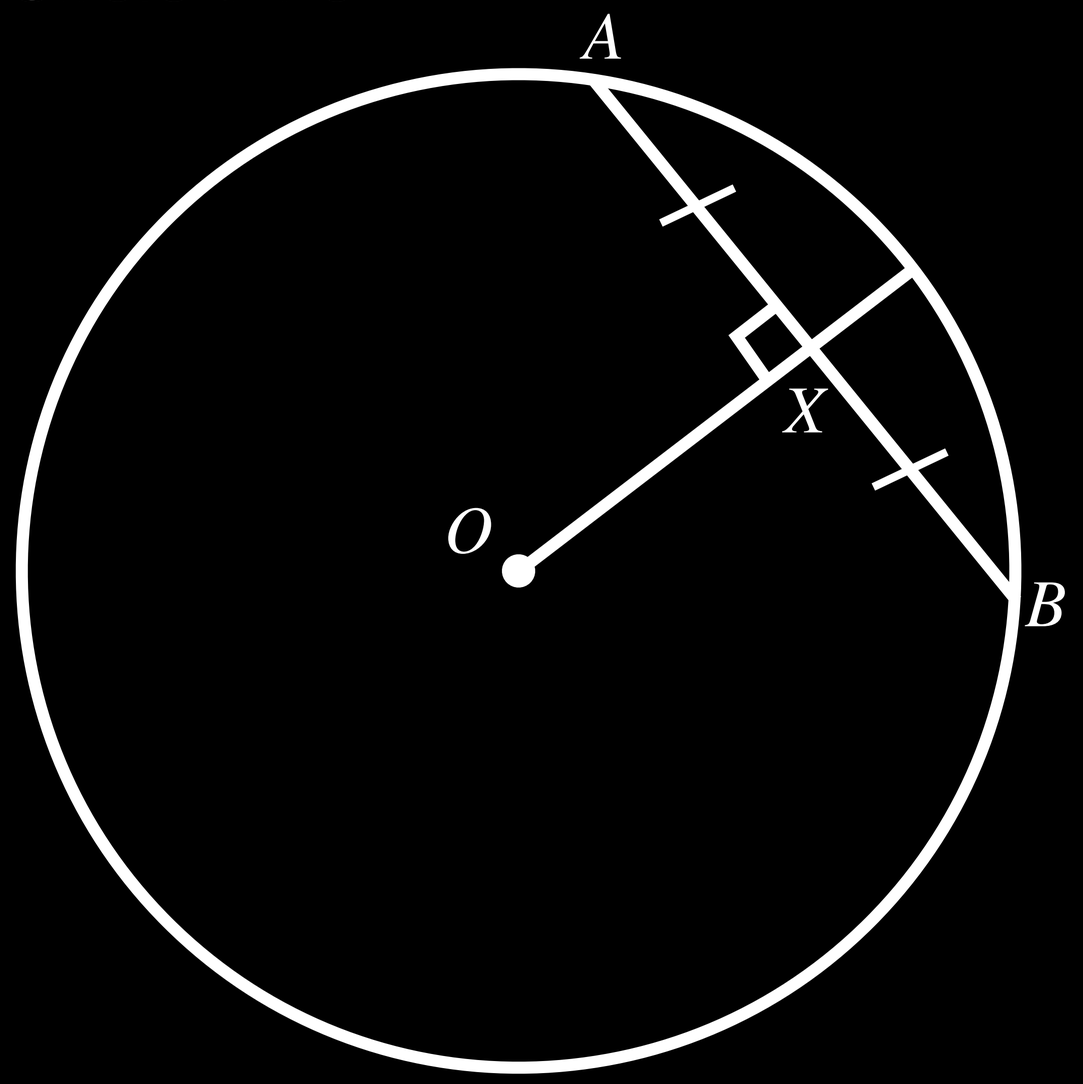
The shaded area is a minor segment
area bounded by minor arc and chord

The unshaded area is a major segment
area bounded by major arc and chord

The dashed line is a tangent
The dotted line is a secant(line that passes through the circle)

Circle Properties

Circle time



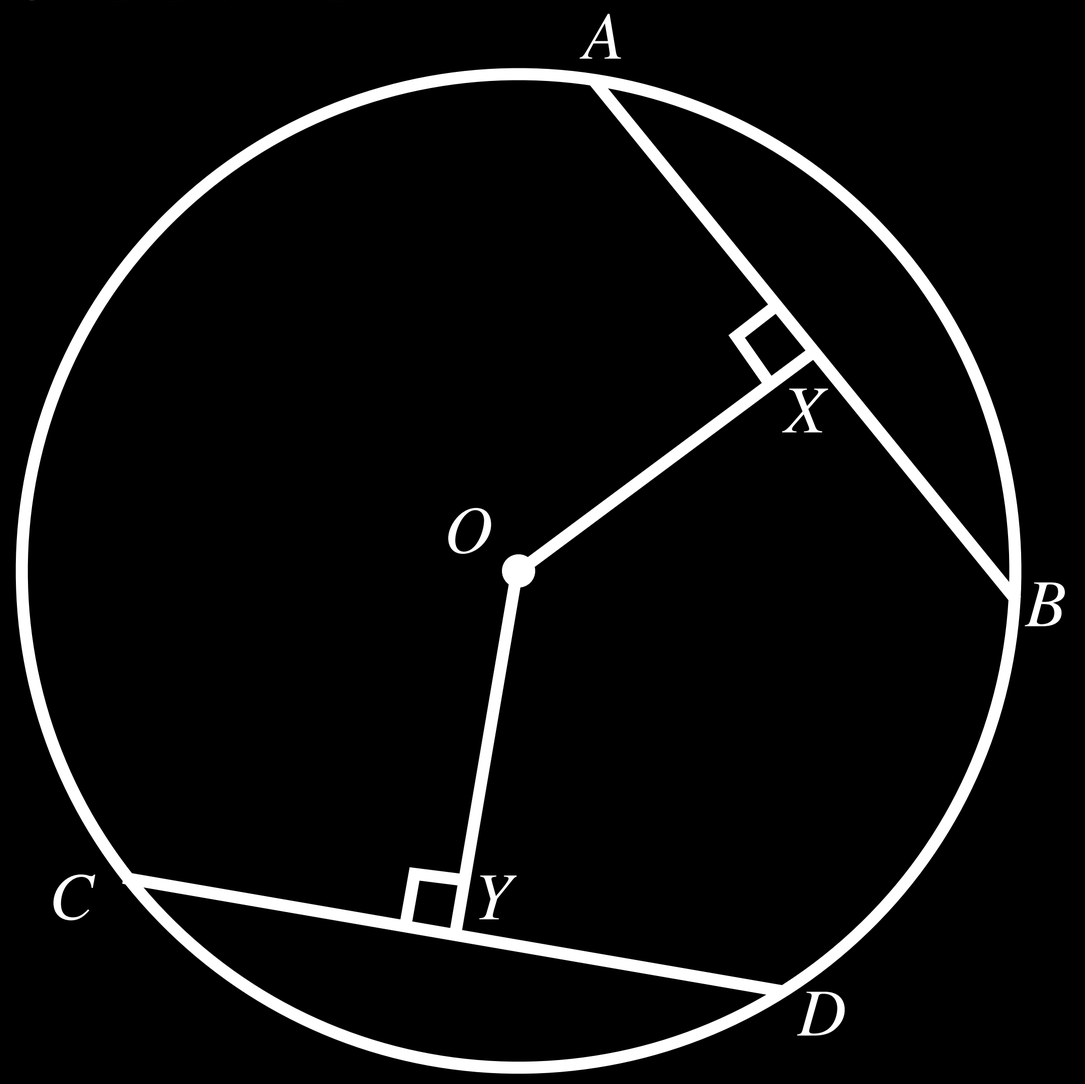
Straight line drawn from center to bisect a chord will be perpendicular to the chord
Perpendicular line from center of circle to chord will bisect chord

i.e: when OX is drawn such that AX = BX, AXO=AXB=90°
When OX is drawn such that AXO=AXB=90°, AX = BX

Abbreviation(what you write): perpendicular bisector of chord passes through center

Circle Properties

Circle time



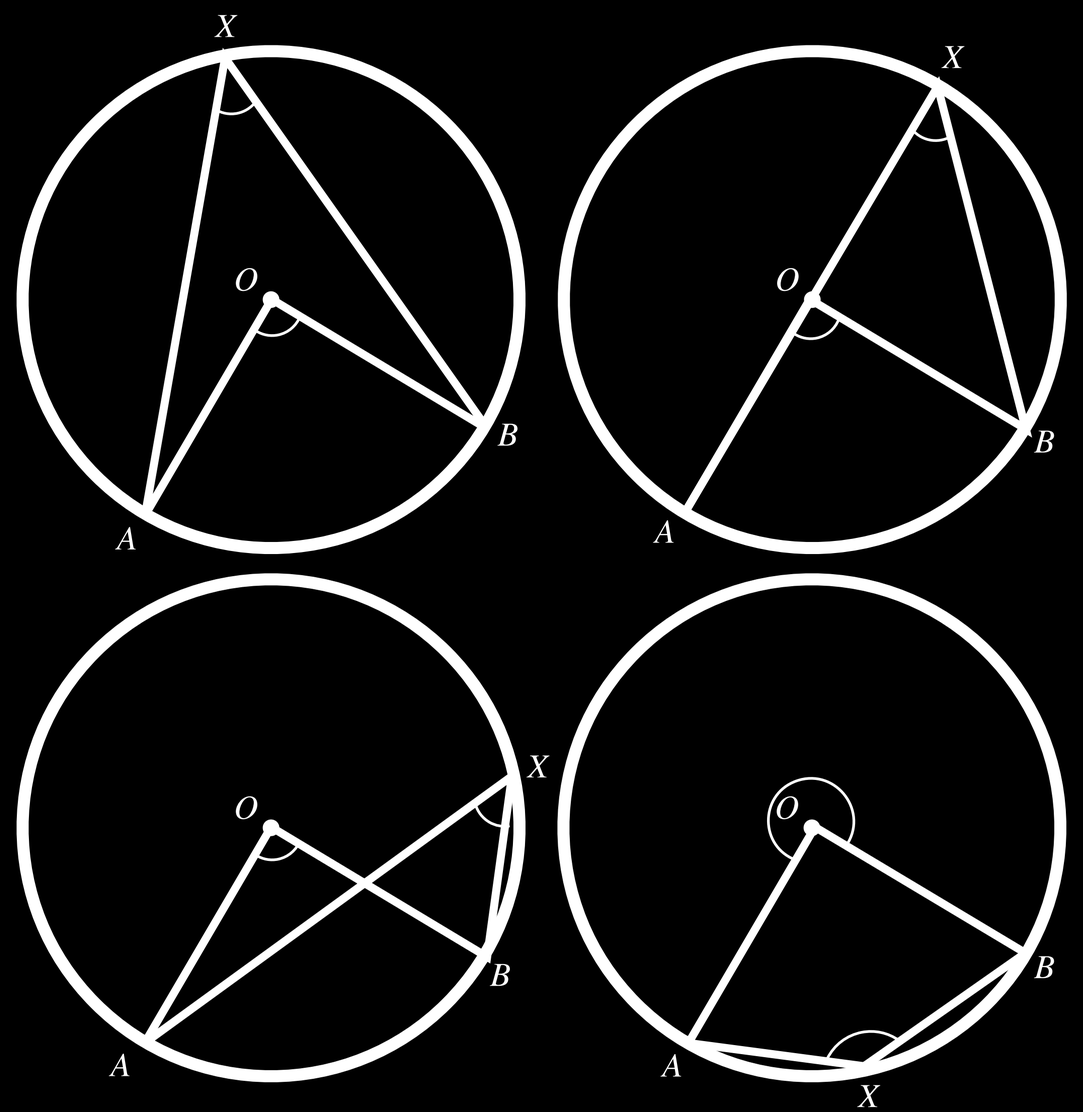
Chords of equal length are the same distance to the center

i.e: When AB = CD, OX = OY

Abbreviation(what you write): equal chords are equidistant to center

Circle Properties

Circle time



Angle at center of circle is twice of angle at circumference

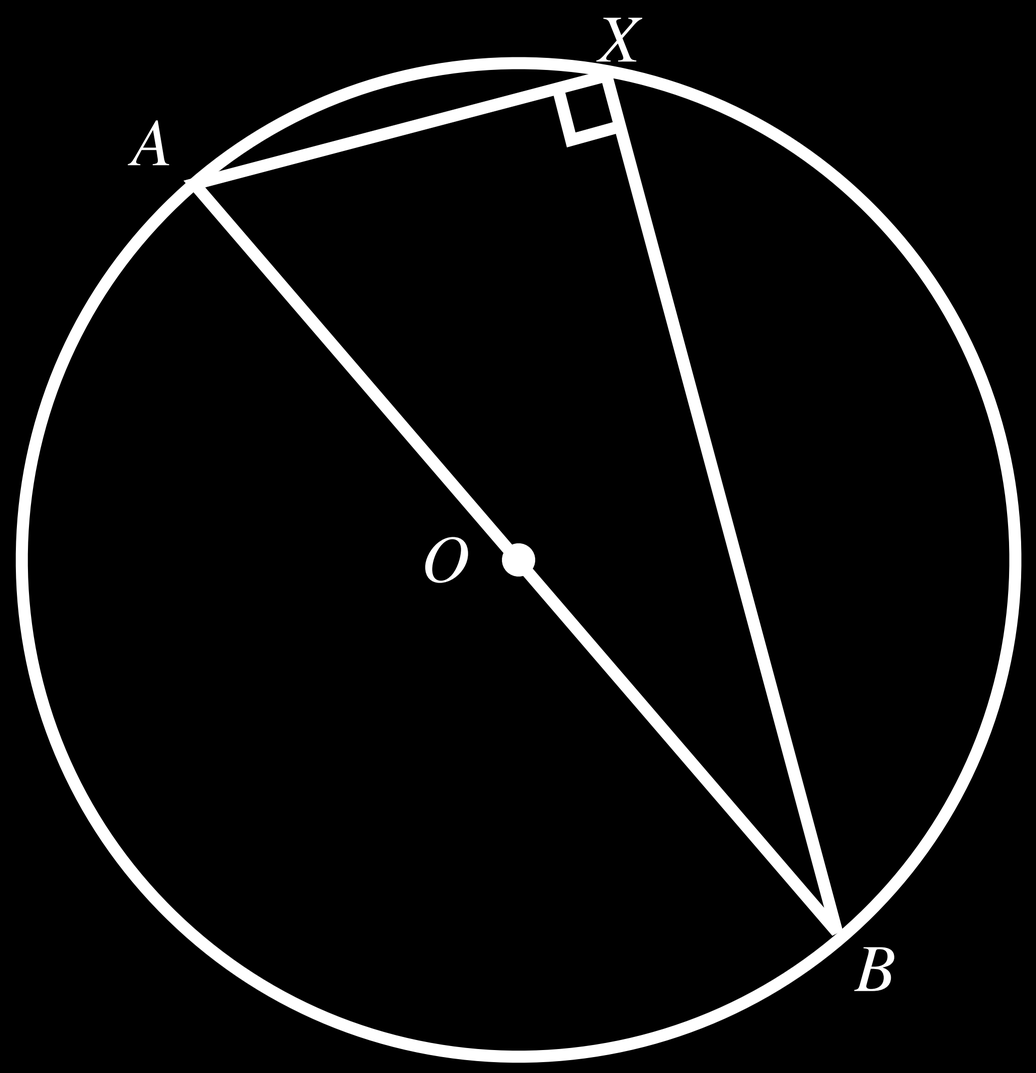
i.e: AOB= 2AXB

Abbreviation(what you write):angle at center is twice angle at circumference

**take note: bottom right case; AOB is now reflex, AXB is not equal to 2(obtuse AOB)

Circle Properties

Circle time



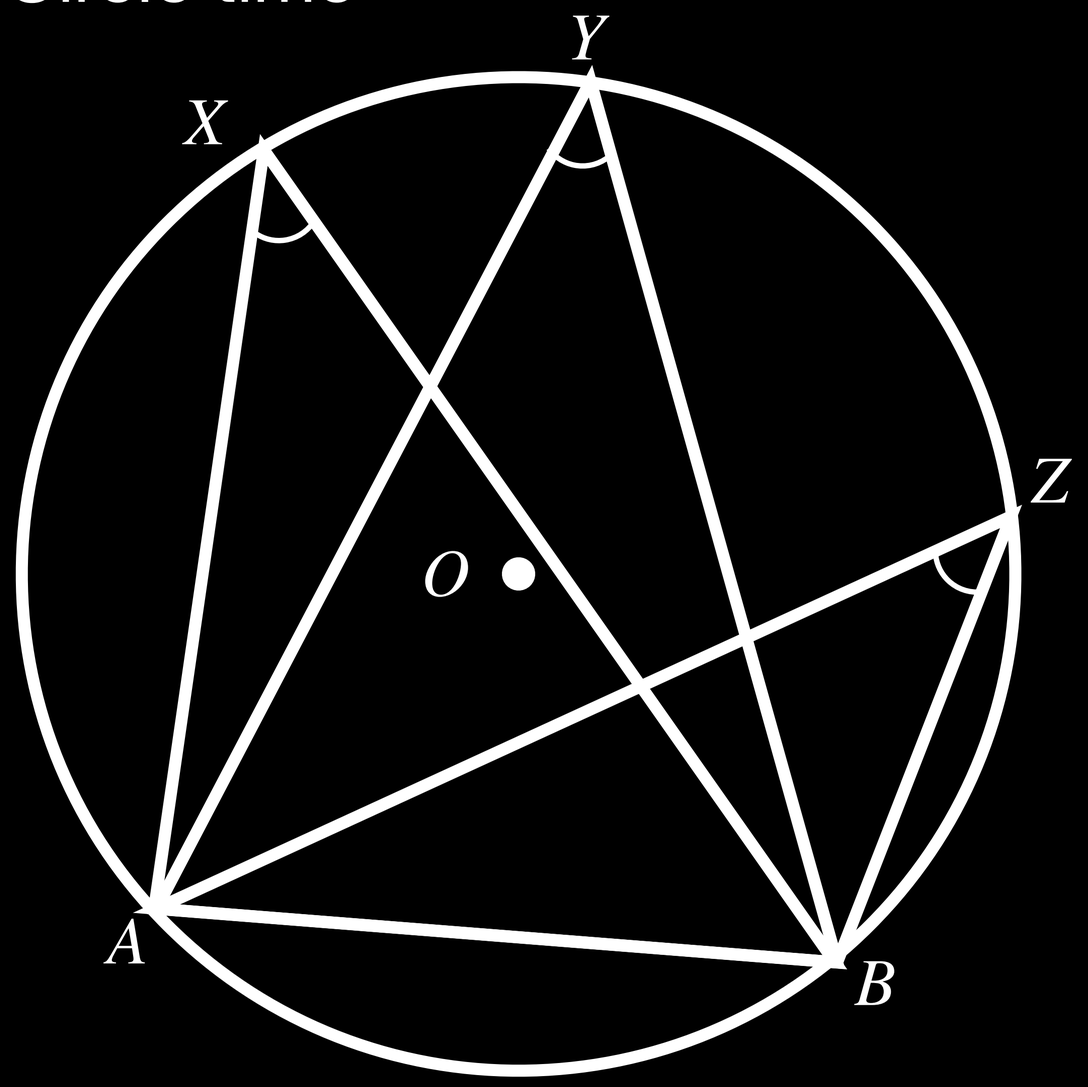
Angle in semicircle is right angle

i.e: When AB passes through O, AXB = 90°

Abbreviation(what you write): angle in semicircle is 90°

Circle Properties

Circle time



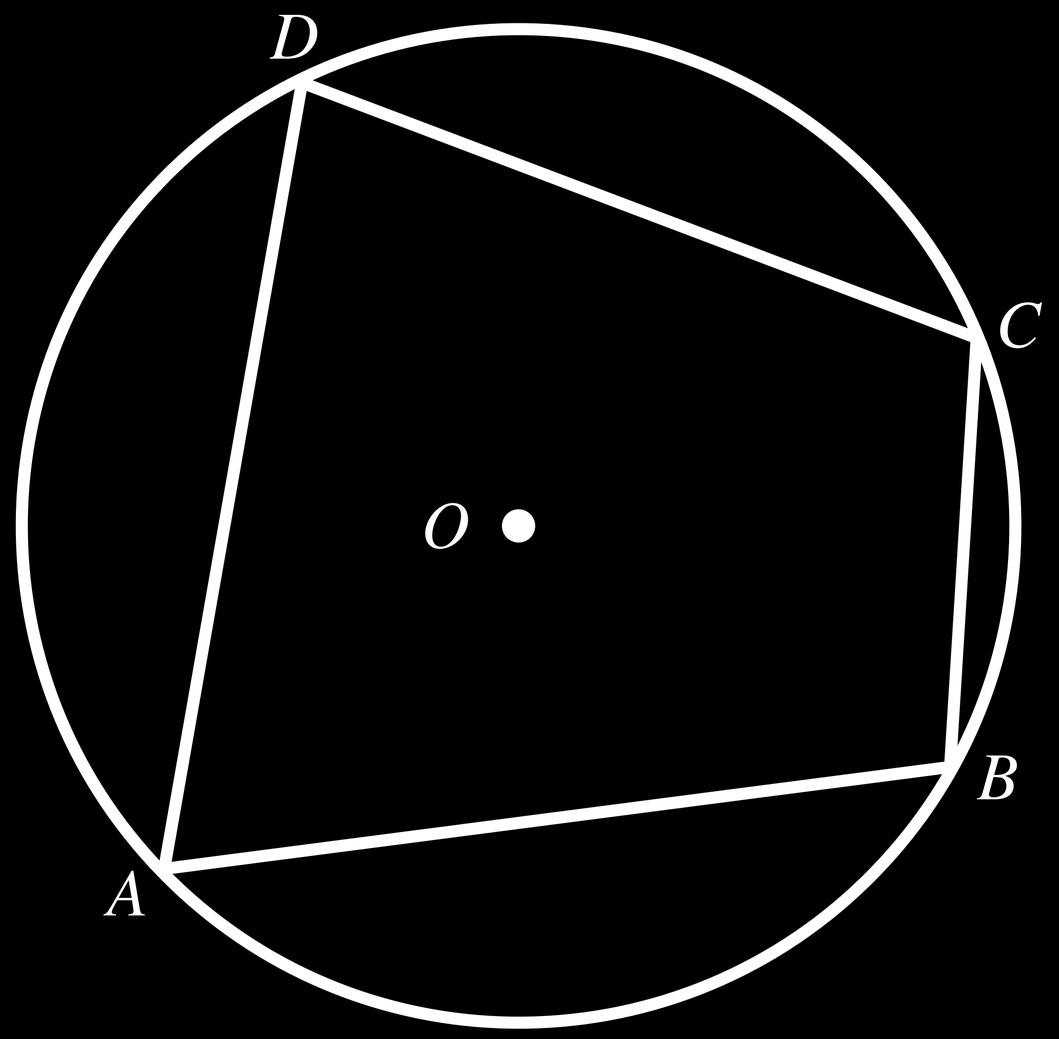
Angles in same segment are equal

i.e: For a segment AB, AXB=AYB=AZB

Abbreviation(what you write): angles in same segment are equal

Circle Properties

Circle time



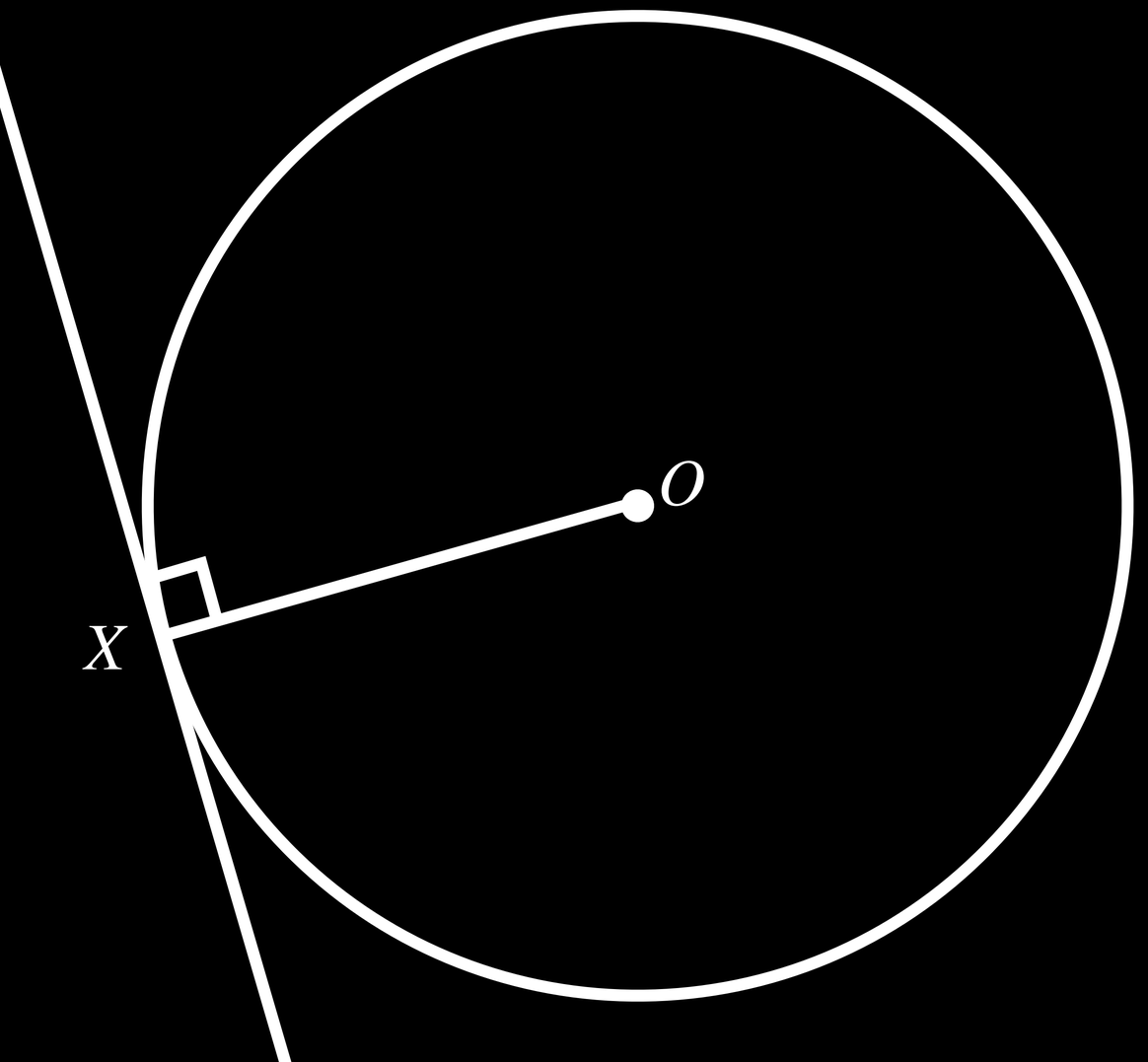
In a cyclic quadrilateral (a quadrilateral where each vertex is on the circumference), opposite angles are supplementary (add to 180°)

i.e: For a cyclic quadrilateral ABCD, ABC+ADC =180° and BAD+DCB=180°

Abbreviation(what you write): angles in opposite segments are supplementary

Circle Properties

Circle time

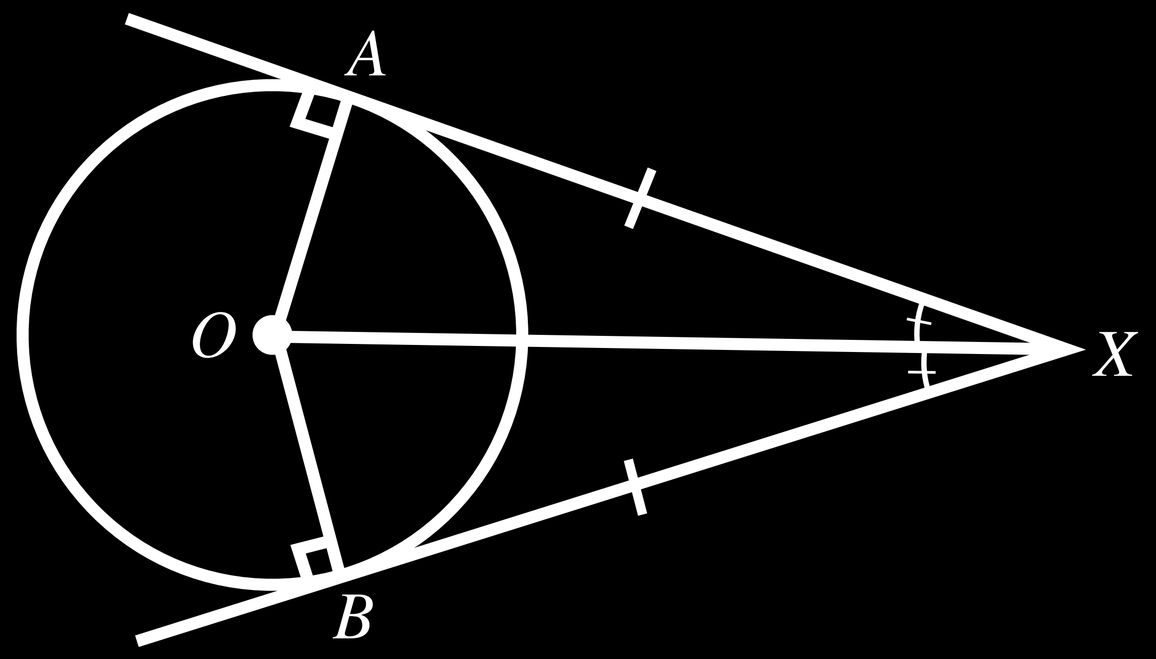


Tangent is perpendicular to radius

Abbreviation(what you write): Tangent is perpendicular to radius

Circle Properties

Circle time



Tangents from an external point are equal in length & Line joining external point to center bisects angle between tangents

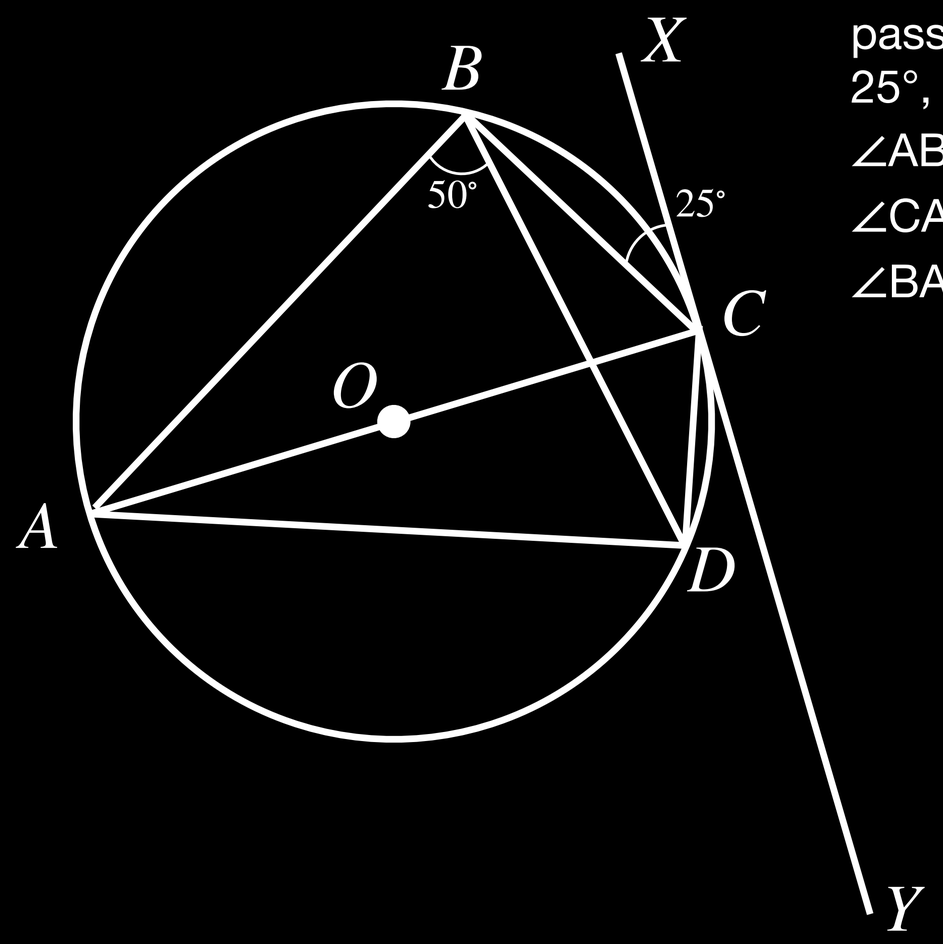
i.e AX = BX, and AXO = BXO

Abbreviation(what you write): tangent from external point are equal lengths

Line joining external point to center bisects angle between tangents

Circle Properties - Example Questions

Circle time



From the figure shown, XY is a tangent at C, and AC is a straight line that passes through O, the center of the circle. Given that ABD = 50° and BCX is 25°, find:

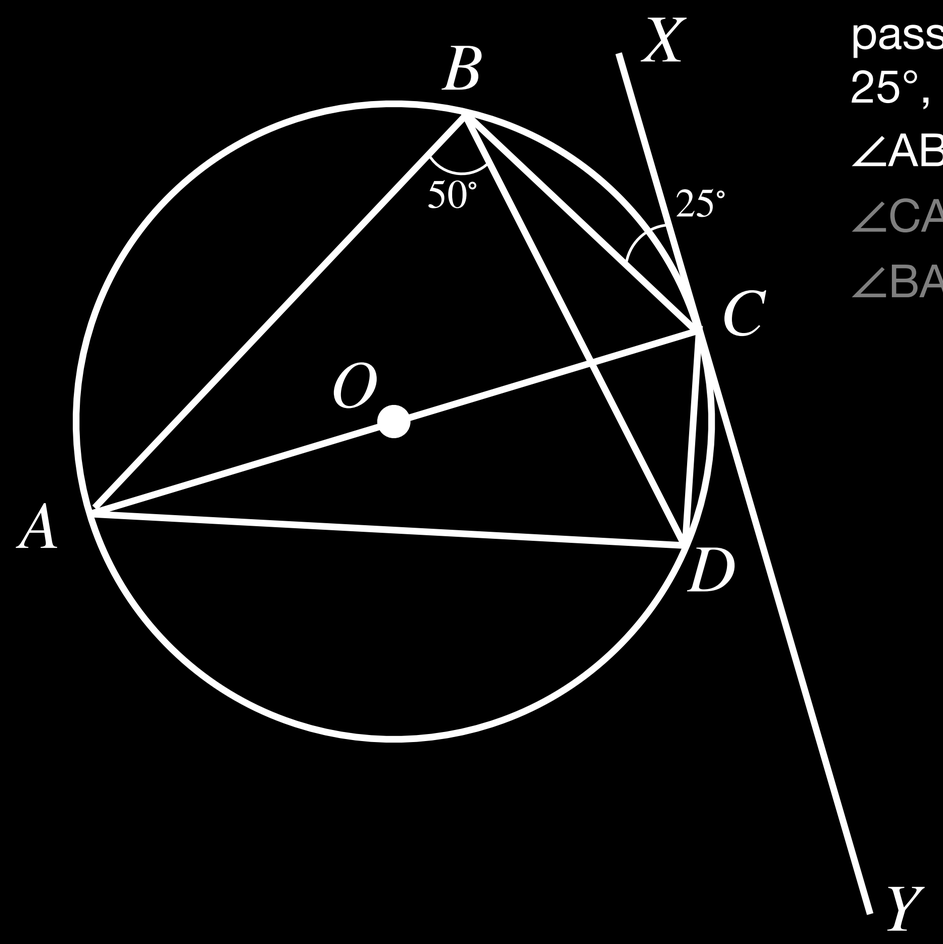
∠ABC

∠CAD

∠BAC

Circle Properties - Example Questions

Circle time



From the figure shown, XY is a tangent at C, and AC is a straight line that passes through O, the center of the circle. Given that ABD = 50° and BCX is 25°, find:

∠ABC

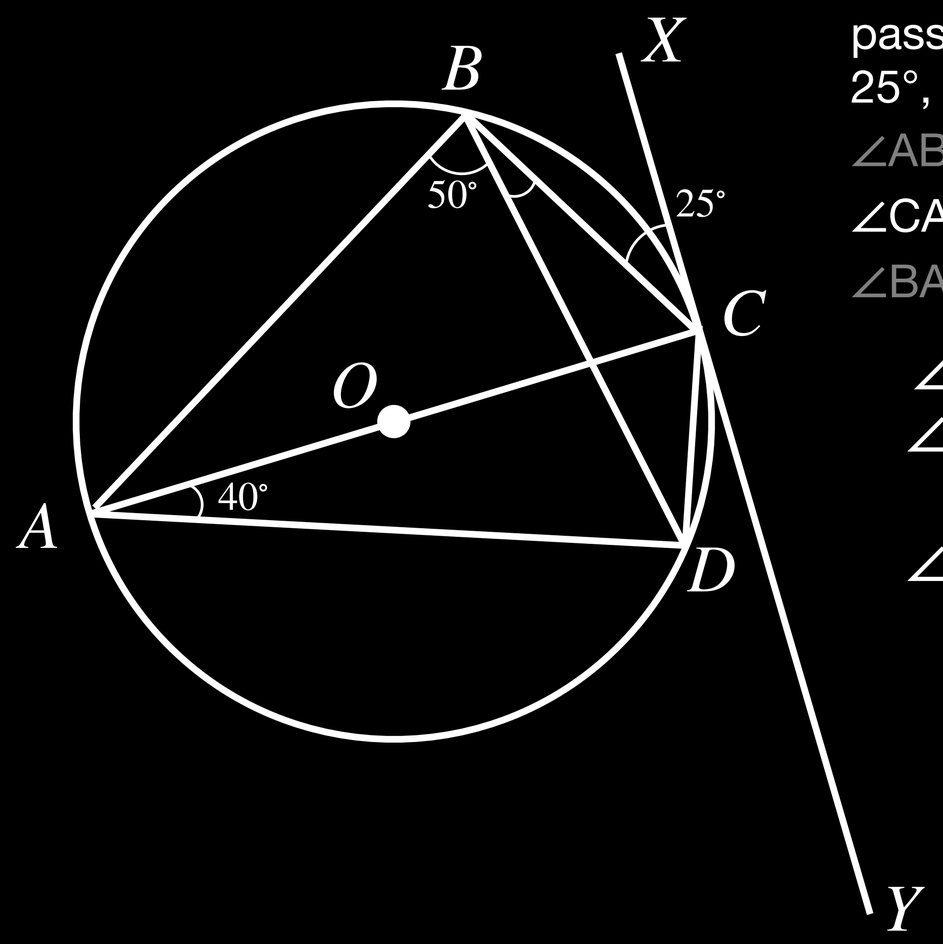
∠CAD

∠BAC

$\angle ABC = 90^\circ$ (angle in semicircle is 90°)

Circle Properties - Example Questions

Circle time



From the figure shown, XY is a tangent at C, and AC is a straight line that passes through O, the center of the circle. Given that ABD = 50° and BCX is 25°, find:

∠ABC

∠CAD

∠BAC

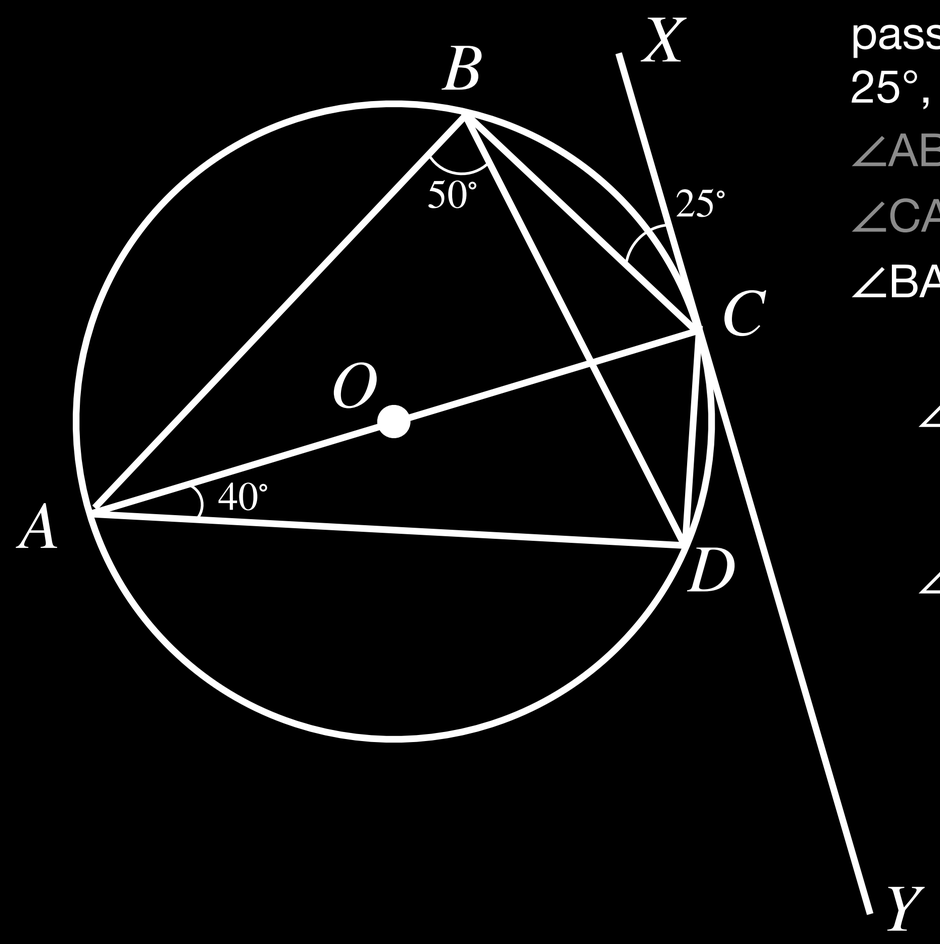
$\angle ABC = 90^\circ$ (angle in semicircle is 90°)

$$\begin{aligned}\angle DBC &= 90^\circ - 50^\circ \\ &= 40^\circ\end{aligned}$$

$\angle DBC = \angle CAD = 40^\circ$ (angles in same segment are equal)

Circle Properties - Example Questions

Circle time



From the figure shown, XY is a tangent at C, and AC is a straight line that passes through O, the center of the circle. Given that ABD = 50° and BCX is 25°, find:

∠ABC

∠CAD

∠BAC

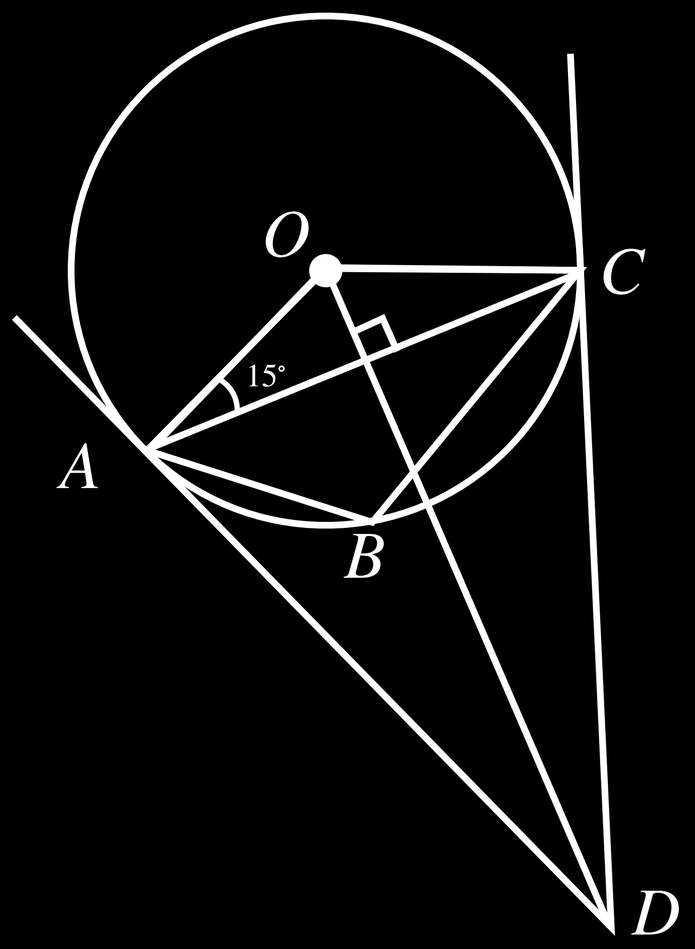
Do note that there are other possible solutions for this

$\angle BCA = 90^\circ - 25^\circ$(tangent is perpendicular to radius)
$= 65^\circ$

$\angle BAC = 90^\circ - 65^\circ$(sum interior angles in right triangle)
$= 25^\circ$

Circle Properties - Example Questions

Circle time



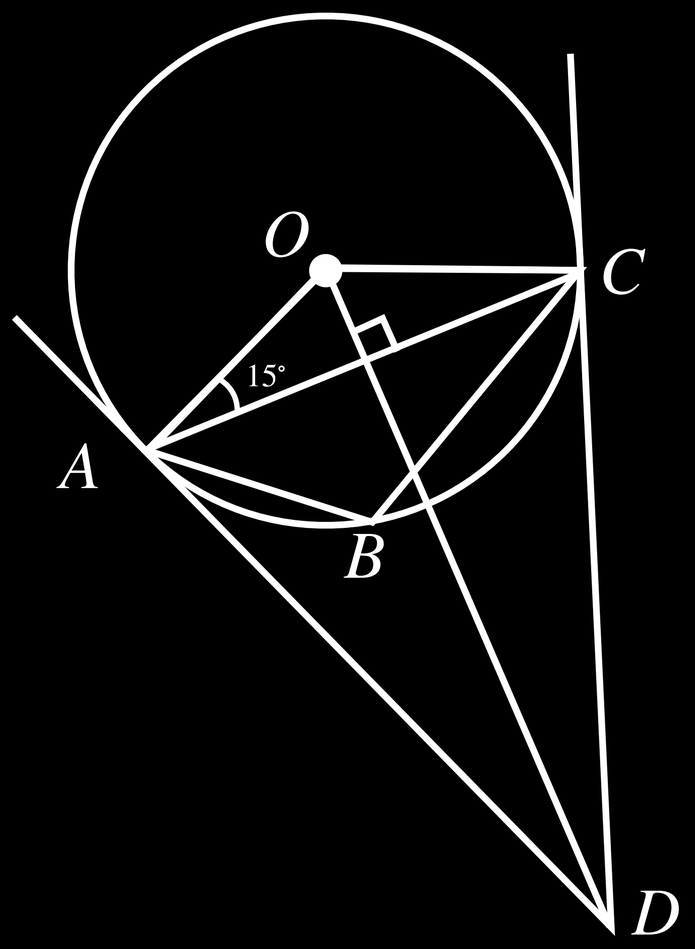
From the figure shown, a circle of center O has tangents AD and CD. Given that angle OAC = 15° and that DO is perpendicular to AC, find:

∠CDA

∠ABC

Circle Properties - Example Questions

Circle time



From the figure shown, a circle of center O has tangents AD and CD. Given that angle OAC = 15° and that DO is perpendicular to AC, find:

CDA

ABC

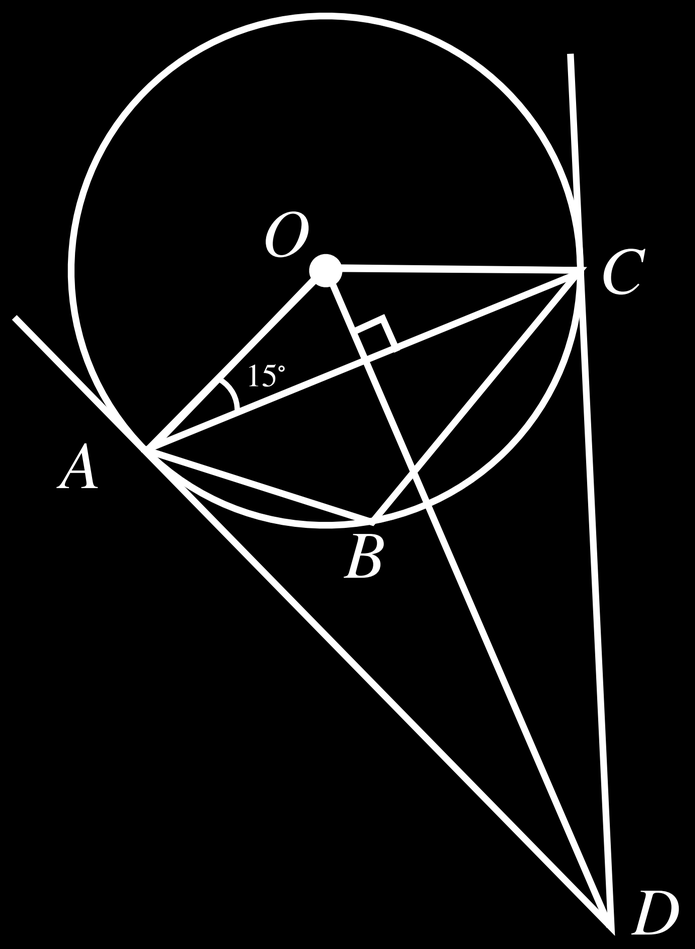
$\angle CAD = 90^\circ - 15^\circ$(tangent is perpedicular to radius)
$\angle CAD = 75^\circ$

$\angle ADO = 90^\circ - 75^\circ$(sum of angles in triangle)
$\angle ADO = 15^\circ$

$\angle CDA = 2(15^\circ)$(line joining external point to center bisects angle between tangents)
$\angle CDA = 30^\circ$

Circle Properties - Example Questions

Circle time



From the figure shown, a circle of center O has tangents AD and CD. Given that angle OAC = 15° and that DO is perpendicular to AC, find:

CDA

ABC

$\angle DOA = 90^\circ - 15^\circ$(sum angles in triangle)
$\angle DOA = 75^\circ$

$\angle AOC = 2(75^\circ)$(perpedicular bisector passes through center, thus bisects angle AOC)
$\angle AOC = 150^\circ$

Reflex $\angle AOC = 360^\circ - 150^\circ$
Reflex $\angle AOC = 210^\circ$

$\angle ABC = 0.5(210^\circ)$ (angle at center is twice of angle at circumference)
$\angle ABC = 105^\circ$

Do note that other concepts like trigo and congruency and similairty can be tested with this

Radian Measure

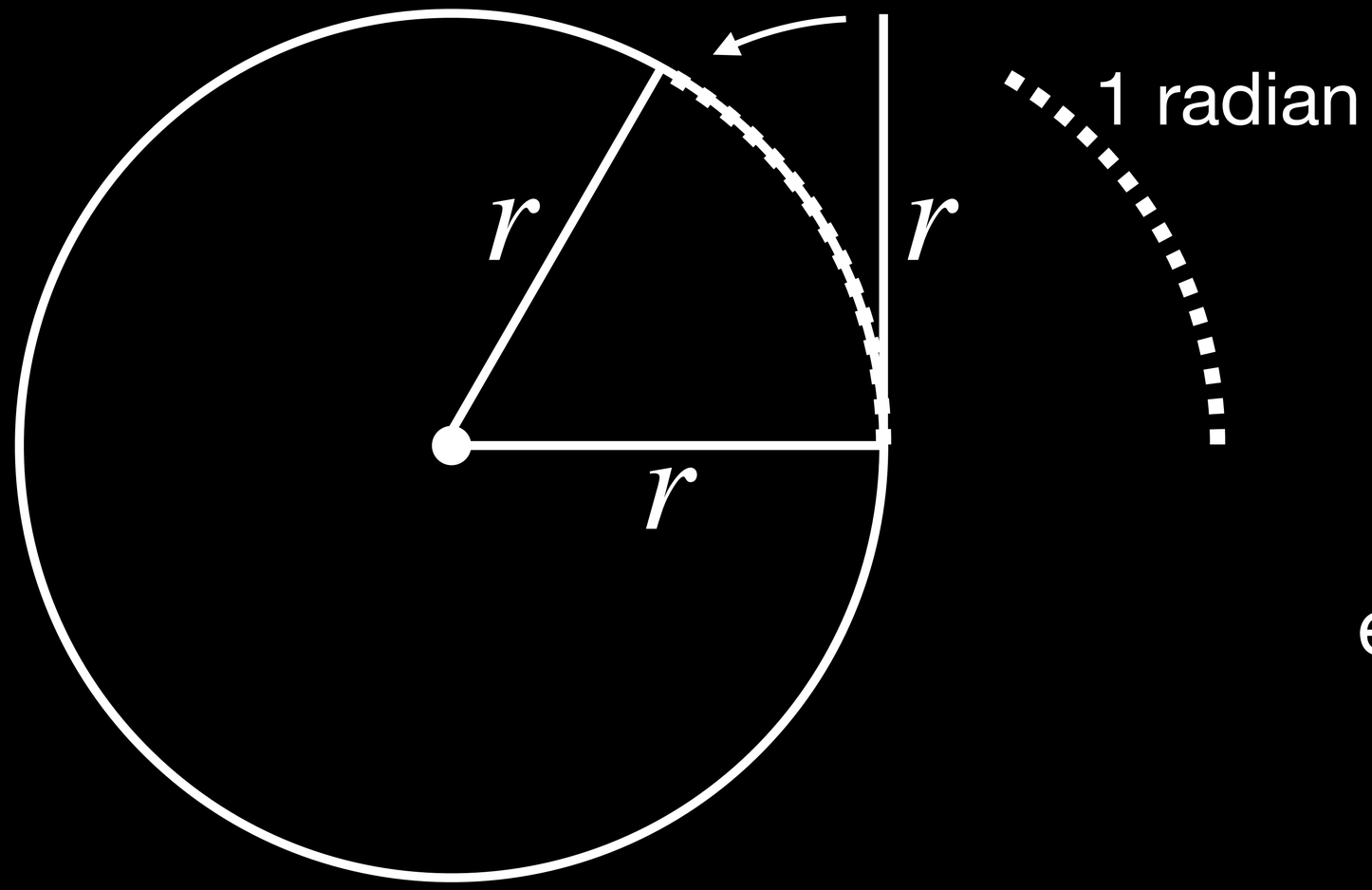
Angle Presentation

cambrige

If the angle is in degrees, you are REQUIRED to put the degrees symbol
Else, it will be regarded as radians

Radians

(Hey guys radians are not really that cool)



$$2\pi\ rad = 360^\circ$$
$$\pi\ rad = 180^\circ$$
$$\therefore 1\ rad = \frac{180}{\pi}$$
$$1^\circ = \frac{\pi}{180}$$

e.g. $52^\circ = 52 * \frac{\pi}{180} rad$

$$52^\circ = 0.908 rad (3.s.f)$$

you do not need to understand this visual

Radians/Degrees Conversion

deez

When to change calculator from degrees to radians? (and vice versa)

only when dealing with trigo functions

Express 91° in radians.

$$91^\circ = 91\left(\frac{\pi}{180}\right)$$

$$91^\circ = 1.59 rad(3.s.f)$$

no switch needed

Find $\sin\left(\frac{\pi}{6}\right)$

$$\sin\left(\frac{\pi}{6}\right) = 0.5$$

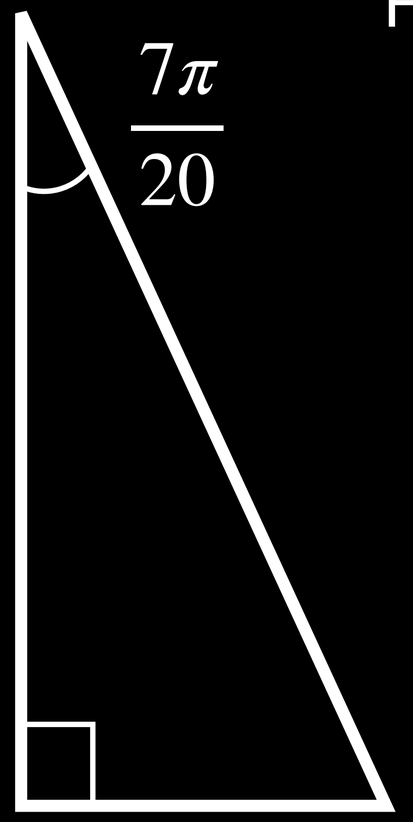
with radians (correct)

$$\sin\left(\frac{\pi}{6}\right) = 0.00914(3.s.f)$$

with degrees (wrong)

Geometric Diagrams and Radians

No way dude



From the diagram shown, what is the remanining angle in radians? (in terms of π)

$$\frac{\pi}{2} - \frac{7\pi}{20} = \frac{10\pi}{20} - \frac{7\pi}{20}$$
$$= \frac{3\pi}{20}$$

$$\frac{7\pi}{20} = 63^\circ$$
$$90^\circ - 63^\circ = 27^\circ$$
$$27^\circ = \frac{27\pi}{180}$$
$$= \frac{3\pi}{20}$$

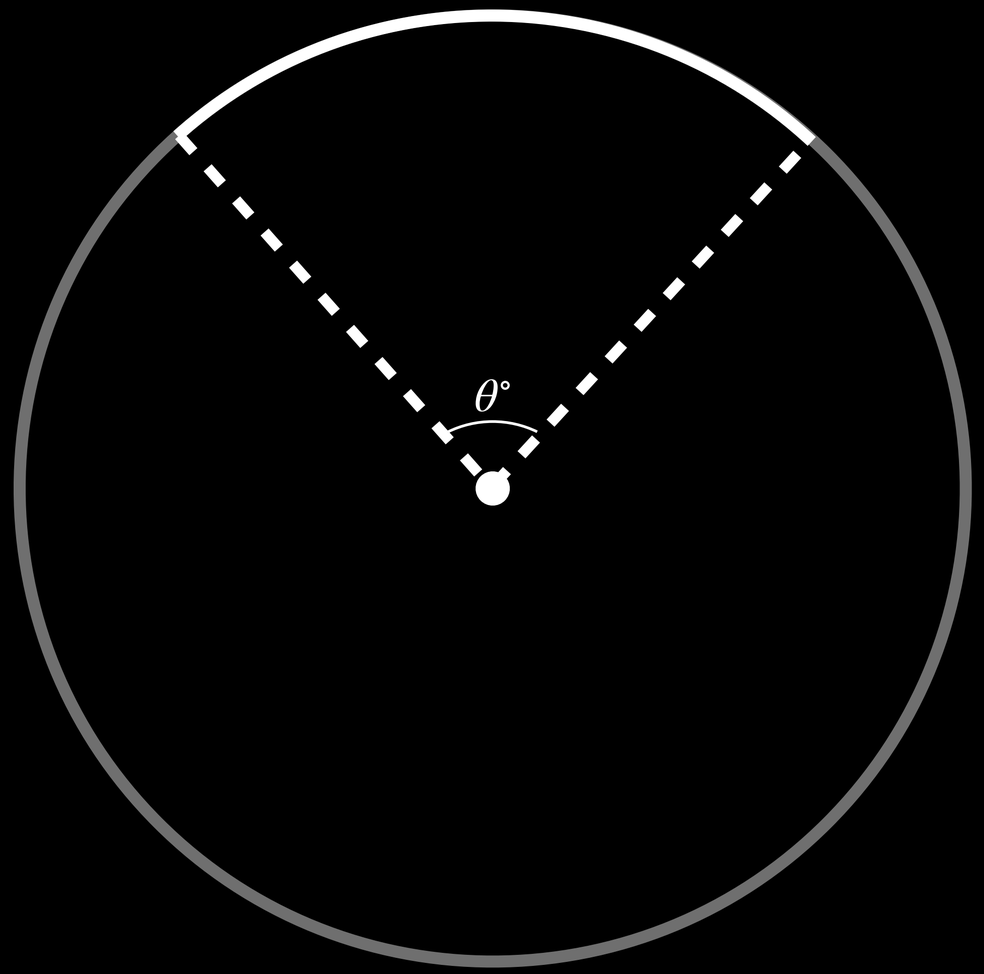
Both methods are viable, however first method is recommended (aka if asked for radians, stay in radians)

Degrees Formulae

deez

$$\text{Arc length} = \frac{\theta}{360^\circ}(2\pi r)$$

Where theta is the angle subtended by the arc



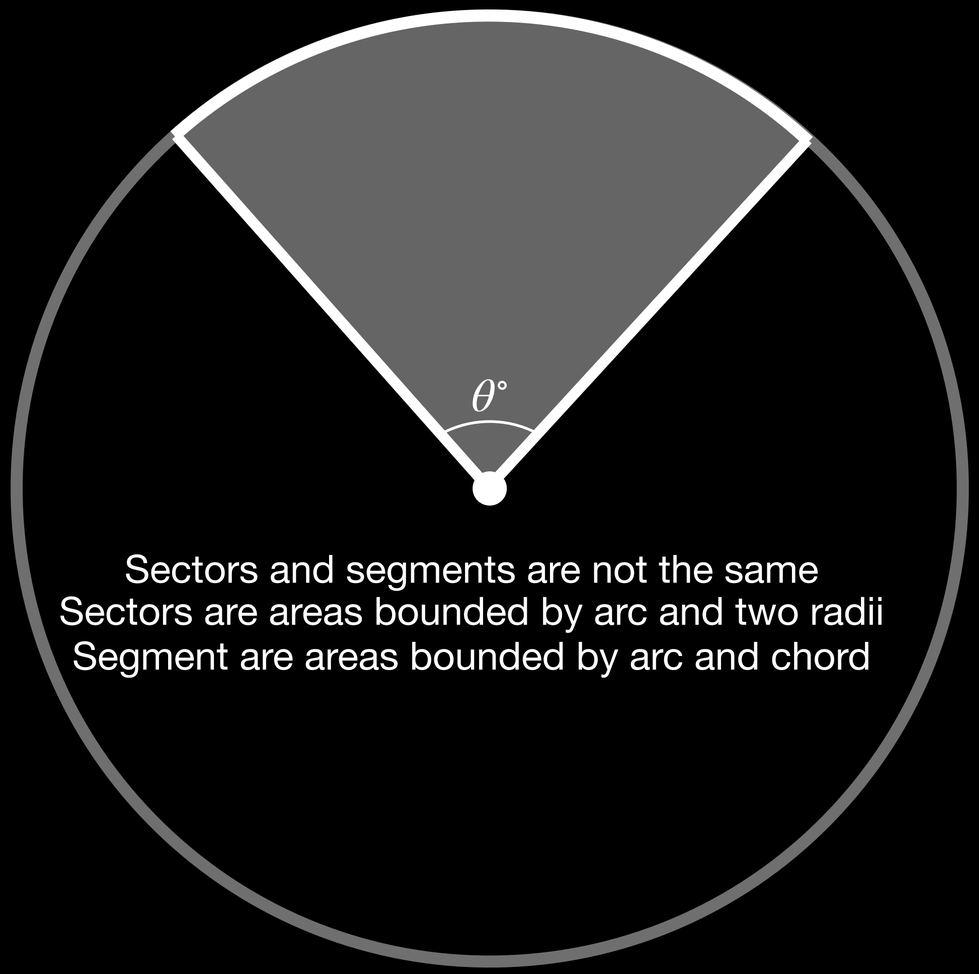
**THIS FORMULA IS NOT GIVEN IN FORMULA SHEET (lmao)

Degrees Formulae

deez

$$\text{Sector Area} = \frac{\theta}{360^\circ}(\pi r^2)$$

Where theta is the angle subtended by the sector



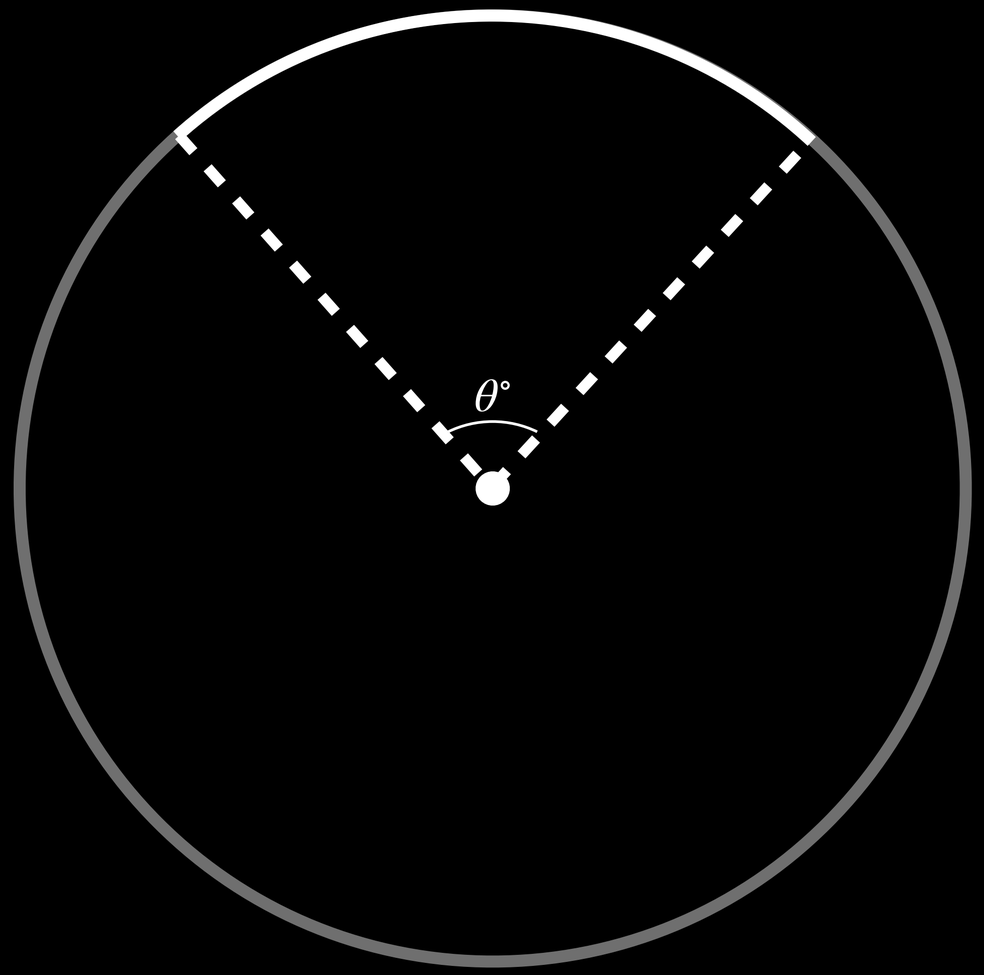
**THIS FORMULA IS NOT GIVEN IN FORMULA SHEET (lmao)

Radian Formulae

ray

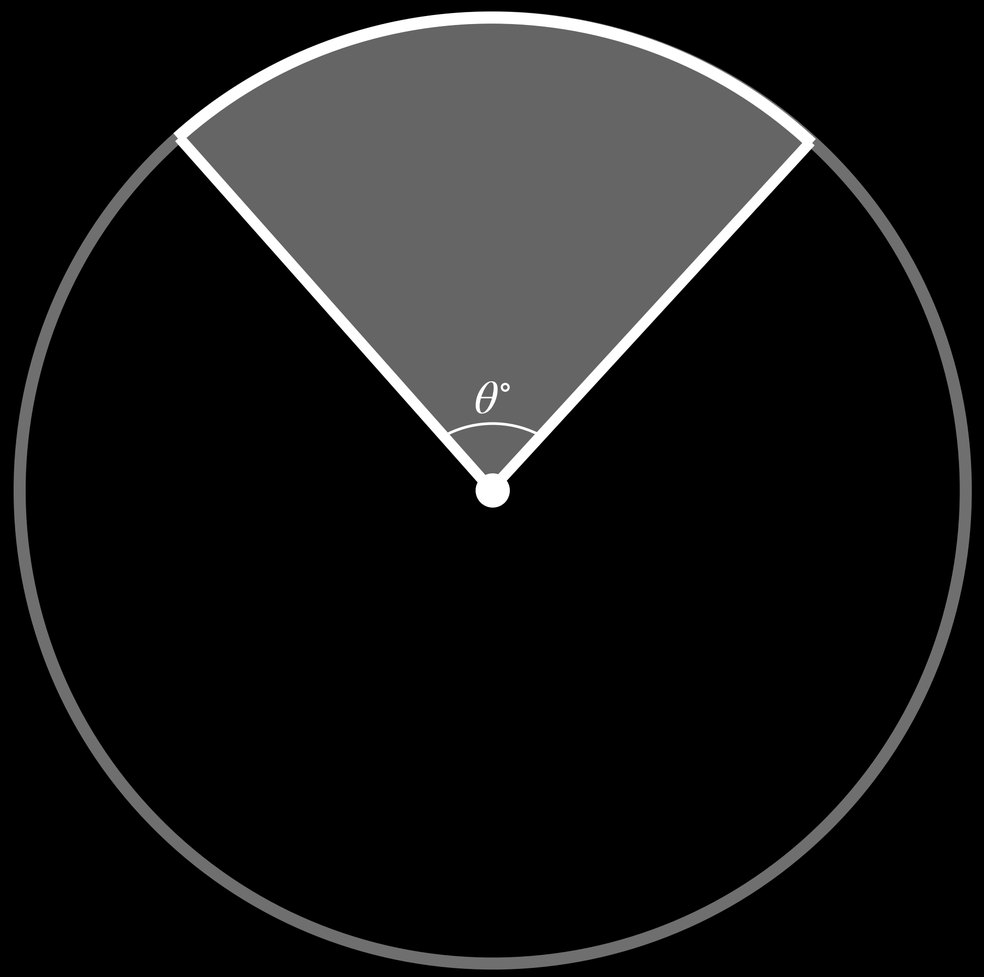
Arc length $= r\theta$

Where theta is the angle subtended by the arc
And angle theta is in radians (does not work for degrees)



Radian Formulae

ray

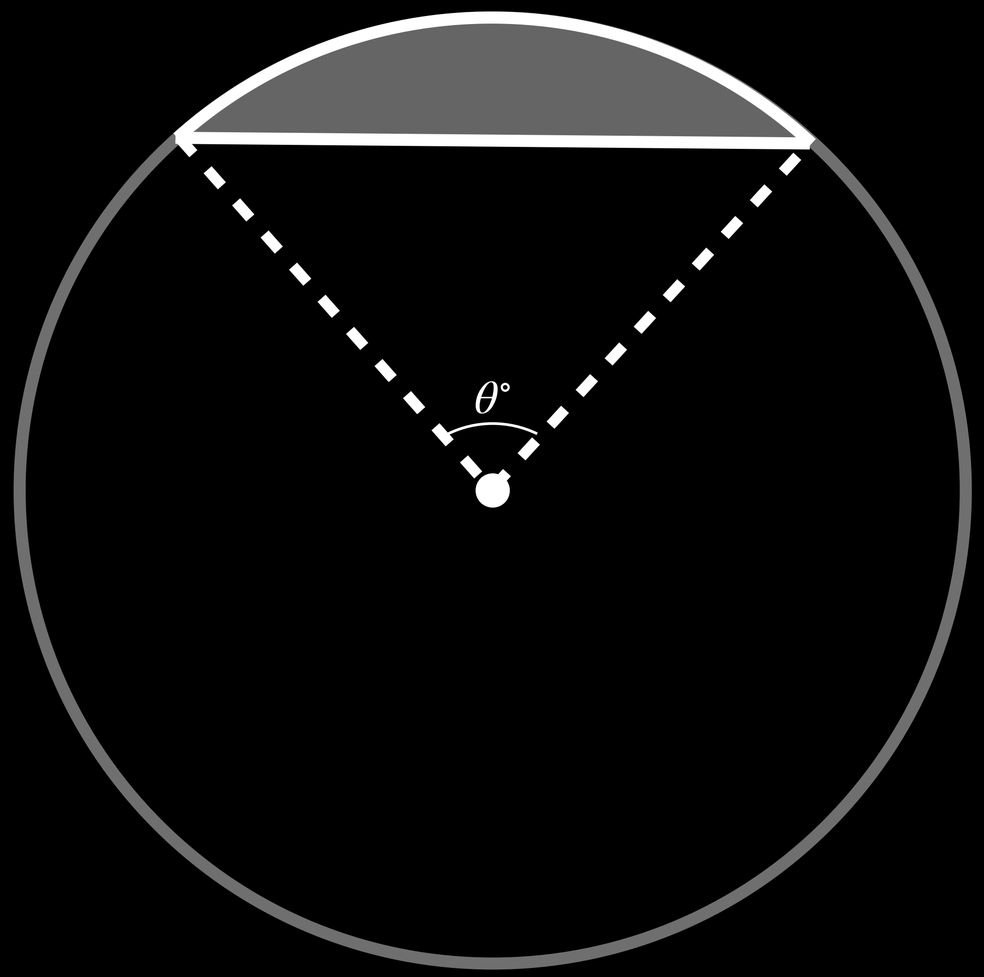


Sector Area $= 0.5r^2\theta$

Where theta is the angle subtended by the sector
And angle theta is in radians (does not work for degrees)

Radian Formulae

ray



$$\text{Segment area} = 0.5r^2(\theta - \sin(\theta))$$
$$= \frac{1}{2}r^2\theta - \frac{1}{2}r^2\sin(\theta)$$

Sector area - Triangle area (0.5absin(C) formula)

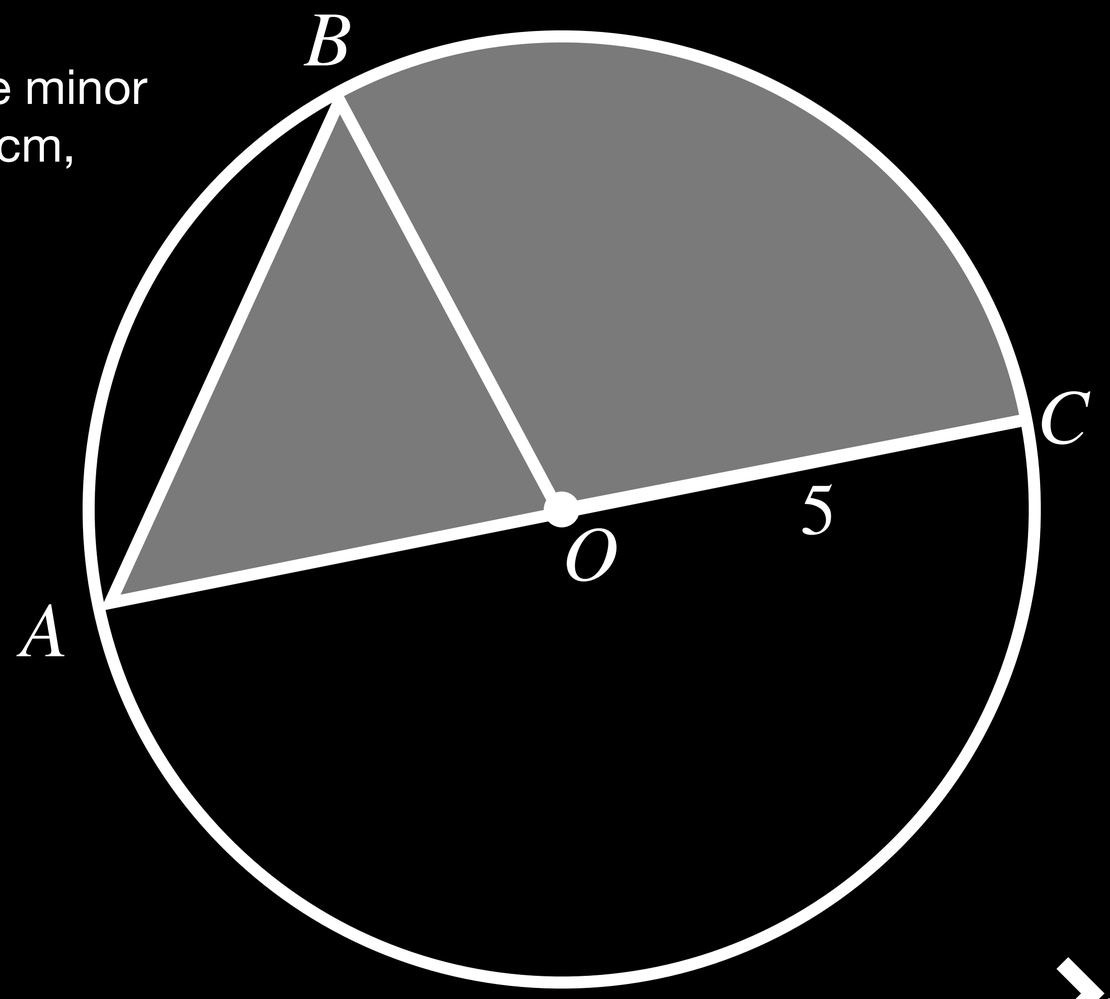
Where theta is the angle subtended by the arc
And angle theta is in radians (does not work for degrees)

**THIS FORMULA IS NOT GIVEN IN FORMULA SHEET (lmao)

Sample Problem

ray

From the figure shown, AB is a diameter of the circle. If the minor arc length of BC is 12.5cm and the circle has a radius of 5cm, what is the area of the shaded region?



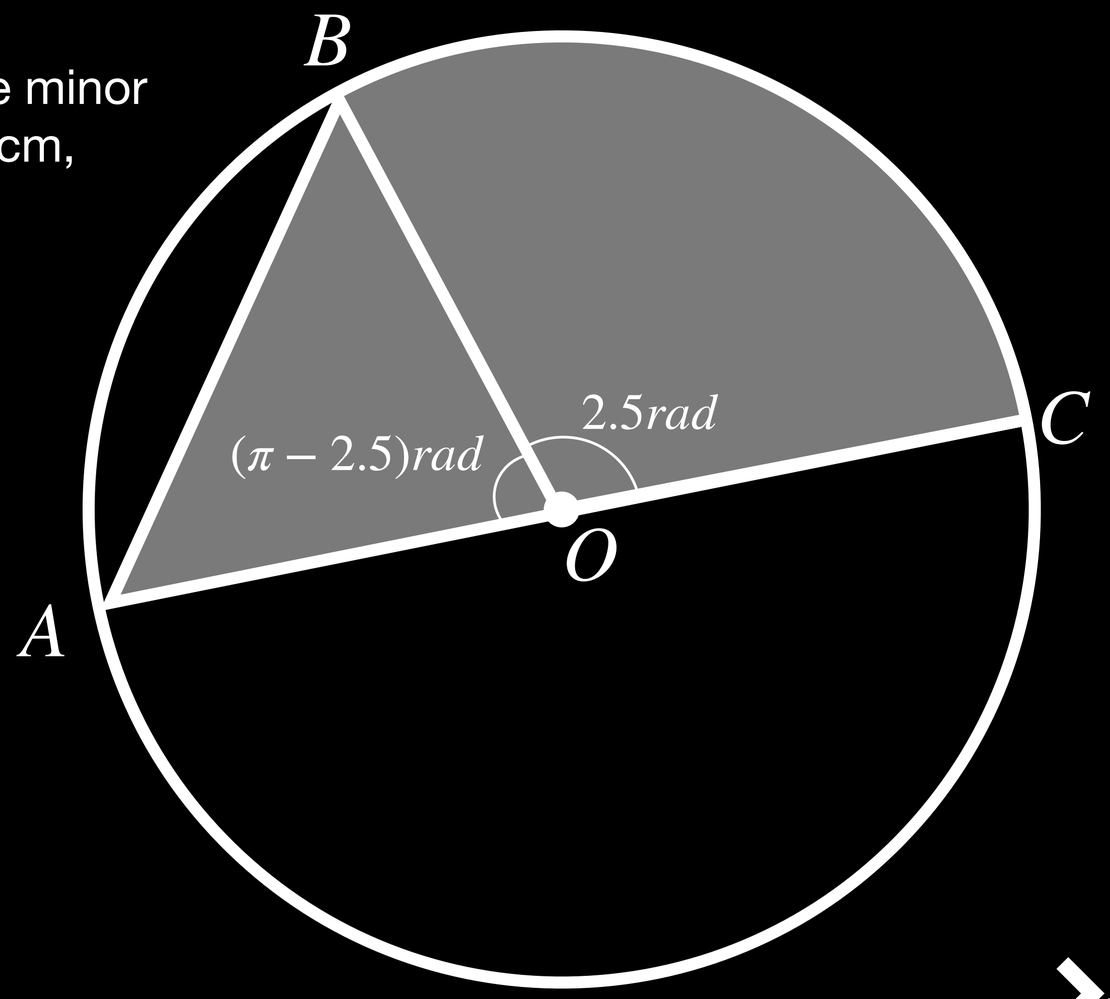
Sample Problem

ray

From the figure shown, AB is a diameter of the circle. If the minor arc length of BC is 12.5cm and the circle has a radius of 5cm, what is the area of the shaded region?

$$s = r\theta$$
$$12.5 = 5(\theta)$$
$$\frac{12.5}{5} = (\theta)$$
$$\theta = 2.5rad$$

May not necessarily ask to solve for arc length, can be asked to solve radius/subtended angle



Angle subtended by segment AB=$(\pi - 2.5)rad$

Shaded Area = Semicircle − Segment AB

$$= 0.5\pi(5)^2 - 0.5(5)^2((\pi - 2.5) - \sin((\pi - 2.5)))$$
$$= 12.5\pi - 12.5((\pi - 2.5) - \sin((\pi - 2.5)))$$
$$= 38.7cm^2$$

Solving for sector area BC + triangle AOB is also a valid solution

Problems in Real World Contexts

Question Overview

October 21st, 2024

PiRWC questions are at the end of paper 2 and involve you to make a judgement based on mathematical evidence
i.e. mini pt
First part may ask you “What is the cost/value of x if [condition],” aka specific conditions are already given, all you need to do is evaluate

Second part usually will ask you for an evaluation, e.g “what would be recommended for [condition/situation]?” Other than using mathematical evidence, you may also need to justify with other info given (that may not need be mathematical)

Questions are not necessarily tied to a specific topic

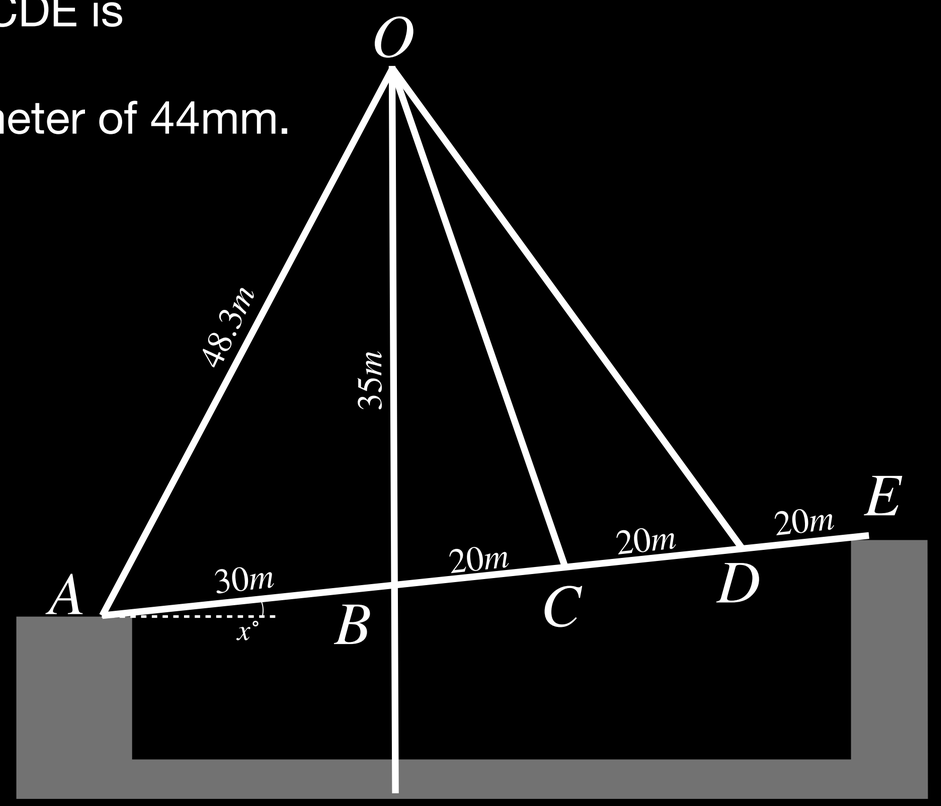
Example Question

October 21st, 2024

The diagram shown represents a cable-stayed footbridge. The deck ABCDE is supported by cables OA, OC and OD, from a vertical mast OB.
Cable OA has a diameter of 64mm, while Cable OC and OD have a diameter of 44mm.

The information about cables used in bridge building is shown here:

Product Code	Cable Diameter/mm	Design Load/kN	Unit mass/kgm^-1
FLC 40	40	973	9.0
FLC 44	44	1179	10.8
FLC 48	48	1403	12.9
FLC 52	52	1667	15.3
FLC 56	56	1933	17.8
FLC 60	60	2218	20.4
FLC 64	64	2524	23.2



[N2021]

Example Question

October 21st, 2024

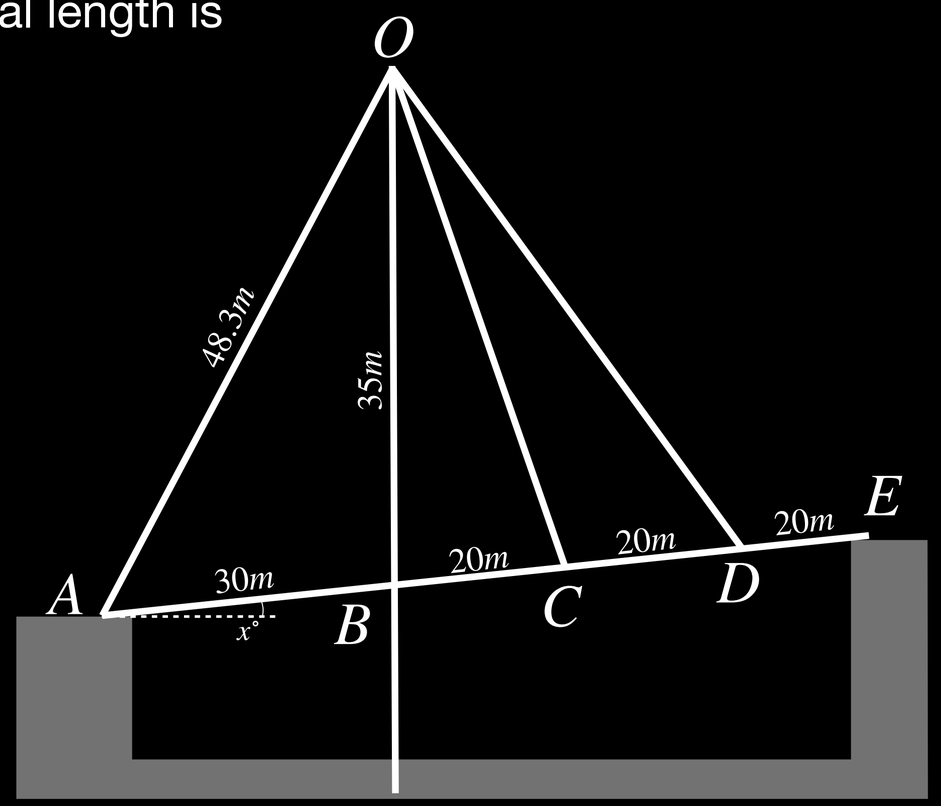
a) given that the ratio of the vertical height of the deck to the horizontal length is 1:10, find x.

$$m = \frac{1}{10}$$

$$\frac{1}{10} = \tan(x)$$

$$x = \tan^{-1}\left(\frac{1}{10}\right)$$

$$x = 5.7°(1.d.p)$$



Example Question

October 21st, 2024

b) show that OC = 38.5m, correct to 1.d.p.

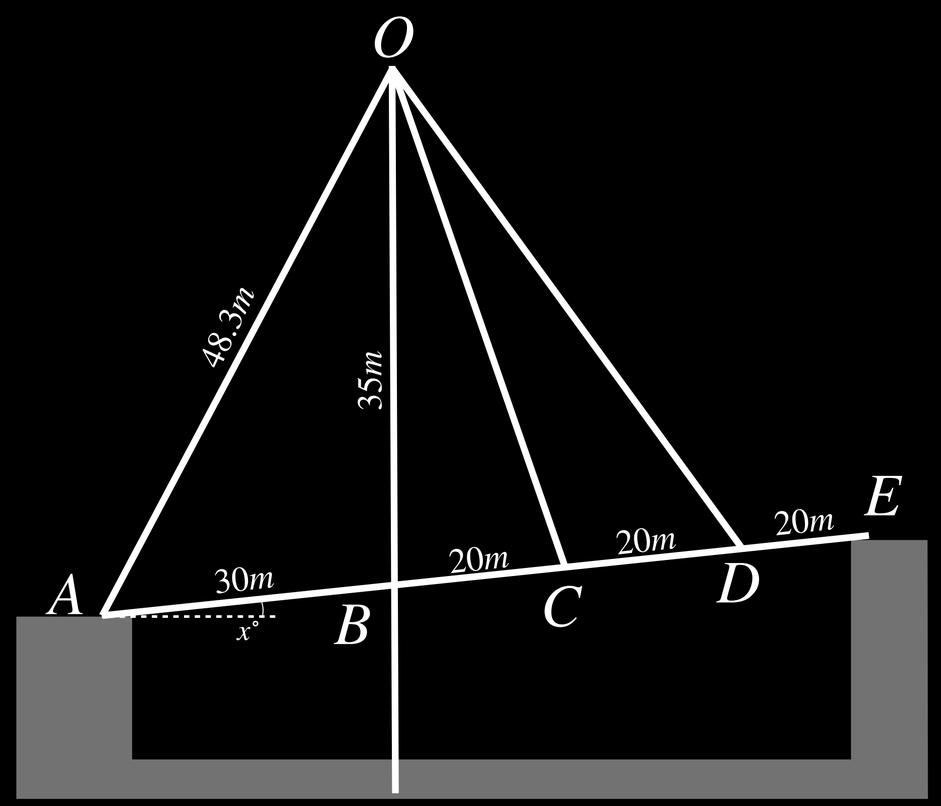
$$\angle OBC = 90^\circ - \tan^{-1}(0.1)$$

$$OC^2 = BC^2 + OB^2 - 2(BC)(OB)\cos(OBC)$$

$$OC = \sqrt{(20)^2 + (35)^2 - 2(20)(35)\cos(90^\circ - \tan^{-1}(0.1))}$$

$$OC = 38.545m(5.s.f)$$

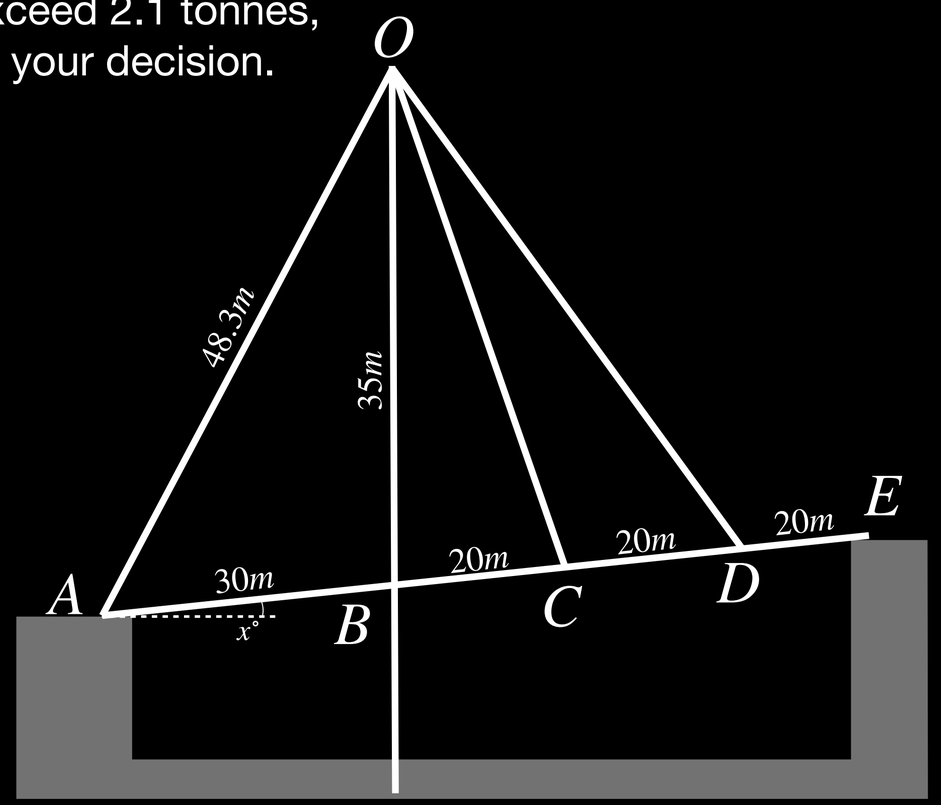
$$OC = 38.5m(1.d.p)$$



Example Question

October 21st, 2024

c) The engineers have specified the total mass of the three cables must not exceed 2.1 tonnes, where 1ton = 1000kg. Do the cables meet the engineers specification? Justify your decision.



$$OD^2 = BD^2 + OB^2 - 2(BD)(OB)\cos(OBC)$$

$$OD = \sqrt{(40)^2 + (35)^2 - 2(40)(35)\cos(90^\circ - \tan^{-1}(0.1))}$$

$$OD = 50.462m(5.s.f)$$

Mass of cable OA $= 23.2kg/m \times 48.3m$
Mass of cable OA $= 1120.56kg$

Mass of cable OC and OD $= 10.8kg/m \times (50.462m + 38.545m)$
Mass of cable OC and OD $= 961.2756kg$

Total mass $= 961.2756kg + 1120.56kg$
Total mass $= 2081.83kg$

$2.1tonnes = 2100kg$

As the total mass of the cables is 2081.83kg, which is less than the 2100 kg limit [2081.83<2100], the cables meet the specifications

Make sure to include an inequality in the final evaluative statement
Questions could ask which cable is suitable for all 3, etc

Financial Math

Profit/Loss

Sdjnbhfjd

amount needed to buy product

amount needed to make product

Profit = Selling - Cost

Loss = Cost - Selling

Profit/Loss is always positive

Profit/Loss

Sdjnbhfjd

however, this formula is much more common to be used

$$\frac{\text{Profit/Loss}}{\text{Cost}} \times 100\,\%$$

$$\frac{\text{Profit/Loss}}{\text{Selling}} \times 100\,\%$$

Both formulas shown express the profit as a percentage
the denominator is determined by the qn, e.g. "of the selling price" or "from the cost price"

Profit/Loss

Sdjnbhfjd

The cost of manufacturing a product is $12000. The product is sold to a customer at at profit of 9% the cost. Find how much the customer paid for the product.

$$\$12000 \times 109\,\% = \$13080$$

or

$$\$12000 \times 1.09 = \$13080$$

Profit/Loss

Sdjnbhfjd

The cost of manufacturing a product is $12000. The product is then sold at a price of $12720. Find the profit margin.

$$\frac{\$12720 - \$12000}{\$12000} \times 100\,\% = 6\,\%$$

Profit/Loss

Sdjnbhfjd

A manufacturer sells a product for $14512.50 and makes a profit of 7.5%. Find the cost price of the product.

this price is 107.5% of the cost price

$$\frac{\$14512.50}{107.5\,\%} \times 100\,\% = \$13500$$

multiply by 100% to find cost price

dividing by 107.5% gives 1% of cost price

or

this price is 1.075 of the cost price

$$\frac{\$14512.50}{1.075} = \$13500$$

dividing by 1.075 gives the cost price, thus no need for 100%

Discount

Sdjnbhfjd

$$\text{Discount} = \underset{\text{original price}}{\text{Marked price}} - \underset{\text{discounted price}}{\text{Sale price}}$$

$$\text{Percentage Discount} = \frac{\text{discount}}{\underset{\text{original price}}{\text{marked price}}} \times 100\,\%$$

Discount

Sdjnbhfjd

A store has a sale where all items have a 15% discount and a membership discount of 20%. A non member buys a product for $170.
a) What is the marked price of the product?

$$\frac{\$170}{100\% - 15\,\%} \times 100\,\% = \$200$$

or

$$\frac{\$170}{1 - 0.15} = \$200$$

Discount

Sdjnbhfjd

A store has a sale where all items have a 15% discount and a membership discount of 20%. A non member buys a product for $170.
b) How much does a member have to pay for the same product?

$$\$200 \times (100\% - 15\%)(100\% - 20\%) = \$200 \times 68\,\% = \$136$$

or

$$\$200 \times (1 - 0.15)(1 - 0.2) = \$200 \times 0.68 = \$136$$

Simple/Compound Interest

Sdjnbhfjd

Interest is the extra amount of money you need to pay after making a large payment/loan/ etc.

$$I = \frac{PRT}{100}$$

simple interest

$$I = P\left(1 + \frac{r}{100}\right)^n - P$$

compound interest

Simple/Compound Interest

Sdjnbhfjd

$$I = \frac{PRT}{100}$$

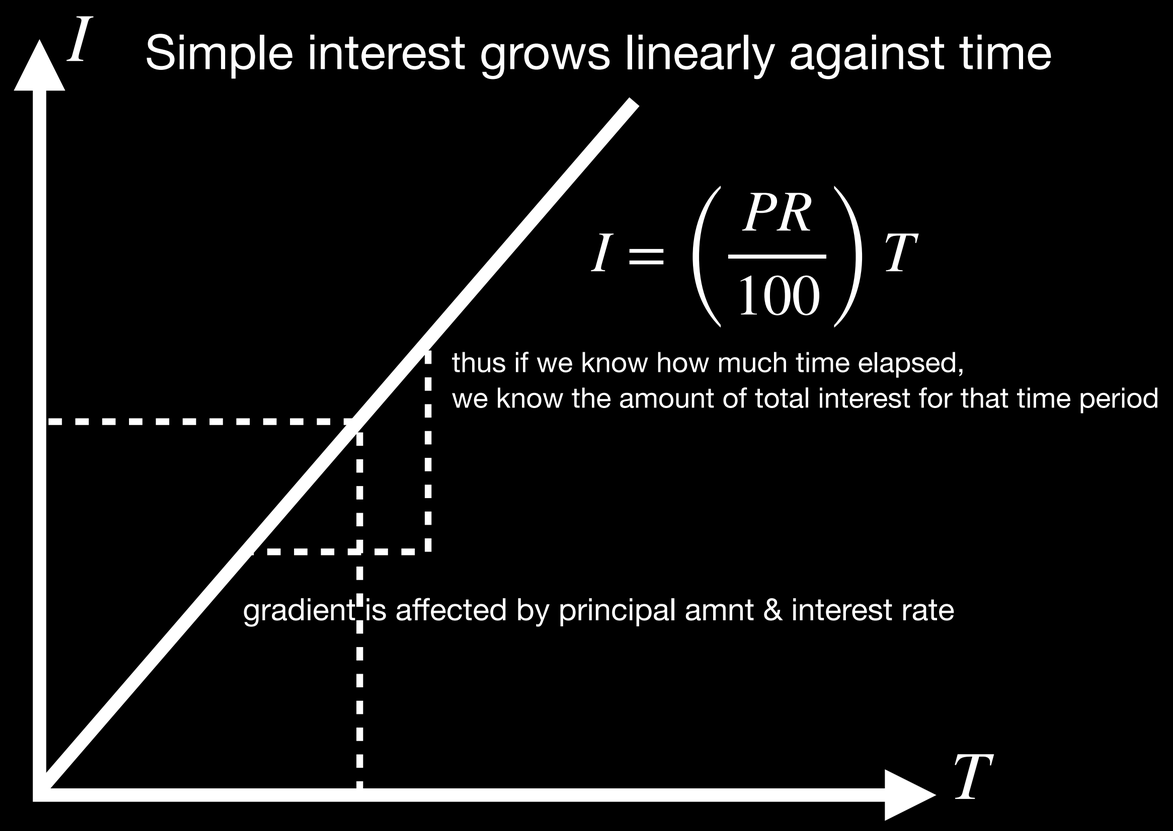
P = principal amount (original amount)

R = rate of interest per annum (per year)

T = period of time in years

simple interest

(formula not provided)



**this visualisation is not neccesary for understanding

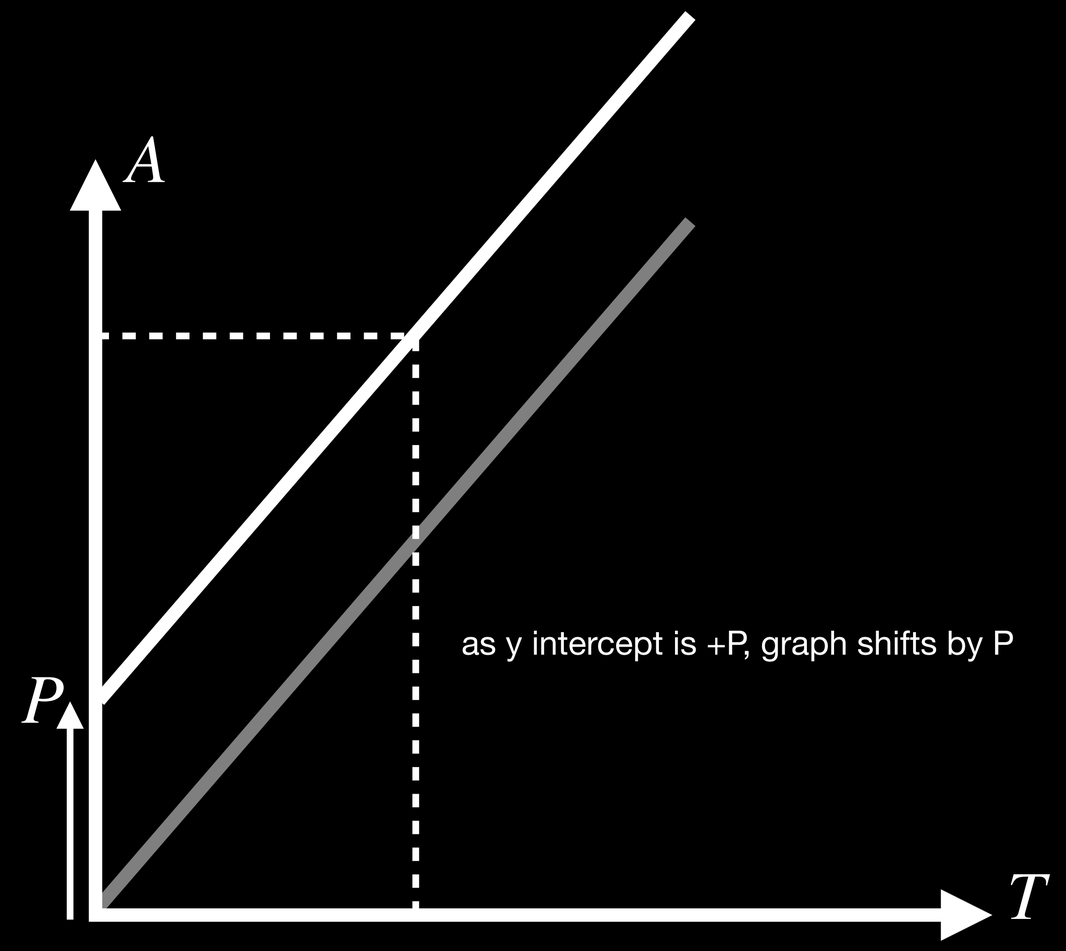
Simple/Compound Interest

Sdjnbhfjd

$$A = P + \frac{PRT}{100}$$

simple interest(total value)

(formula not provided)



**this visualisation is not neccesary for understanding

Simple/Compound Interest

Sdjnbhfjd

$$A = P\left(1 + \frac{r}{100}\right)^n$$

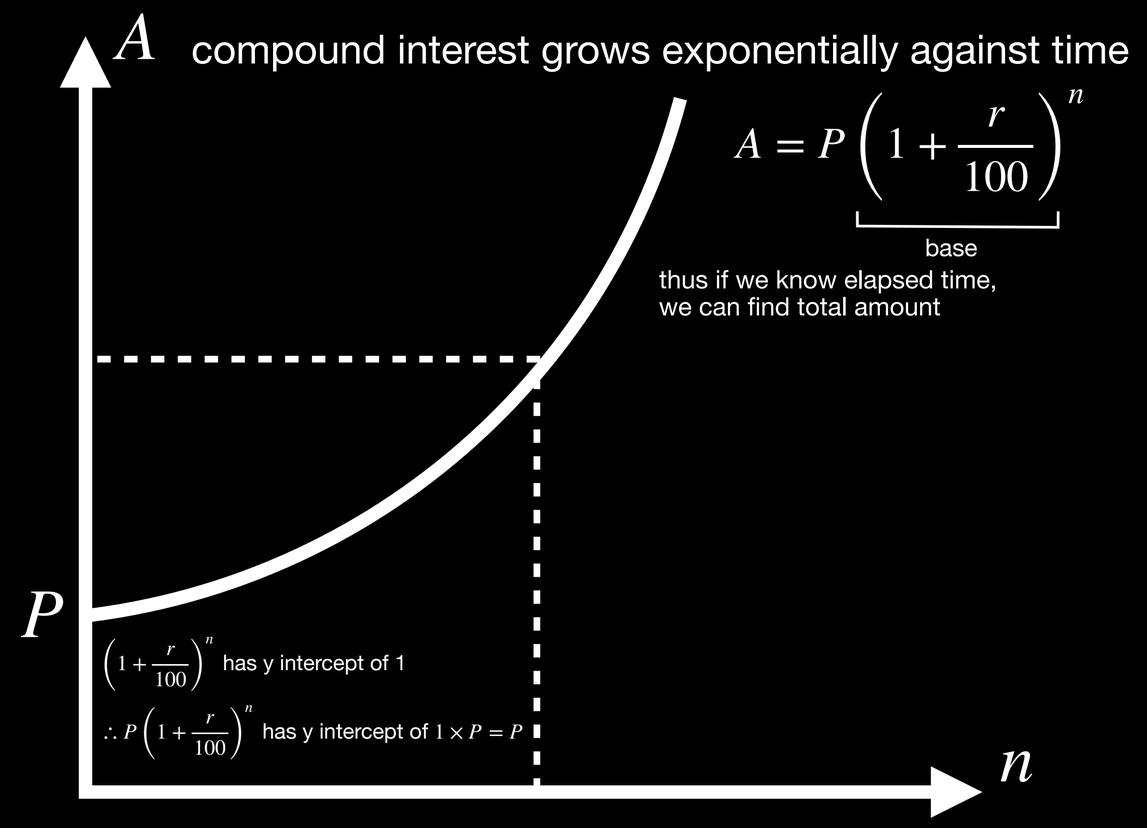
compound interest(total value)

(formula provided)

$P =$ principal amount (original amount)

$r =$ rate of interest per annum (per year)

$n =$ period of time in years



**this visualisation is not neccesary for understanding

Simple/Compound Interest

Sdjnbhfjd

$$A = P\left(1 + \frac{r}{100}\right)^n - P$$

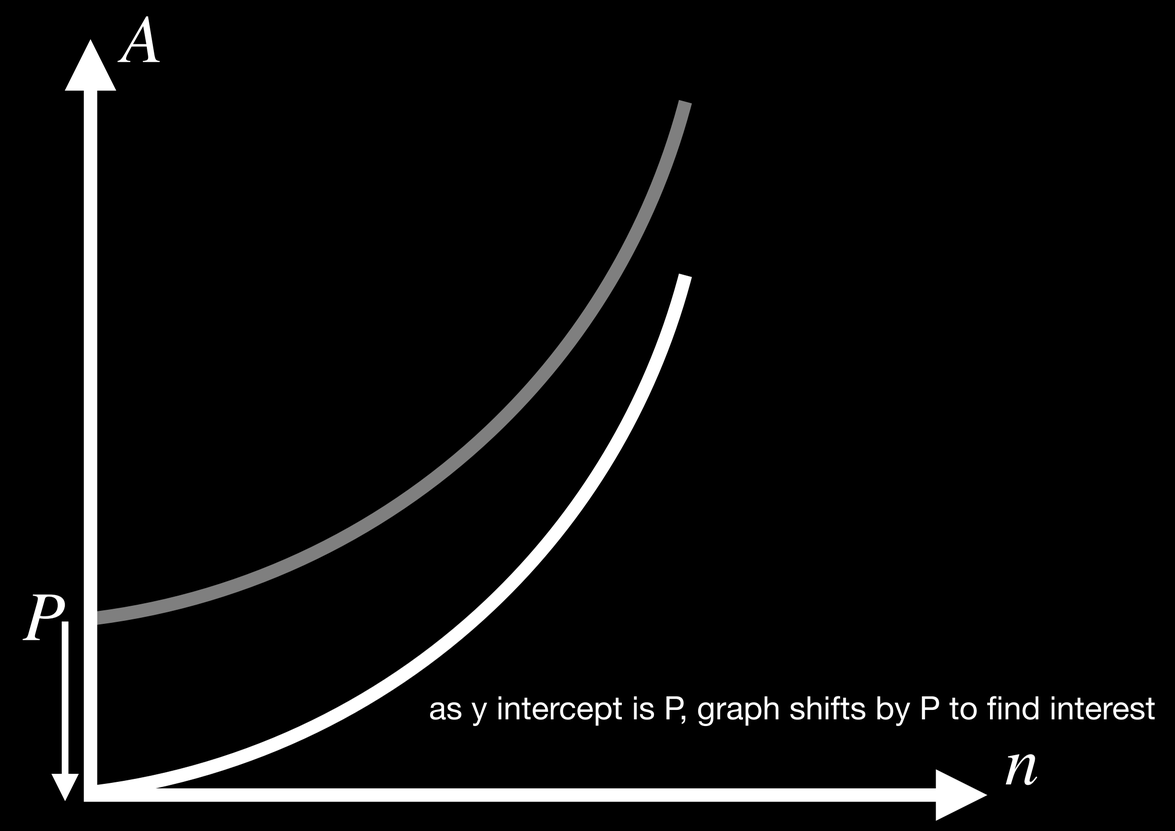
compound interest

(formula not provided)

P = principal amount (original amount)

r = rate of interest per annum (per year)

n = period of time in years



**this visualisation is not neccesary for understanding

Simple/Compound Interest

Sdjnbhfjd

A person borrows $3100 from a bank at an simple interest of 8% per annum. How much interest is made after a)2 years b)8 months

$$\frac{(3100)(8)(2)}{100} = \$496$$

$$\frac{(3100)(8)\left(\frac{8}{12}\right)}{100} = \$165.33\text{(nearest cent)}$$

**the interest rate excludes the percent sign, as we already divide by 100 which accounts for the percentage

The time period must also be in terms of years, for example 8 months = (8/12) years

Simple/Compound Interest

Sdjnbhfjd

A person borrows money for 20 months at a 9% simple interest rate per annum, and repays $1840. How much money did they borrow?

$$\frac{P(9)\left(\frac{20}{12}\right)}{100} + P = \$1840$$

$$0.15P + P = \$1840$$

$$1.15P = \$1840$$

$$P = \$1600$$

Other questions may ask you to solve for time elapsed or interest rate

Simple/Compound Interest

Sdjnbhfjd

Find the total interest earned on $7000 for 3 years at 4% per annum compounded annually.

$$7000\left(1+\frac{4}{100}\right)^{3}-7000=\$874.05\text{(nearest cent)}$$

Percentage sign for interest is omitted as divided by 100

Simple/Compound Interest

Sdjnbhfjd

Find the total interest earned on $7000 for 3 years at 4% per annum compounded half-yearly.

$$7000\left(1+\frac{\frac{4}{2}}{100}\right)^{3\times 2}-7000$$

$$= \$883.14 \text{(nearest cent)}$$

When interest is compounded non annually, both the interest rate and the elapsed time need to be changed:

$$A = P\left(1+\frac{\frac{r}{f}}{100}\right)^{nf}$$

where p is the compound frequency, i.e how many times its compounded in one year

Half yearly—> f=2, monthly —> f=12

Elapsed time is multiplied as we convert into the smaller time intervals, i.e.3 years = 6 half-years

Rate is divided as rate is per annum, i.e. 4% is applied across 1 year, therefore 2% will be applied after half a year

Simple/Compound Interest

Sdjnbhfjd

Find the total interest earned on $7000 for 3 years at 4% per annum compounded monthly.

$$7000\left(1+\frac{\frac{4}{12}}{100}\right)^{3\times 12}-7000=\$890.90\text{(nearest cent)}$$

Simple/Compound Interest

Sdjnbhfjd

Find the total amount to be repaid on $6400 for 20 months at 4% per annum compounded quarterly.

$$6400\left(1+\frac{\frac{4}{4}}{100}\right)^{\frac{20}{12}\times 4}$$

$$= \$6838.95 \text{(nearest cent)}$$

For time periods which are not in years, we need to convert them in terms of years before multiplying by the compounding frequency.
(In some cases, like when compounded monthly, there is no need to divide by 12 to convert to years, as 20/12*12 = 20, thus 20 can just be written)

Simple/Compound Interest

Sdjnbhfjd

Some money was invested in a saving account compounded annually at 5% per annum and $2646 were received after 2 years. Find the original amount of money in the account

$$P\left(1+\frac{5}{100}\right)^2 = \$2646$$

$$P(1.05)^2 = \$2646$$

$$1.1025P = \$2646$$

$$P = \$2400$$

Other questions may ask you to solve for time elapsed or interest rate

Hire Purchase

Sdjnbhfjd

Hire purchase is a payment type that involves paying across a time period in amounts called instalments, where each installment is equal
Hire purchase is always higher than the original price of the product due to interest

$$\text{Total Hire Purchase} = \boxed{\text{Cash price}} + \text{Interest}$$

Total amount payed after the time period — Original price — Simple interest of cash price

$$= \boxed{\text{Deposit/Down payment} + \text{Outstanding balance}} + \text{Interest}$$

amount payed before the time period — cash price - down payment — Simple interest of outstanding balance

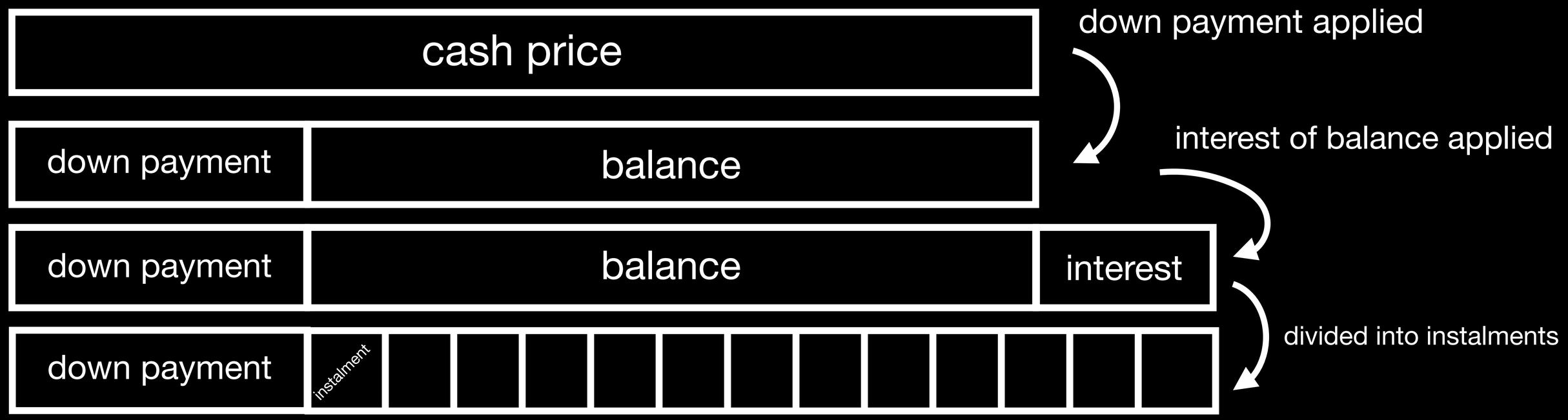
$$= \text{Deposit/Down payment} + (\text{No. of instalments} \times \text{Value of each instalment})$$

amount payed before the time period

**interest is usually simple interest

Hire Purchase

Sdjnbhfjd



Hire Purchase

Sdjnbhfjd

A product of cash price $4300 is bought with zero down payment and 24 monthly instalments at 3% per annum. How much is one monthly instalment?

Total Hire purchase $= 4300 + \dfrac{(4300)(3)(\frac{24}{12})}{100}$ find total from interest

$= \$4558$

Monthly installment $= \dfrac{4558}{24}$ divide by no. of installments

Monthly installment $= \$189.92$(nearest cent)

Hire Purchase

Sdjnbhfjd

A product of cash price $3700 is bought with 25% down payment and 24 monthly instalments at 4% per annum. How much is paid during one instalment?

Outstanding Balance $= 75\% \times 3700$

$= \$2775$

money paid by installments $= 2775 + \dfrac{(2775)(4)(\frac{24}{12})}{100}$

$= \$2997$

one instalment $= \dfrac{2997}{24} = \$124.88$(nearest cent)

Hire Purchase

Sdjnbhfjd

A product of cash price $3500 is bought at a hire purchase of $3628 with 20% down payment and 24 monthly instalments. How much is paid during one instalment?

$$\text{Hire purchase} = 3500 \times 20\% + \text{instalment} \times 24$$

$$3628 = 3500 \times 20\% + \text{instalment} \times 24$$

$$3628 = 700 + \text{instalment} \times 24$$

$$2928 = \text{instalment} \times 24$$

$$\text{instalment} = \$122$$

Exchange Rate

Sdjnbhfjd

Exchange rate refers to the relationship between different currencies.
For example, given that 1 SGD = 3.25 MYR(Malaysian ringgit),
find 45 SGD in MYR and 45 MYR in SGD

$$1\ SGD = 3.25\ MYR$$

$$45\ SGD = 3.25\ MYR \times 45$$

$$45\ SGD = 146.25\ MYR$$

$$1\ SGD = 3.25\ MYR$$

$$1\ SGD \times \frac{45}{3.25} = 45\ MYR$$

$$13.85\ SGD = 45\ MYR$$

Exchange rate is basically a ratio of the two currencies

Bank Exchange Rate

Sdjnbhfjd

Currency	Bank Sell	Bank Buy
USD	1.339	1.359
CAD	0.993	1.014
GBP	1.634	1.674

Bank has foreign currency while you either have local currency or said foreign currency

to convert local to foreign, you want to get foreign currency from the bank thus the bank is selling the foreign currency to you(think of bank's perspective)

to convert foreign to local, you want to give the foreign currency to the bank, thus the bank is buying the foreign currency to you

Bank Exchange Rate

Sdjnbhfjd

Currency	Bank Sell	Bank Buy
USD	1.339	1.359
CAD	0.993	1.014
GBP	1.634	1.674

For bank buy(foreign to local):
The numbers shown represent how much of the foreign currency you will receive when you give one unit of the local currency. For example, when 1 CAD is given, you receive 1.014 SGD.
Thus, if we were to convert 100 CAD to SGD:

$$100\ CAD \times 1.014 = 101.40\ SGD$$

Bank Exchange Rate

Sdjnbhfjd

Currency	Bank Sell	Bank Buy
USD	1.339	1.359
CAD	0.993	1.014
GBP	1.634	1.674

For bank sell:(local to foreign):
The numbers shown represent how much of the local currency you need for one unit of the foreign currency. For example, in order to get 1USD, you must give 1.339 SGD.
Thus, if we were to convert 100 SGD to USD:

$$\frac{100\ SGD}{1.339} = 74.68\ USD$$

**all numbers in the exchange table are in terms of the local currency

Bank Exchange Rate

Sdjnbhfjd

Currency	Bank Sell	Bank Buy
USD	1.359	1.339
CAD	1.014	0.993
GBP	1.674	1.634

Alternatively, you can convert the table into statements as so:
Bank sells 1 USD to you for 1.359SGD
Thus this is SGD to USD

Bank buys 1 USD from you for 1.339SGD
Thus this is USD to SGD

Bank Exchange Rate

Sdjnbhfjd

Currency	Amount	Bank Sell	Bank Buy
USD	US$1	1.359	1.339
CAD	CA$1	1.014	0.993
HKD	HK$100	17.55	17.13

Some conversion tables may have an extra column called “amount”, which changes as so:
For bank sell: to get 100 HKD, you need to give 17.55SGD

For bank buy: when giving 100HKD , you receive back 17.13 SGD

The foreign currency is now in terms of 100 instead of 1

Bank Exchange Rate

Sdjnbhfjd

Currency	Amount	Bank Sell	Bank Buy
USD	US$1	1.359	1.339
CAD	CA$1	1.014	0.993
HKD	HK$100	17.55	17.13

The table shows a bank conversion rate in Singapore.

a) convert 460 USD to SGD.
b) How much SGD is needed to receive 3700 HKD?
c) A person converts 2000 SGD to CAD for a vacation and spends 300 CAD during the trip. They then convert the remaining money back to SGD. Find how much SGD was lost in the process to the nearest dollar.

Bank Exchange Rate

Sdjnbhfjd

Currency	Amount	Bank Sell	Bank Buy
USD	US$1	1.359	1.339
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The table shows a bank conversion rate in Singapore.

a) convert 460 USD to SGD.
b) How much SGD is needed to receive 3700 HKD?
c) A person converts 2000 SGD to CAD for a vacation and spends 300 CAD during the trip. They then convert the remaining money back to SGD. Find how much SGD was lost in the process to the nearest dollar.

$a)$

$$460\ USD \times 1.339 = 615.94\ SGD$$

Bank Exchange Rate

Sdjnbhfjd

Currency	Amount	Bank Sell	Bank Buy
USD	US$1	1.359	1.339
CAD	CA$1	1.014	0.993
HKD	HK$100	17.55	17.13

The table shows a bank conversion rate in Singapore.

a) convert 460 USD to SGD.
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c) A person converts 2000 SGD to CAD for a vacation and spends 300 CAD during the trip. They then convert the remaining money back to SGD. Find how much SGD was lost in the process to the nearest dollar.

$b)$

$$\frac{100x}{17.55} = 3700\ HKD$$

$$100x = 3700\ HKD \times 17.55$$

$$x = \frac{3700\ HKD \times 17.55}{100} = 649.35\ SGD$$

Bank Exchange Rate

Sdjnbhfjd

Currency	Amount	Bank Sell	Bank Buy
USD	US$1	1.359	1.339
CAD	CA$1	1.014	0.993
HKD	HK$100	17.55	17.13

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a) convert 460 USD to SGD.

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c) A person converts 2000 SGD to CAD for a vacation and spends 300 CAD during the trip. They then convert the remaining money back to SGD. Find how much SGD was lost in the process to the nearest dollar.

$c)$

$$\frac{2000SGD}{1.014} = 1972.38CAD$$

$$1972.38CAD - 300CAD = 1672.38CAD$$

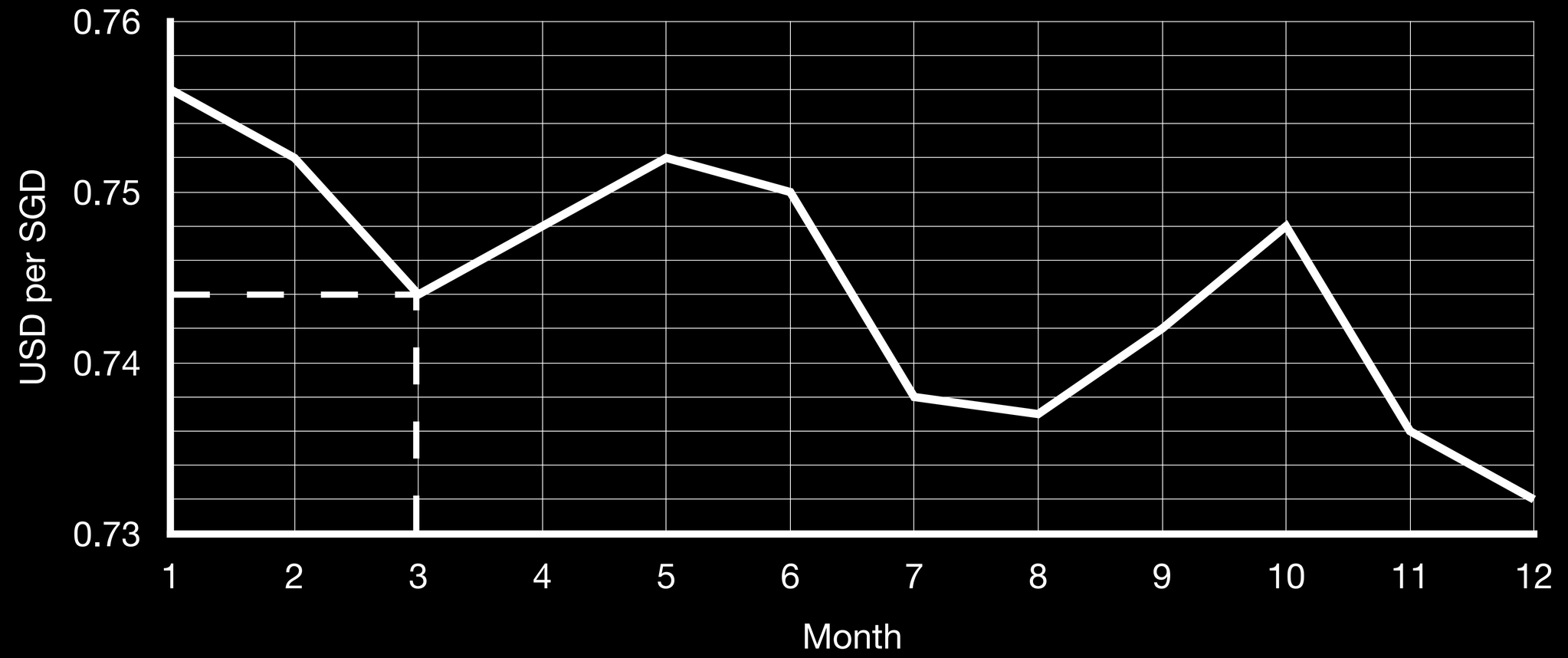
$$1672.38CAD \times 0.993 = 1660.67\ SGD$$

$$2000\ SGD - 1660.67\ SGD = 339\ SGD \text{(nearest dollar)}$$

Conversion Graphs

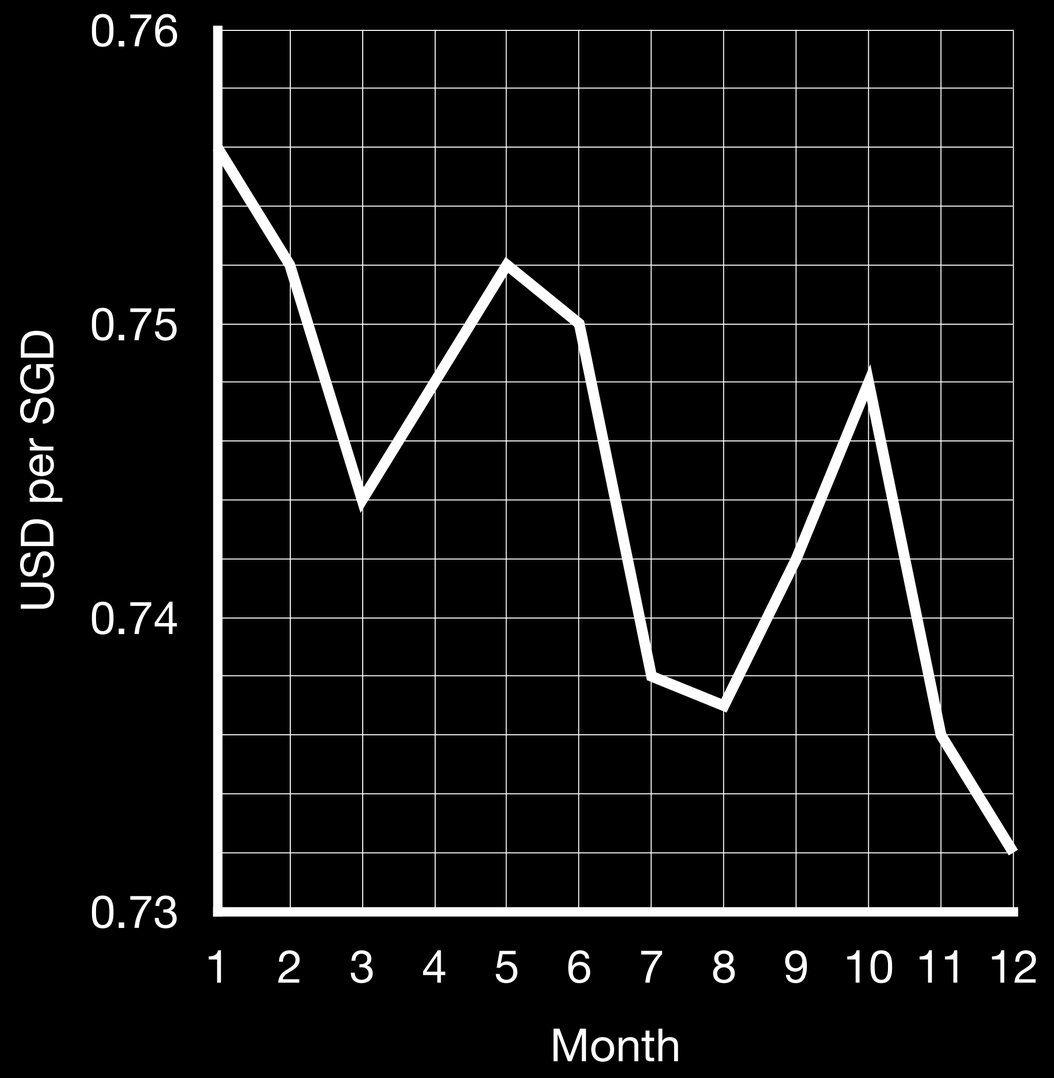
Sdjnbhfjd

Conversion graphs plots out how the conversion rate changes over time: in this case the conversion rate between USD and SGD for each month. The term “USD per SGD” means that for every 1 SGD on a certain month, it will give an amount of USD depending on what month it is.(like USD per one SGD)
For example, if I want to exchange 1 SGD to USD on the third month, I will get 0.744 USD.



Conversion Graphs

Sdjnbhfjd



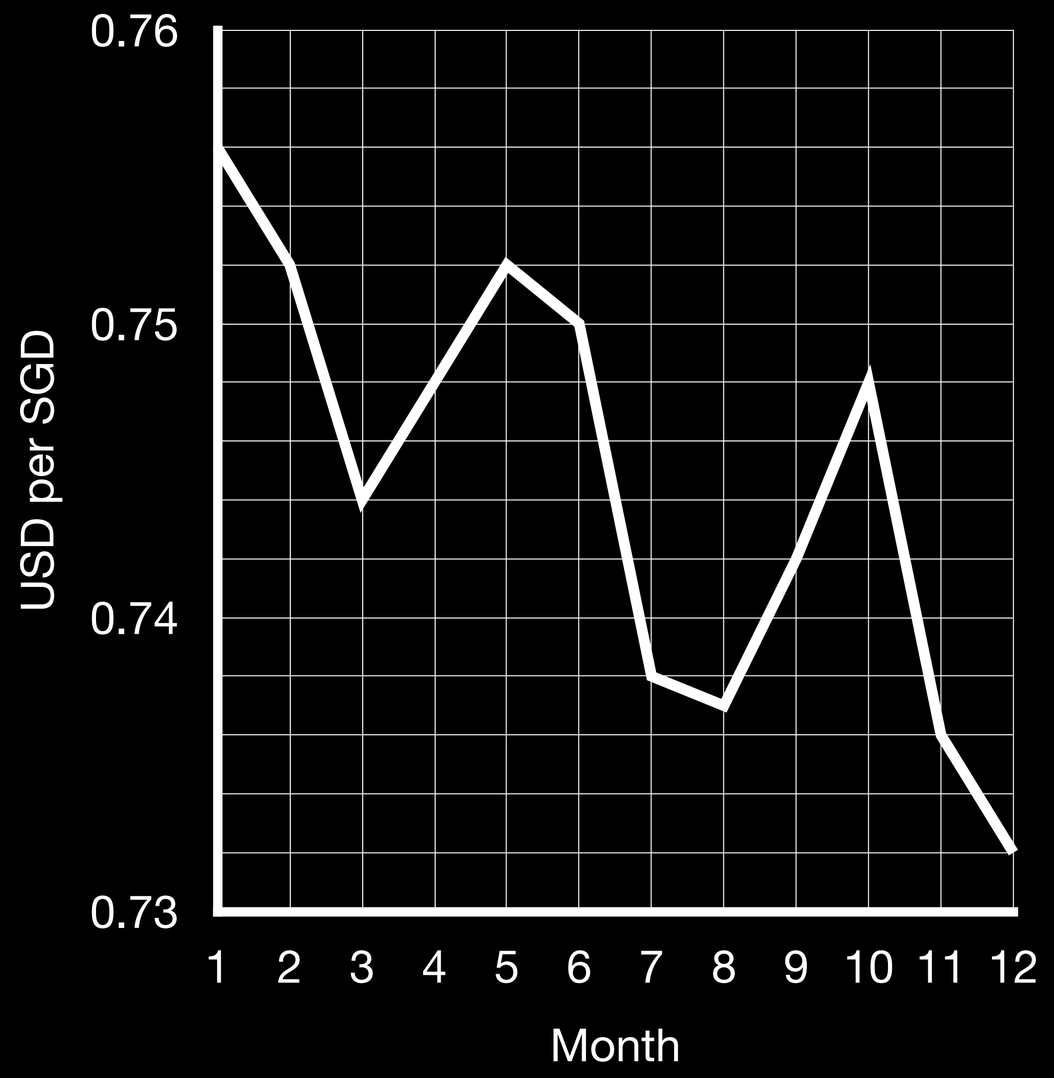
The conversion graph shows the exchange rate of the past year between SGD and USD.

a)find how much USD would be received from 140 SGD on April.

b)find the difference between the greatest and least possible values of USD when converting 2100 SGD

Conversion Graphs

Sdjnbhfjd



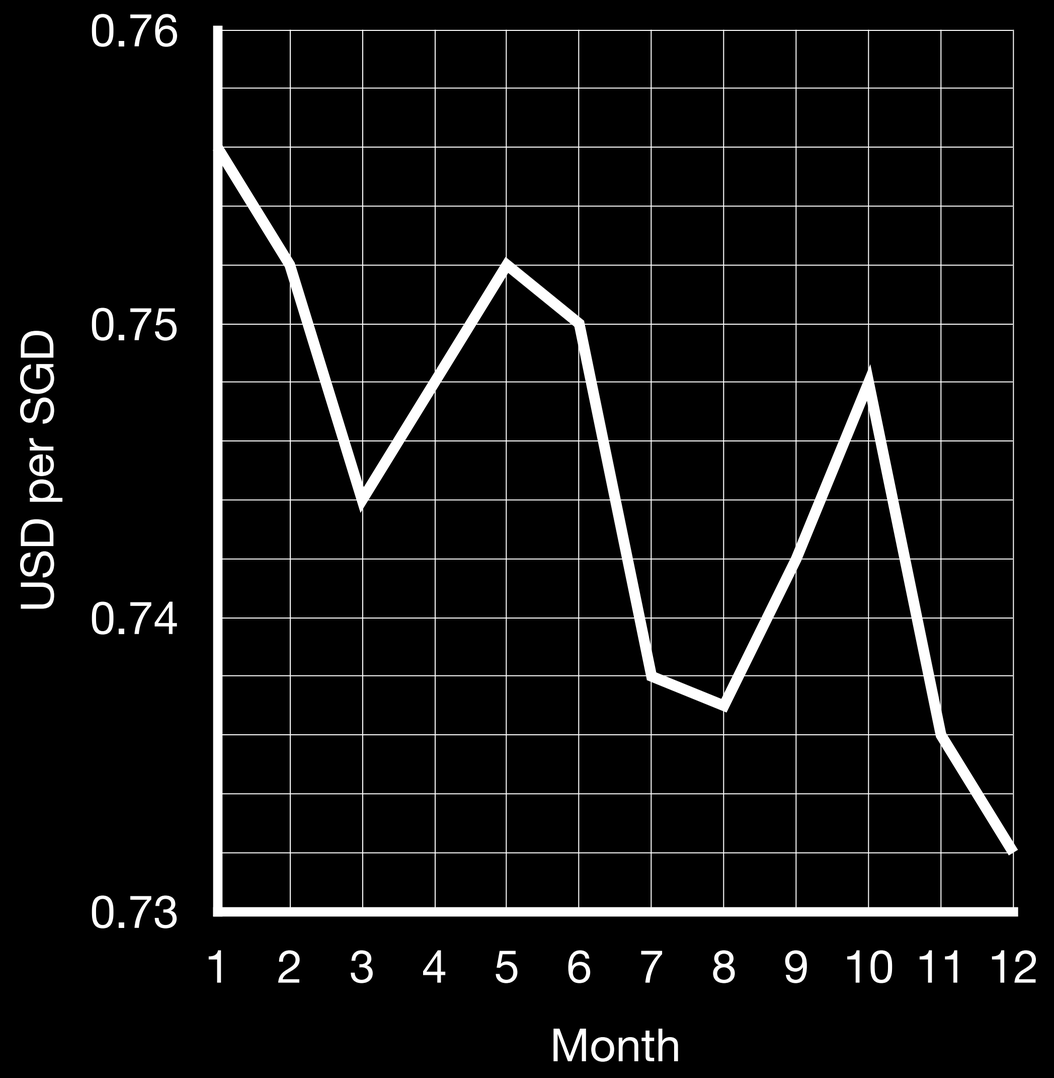
The conversion graph shows the exchange rate of the past year between SGD and USD.
a)find how much USD would be received from 140 SGD on April.
b)find the difference between the greatest and least possible values of USD when converting 2100 SGD

$a)$

$$140\ SGD \times 0.748 = 104.72\ USD$$

Conversion Graphs

Sdjnbhfjd



The conversion graph shows the exchange rate of the past year between SGD and USD.
a)find how much USD would be received from 140 SGD on April.
b)find the difference between the greatest and least possible values of USD when converting 2100 SGD

$b)$

$$2100\ SGD \times 0.756 = 1587.60\ USD$$
$$2100\ SGD \times 0.732 = 1537.20\ USD$$

$$1587.60\ USD - 1537.20\ USD = 50.4\ USD$$

Taxation - Service Charge & GST

Sdjnbhfjd

Service charge and GST are extra charges of the original price of goods or services, where service charge is usually 10% and GST is usually 8%.
Service change is calculated from the original price of the service, followed by GST which is calculated from the original price and service charge

$$\text{original price} \times 110\% \times 108\,\% = \text{total price}$$

service charge

GST

Due to the percentages being multiplied, the order of which percentage is applied first does not matter
**Service charge is only subject to services, like dining at restaurants and not buying products, while GST is for all purchases

Taxation - Service Charge & GST

Sdjnbhfjd

The cost of a meal for 5 people at a restaurant is $78.60 before service charge and GST. If they decide to split the bill evenly after service charge and GST, find how much one person pays after the cost of the meal is subject to 10% service and 7%

$$\$78.60 \times 110\% \times 107\,\% = \$92.5122$$

$$\frac{\$92.5122}{5} = \$18.50 \text{(nearest cent)}$$

Taxation - Income Tax

Sdjnbhfjd

Income tax is an annual tax based on the income of an individual and other things the individual has done. Calculating income tax has a lot of terminology shown:

Gross income: the original amount of money someone makes monthly or yearly before tax
Assessable income: Gross income - Allowed Donations and expenses
Reliefs: Deductibles based on the individuals conditions, like caregiving for ones parents
Rebate: Deductibles usually based on the individuals age
Chargeable Income/Taxable Income: Assessable Income - Reliefs; this is the value where tax is applied
Income tax: the amount of taxes an individual has to pay, do not get confused with taxable income

In summary:

Chargable Income(value that tax is calculated from) =
Gross Income(original money you make) – Approved discounts/expenses – Reliefs(deductibles)

Assessable Income(shortcut label of this)

Taxation - Income Tax

Sdjnbhfjd

In one year, a person receives a salary of $7100 every month. They donate 2% of their income to charity monthly and is entitled to an annual relief amount of $11000 for looking after their parents. Calculate their chargeable income for that year.

$$\text{Annual Salary} = \$7100 \times 12$$
$$\text{Annual Salary} = \$85200$$

$$\text{Chargable Income} = \overset{\text{Assessable Income}}{\$85200 \times 98\%} - \overset{\text{Relief}}{\$11000}$$
$$\text{Chargable Income} = \$72496$$

Taxation - Income Tax

Sdjnbhfjd

The income tax is calculated using a table like so: where it calculates your tax from your taxable income

Taxable Income	Rate (%)	Gross Tax Payable($)
First $20 000 Next $10 000	0 2	0 200
First $30 000 Next $10 000	- 3.5	200 350
First $40 000 Next $40 000	- 7	550 2800
First $80 000 Next $40 000	- 11.5	3350 4600

Taxation - Income Tax

Sdjnbhfjd

Taxable Income	Rate (%)	Gross Tax Payable($)
First $20 000 Next $10 000	0 2	0 200
First $30 000 Next $10 000	- 3.5	200 350
First $40 000 Next $40 000	- 7	550 2800
First $80 000 Next $40 000	- 11.5	3350 4600

10% rebate for taxpayers above 50

To use the table, the steps are:

1. Check which row your tax is calculated in by seeing which row has a higher starting taxable income, and picking the one above it
2. Subtract the first gross tax payable from the taxable income and apply the “next” rate
3. Add the first gross tax payable to the remaining taxable income with the applied rate to get the final income tax.
4. Deduct any rebates if there is any

Read example on next slide if this is unclear

Taxation - Income Tax

Sdjnbhfjd

Taxable Income	Rate (%)	Gross Tax Payable($)
First $20 000 Next $10 000	0 2	0 200
First $30 000 Next $10 000	- 3.5	200 350
First $40 000 Next $40 000	- 7	550 2800
First $80 000 Next $40 000	- 11.5	3350 4600

10% rebate for taxpayers above 50

For example, let's say an individual of 52 years old has a taxable income of $37 230.

We know that we have to use the third row as
$37 230 is below $40 000 and thus we use the row above it.

Next, we deduct $30 000 from $33507 to find the remaining taxable income.
$[\$37\ 230 - \$30\ 000 = \$7\ 230]$
Then multiply it by the rate along the "next $10 000" row
$[3.5\% \times \$7\ 230 = \$253.05]$

Then, we take the first gross tax payable [$200] and add it to $253.05 to get the final income tax. $[\$200 + \$253.05 = \$453.05]$

Finally, apply the rebate as so $[90\% \times \$453.05 = \$407.75(\text{nearest cent})]$

**The gross tax payable of the next $10 000 is when you take 3.5% of $10 000, aka with this method it can be ignored as it will be calculated as such anyway

Taxation - Income Tax

Sdjnbhfjd

Taxable Income	Rate (%)	Gross Tax Payable($)
First $20 000 Next $10 000	0 2	0 200
First $30 000 Next $10 000	- 3.5	200 350
First $40 000 Next $40 000	- 7	550 2800
First $80 000 Next $40 000	- 11.5	3350 4600

10% rebate for taxpayers above 50

Find how much income tax needs to be paid for a 56 year old individual with a taxable income of $72496.

$$\$550 + 7\,\%\,(\$72\,496 - \$40\,000) = \$2824.72$$
$$90\% \times \$2824.72 = \$2542.248$$

Utility Bills

Sdjnbhfjd

Current month Charges	Usage	Rate($)	Amount($)
Electricity	699 kWh	0.2413	168.67
Gas	24 kWh	0.1842	4.42
Water	15.5 Cu M	1.25	19.38
Waterborne Fee	15.5 Cu M	0.30	4.65
Sanitary Appliance Fee	3 fittings	3	9.00
Refuse Removal	1 Qty	8.23	8.23
Water Conservation Tax	$19.38	30%	5.81
GST Tax	$206.51	7%	14.46
Total Charges:			234.62

Utility bills keep track of the monthly payment needed for utilities

Different utilities have different consumption units, like kWh(kilowatts hour), Cu M(cubic meters), etc.

There is also a rate for each utility, called the tariff rate, which when multiplied by the rate gives the consumption charge; how much needs to be paid for that utility.

For example, for gas:

$24\ kWh \times 0.1842 = \$4.42$(nearest cent)

Utility Bills

Sdjnbhfjd

Current month Charges	Usage	Rate($)	Amount($)
Electricity	699 kWh	0.2413	168.67
Gas	24 kWh	0.1842	4.42
Water	15.5 Cu M	1.25	19.38
Waterborne Fee	15.5 Cu M	0.30	4.65
Sanitary Appliance Fee	3 fittings	3	9.00
Refuse Removal	1 Qty	8.23	8.23
Water Conservation Tax	$19.38	30%	5.81
GST Tax	$206.51	7%	14.46
Total Charges:			234.62

There are two extra taxes for utility bills:
Water consumption Tax:
A tax by the government to encourage water consumption by adding a 30% tax to the amount of money spent on water. However, for water consumption exceeding 40 cubic meters, the tax rate is 45%

Utility Bills

Sdjnbhfjd

Current month Charges	Usage	Rate($)	Amount($)
Electricity	699 kWh	0.2413	168.67
Gas	24 kWh	0.1842	4.42
Water	15.5 Cu M	1.25	19.38
Waterborne Fee	15.5 Cu M	0.30	4.65
Sanitary Appliance Fee	3 fittings	3	9.00
Refuse Removal	1 Qty	8.23	8.23
Water Conservation Tax	$19.38	30%	5.81
GST Tax	$206.51	7%	14.46
Total Charges:			234.62

GST Tax: a 7% tax on the sum of each consumption charge except for waterborne and sanitary appliance fees(aka only the ones highlighted)

Utility Bills

Sdjnbhfjd

multiply

Current month Charges	Usage	Rate($)	Amount($)
Electricity	699 kWh	0.2413	
Gas	24 kWh	0.1842	
Water	15.5 Cu M	1.25	
Waterborne Fee	15.5 Cu M	0.30	
Sanitary Appliance Fee	3 fittings	3	
Refuse Removal	1 Qty	8.23	
Water Conservation Tax		30%	
GST Tax		7%	
Total Charges:			

sum

Questions usually ask you to fill in the rest of the utility bill and finding the total charges, which you solve by multiplying accordingly and summing each utilities' charges

Statistics & Data Analysis

Misleading Graphs

!!!!!!!

Questions may ask you to answer why a graph may be misleading
Most cases of misleading graphs are:

Misleading Feature	Viewer Interpretation
Unrepresentative/biased title e.g.“huge increase in X”	Does not allow reader to make their own judgement
Vertical scale does not start from 0	Exaggerates the difference in [measured data]
Pictogram pictures are non proportional	Unclear if height or width of picture represents [dataset], exaggerates the difference of [measured data]
Horizontal axis has inconsistent scale (e.g. 2002,2003,2004,2006,2007, etc.)	Misrepresents trend over years as missed out [x value] could be high

e.g: As the pictogram pictures are non proportional, it is unclear if the height or the width of the picture represents the number of hours slept, which exaggerates the difference of the number of hours slept

Mean, Median, Mode (Central Tendency)

!!!!!!!

Given the list $6,3,4,5,7,5,2,9,8$:

Mean:

Sum of values in list divided by length of list [Used for a general method to find central tendency]

$$\frac{x_1 + x_2 + \ldots + x_n}{n} = \frac{6+3+4+5+7+5+2+9+8}{9} = 5.44(3.s.f)$$

Median:

The middle value that divides the data into two halves [Used when there are very extreme values]

$6,3,4,5,7,5,2,9,8$ $2,3,4,5,\boxed{5},6,7,8,9$ For lists with an even number of terms, take the mean of the middle two terms

Mode:

The most frequent value in the list [Used when mean and median are inaccurate]

$6,3,4,\boxed{5},7,\boxed{5},2,9,8$ If no term repeats, there is no mode there can also be more than one mode

Discrete Data

Not grouped!!!!!!

Discrete data is where numbers have definite values (e.g. whole numbers)
for example, given the discrete data $[1,3,4,3,2,3,4,2,2,4,3,2,1]$, we can represent it as

Number	1	2	3	4
Frequency	2	4	4	3

this allows us to find the mean/median/mode as so:

mean:

$$\text{mean} = \frac{\text{sum of data} \times \text{frequencies}}{\text{sum of frequencies}} \boxed{= \frac{\Sigma fx}{\Sigma f}}$$

provided formula

$$\text{mean} = \frac{2(1) + 4(2) + 4(3) + 3(4)}{2 + 4 + 4 + 3}$$

$$\text{mean} = 2.62$$

median:

take the sum of frequencies and divide by two
from left, count until you have reached the middle

$$\frac{\Sigma f}{2} = \frac{2 + 4 + 4 + 3}{2} = 7.5$$

Number	1	2	3	4
Frequency	2	4	4	3

$7.5 \xrightarrow{-2} 5.5 \xrightarrow{-4} 1.5$

as 1.5 is less than 4,
median has to be 3

if the value here is less than 1,
take the two numbers at the boundary and take the mean
**clarification in examples

$$\text{median} = 3$$

mode:

data with highest frequency

$$\text{mode} = 2,3$$

Discrete Data

Not grouped!!!

Find the mean, median and mode of the data:

Number	1	2	3	4	5
Frequency	2	3	7	6	2

$$\text{mean} = \frac{2(1) + 3(2) + 7(3) + 6(4) + 2(5)}{2 + 3 + 7 + 6 + 2}$$

$$\text{mean} = 3.15$$

$$\text{mode} = 3$$

$$\frac{\Sigma f}{2} = \frac{2 + 3 + 7 + 6 + 2}{2} = 10$$

Number	1	2	3	4	5
Frequency	2	3	7	6	2

median is here

$$10 \xrightarrow{-2} 8 \xrightarrow{-3} 5$$

$$1,1,2,2,2,3,3,3,3,\boxed{3,3},3,4,4,4,4,4,4,5,5$$

$$2 \quad 3 \quad 5$$

median is here

$$\text{median} = 3$$

Discrete Data

Not grouped!!!

The data shows how many hours of sleep a group of students get

Hours	6	7	8	9	10
Students	4	4	8	x	2

a) Given that the mode is 9, find the smallest value of x.
b) Given that the mean is 8, find x.
c) If the median is 8, find range of values for x

Discrete Data

Not grouped!!!

The data shows how many hours of sleep a group of students get

Hours	6	7	8	9	10
Students	4	4	8	x	2

a) Given that the mode is 9, find the smallest value of x.

Mode has to be highest frequency

Thus $x = 8 + 1 = 9$

Discrete Data

Not grouped!!!

The data shows how many hours of sleep a group of students get

Hours	6	7	8	9	10
Students	4	4	8	x	2

b) Given that the mean is 8, find x.

$$\frac{4(6)+4(7)+8(8)+9x+2(10)}{4+4+8+x+2}=8$$

$$\frac{136+9x}{18+x}=8$$

$$136+9x=8(18+x)$$

$$136+9x=144+8x$$

$$9x-8x=144-136$$

$$x=8$$

Discrete Data

Not grouped!!!

The data shows how many hours of sleep a group of students get

Hours	6	7	8	9	10
Students	4	4	8	x	2

$\boxed{8}$,8,8,8,8,8,8,$\boxed{8}$

c) If the median is 8, find the range of values for x.

For the median to be 8,
we know that the middle value must be in the 8 category

If the median is the first 8 in the data set,
data on left side = data on right side

$4 + 4 = 7 + x + 2$

$8 = 9 + x$

$x = -1$ As x cannot be negative, minimum of x has to be 0

If the median is the last 8 in the data set,
data on left side = data on right side

$4 + 4 + 7 = x + 2$

$15 = x + 2$

$x = 13$

thus range of x is $0 \leq x \leq 13$

Discrete Data - Stem and Leaf

Not grouped!!!

The stem and leaf diagram shows the ages of workers in a company.

3	0	2	2	3	6	7
4	2	3	3	3		
5	4	7				
6	2					

Find the mean, median and mode

Stem and leaf diagrams group numbers by the tens position

For example, given the data $[6,7,12,12,23]$

the stem and leaf diagram is as so:

0	6	7
1	2	2
2	3	

this represents 6

this represents 12

Discrete Data - Stem and Leaf

Not grouped!!!

The stem and leaf diagram shows the ages of workers in a company.

3	0 2 2 3 6 7
4	2 3 3 3
5	4 7
6	2

mode

length of data set = 13

median

$$\text{mean} = \frac{30 + 32 + 32 + 33 + 36 + 37 + 42 + 43 + 43 + 43 + 54 + 57 + 62}{13}$$

$$\text{mean} = 41.8(3.s.f)$$

$$\text{median} = 42$$

$$\text{mode} = 43$$

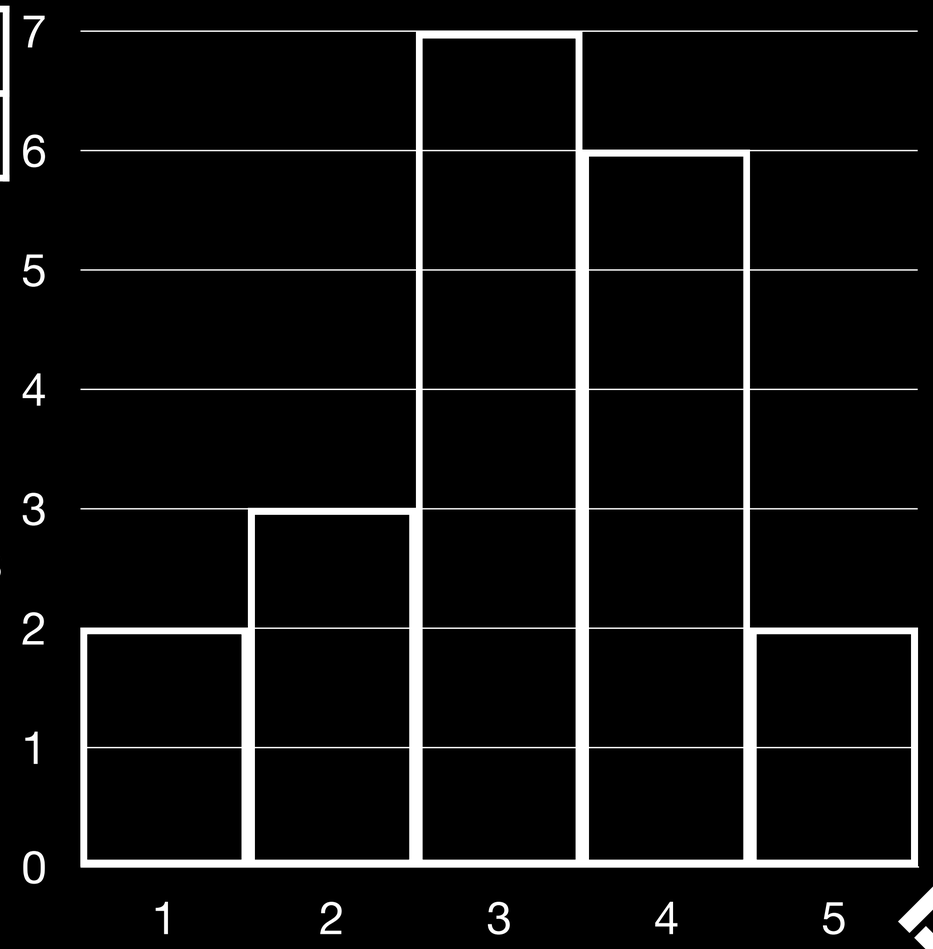
Discrete Data - Histogram

Not grouped!!!

Histograms represent the frequency of discrete data similar to bar graphs

Number	1	2	3	4	5
Frequency	2	3	7	6	2

For example, the table above can be represented by this histogram:



Histograms have no gaps between bars while bar graphs do
You will NOT be required to draw histograms for o lvls

You will NOT be required to ANY STATS DIAGRAMS for o lvls

Grouped Data

grouped!!!

When the data is continuous (e.g all real numbers), the data can be grouped into class intervals:
For example, here is a sample dataset:

Number/x	$0 < x \leq 10$	$10 < x \leq 20$	$20 < x \leq 30$	$30 < x \leq 40$	$40 < x \leq 50$
Frequency	1	2	4	6	3

class interval

As the data is grouped, we lose precision as the data will not be exact
instead of referring to a specific value in our data, we refer to the class interval

Grouped Data

grouped!!!

We can find the mean, median and mode as so:

Number/x	$0 < x \leq 10$	$10 < x \leq 20$	$20 < x \leq 30$	$30 < x \leq 40$	$40 < x \leq 50$
Frequency	1	2	4	6	3

estimated mean:

find the mid interval value
by taking the mean of the bounds of each class

for example, mid interval value of $0 < x \leq 10$ is $\frac{0+10}{2} = 5$

then apply the $\frac{\Sigma fx}{\Sigma f}$ formula as so:

$$\frac{1(5) + 2(15) + 4(25) + 6(35) + 3(45)}{1 + 2 + 4 + 6 + 3} = 30$$

median class:

class interval with the middle value
same as median for discrete data

$$\frac{\Sigma f}{2} = 8$$

median class is $30 < x \leq 40$

modal class:

class interval with highest frequency

modal class is $30 < x \leq 40$

modal class $\neq$ actual mode

estimated mean $\neq$ actual mean

median class $\neq$ actual median

Grouped Data

grouped!!!

The marks of a group of students for a test are shown below:

Number/x	$70 < x \leq 75$	$75 < x \leq 80$	$80 < x \leq 85$	$85 < x \leq 90$	$90 < x \leq 95$	$95 < x \leq 100$
Frequency	3	7	8	5	4	1

a) Find the modal class
b) Calculate an estimated mean of the marks
c) What percentage of the students scored over 85 marks?
d) If the data was represented in a pie chart, what is the angle of the median class?

Grouped Data

grouped!!!

The marks of a group of students for a test are shown below:

Number/x	$70 < x \leq 75$	$75 < x \leq 80$	$80 < x \leq 85$	$85 < x \leq 90$	$90 < x \leq 95$	$95 < x \leq 100$
Frequency	3	7	8	5	4	1

a) Find the modal class

$80 < x \leq 85$

Grouped Data

grouped!!!

The marks of a group of students for a test are shown below:

Number/x	$70 < x \leq 75$	$75 < x \leq 80$	$80 < x \leq 85$	$85 < x \leq 90$	$90 < x \leq 95$	$95 < x \leq 100$
Frequency	3	7	8	5	4	1

b) Calculate an estimated mean of the marks

estimated mean $= \dfrac{3(72.5) + 7(77.5) + 8(82.5) + 5(87.5) + 4(92.5) + 1(97.5)}{3 + 7 + 8 + 5 + 4 + 1}$

estimated mean $= 83.0(3.s.f)$

Grouped Data

grouped!!!

The marks of a group of students for a test are shown below:

Number/x	$70 < x \leq 75$	$75 < x \leq 80$	$80 < x \leq 85$	$85 < x \leq 90$	$90 < x \leq 95$	$95 < x \leq 100$
Frequency	3	7	8	5	4	1

c) What percentage of the students scored over 85 marks?

$$\frac{5+4+1}{3+7+8+5+4+1} \times 100\,\% = 35.7\,\%$$

Grouped Data

grouped!!!

The marks of a group of students for a test are shown below:

Number/x	$70 < x \leq 75$	$75 < x \leq 80$	$80 < x \leq 85$	$85 < x \leq 90$	$90 < x \leq 95$	$95 < x \leq 100$
Frequency	3	7	8	5	4	1

d) If the data was represented in a pie chart, what is the angle of the median class' sector?

median class $= 80 < x \leq 85$

$$\frac{8}{3+7+8+5+4+1} \times 360^\circ = 102.9^\circ$$

use proportionality; 100% of the pie chart has an angle of 360°

Measures of Spread - Range

w i d e

The range of data refers to the difference between the largest and smallest values
For grouped data, the range is the difference between the largest value from the last class interval and the smallest value from the first class interval

Number/x	$70 < x \leq 75$	$75 < x \leq 80$	$80 < x \leq 85$	$85 < x \leq 90$	$90 < x \leq 95$	$95 < x \leq 100$
Frequency	3	7	8	5	4	1

For example, the range of this dataset is $100 - 70 = 30$

However, the range is not very accurate in finding how spread the data is as extreme values are included in the range.

Measures of Spread - Quartiles

w i d e

Quartiles are specific points in the data that split the data into different parts. For example, the median divides the data into two equal parts; i.e. it is at the 50% mark. We also call the median as the second quartile.

If we find the data at the 25% mark, that is referred to as the first/lower quartile

In this case, it is $\frac{13+18}{2} = 15.5$

If we find the data at the 75% mark, that is referred to as the third/upper quartile

In this case, it is $\frac{36+42}{2} = 39$

$$[12, 13, 18, 19, \boxed{24}, 31, 36, 42, 43]$$

Median,second quartile

$$IQR = 39 - 15.5 = 23.5$$

We can also represent the first, second and third quartile as Q_1, Q_2, Q_3 respectively.

The difference between the third and first quartile is called the interquartile range (IQR, but acronym cannot be used in explanations). This can be used to see how consistent the dataset is, without being affected by extreme values. A smaller IQR represents a more consistent dataset, and vice versa.

Measures of Spread - Quartiles

w i d e

Find the median, lower quartile, upper quartile, and interquartile range of the dataset
$[5,7,9,14,19,21,23,26,32,41]$

$$\text{median} = \frac{19+21}{2} = 20$$

lower quartile $= 9$
upper quartile $= 26$

interquartile range $= 26 - 9$
interquartile range $= 17$

median of this = Q_1 | median of this = Q_3

$$[5,7,9,14,19,21,23,26,32,41]$$

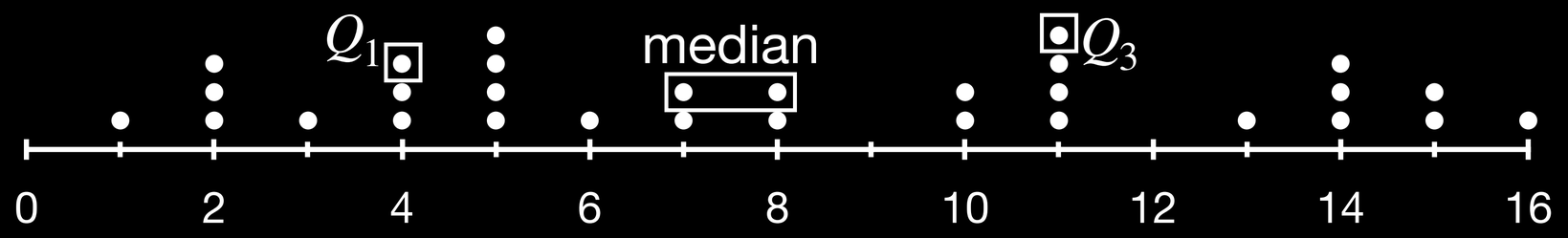
median

First quartile is the median between the lowest value and the median
Third quartile is the median between the median and the highest value
**for datasets with an odd number of data, DO NOT include the median when finding Q_1 and Q_3

Measures of Spread - Quartiles

w i d e

Find the median, lower quartile, upper quartile and interquartile range of the following:



Dot diagrams are basically just histograms but instead of bars dots are used

median $= \dfrac{7+8}{2} = 7.5$

lower quartile $= 4$

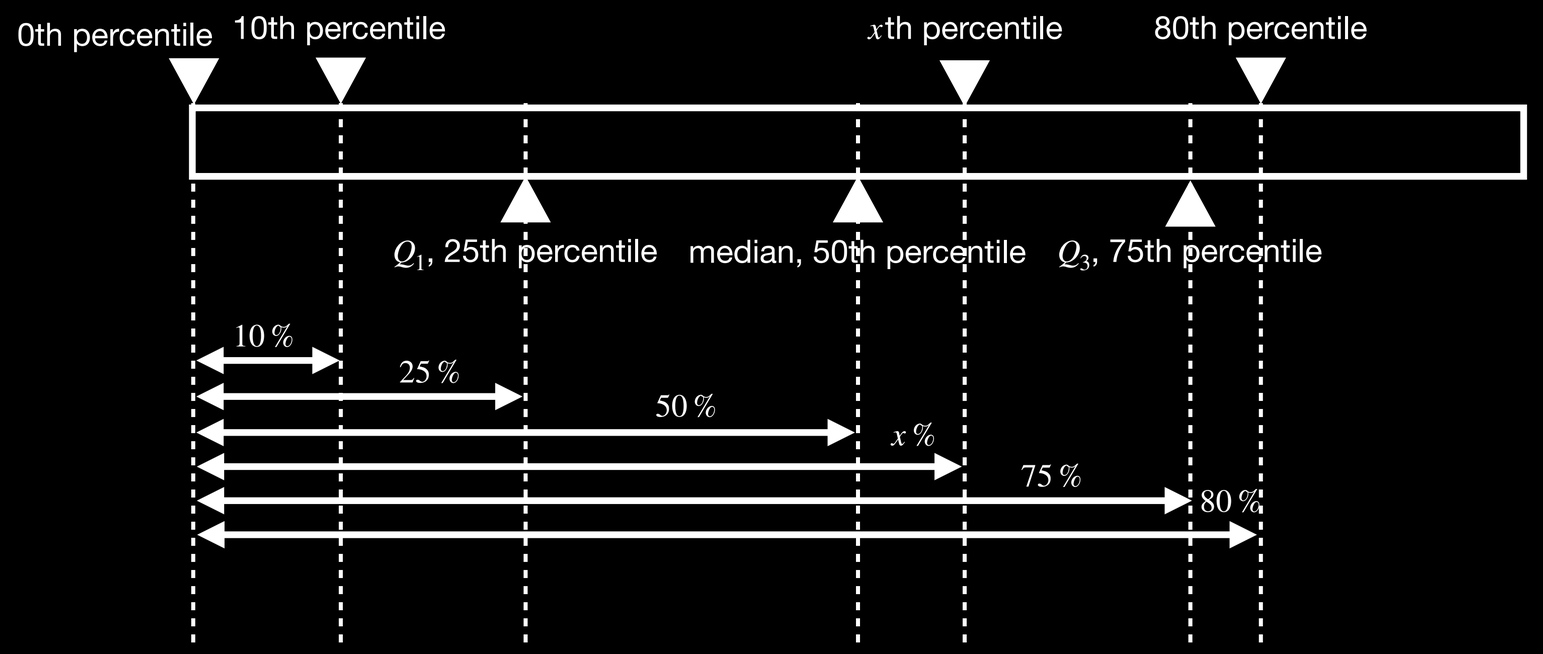
upper quartile $= 11$

interquartile range $= 11 - 4 = 7$

Measures of Spread - Percentiles

w i d e

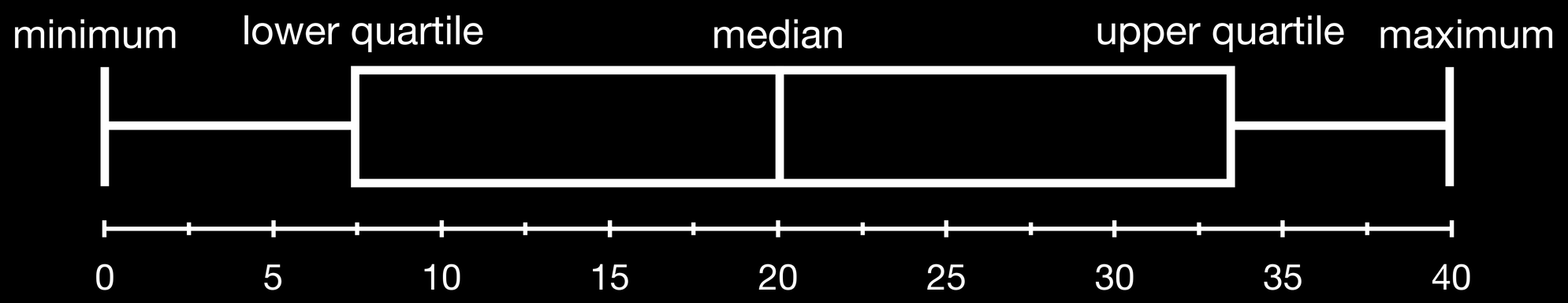
Percentiles are similar to quartiles, where they split the dataset into different parts.
Given the xth percentile, the xth percentile has $x\,\%$ of the data below it. (Not above!!!!!)
For example, the median is also the 50th percentile, as it is the value that has 50% of the data below it. Likewise, the upper quartile is the 75th percentile, as it has 75% of the data below it.



Box-and-Whisker Plots

Truly box

Box and whisker plots (or box plots) display the minimum and maximum of the data, the median, and the upper and lower quartiles in one diagram.

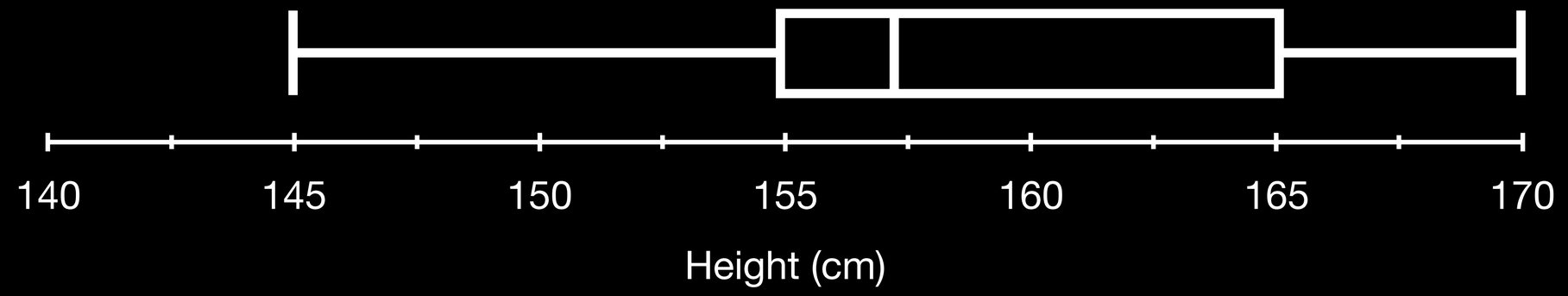


Box plots are always drawn to scale, and either have labels at each part of the box plot, or have a scale next to it(as shown)
You will NOT be tested on drawing box plots

Box-and-Whisker Plots

Truly box

The box plot diagram shows the heights of some students.

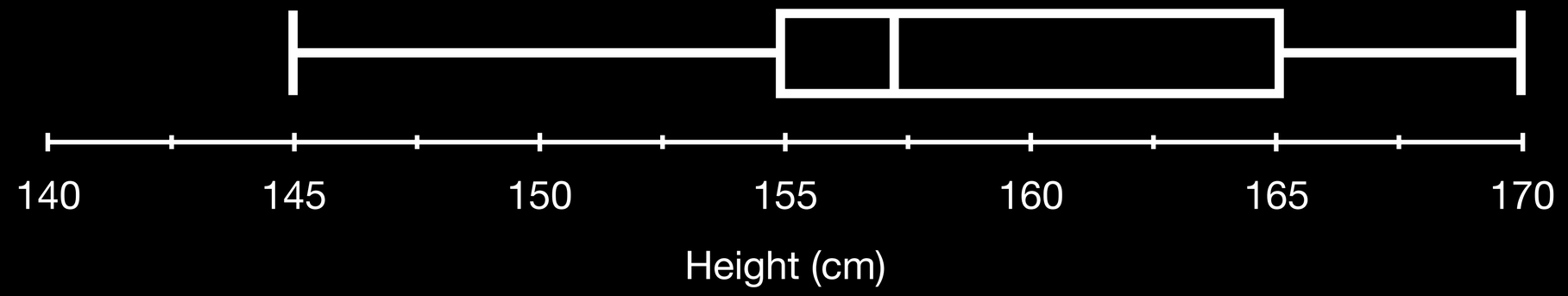


a) Find the interquartile range of the data.
b) What percentage of the students are at least 155cm tall?

Box-and-Whisker Plots

Truly box

The box plot diagram shows the heights of some students.



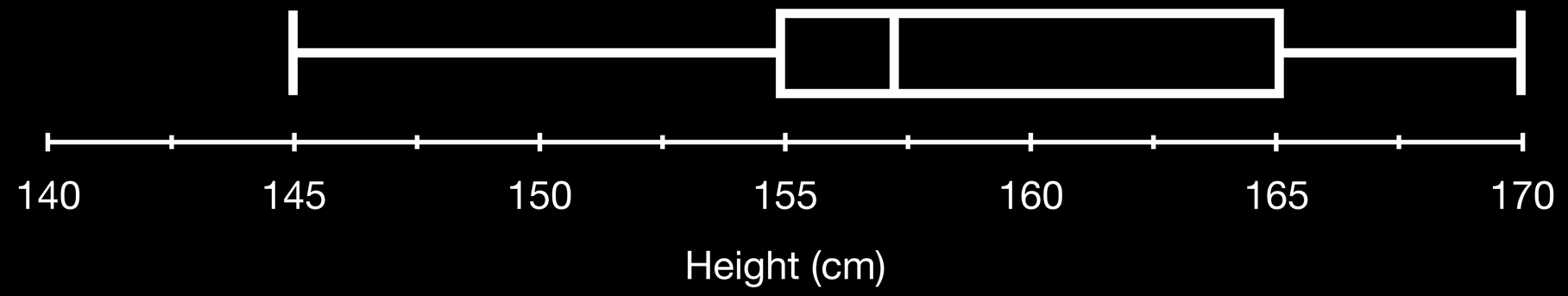
a) Find the interquartile range of the data.

interquartile range $= 165cm - 155cm = 10cm$

Box-and-Whisker Plots

Truly box

The box plot diagram shows the heights of some students.



b) What percentage of the students are at least 155cm tall?

as 155 is the lower quartile, 75% of students will be above the lower quartile

thus 75% of the students are at least 155cm tall

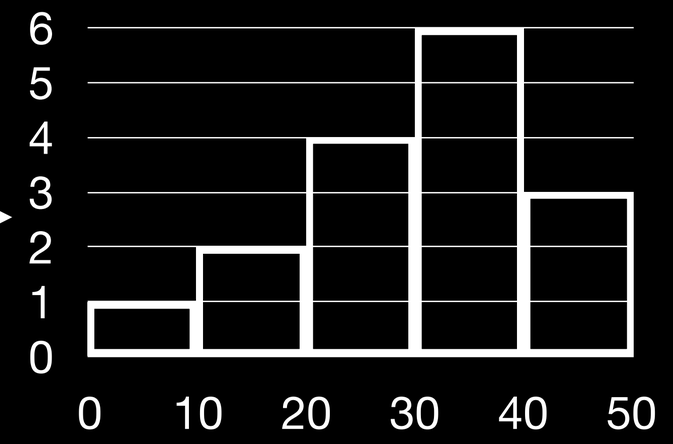
Cumulative Frequency

AAAA

Cumulative frequency is found by adding each frequency to the sum of the previous frequencies
For example, instead of looking at the frequency of each class interval, we look at how much of the data is below a certain number as so:

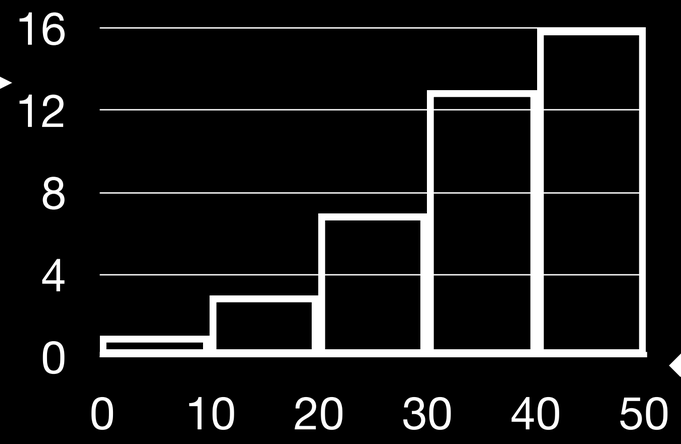
frequency table

Number/x	$0 < x \leq 10$	$10 < x \leq 20$	$20 < x \leq 30$	$30 < x \leq 40$	$40 < x \leq 50$
Frequency	1	2	4	6	3



cumulative frequency table

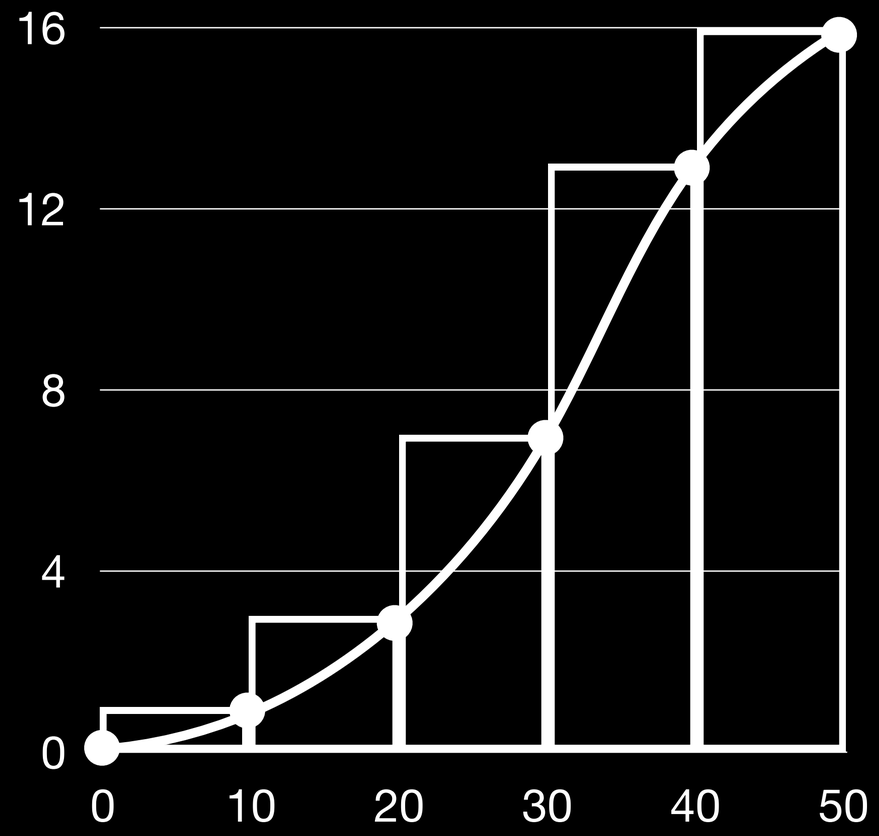
Number/x	$x \leq 10$	$x \leq 20$	$x \leq 30$	$x \leq 40$	$x \leq 50$
Frequency	1	3	7	13	16

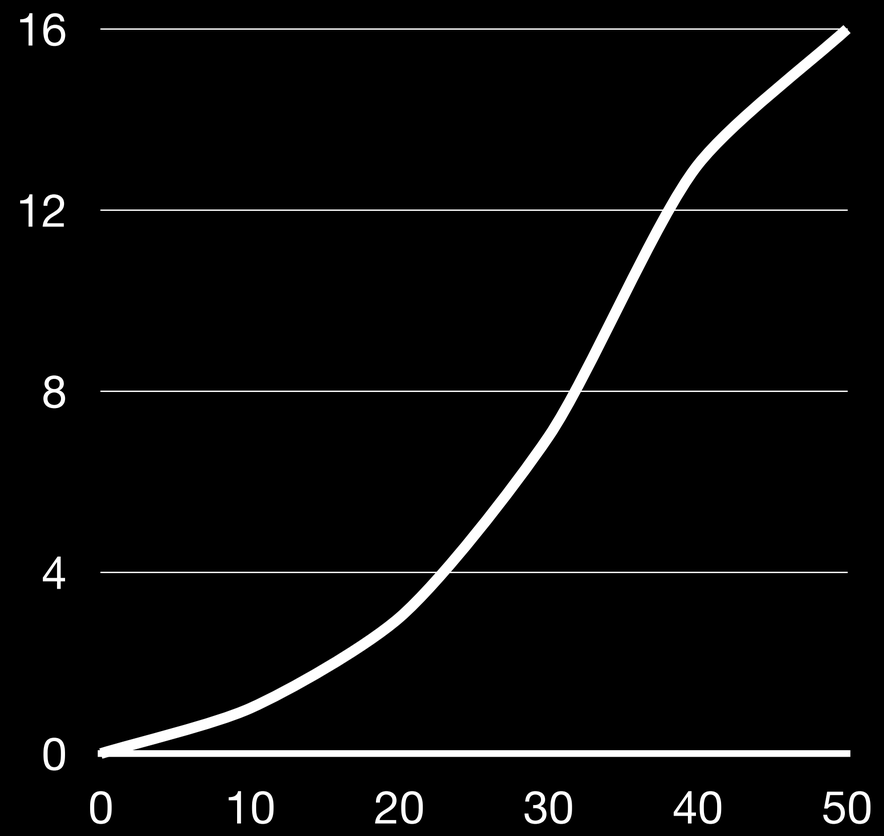


Cumulative Frequency

AAAA

If we join the end points on the graph with a curve, we get a cumulative frequency curve. This curve allows us to estimate other aspects of our data, like percentiles.



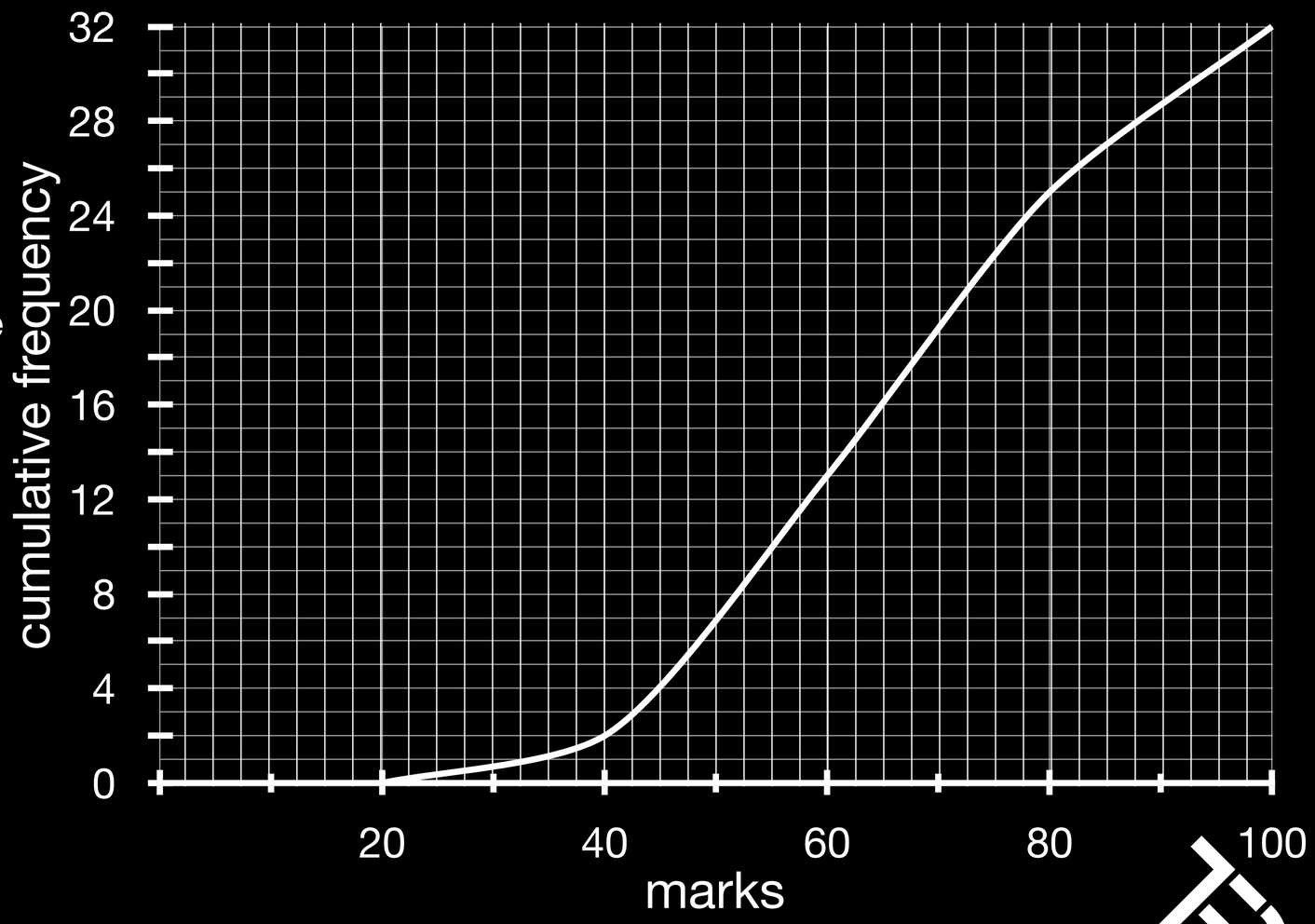


Cumulative Frequency

AAAA

Shown below is a cumulative frequency curve of the marks scored in a test

a) Find the range of the marks scored
b) Using the curve, estimate (i)the median mark, (ii)the interquartile range, (iii)the number of students with less than 45 marks and (iv) the 60th percentile.
c) Given that the passing mark was 60 marks, was the test too difficult? Explain.



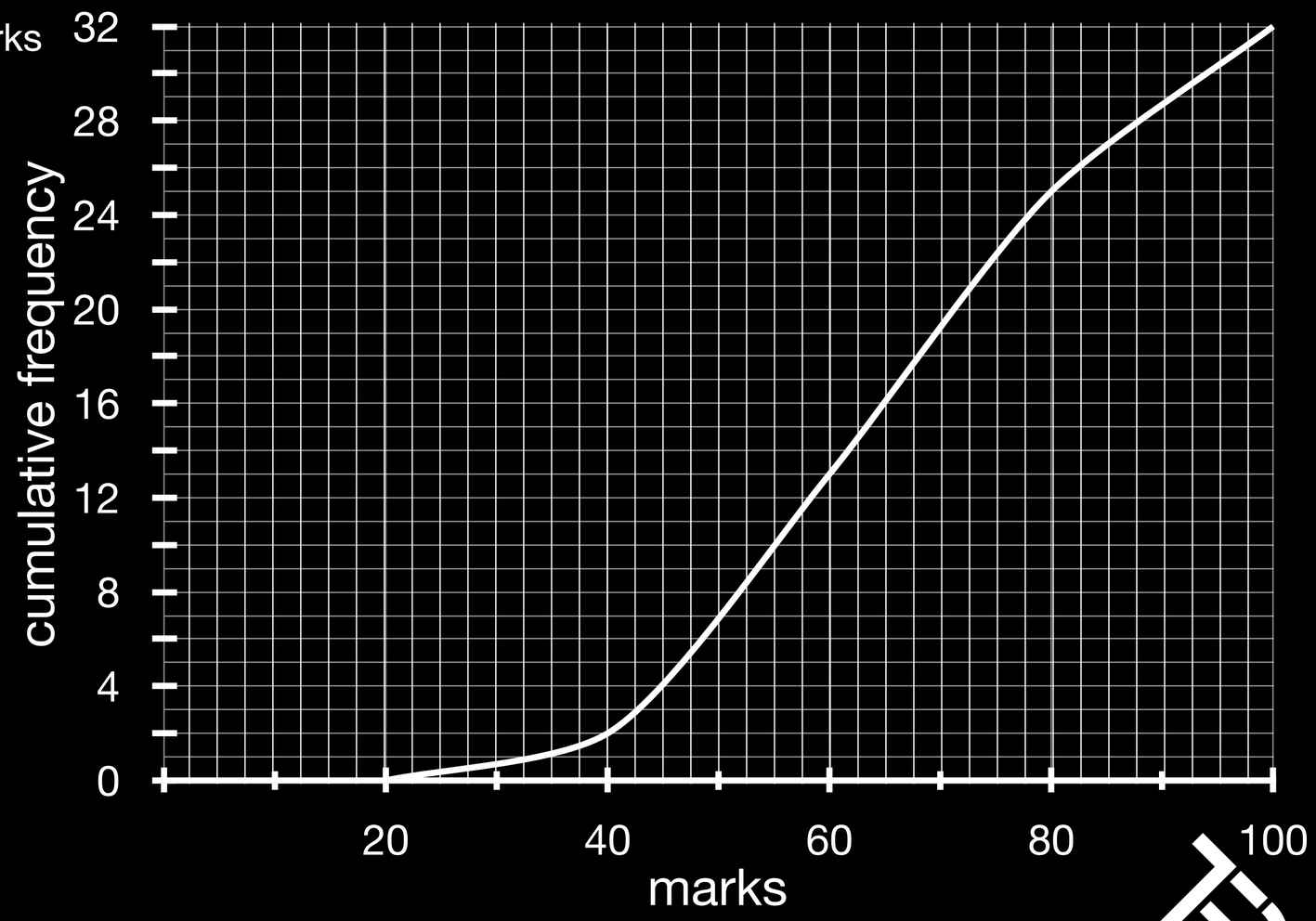
Cumulative Frequency

AAAA

Shown below is a cumulative frequency curve of the marks scored in a test

a) Find the range of the marks scored

range $= 100 - 20 = 80$ marks



Cumulative Frequency

AAAA

Shown below is a cumulative frequency curve of the marks scored in a test

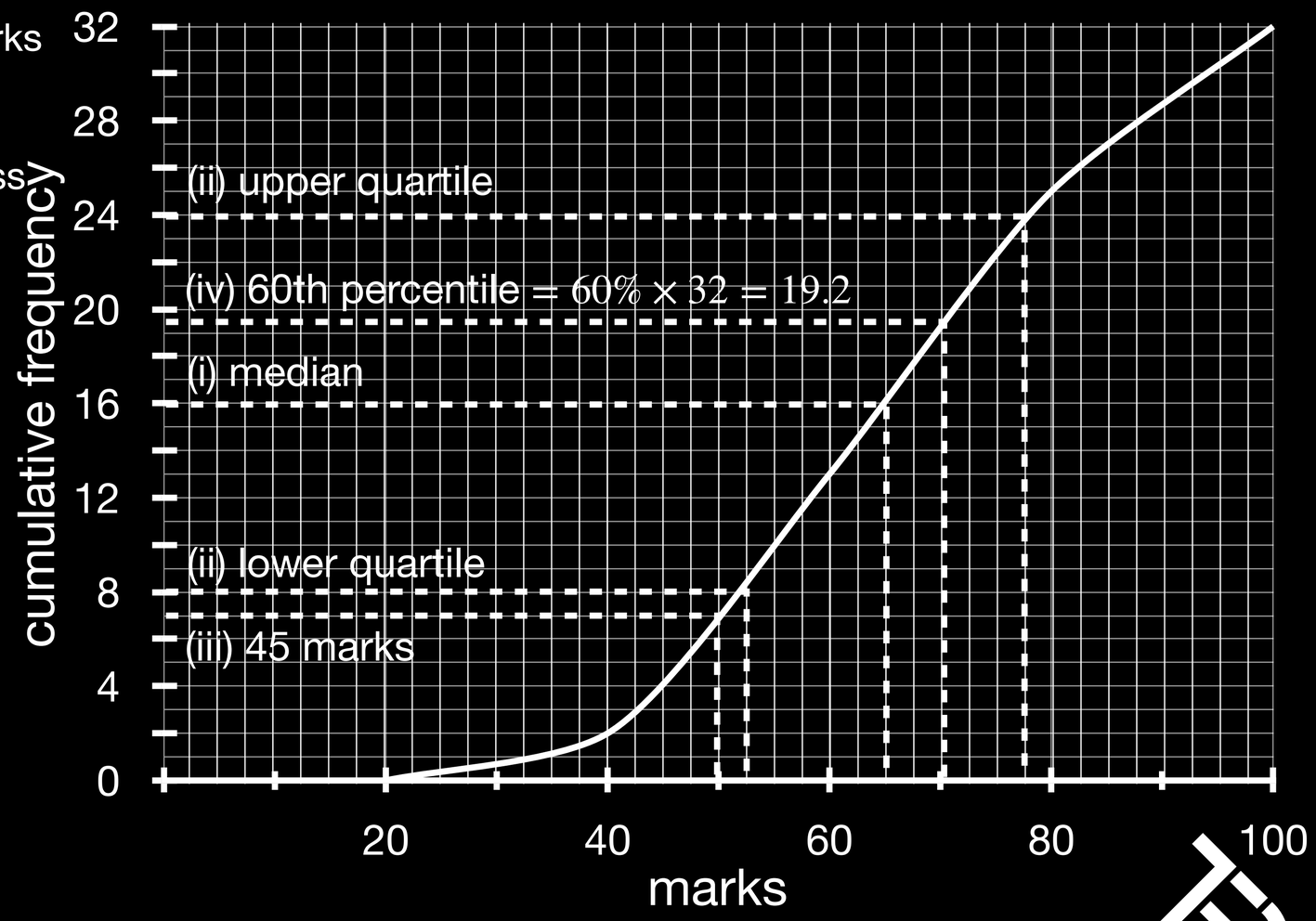
b) Using the curve, estimate (i)the median mark, (ii)the interquartile range, (iii)the number of students with less than 45 marks and (iv) the 60th percentile.

(i) median $= 65$ marks

(ii) interquartile range $= 77.5 - 52.5$
$= 25$ marks

(iii) no. of students <45 marks $= 7$

(iv) 60th percentile $= 70$ marks



Cumulative Frequency

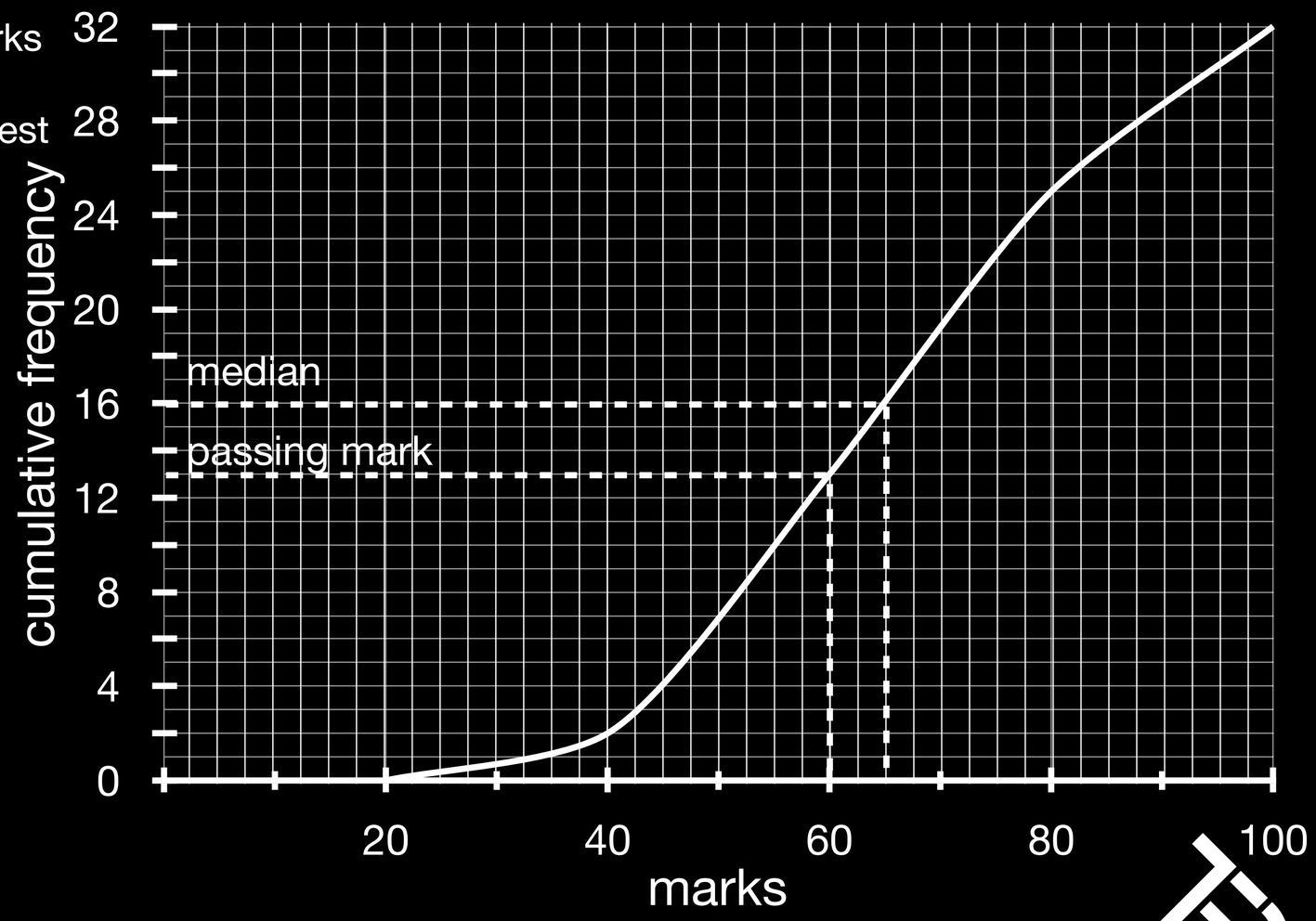
AAAA

Shown below is a cumulative frequency curve of the marks scored in a test

c) Given that the passing mark was 60 marks, was the test too difficult? Explain.

median $= 65$

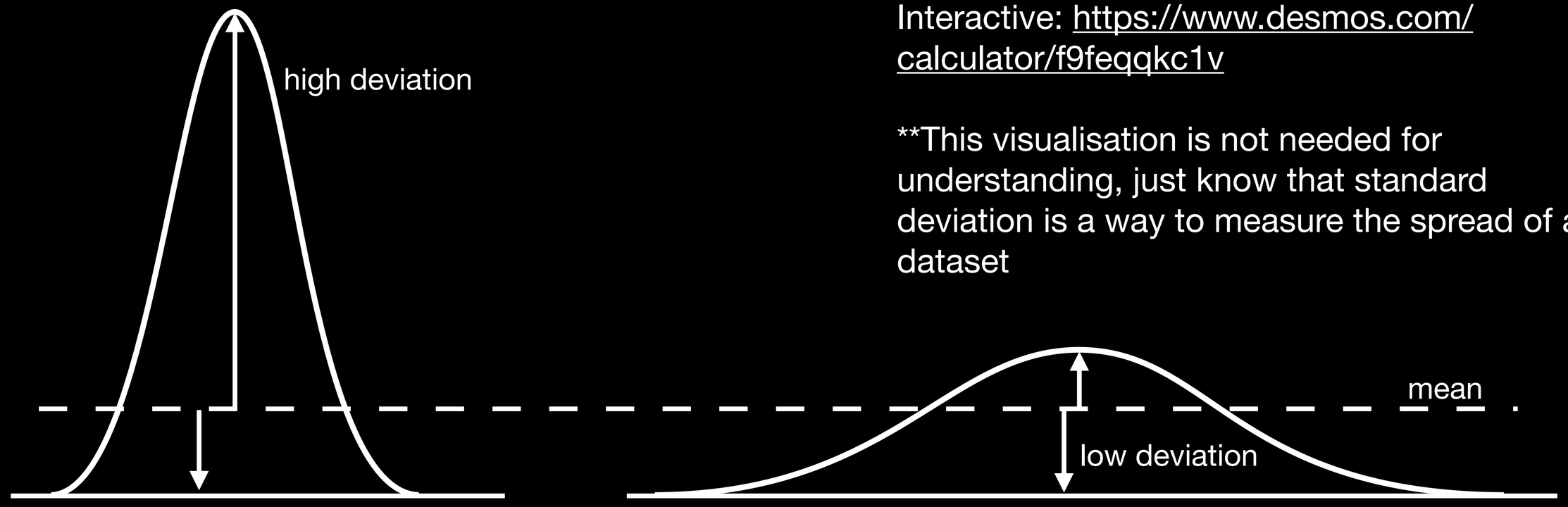
No. The median of the marks scored is 65, which is greater than the passing mark of 60, thus enough people have passed the test, making it sufficiently difficult



Standard Deviation

sigma (!!!!!!!)

Standard deviation is another measurement on the spread of the data by calculating how much the dataset deviates from the mean.



Interactive: https://www.desmos.com/calculator/f9feqqkc1v

**This visualisation is not needed for understanding, just know that standard deviation is a way to measure the spread of a dataset

Standard Deviation

sigma (!!!!!!!)

ungrouped grouped(provided)

$$S.D, \sigma = \sqrt{\frac{\Sigma(x-\bar{x})^2}{\underset{\text{no. of data entries}}{N}}} = \sqrt{\frac{\Sigma fx^2}{\Sigma f} - \left(\frac{\Sigma fx}{\Sigma f}\right)^2}$$

standard deviation

However, most of the time, this formula is not necessary as your calculator can compute the mean and standard deviation of a dataset:
Videos on how to find mean and standard deviation for each calculator model:
SHARP EL-509WS CASIO fx-95SG PLUS or fx-96SG PLUS CASIO fx-97SG X

Standard Deviation

sigma (!!!!!!!)

Find the mean and standard deviation of $[13,18,20,15,14,17,15,16,12,15]$.

mean $= 15.5$, $\sigma = 2.37(3.s.f)$

Find the mean and standard deviation of the grouped data below:

Mid-interval value	5	15	25	35	45

use mid interval value

Number/x	$0 < x \leq 10$	$10 < x \leq 20$	$20 < x \leq 30$	$30 < x \leq 40$	$40 < x \leq 50$
Frequency	3	4	9	2	1

mean $= 21.8(3.s.f)$, $\sigma = 10.3(3.s.f)$

Standard Deviation

sigma (!!!!!!!)

Given a dataset of numbers, if each value x is changed by $ax + b$, where a and b are constants, the mean will be $a(\text{mean}) + b$ and the standard deviation will be $a(\text{standard deviation})$

For example, given the dataset $[13,18,20,15,14,17,15,16,12,15]$, if we were to add 3 to each value or double each value, the mean and standard deviation would also increase by 3 and double respectively

$$[13,18,20,15,14,17,15,16,12,15], \text{ mean} = 15.5, \sigma = 2.37$$

$$\xrightarrow{+3} [16,21,23,18,17,20,18,19,15,18], \text{ mean} = 18.5, \sigma = 2.37$$

$$\xrightarrow{\times 2} [26,36,40,30,28,34,30,32,24,30], \text{ mean} = 31, \sigma = 4.74$$

Standard Deviation

sigma (!!!!!!!)

Given the dataset [19,13,15,12,16,13,15,17,13,12,10,13],

a) find the mean and standard deviation.
b) If each value is subtracted by 2, find the new mean of the data
c) If each value is halved, find the new standard deviation of the data.

Standard Deviation

sigma (!!!!!!!)

Given the dataset $[19,13,15,12,16,13,15,17,13,12,10,13]$,

a) find the mean and standard deviation.
b) If each value is subtracted by 2, find the new mean of the data
c) If each value is halved, find the new standard deviation of the data.

$a)$mean $= 14$, $\sigma = 2.4863 = 2.49(3.s.f)$

$b)$mean $= 14 - 2 = 12$

$c)\sigma = 0.5 * 2.4863 = 1.24$

Data Commentary & Comparison

the c

Questions may often ask you to comment on a dataset or compare two datasets that may have different presentation methods.
In summary, when asked to comment on a dataset, we can find an aspect of the data and draw conclusions like so:

Data Aspect	Used for:	Conclusion	When to use?
Mean	Central tendency	Depending on scenario, the central value tells us what value represents the dataset best e.g. how well a class did <u>overall</u>	General purpose (aka mostly reliable)
Median			Data has extreme values
Mode			Has extreme values/Non-numerical data/value needs to be part of dataset
Range	Data Spread/consistency	Higher range/IQR/SD = less consistent or more spread out Lower range/IQR/SD = more consistent or less spread out	No extreme values (and rarely used)
IQR			Data has extreme values
S.D.			Data has NO extreme values

For all explanations (not just stats), mathematical evidence/comparison must be presented, i.e. calculating mean of each data and comparing them to each other
You should explain with: point + math evidence with comparison + conclusion & relate back to scenario

Data Commentary & Comparison

the c

The stem and leaf diagram below represents the marks of two classes, A and B.

a)Find the mean and standard deviation of i)class A ii)class B

b)Which class did better overall? Explain.

c)Which class had more reliable scores? Explain.

Class A					Class B				
		7	2	4	5	5			
	9	3	0	5	1	4	7	8	
7	6	4	4	6	2	3	7	8	8
8	8	5	1	7	0	6			
		4	3	8	3	7			
			4	9	2				

Dual stem and leaf diagrams:

X			Y	
7	2	1	5	5
	2	2	0	

12, not 21

20

Data Commentary & Comparison

the c

The stem and leaf diagram below represents the marks of two classes, A and B.
a)Find the mean and standard deviation of i)class A ii)class B

Class A					Class B				
		7	2	4	5	5			
	9	3	0	5	1	4	7	8	
7	6	4	4	6	2	3	7	8	8
8	8	5	1	7	0	6			
		4	3	8	3	7			
			4	9	2				

Class A:
mean $= 67.1875, \sigma = 14.6 (3.s.f)$

Class B:
mean $= 65.375, \sigma = 14.0 (3.s.f)$

Data Commentary & Comparison

the c

The stem and leaf diagram below represents the marks of two classes, A and B.
b)Which class did better overall? Explain.

Class A					Class B				
		7	2	4	5	5			
	9	3	0	5	1	4	7	8	
7	6	4	4	6	2	3	7	8	8
8	8	5	1	7	0	6			
		4	3	8	3	7			
			4	9	2				

point

math evidence & comparison

Class A. As the mean of class A's scores is 67.1875, which is greater than class B's mean score of 65.375, class A performed better than class B overall.

conclusion & relate back to scenario(class performance)

Data Commentary & Comparison

the c

The stem and leaf diagram below represents the marks of two classes, A and B.
c)Which class had more reliable scores? Explain.

Class A		Class B
7 2	4	5 5
9 3 0	5	1 4 7 8
7 6 4 4	6	2 3 7 8 8
8 8 5 1	7	0 6
4 3	8	3 7
4	9	2

point

math evidence & comparison

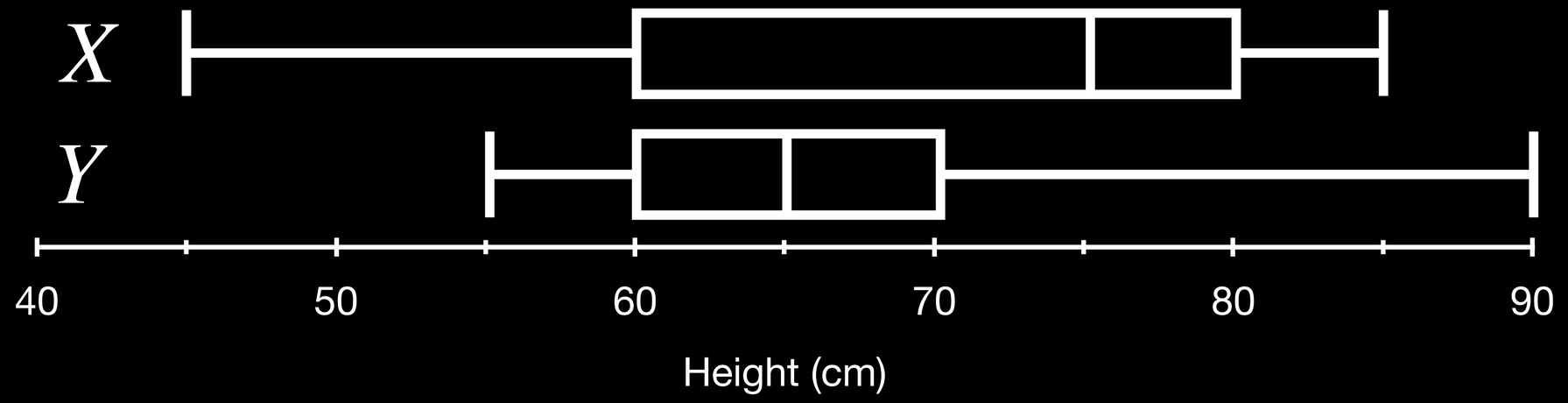
Class B. As the standard deviation of class B's scores is 14.0, which is less than class A's standard deviation of 14.6, class B had more reliable scores than class A.

conclusion & relate back to scenario(reliability)

Data Commentary & Comparison

the c

The box plots shows the distribution of marks from two classes X and Y.

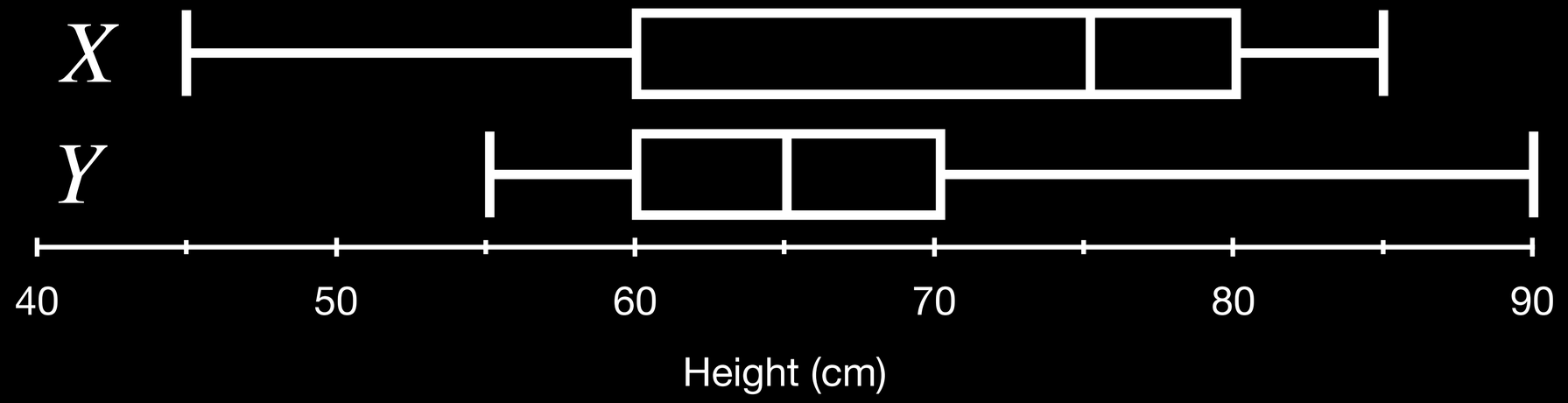


a) Which class had a better result? Explain.
b) Which class was more consistent? Explain.

Data Commentary & Comparison

the c

The box plots shows the distribution of marks from two classes X and Y.



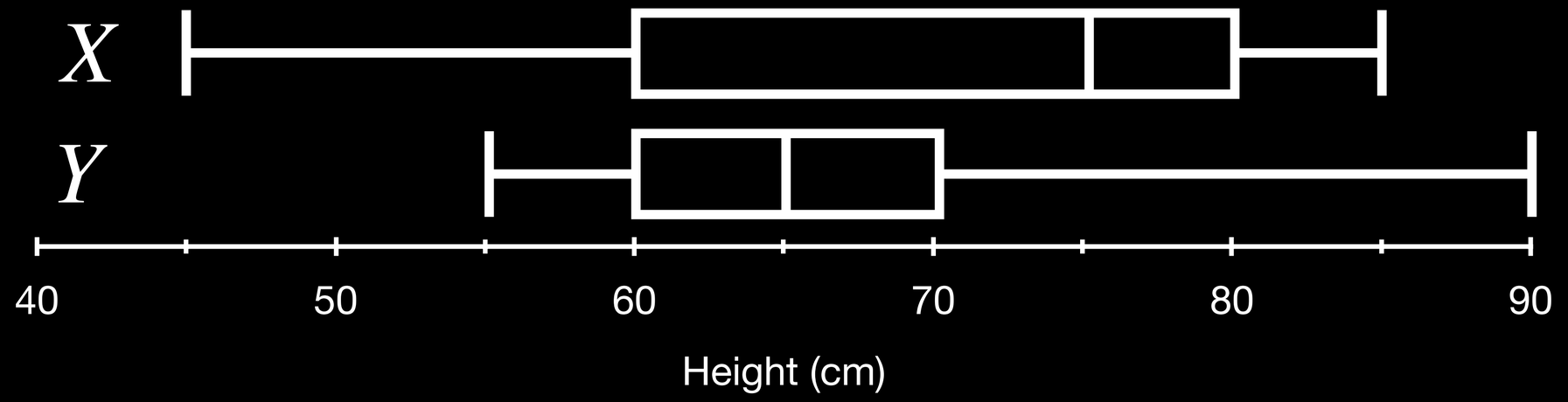
a) Which class had a better result? Explain.

Class X, as it has a median mark of 75 marks while class Y has a lower median mark of 65 marks, thus class X did better overall.

Data Commentary & Comparison

the c

The box plots shows the distribution of marks from two classes X and Y.



b) Which class was more consistent? Explain.

Class Y, as it has an interquartile range of 10, while class X has a higher interquartile range of 20, thus class Y was more consistent

Data Commentary & Comparison

the c

The data shown represents the time taken for some workers to complete a task.

Time taken in seconds/x	$10 \leq x < 20$	$20 \leq x < 30$	$30 \leq x < 40$	$40 \leq x < 50$	$50 \leq x < 60$
Frequency	2	11	7	5	2

a) Find the estimated mean, median class and modal class of the data.
b) Between the mean and median, which is a better measure of central tendency?
c) Between the median and mode, which is a better measure of central tendency?

Data Commentary & Comparison

the c

The data shown represents the time taken for some workers to complete a task.

Time taken in seconds/x	$10 \leq x < 20$	$20 \leq x < 30$	$30 \leq x < 40$	$40 \leq x < 50$	$50 \leq x < 60$
Frequency	2	11	7	5	2

a) Find the estimated mean, median class and modal class of the data.

Estimated mean = 32.7(3.s.f)
Median class is $30 \leq x < 40$
Modal class is $20 \leq x < 30$

Data Commentary & Comparison

the c

The data shown represents the time taken for some workers to complete a task.

Time taken in seconds/x	$10 \leq x < 20$	$20 \leq x < 30$	$30 \leq x < 40$	$40 \leq x < 50$	$50 \leq x < 60$
Frequency	2	11	7	5	2

b) Between the mean and median, which is a better measure of central tendency?

Estimated mean = 32.7(3.s.f)

Median class is $30 \leq x < 40$

Median. As there are outliers in the data, the mean will affected by the outliers and will be an inaccurate measure of central tendency, thus the median would be more appropriate.

Data Commentary & Comparison

the c

The data shown represents the time taken for some workers to complete a task.

Time taken in seconds/x	$10 \leq x < 20$	$20 \leq x < 30$	$30 \leq x < 40$	$40 \leq x < 50$	$50 \leq x < 60$
Frequency	2	11	7	5	2

c) Between the median and mode, which is a better measure of central tendency?

Median class is $30 \leq x < 40$

Modal class is $20 \leq x < 30$

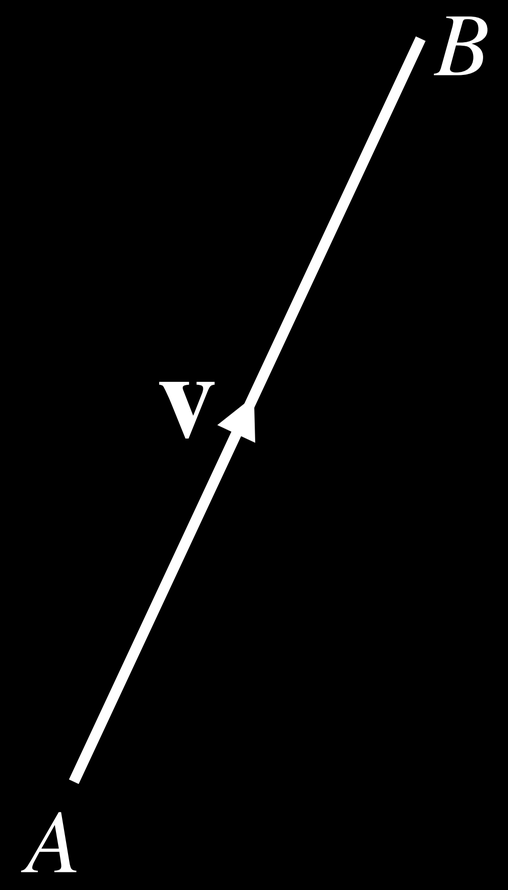
Mode. As the mode is much more significantly larger than the other values, the mode would be much more accurate compared to the median, thus the mode would be more appropriate.

**this is more of a special case, but other cases where mode is more useful is when median is not part of dataset, i.e. median of number of children in household is 2.5 (not possible for 2.5 children, thus mode is better)

Vectors

Vector Notation

physics jumpscare!!!!!!!!!



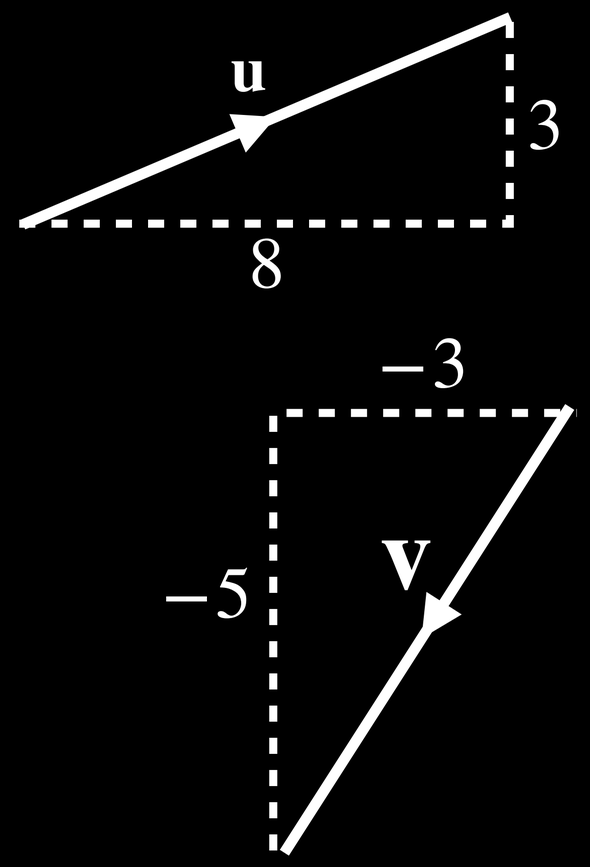
$$\overrightarrow{AB} = \mathbf{v} = \underset{\sim}{v}$$

Vectors have both magnitude (length of vector, denoted as AB, $|\overrightarrow{AB}|$ or $|\mathbf{v}|$) and direction

Vectors are written as a pair of letters with and arrow or a letter that is bolded(in printing) or a letter with a tilde at the bottom (in writing)

Vector Notation

physics jumpscare!!!!!!!!!



Column vectors are notated with the horizontal direction from its start on the top, and the vertical direction from its start at the bottom, like so:

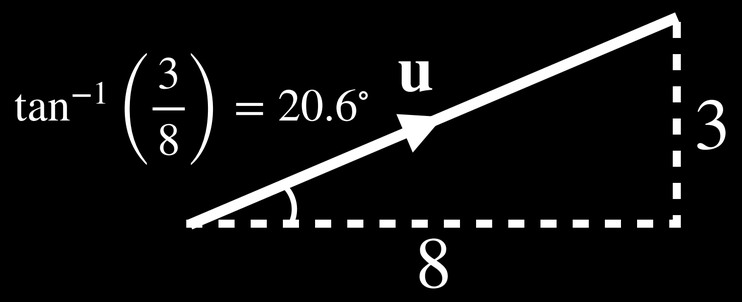
$$\mathbf{u} = \begin{pmatrix} 8 \\ 3 \end{pmatrix}, \mathbf{v} = \begin{pmatrix} -5 \\ -3 \end{pmatrix}$$

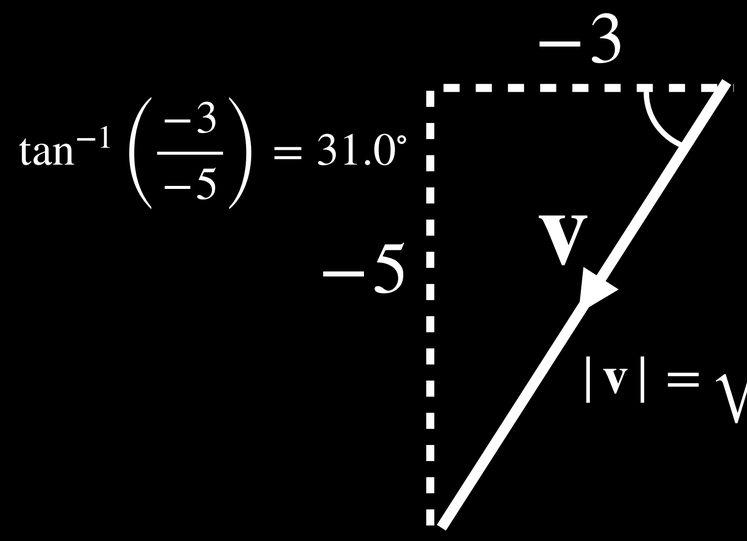
Two vectors are equal when they have the same magnitude and direction

Vector Properties

physics jumpscare!!!!!!!!!

$$|\mathbf{u}| = \sqrt{3^2 + 8^2} = \sqrt{73}$$





$$|\mathbf{v}| = \sqrt{(-3)^2 + (-5)^2} = \sqrt{34}$$

The magnitude of a vector is calculated as so:

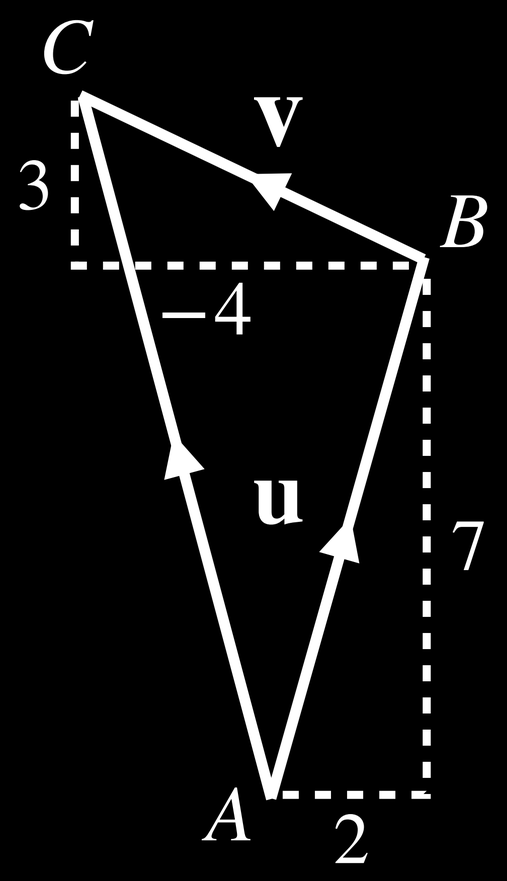
Given that $\overrightarrow{AB} = \begin{pmatrix} a \\ b \end{pmatrix}$, $|\overrightarrow{AB}| = \sqrt{a^2 + b^2}$ [distance formula]

The angle a vector makes with the horizontal from where the vector starts may also be tested (more under trigo)

Angle $\overrightarrow{AB}$ makes with the horizontal $= \tan^{-1}\left(\frac{b}{a}\right)$

Vector Operations - Addition

physics jumpscare!!!!!!!!!



Two vectors can be added as so:

Given that $\overrightarrow{AB} = \mathbf{u} = \begin{pmatrix} 2 \\ 7 \end{pmatrix}, \overrightarrow{BC} = \mathbf{u} = \begin{pmatrix} -4 \\ 3 \end{pmatrix}$

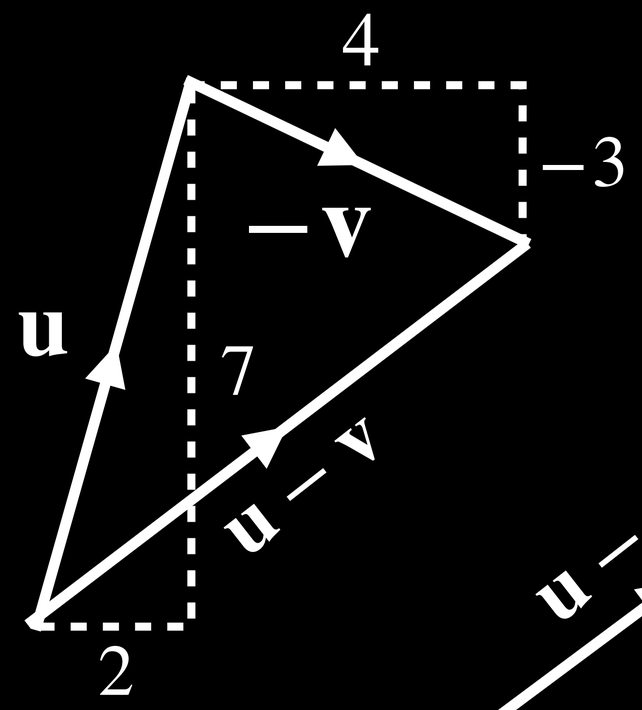
$\overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{AC}$ and

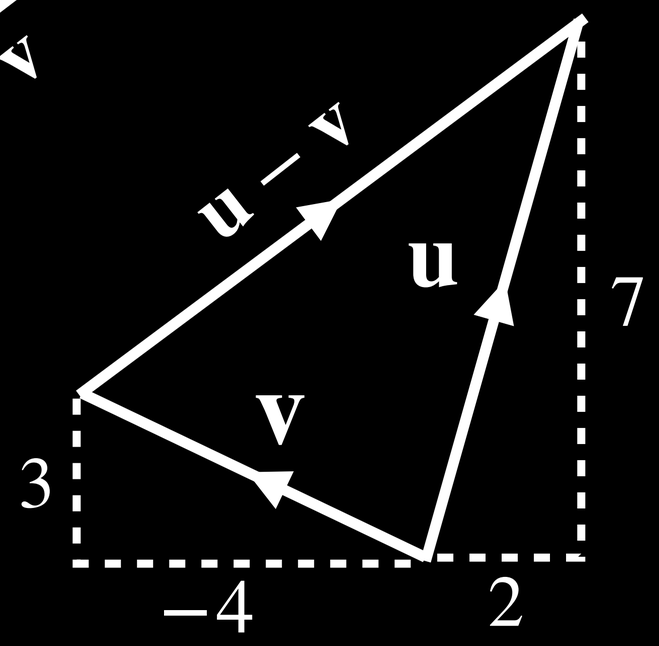
$$\mathbf{u} + \mathbf{v} = \begin{pmatrix} 2 \\ 7 \end{pmatrix} + \begin{pmatrix} -4 \\ 3 \end{pmatrix} = \begin{pmatrix} 2-4 \\ 7+3 \end{pmatrix} = \begin{pmatrix} -2 \\ 10 \end{pmatrix}$$

Note that $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$

Vector Operations - Subtraction

physics jumpscare!!!!!!!!!





Likewise, vectors can be subtracted as so:
Make a vector negative and add them together
$\mathbf{u} - \mathbf{v} = \mathbf{u} + (-\mathbf{v})$ (When a vector is negative, it has the same magnitude but its direction is opposite)

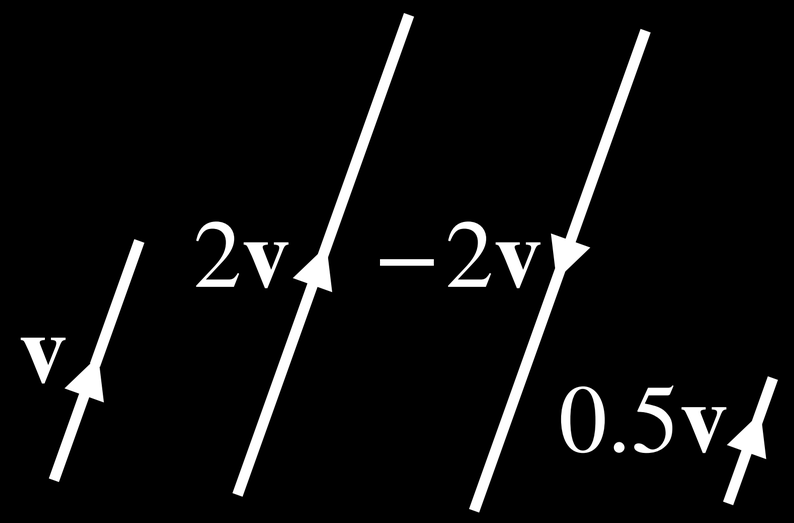
The vector drawn from the end point of $\mathbf{v}$ to the end point of $\mathbf{u}$ also represents $\mathbf{u} - \mathbf{v}$ (when both vectors start from the same point) [memorise as the vector it leaves from is behind the minus sign]

The vector can be found as so:

$$\mathbf{u} - \mathbf{v} = \begin{pmatrix} 2 \\ 7 \end{pmatrix} - \begin{pmatrix} -4 \\ 3 \end{pmatrix} = \begin{pmatrix} 2 - -4 \\ 7 - 3 \end{pmatrix} = \begin{pmatrix} 6 \\ 4 \end{pmatrix}$$

Vector Operations - Scalars

physics jumpscare!!!!!!!!!



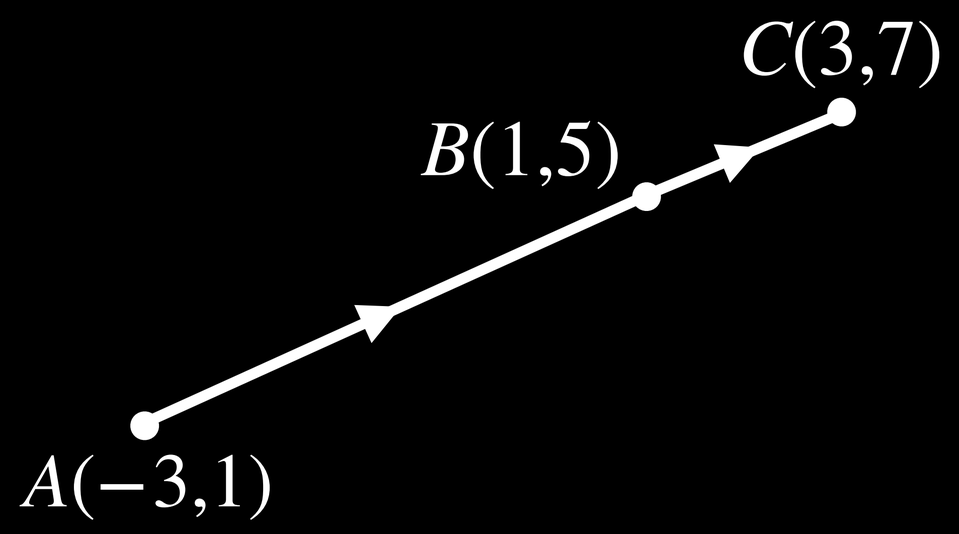
A vector $\mathbf{v}$ can be multiplied by a scalar k which multiplies the magnitude of the vector by k. The direction of the vector will either stay the same (when k is positive) or will flip (when k is negative).

The vector $\mathbf{v}$ will always be parallel with $k\mathbf{v}$/ lie on the same line as $k\mathbf{v}$, but not necessarily in the same direction.

Given that $\mathbf{v} = \begin{pmatrix} a \\ b \end{pmatrix}$, $k\mathbf{v} = k\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} ka \\ kb \end{pmatrix}$ (likewise, both numbers in the vector can be divided by a common number to make a new vector multiplied to a scalar)

Vector Operations - Collinearity

physics jumpscare!!!!!!!!!



We can use scalars to show that 3 points are collinear
To prove that 3 points are collinear, we must show that $\overrightarrow{AB} = k\overrightarrow{BC}$, where k is a scalar and one point is common:

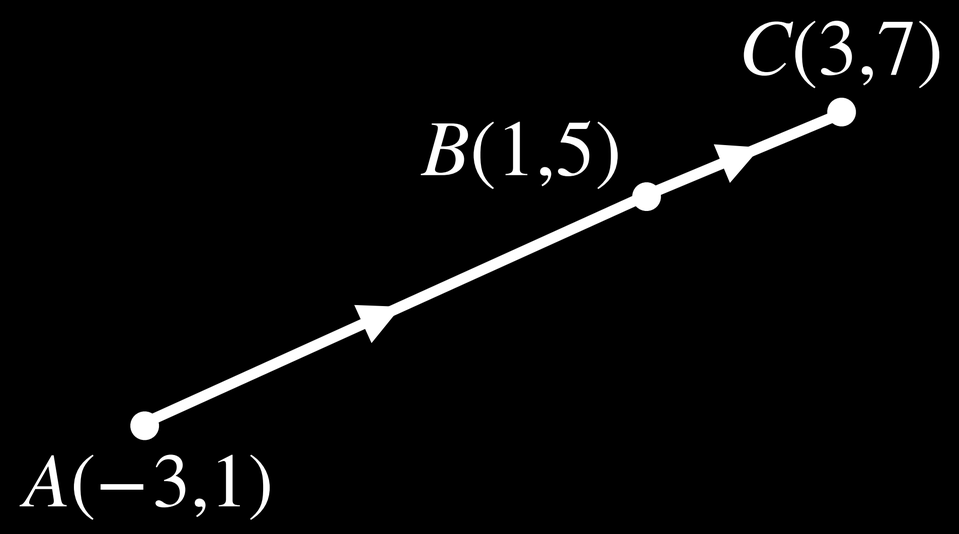
$$\overrightarrow{BC} = \begin{pmatrix} 3-1 \\ 7-5 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \end{pmatrix}$$

$$\overrightarrow{AB} = \begin{pmatrix} 1--3 \\ 5-1 \end{pmatrix} = \begin{pmatrix} 4 \\ 4 \end{pmatrix} = 2\begin{pmatrix} 2 \\ 2 \end{pmatrix} = 2\overrightarrow{BC}$$

As B is a common point, the points A, B and C are collinear.

Vector Operations - Collinearity

physics jumpscare!!!!!!!!!



Questions will commonly ask about facts about vectors/ points and if they are parallel / are collinear /
This outline should be followed:

1. As $\overrightarrow{AB} = k\overrightarrow{BC}$,
2. $\overrightarrow{AB} // \overrightarrow{BC}$
3. $|\overrightarrow{AB}| = k|\overrightarrow{BC}|$
4. As B is a common point, A, B and C are collinear.

e.g. Write down two facts about $\overrightarrow{AB}$ and $\overrightarrow{BC}$.
As $\overrightarrow{AB} = 2\overrightarrow{BC}$, $\overrightarrow{AB} // \overrightarrow{BC}$ and $|\overrightarrow{AB}|$ is two times of $|\overrightarrow{BC}|$.

e.g. Write down a fact about points A, B and C.
As $\overrightarrow{AB} = 2\overrightarrow{BC}$, $\overrightarrow{AB} // \overrightarrow{BC}$. As B is a common point, A, B and C are collinear.

Vector Operations

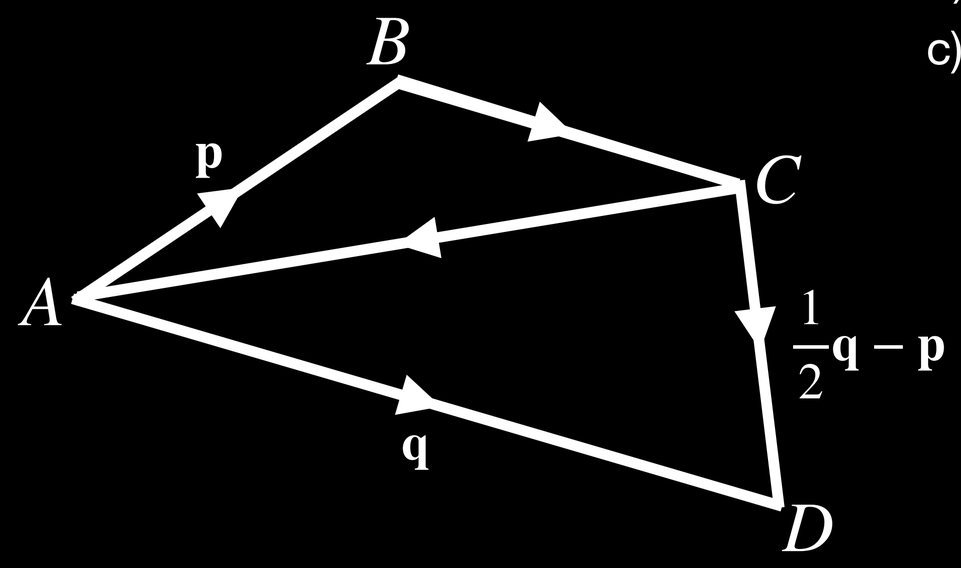
physics jumpscare!!!!!!!!!

The figure shows $\overrightarrow{AB} = \mathbf{p}$, $\overrightarrow{AD} = \mathbf{q}$ and $\overrightarrow{CD} = \frac{1}{2}\mathbf{q} - \mathbf{p}$.

a) Express $\overrightarrow{CA}$ in terms of $\mathbf{p}$ and $\mathbf{q}$

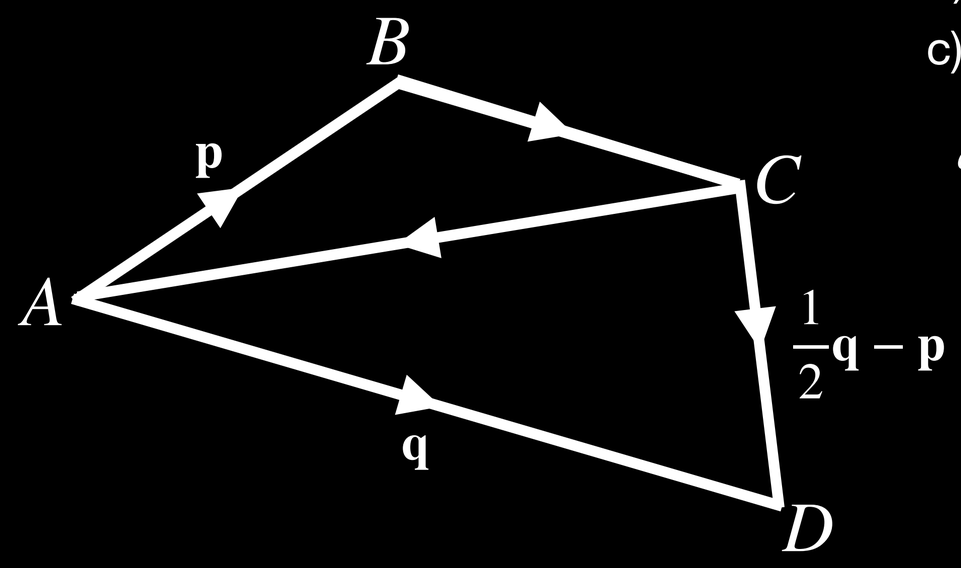
b) Prove that $\overrightarrow{BC}$ is parallel to $\overrightarrow{AD}$. Write another fact about $\overrightarrow{BC}$ and $\overrightarrow{AD}$.

c) $\overrightarrow{AB}$ and $\overrightarrow{DC}$ are produced to meet at X. Express $\overrightarrow{XA}$ in terms of $\mathbf{p}$.



Vector Operations

physics jumpscare!!!!!!!!!



The figure shows $\overrightarrow{AB}=\mathbf{p}$, $\overrightarrow{AD}=\mathbf{q}$ and $\overrightarrow{CD}=\frac{1}{2}\mathbf{q}-\mathbf{p}$.

a) Express $\overrightarrow{CA}$ in terms of $\mathbf{p}$ and $\mathbf{q}$

b) Prove that $\overrightarrow{BC}$ is parallel to $\overrightarrow{AD}$. Write another fact about $\overrightarrow{BC}$ and $\overrightarrow{AD}$.

c) $\overrightarrow{AB}$ and $\overrightarrow{DC}$ are produced to meet at X. Express $\overrightarrow{XA}$ in terms of $\mathbf{p}$.

$a)\ \overrightarrow{CD}=\overrightarrow{CA}+\overrightarrow{AD}$

$$\frac{1}{2}\mathbf{q}-\mathbf{p}=\overrightarrow{CA}+\mathbf{q}$$

$$\frac{1}{2}\mathbf{q}-\mathbf{p}-\mathbf{q}=\overrightarrow{CA}$$

$$\overrightarrow{CA}=-\frac{1}{2}\mathbf{q}-\mathbf{p}$$

form eqn and
make CA the subject

$b)\ \overrightarrow{BC}=\overrightarrow{BA}+\overrightarrow{AD}+\overrightarrow{DC}$ find sum of vectors that are equal to $\overrightarrow{BC}$

$$=-\mathbf{p}+\mathbf{q}-\left(\frac{1}{2}\mathbf{q}-\mathbf{p}\right)$$

AB and CD need to be flipped (make negative)

$$=-\mathbf{p}+\mathbf{q}-\frac{1}{2}\mathbf{q}+\mathbf{p}$$

$$=\mathbf{q}-\frac{1}{2}\mathbf{q}$$

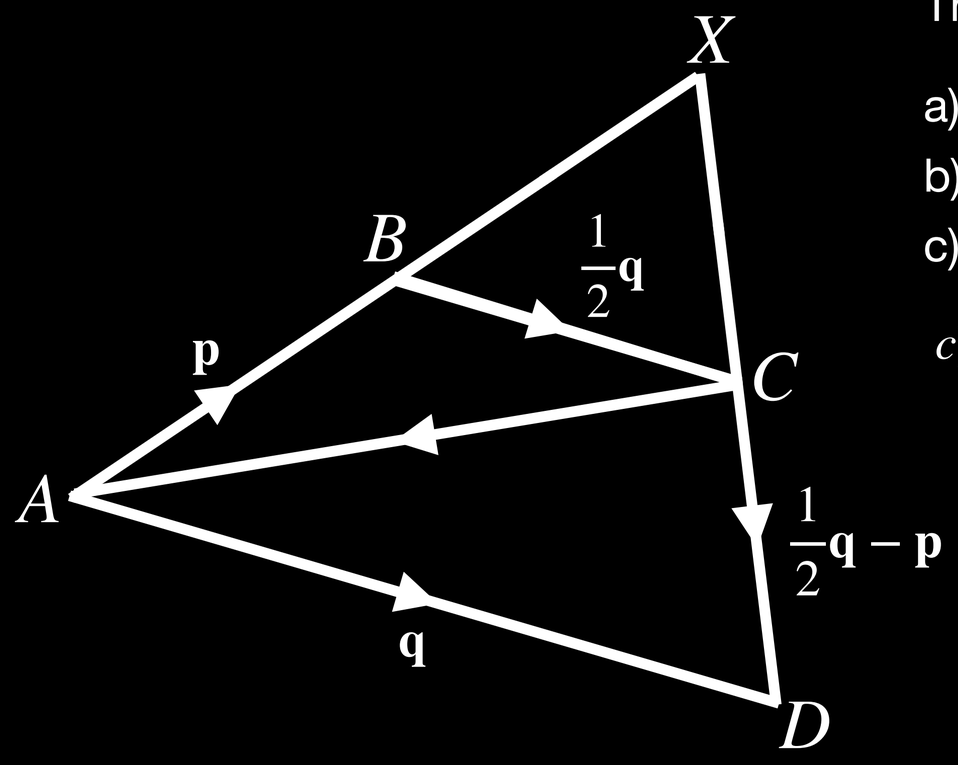
$$=\frac{1}{2}\mathbf{q}$$

As $\frac{1}{2}\overrightarrow{AD}=\overrightarrow{BC}$, $\overrightarrow{BC}//\overrightarrow{AD}$.

As $\frac{1}{2}\overrightarrow{AD}=\overrightarrow{BC}$, $|\overrightarrow{BC}|$ is $\frac{1}{2}$ times of $|\overrightarrow{AD}|$

Vector Operations

physics jumpscare!!!!!!!!!



The figure shows $\overrightarrow{AB} = \mathbf{p}$, $\overrightarrow{AD} = \mathbf{q}$ and $\overrightarrow{CD} = \frac{1}{2}\mathbf{q} - \mathbf{p}$.

a) Express $\overrightarrow{CA}$ in terms of $\mathbf{p}$ and $\mathbf{q}$

b) Prove that $\overrightarrow{BC}$ is parallel to $\overrightarrow{AD}$. Write another fact about $\overrightarrow{BC}$ and $\overrightarrow{AD}$.

c) $\overrightarrow{AB}$ and $\overrightarrow{DC}$ are produced to meet at X. Express $\overrightarrow{XA}$ in terms of $\mathbf{p}$.

$c)$ By similar triangles,

$$\frac{XB}{XA} = \frac{BC}{AD}$$

$$\frac{XB}{XA} = \frac{|\frac{1}{2}\mathbf{q}|}{|\mathbf{q}|}$$

$$\frac{XB}{XA} = \frac{1}{2}$$

clarification on how XA is derived from BA

$\therefore XB : XA = 1 : 2$

as $BA = XA - XB$

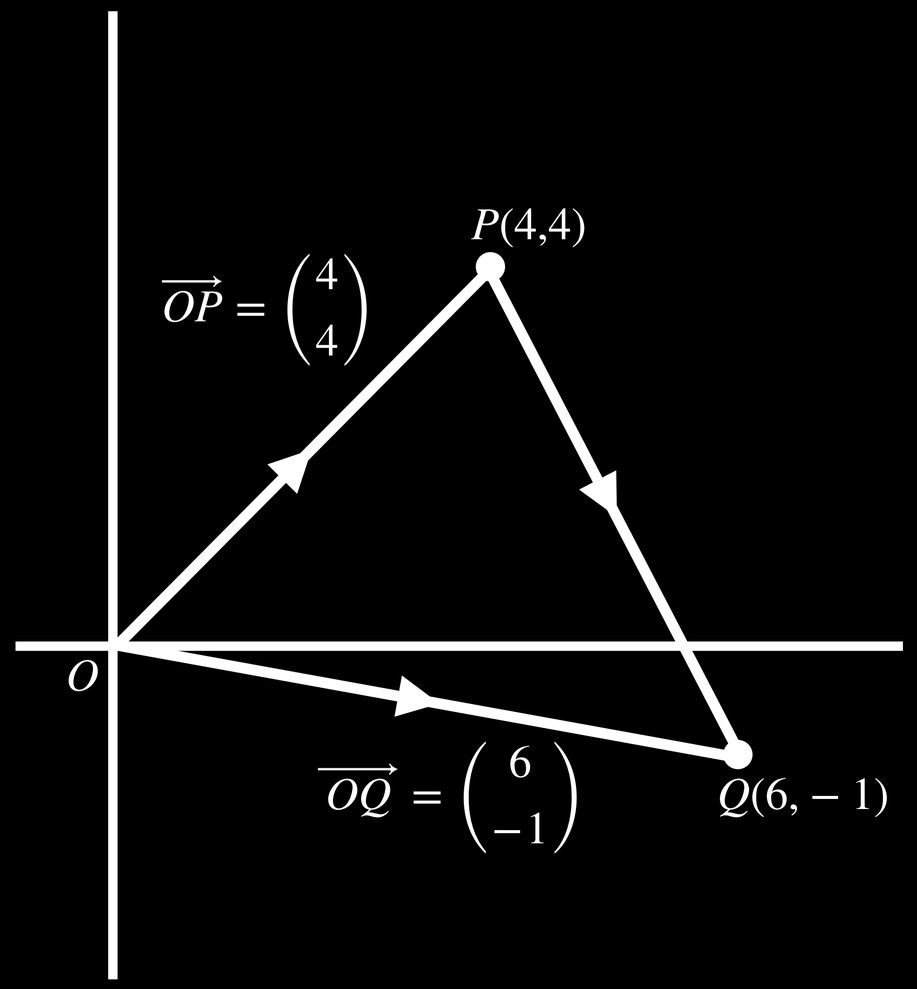
$$\begin{aligned} BA : XA &= XA - XB : XA \\ &= 2 - 1 : 2 \\ &= 1 : 2 \end{aligned}$$

as $\overrightarrow{AB} = \mathbf{p}$, $\overrightarrow{BA} = -\mathbf{p}$

$\therefore \overrightarrow{XA} = -2\mathbf{p}$

Position Vectors

physics jumpscare!!!!!!!!!



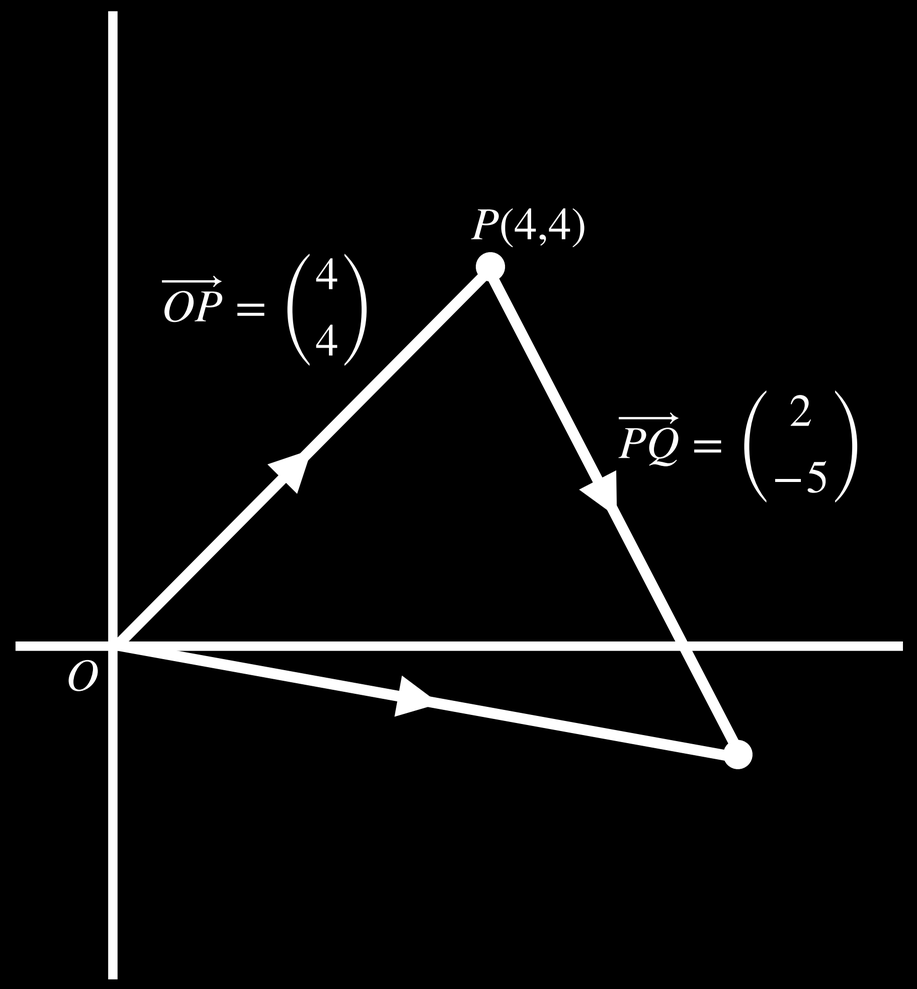
If we have a point $P(4,4)$ that lies on the cartesian plane, the vector from the origin of the plane O to the point P is $\begin{pmatrix} 4 \\ 4 \end{pmatrix}$, which is called the position vector. Thus, if we have another point $Q(6, -1)$, its position vector is $\begin{pmatrix} 6 \\ -1 \end{pmatrix}$.

Thus, if we want to find $\overrightarrow{PQ}$, we can just use vector subtraction as so:

$$\overrightarrow{PQ} = \overrightarrow{OQ} - \overrightarrow{OP} = \begin{pmatrix} 6 \\ -1 \end{pmatrix} - \begin{pmatrix} 4 \\ 4 \end{pmatrix} = \begin{pmatrix} 6-4 \\ -1-4 \end{pmatrix} = \begin{pmatrix} 2 \\ -5 \end{pmatrix}$$

Position Vectors

physics jumpscare!!!!!!!!!



Position vectors can be used to find the coordinates of other points:

Lets say we only know point P and $\overrightarrow{PQ}$, and that we want to find the coordinates of Q.

As $P(4,4)$, $\overrightarrow{OP} = \begin{pmatrix} 4 \\ 4 \end{pmatrix}$

$\overrightarrow{OQ} = \overrightarrow{OP} + \overrightarrow{PQ}$ [can be derived from $\overrightarrow{PQ} = \overrightarrow{OQ} - \overrightarrow{OP}$]

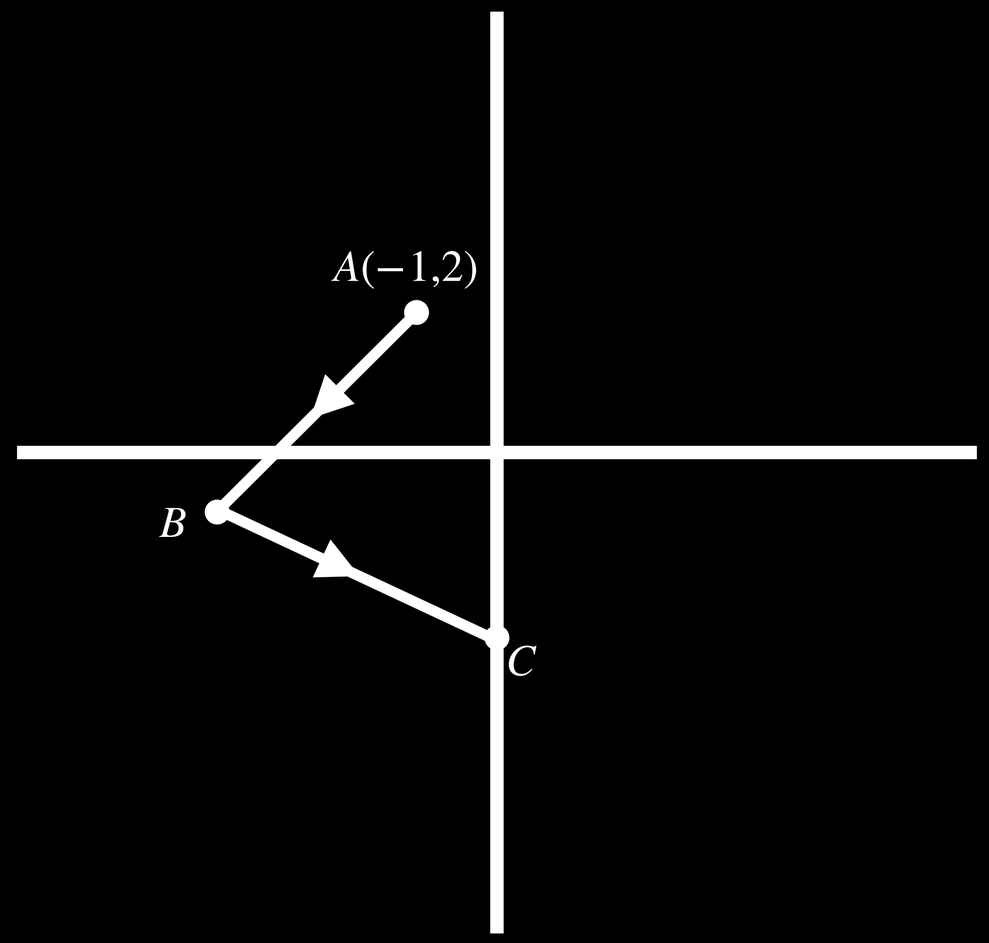
$$\overrightarrow{OQ} = \begin{pmatrix} 4 \\ 4 \end{pmatrix} + \begin{pmatrix} 2 \\ -5 \end{pmatrix} = \begin{pmatrix} 4+2 \\ 4-5 \end{pmatrix} = \begin{pmatrix} 6 \\ -1 \end{pmatrix}$$

$\therefore Q(6, -1)$

Vector PQ also describes the translation of point P to Q (point P moves along $\overrightarrow{PQ}$ to get to Q)

Position Vectors

physics jumpscare!!!!!!!!!



Given that point A is $(-1,2)$, $\overrightarrow{AB} = \begin{pmatrix} -3 \\ -3 \end{pmatrix}$ and $\overrightarrow{BC} = \begin{pmatrix} 4 \\ -2 \end{pmatrix}$,

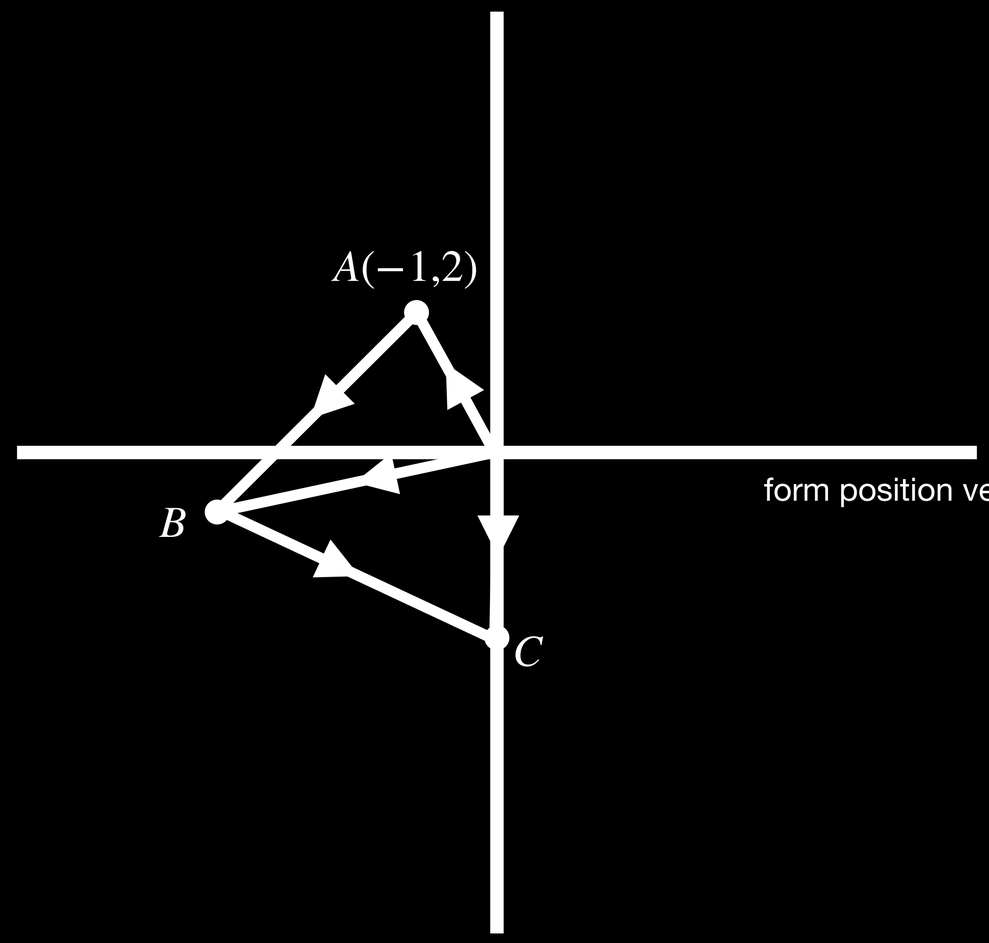
a) find the coordinates of B and C

b) Another point X is given as $(-2, -2)$. State a fact about points B, X and C.

c) Another point D is added such that ABCD is a parallelogram. Find the coordinates of D.

Position Vectors

physics jumpscare!!!!!!!!!



form position vector eqn

Given that point A is $(-1,2)$, $\overrightarrow{AB} = \begin{pmatrix} -3 \\ -3 \end{pmatrix}$ and $\overrightarrow{BC} = \begin{pmatrix} 4 \\ -2 \end{pmatrix}$,

a) find the coordinates of B and C
b) Another point X is given as $(-2, -2)$. State a fact about points B, X and C.
c) Another point D is added such that ABCD is a parallelogram. Find the coordinates of D.

$a)\ \overrightarrow{OA} = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$

$\overrightarrow{OB} = \overrightarrow{OA} + \overrightarrow{AB}$

$\overrightarrow{OB} = \begin{pmatrix} -1 \\ 2 \end{pmatrix} + \begin{pmatrix} -3 \\ -3 \end{pmatrix}$

$\overrightarrow{OB} = \begin{pmatrix} -1-3 \\ 2-3 \end{pmatrix}$

$\overrightarrow{OB} = \begin{pmatrix} -4 \\ -1 \end{pmatrix}$

$\therefore B(-4, -1)$

$\overrightarrow{OC} = \overrightarrow{OB} + \overrightarrow{BC}$ form position vector eqn

$\overrightarrow{OC} = \begin{pmatrix} -4 \\ -1 \end{pmatrix} + \begin{pmatrix} 4 \\ -2 \end{pmatrix}$

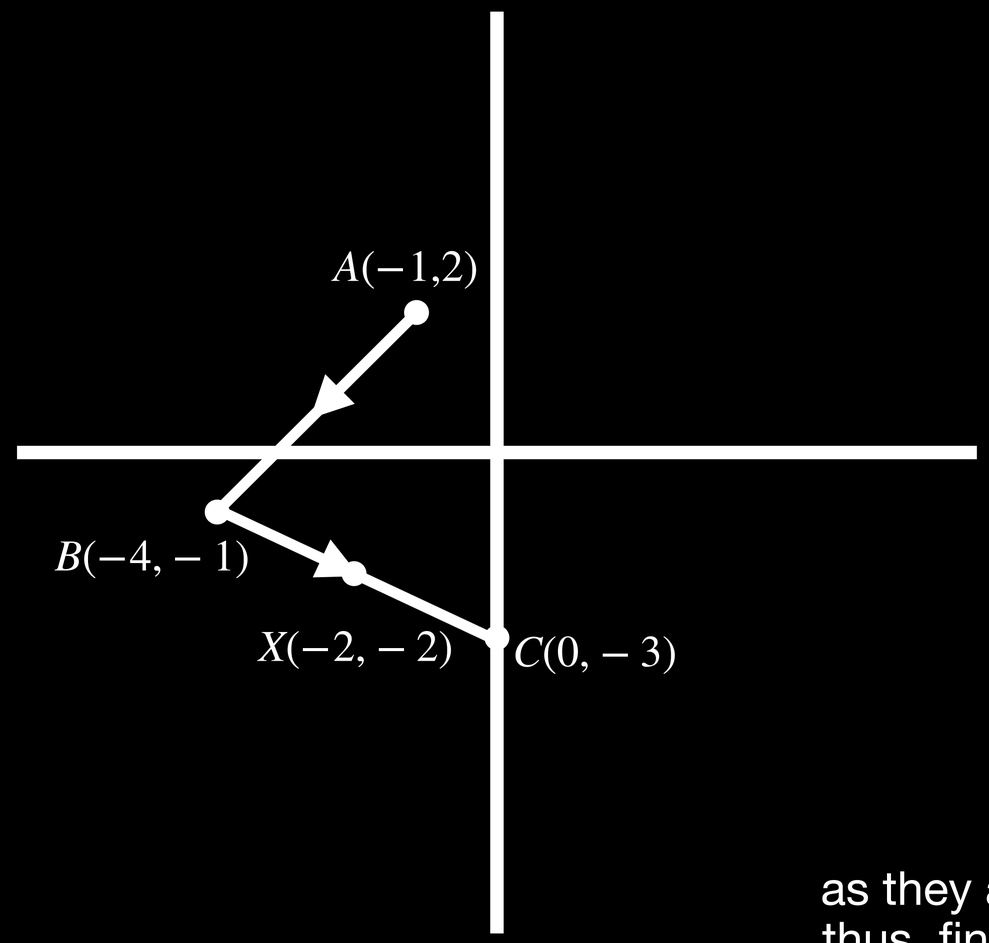
$\overrightarrow{OC} = \begin{pmatrix} -4+4 \\ -1-2 \end{pmatrix}$

$\overrightarrow{OC} = \begin{pmatrix} 0 \\ -3 \end{pmatrix}$

$\therefore C(0, -3)$

Position Vectors

physics jumpscare!!!!!!!!!



Given that point A is $(-1,2)$, $\overrightarrow{AB} = \begin{pmatrix} -3 \\ -3 \end{pmatrix}$ and $\overrightarrow{BC} = \begin{pmatrix} 4 \\ -2 \end{pmatrix}$,

a) find the coordinates of B and C

b) Another point X is given as $(-2,-2)$. State a fact about points B, X and C.

c) Another point D is added such that ABCD is a parallelogram. Find the coordinates of D.

$$b)\ \overrightarrow{BX} = \overrightarrow{OX} - \overrightarrow{OB}$$
$$= \begin{pmatrix} -2 \\ -2 \end{pmatrix} - \begin{pmatrix} -4 \\ -1 \end{pmatrix}$$
$$= \begin{pmatrix} -2-(-4) \\ -2-(-1) \end{pmatrix}$$
$$= \begin{pmatrix} 2 \\ -1 \end{pmatrix}$$

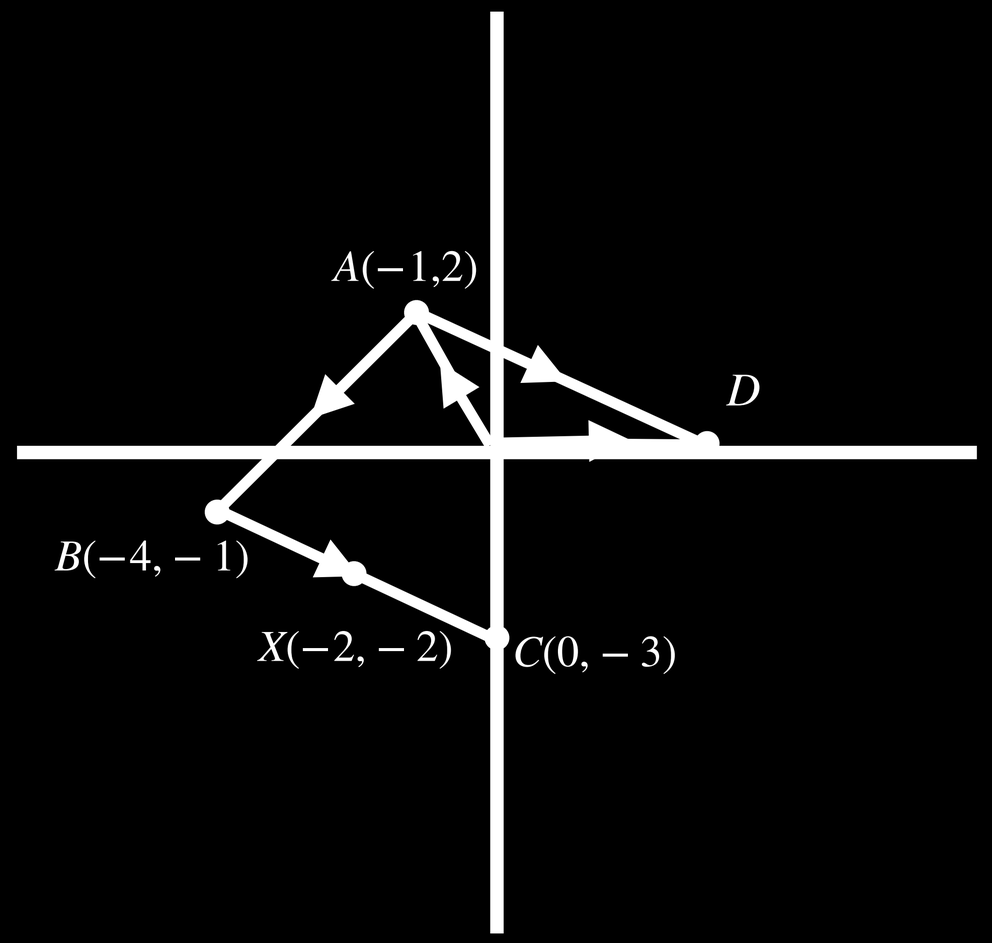
$$\overrightarrow{BC} = \begin{pmatrix} 4 \\ -2 \end{pmatrix}$$
$$= 2\begin{pmatrix} 2 \\ -1 \end{pmatrix}$$
$$= 2\overrightarrow{BX}$$

As $\overrightarrow{BC} = 2\overrightarrow{BX}$,
$\overrightarrow{BC} // \overrightarrow{BX}$.
Since B is a common point,
B, X and C are collinear points

as they ask for state a fact, B,X, and C are probably collinear
thus, find either BX or CX and compare with BC

Position Vectors

physics jumpscare!!!!!!!!!



Given that point A is $(-1,2)$, $\overrightarrow{AB} = \begin{pmatrix} -3 \\ -3 \end{pmatrix}$ and $\overrightarrow{BC} = \begin{pmatrix} 4 \\ -2 \end{pmatrix}$,

a) find the coordinates of B and C

b) Another point X is given as $(-2,-2)$. State a fact about points B, X and C.

c) Another point D is added such that ABCD is a parallelogram. Find the coordinates of D.

$c)$ As ABCD is a parallelogram, $\overrightarrow{BC} = \overrightarrow{AD}$

make use of the info that ABCD is a parallelogram

$\therefore \overrightarrow{AD} = \begin{pmatrix} 4 \\ -2 \end{pmatrix}$

Thus, $\overrightarrow{OD} = \overrightarrow{OA} + \overrightarrow{AD}$ form position vector eqn

$$\overrightarrow{OD} = \begin{pmatrix} -1 \\ 2 \end{pmatrix} + \begin{pmatrix} 4 \\ -2 \end{pmatrix}$$

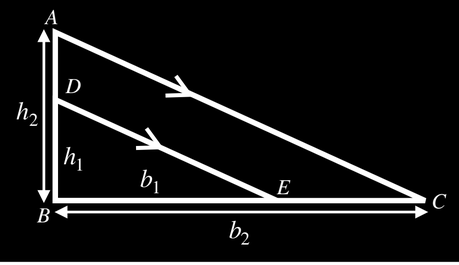
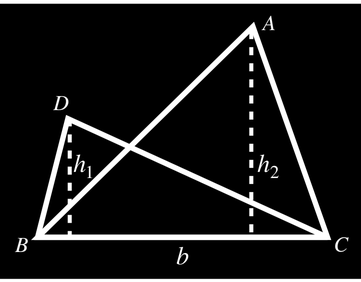
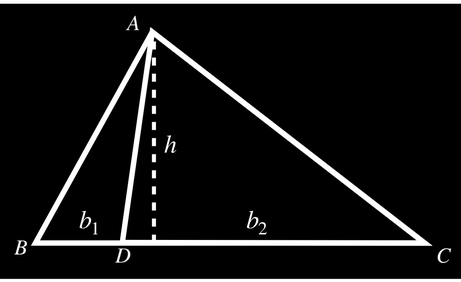
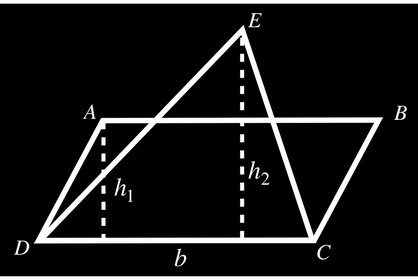
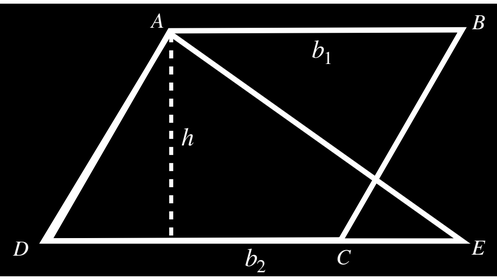
$$= \begin{pmatrix} -1+4 \\ 2-2 \end{pmatrix}$$

$$= \begin{pmatrix} 3 \\ 0 \end{pmatrix} \therefore D(3,0)$$

Similar Figures & Common Lengths

recap

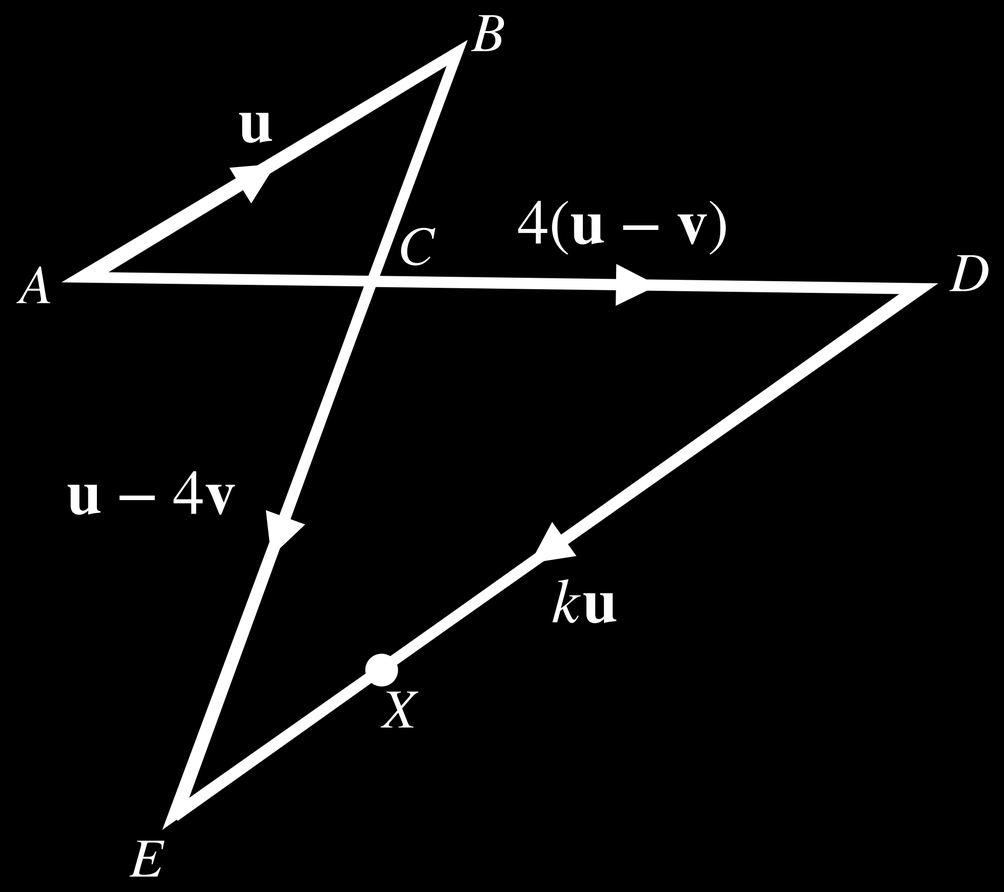
Make sure that the non-common length is a vector for both shapes, otherwise ratio cannot be formed

	Similar figures	Same base	Same height
Triangles only			
	$\frac{\Delta BDE}{\Delta BAC} = \left(\frac{h_1}{h_2}\right)^2 = \left(\frac{b_1}{b_2}\right)^2$	$\frac{\Delta DBC}{\Delta ABC} = \frac{\cancel{0.5}\cancel{b}h_1}{\cancel{0.5}\cancel{b}h_2} = \frac{h_1}{h_2}$	$\frac{\Delta ABD}{\Delta ADC} = \frac{\cancel{0.5}b_1\cancel{h}}{\cancel{0.5}b_2\cancel{h}} = \frac{b_1}{b_2}$
Triangle + Quadrilateral	Nil		
		Parallelogram $ABCD$ / ΔECD $= \frac{\cancel{b}h_1}{0.5\cancel{b}h_2} = \frac{h_1}{0.5h_2}$	Parallelogram $ABCD$ / ΔAED $= \frac{b_1\cancel{h}}{0.5b_2\cancel{h}} = \frac{b_1}{0.5b_2}$

for different quadrilaterals (e.g. trapeziums), use the respective formula

Similar Figures & Common Lengths

recap



The figure shows $\overrightarrow{AB} = \mathbf{u}$, $\overrightarrow{CD} = 4(\mathbf{u} - \mathbf{v})$, and $\overrightarrow{CE} = \mathbf{u} - 4\mathbf{v}$. Point X lies on DE such that $\overrightarrow{DX} = \frac{2}{3}\overrightarrow{DE}$. ACD and BCE are straight lines

a) Given that $\overrightarrow{DE} = k\mathbf{u}$, where k is a scalar, find k

b) Hence, show that ΔABC is similar to ΔDEC.

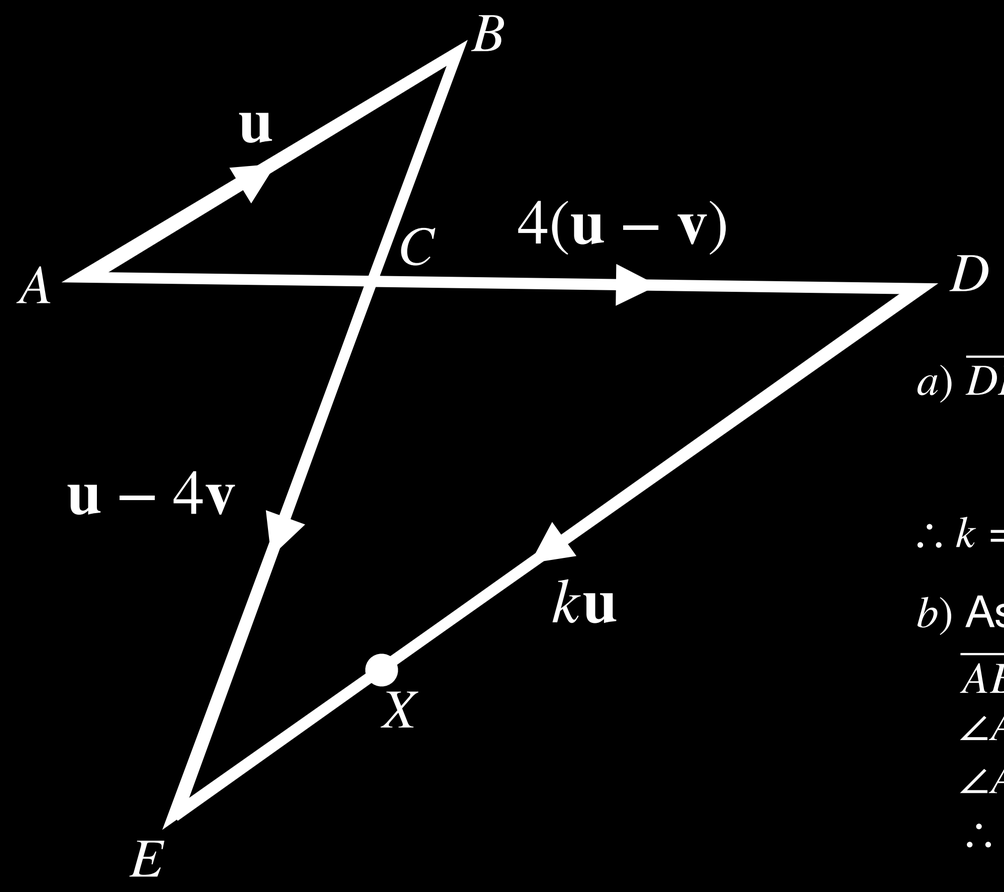
c) Find $\dfrac{\text{area of } \Delta ABC}{\text{area of } \Delta DEC}$.

d) Express $\overrightarrow{CX}$ in terms of $\mathbf{u}$ and $\mathbf{v}$.

e) Find $\dfrac{\text{area of } \Delta CEX}{\text{area of } \Delta CDX}$.

Similar Figures & Common Lengths

recap



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d) Express $\overrightarrow{CX}$ in terms of $\mathbf{u}$ and $\mathbf{v}$.

e) Find $\dfrac{\text{area of } \Delta CEX}{\text{area of } \Delta CDX}$.

$a)$ $\overrightarrow{DE} = \mathbf{u} - 4\mathbf{v} - 4(\mathbf{u} - \mathbf{v})$
$= \mathbf{u} - 4\mathbf{v} - 4\mathbf{u} + 4\mathbf{v}$
$= -3\mathbf{u}$ vector subtraction

$\therefore k = -3$

$b)$ As $-3\overrightarrow{AB} = \overrightarrow{DE}$ bowtie similar triangles $\overrightarrow{AB} // \overrightarrow{DE}$

$\overrightarrow{AB} // \overrightarrow{DE}$

$\angle ACB = \angle ECD$(vertically opposite angles)

$\angle ABC = \angle CED$(alternate angles, AB//DE)

$\therefore \Delta ABC$ is similar to ΔDEC

$c)$ $\dfrac{\text{area of } \Delta ABC}{\text{area of } \Delta DEC} = \left(\dfrac{|\overrightarrow{AB}|}{|\overrightarrow{DE}|}\right)^2$ similar triangle ratio

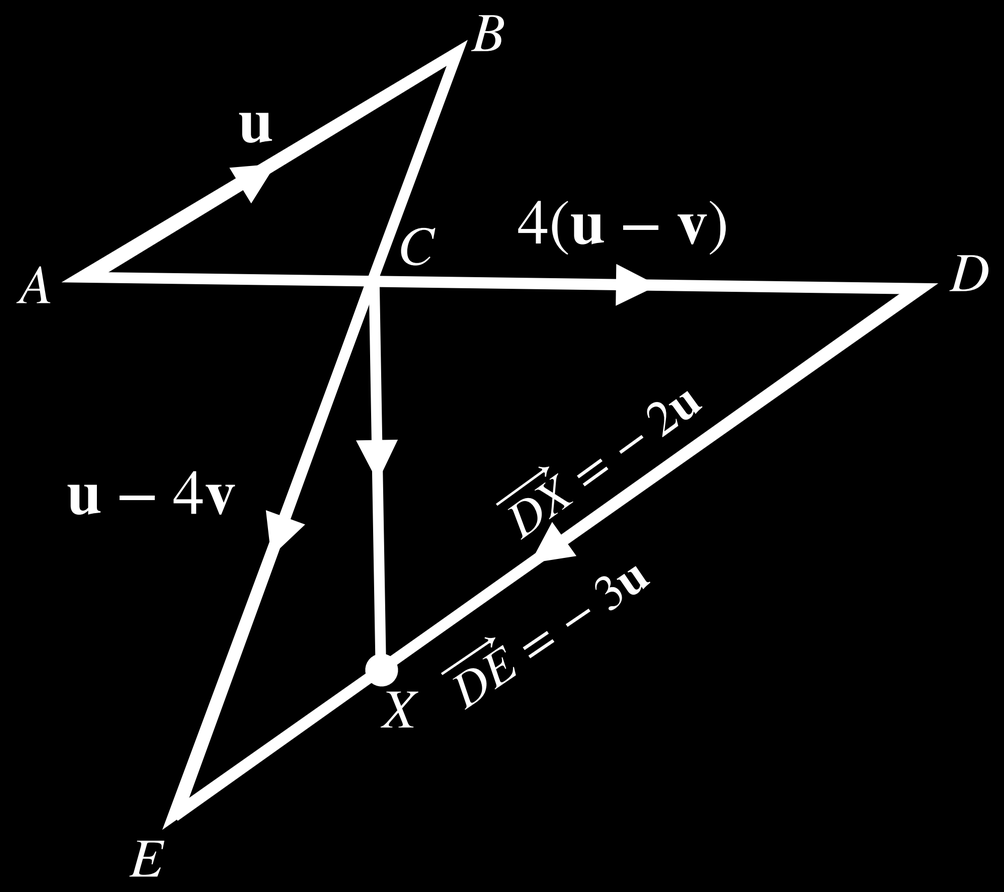
$= \left(\dfrac{|\mathbf{u}|}{|-3\mathbf{u}|}\right)^2$

$= \left(\dfrac{1}{3}\right)^2$

$= \dfrac{1}{9}$

Similar Figures & Common Lengths

recap



The figure shows $\overrightarrow{AB} = \mathbf{u}$, $\overrightarrow{CD} = 4(\mathbf{u} - \mathbf{v})$, and $\overrightarrow{CE} = \mathbf{u} - 4\mathbf{v}$. Point X lies on DE such that $\overrightarrow{DX} = \frac{2}{3}\overrightarrow{DE}$. ACD and BCE are straight lines

a) Given that $\overrightarrow{DE} = k\mathbf{u}$, where k is a scalar, find k

b) Hence, show that ΔABC is similar to ΔDEC.

c) Find $\dfrac{\text{area of } \Delta ABC}{\text{area of } \Delta DEC}$.

d) Express $\overrightarrow{CX}$ in terms of $\mathbf{u}$ and $\mathbf{v}$.

e) Find $\dfrac{\text{area of } \Delta CEX}{\text{area of } \Delta CDX}$.

$$d)\ \overrightarrow{DX} = \frac{2}{3}\overrightarrow{DE}$$
$$= \frac{2}{3}(-3\mathbf{u})$$
$$= -2\mathbf{u}$$

$$\therefore \overrightarrow{CX} = \overrightarrow{CD} + \overrightarrow{DX}$$
$$= 4(\mathbf{u} - \mathbf{v}) + (-2\mathbf{u})$$
$$= 4\mathbf{u} - 4\mathbf{v} - 2\mathbf{u}$$
$$= 2\mathbf{u} - 4\mathbf{v}$$

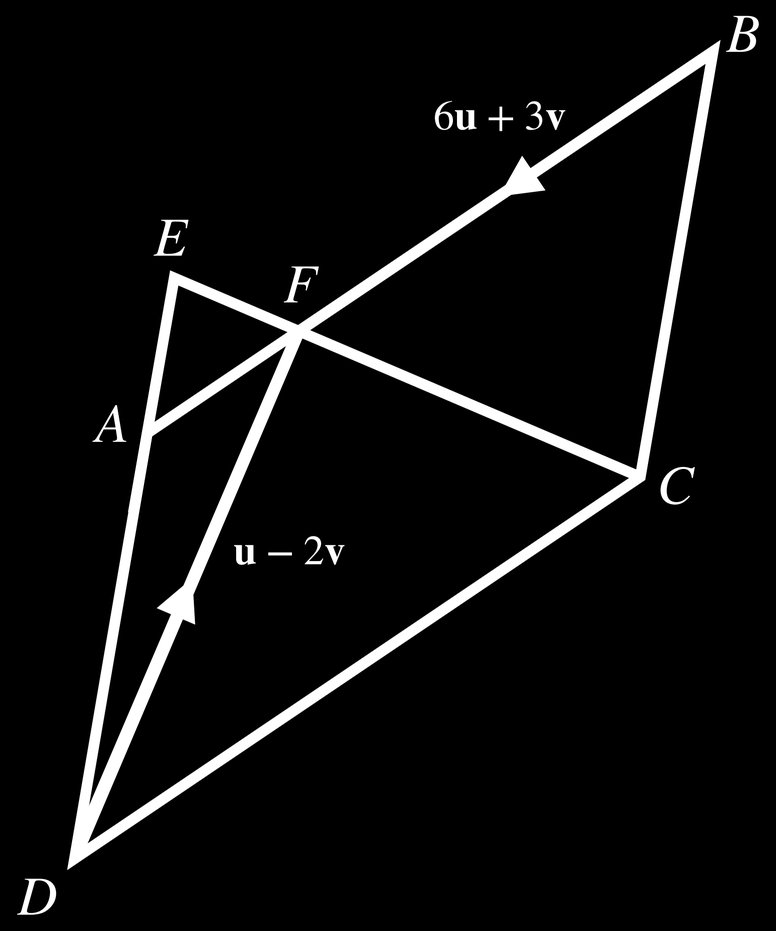
$$e)\ \overrightarrow{DE} - \overrightarrow{DX} = \overrightarrow{XE}$$
$$-3\mathbf{u} - (-2\mathbf{u}) = \overrightarrow{XE}$$
$$\overrightarrow{XE} = -\mathbf{u}$$

$$\frac{\text{area of } \Delta CEX}{\text{area of } \Delta CDX} = \frac{|\overrightarrow{XE}|}{|\overrightarrow{DX}|}$$
$$= \frac{|-\mathbf{u}|}{|-2\mathbf{u}|}$$
$$= \frac{1\cancel{|\mathbf{u}|}}{2\cancel{|\mathbf{u}|}} = \frac{1}{2}$$

We use common height and not common base of CD as the heights of the triangles are not vectors in our figure

Similar Figures & Common Lengths

recap



ABCD is a parallelogram and DA and CF are produced to meet at E. It is given that $\overrightarrow{BF} = 6\mathbf{u} + 3\mathbf{v}$, $\overrightarrow{DF} = \mathbf{u} - 2\mathbf{v}$, and $FB : AB = 3 : 4$.

a) Express the vectors i) $\overrightarrow{FA}$, ii) $\overrightarrow{DA}$, iii) $\overrightarrow{CD}$ and iv) $\overrightarrow{AE}$ in terms of $\mathbf{u}$ and $\mathbf{v}$.

b) Find $\dfrac{\text{area of } \Delta AEF}{\text{area of } \Delta DEC}$

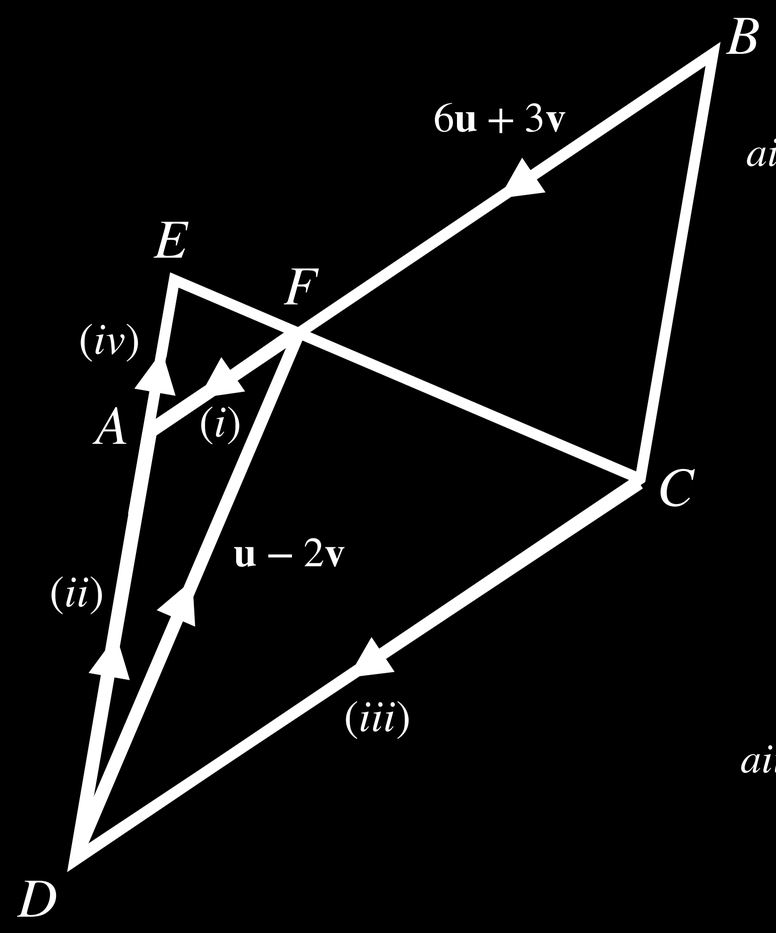
c) Find $\dfrac{\text{area of } \Delta AFD}{\text{area of } \Delta DFC}$

d) Find $\dfrac{\text{area of } \Delta AFD}{\text{area of parallelogram } ABCD}$

e) Hence, Find $\dfrac{\text{area of } \Delta DFC}{\text{area of parallelogram } ABCD}$

Similar Figures & Common Lengths

recap



ABCD is a parallelogram and DA and CF are produced to meet at E. It is given that $\overrightarrow{BF} = 6\mathbf{u} + 3\mathbf{v}$, $\overrightarrow{DF} = \mathbf{u} - 2\mathbf{v}$, and $BF : BA = 3 : 4$.

a) Express the vectors i) $\overrightarrow{FA}$, ii) $\overrightarrow{DA}$, iii) $\overrightarrow{CD}$ and iv) $\overrightarrow{AE}$ in terms of $\mathbf{u}$ and $\mathbf{v}$.

$ai)$

As $BF : BA = 3 : 4$,

$$BF : FA = 3 : 4 - 3_{[AB - FB]}$$
$$= 3 : 1$$

as BF is given, we use BF:BA to find FA

$$\therefore \overrightarrow{FA} = \frac{1}{3}\overrightarrow{BF}$$
$$= \frac{1}{3}(6\mathbf{u} + 3\mathbf{v})$$
$$= 2\mathbf{u} + \mathbf{v}$$

$aii)$

$$\overrightarrow{DA} = \overrightarrow{DF} + \overrightarrow{FA}$$
$$= \mathbf{u} - 2\mathbf{v} + 2\mathbf{u} + \mathbf{v}$$
$$= 3\mathbf{u} - \mathbf{v}$$

$aiii)$

As ABCD is a parallelogram, $\overrightarrow{BA} = \overrightarrow{CD}$

$$\overrightarrow{BA} = \overrightarrow{BF} + \overrightarrow{FA}$$
$$= 6\mathbf{u} + 3\mathbf{v} + 2\mathbf{u} + \mathbf{v}$$
$$= 8\mathbf{u} + 4\mathbf{v}$$
$$\therefore \overrightarrow{CD} = 8\mathbf{u} + 4\mathbf{v}$$

$aiv)$

As ABCD is a parallelogram, $\overrightarrow{DA} = \overrightarrow{CB}$

$$\therefore \overrightarrow{CB} = 3\mathbf{u} - \mathbf{v}$$

As $BF : FA = 3 : 1$, by similar triangles,

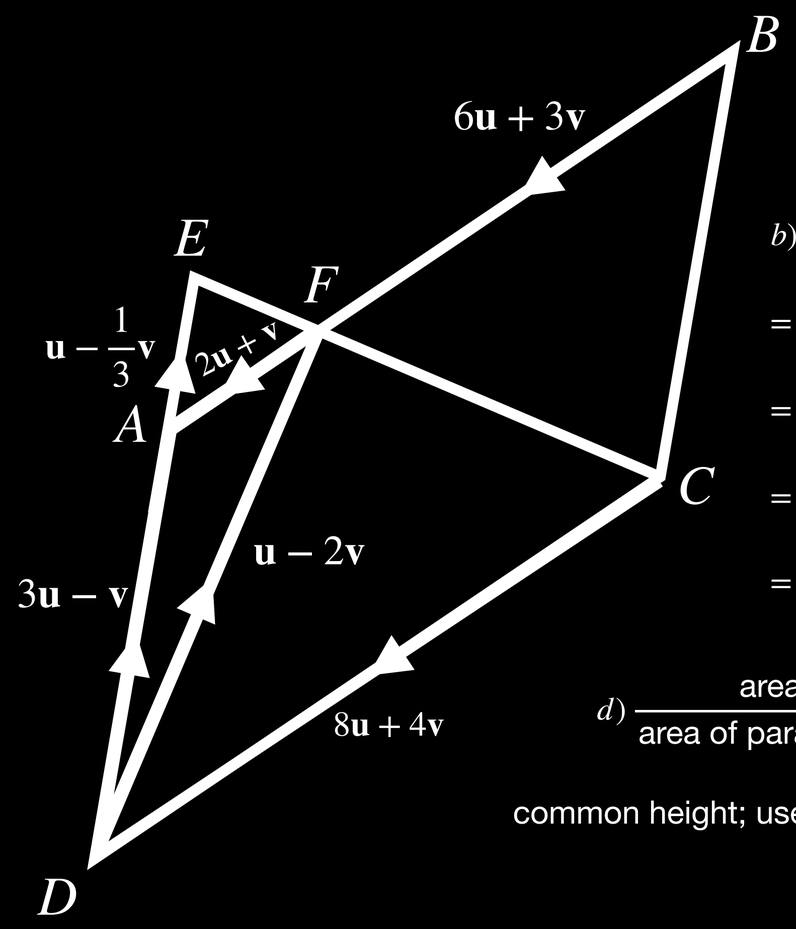
$$CB : AE = 3 : 1$$

this solution uses ΔAEF and ΔBCF to be similar

$$\therefore \overrightarrow{AE} = \frac{1}{3}\overrightarrow{CB}$$
$$= \frac{1}{3}(3\mathbf{u} - \mathbf{v}) = \mathbf{u} - \frac{1}{3}\mathbf{v}$$

Similar Figures & Common Lengths

recap



ABCD is a parallelogram and DA and CF are produced to meet at E. It is given that $\overrightarrow{BF}=6\mathbf{u}+3\mathbf{v}$, $\overrightarrow{DF}=\mathbf{u}-2\mathbf{v}$, and $FB:AB=3:4$.

b) Find $\dfrac{\text{area of } \Delta AEF}{\text{area of } \Delta DEC}$ c) Find $\dfrac{\text{area of } \Delta AFD}{\text{area of } \Delta DFC}$ d) Find $\dfrac{\text{area of } \Delta AFD}{\text{area of parallelogram } ABCD}$

e) Hence, Find $\dfrac{\text{area of } \Delta DFC}{\text{area of parallelogram } ABCD}$

$$b)\ \frac{\text{area of } \Delta AEF}{\text{area of } \Delta DEC}=\left(\frac{AF}{DC}\right)^2$$
$$=\left(\frac{|2\mathbf{u}+\mathbf{v}|}{|8\mathbf{u}+4\mathbf{v}|}\right)^2$$
$$=\left(\frac{|2\mathbf{u}+\mathbf{v}|}{4|2\mathbf{u}+\mathbf{v}|}\right)^2$$
$$=\left(\frac{1}{4}\right)^2 \text{ similarity ratio}$$
$$=\frac{1}{16}$$

$$c)\ \frac{\text{area of } \Delta AFD}{\text{area of } \Delta DFC}=\frac{AF}{DC}$$
$$=\frac{|2\mathbf{u}+\mathbf{v}|}{|8\mathbf{u}+4\mathbf{v}|} \text{ common height; use ratio of bases}$$
$$=\frac{|2\mathbf{u}+\mathbf{v}|}{4|2\mathbf{u}+\mathbf{v}|}$$
$$=\frac{1}{4}$$

product of two ratios to form new similarity ratio is commonly tested
sometimes, this new similairty ratio is not possible from common base/height

denominator and numerator will cancel to form new ratio

$$d)\ \frac{\text{area of } \Delta AFD}{\text{area of parallelogram } ABCD}=\frac{0.5(AF)\cancel{(\text{height})}}{(DC)\cancel{(\text{height})}}$$

common height; use ratio of bases

$$=\frac{0.5(AF)}{DC}$$
$$=\frac{0.5(1)}{4} \text{ (shown in part (c))}$$
$$=\frac{1}{8}$$

$$e)\ \frac{\text{area of } \Delta DFC}{\text{area of parallelogram } ABCD}=\frac{\text{area of } \Delta DFC}{\text{area of } \Delta AFD}\times\frac{\text{area of } \Delta AFD}{\text{area of parallelogram } ABCD}$$
$$=\left(\frac{\text{area of } \Delta AFD}{\text{area of } \Delta DFC}\right)^{-1}\times\frac{1}{8}$$
$$=\left(\frac{1}{4}\right)^{-1}\times\frac{1}{8}$$
$$=4\times\frac{1}{8}=\frac{1}{2}$$

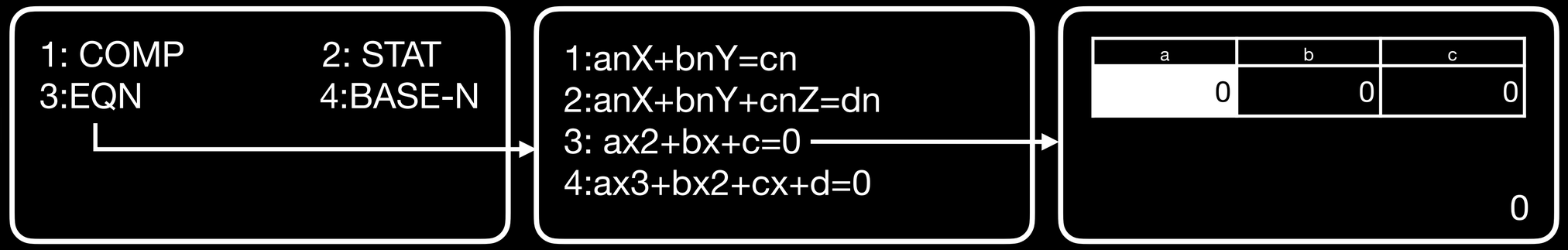
Exam Techniques & Answer Presentation

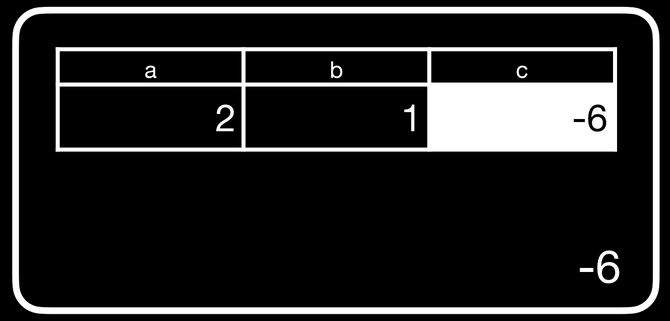
Quick Factorisation

quick++

When dealing with quadratics(and also cubics), your calculator can help to evaluate the solutions of such equations like so
[mode] —> [3](or whatever number is under “equation”)—>[3](aka pick polynomial of desired degree)

[mode]





Enter the coefficients of each term by entering them and pressing equals [=] to enter the next coefficient.
For example, these coefficients refer to the eqn: $2x^2 + x - 6 = 0$

Quick Factorisation

quick++

After entering the coefficients of the equation, pressing equals will show you the solutions of the equation

a	b	c
2	1	-6

-6

X1=

$\frac{3}{2}$

X2=

-2

Thus, the solution of the eqn $2x^2 + x - 6 = 0$ has the solutions -2 and 1.5. However, when presenting solutions of quadratics, you must either write the quadratic formula or show that you have factorised the quadratic.
If the solutions of the quadratic are rational, it is easier to write the factorised form instead of the quadratic formula.

Quick Factorisation

quick++

X1=	X2=
$\frac{3}{2}$	-2

We know the second solution is an integer, thus we just need to flip the sign to write its corresponding factor$[(x+2)]$
For fractions, we cannot write $(x-1.5)$ as this would not be possible by factorisation. Instead, we make the coefficient of x be the denominator $[3x]$ and then subtract that by the numerator to get $3x-2$ (or let $x=1.5$ and make one side equal to 0)
Thus, the factorisation of the quadratic is $(x+2)(3x-2)$.
This allows us to write our working as:

$$2x^2+x-6=0$$
$$(x+2)(3x-2)=0$$
$$x=-2,\ 1.5$$

**note: quadratics like $4x^2+2x-12=0$ have the same roots as $2x^2+x-6=0$, however, the factorised forms are not the same
Thus, make sure to remove any common terms as so:
$4x^2+2x-12=2(2x^2+x-6)$
then factorise the inner quadratic:
$2(2x^2+x-6)=2(x+2)(3x-2)$

Quick Factorisation

quick++

X1=	X2=	X3=
$\frac{3}{2}$	-2	$-\frac{1}{2}$

For factorising cubics, we only need one integer solution to “fake our guess”, in this case we let x= -2 and divide the cubic by the respective factor of (x+2)

$f(x) = 4x^3 + 4x^2 - 11x - 6$

Let $x = -2$

$f(-2) = 4(-2)^3 + 4(-2)^2 - 11(-2) - 6$

$f(-2) = 0$

$\therefore (x+2)$ is a factor of $f(x)$

$f(x) = 4x^3 + 4x^2 - 11x - 6$

$f(x) = (2x+1)\left(2x^2 + x - 6\right)$

$f(x) = (x+2)(2x-3)(2x+1)$

Quick Factorisation

quick++

a	b	c
1	2	5

5

X1=

-1+2**i**

X2=

-1-2**i**

If your calculator outputs solutions with the letter "i", it means your equation has no real solutions, so you should check back that your equation is right/you inputted the right coefficients

**different calculator models will have different ways of changing to factorisation mode, refer to calculator manual if necessary

Time Conversion

t

Your calculator can convert hours into hours, minutes and seconds, by using the [° ’ ”] button.
Type in the number of hours in decimals then followed by the [° ’ ”] button to type out a °.
The calculator will output the time in hours° minutes’ seconds” (so for example, the time shown on the calculator is 4h, 13min and 48 seconds)

Substitution For Checking

Czech Republic

Almost all algebraic questions can be checked via substitution, where you replace variables with your answer/ arbituary values to let your calculator check if your answer is right or wrong

Simplify $\frac{3^{x+2}}{(2^x)(6^x)}$ in positive index form

$$\frac{3^{x+2}}{(2^x)(6^x)} = \frac{(3^x)(3^2)}{(2^x)(6^x)}$$
$$= \frac{9(3^x)}{12^x}$$
$$= \frac{9}{4^x}$$

checking:

Let $x = 3$

$$\frac{3^{x+2}}{(2^x)(6^x)} = \frac{3^{3+2}}{(2^3)(6^3)} = \frac{9}{64}$$

$$\frac{9}{4^x} = \frac{9}{4^3} = \frac{9}{64}$$

Make a the subject of $ac^2 = b^2c - ab$

$$ac^2 + ab = b^2c$$
$$a(c^2 + b) = b^2c$$
$$a = \frac{b^2c}{c^2 + b}$$

checking:

Let b=2,c=3

$$a = \frac{(2)^2(3)}{(3)^2 + (2)}$$
$$a = \frac{12}{11}$$

$$ac^2 + ab = b^2c$$
$$LHS = \frac{12}{11}(3)^2 + \frac{12}{11}(2)$$
$$LHS = 12$$
$$RHS = (2)^2(3)$$
$$RHS = 12$$

checking does not need to be written down, just type in your calculator

EM Specific Tips

Czech Republic

If you find yourself needing to use AM topics (e.g. logarithms in indices, shoelace formula/ midpoint in coordinate geometry), check your steps again/redo your steps to use only EM concepts

EM allows non-exact values for final answers, so you do not need to simplify expressions and can just evaluate to 3.s.f or 5.s.f for intermediate steps. (e.g. for quadratic formula, it is better to simplify in AM for intermediate steps but EM does not need you to simplify)

Answer Presentation

p

Labelling Calculations
Make sure that your calculations are labelled with what the value is/why you are doing said calculation: e.g. if qn asks for duration, write Duration $= a - b = c$, instead of just $a - b = c$

Rejecting answers
Make sure to write why certain answers are rejected, either if the value is strictly positive or negative, or if having the answer be 0 is illogical, or if the answer does not give any real solutions: e.g. For $x^2 - 4 = 0$ where $x > 0$, $x = \sqrt{4}$, not $\pm\sqrt{4}$.

Show/Prove/Explain Questions
Questions that need you to show or prove a statement need you to agree with the statement without using the statement you need to prove: e.g. if you need to prove that $x = 3$, you cannot use that $x = 3$ to show that $x = 3$
Explain questions need you to state your reasoning for a statement (e.g. As 0.5 cycles occur from 0 to π, period = 2π)

Accuracy & Precision
REMEMBER: 3.s.f for final, 5.s.f for non exact intermediate, 1.d.p for degrees, other precisions used when asked

"Using math concepts is like peeling an onion: you have to deal with each layer at a time, and if it is too big you will start crying."

me (A humorous professor)

kah